# Breedbase User Manual

Breedbase team

2026-07-16

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# Introduction

Welcome to the Breedbase manual!

This manual is intended for database users.
If you are a developer looking for software implementation details, please visit the developer wiki instead: https://github.com/solgenomics/sgn/wiki

# Chapter 1

# Basic Website Usage

Breedbase is usually hosted on the cloud and is entirely web-based, with only a browser required to access and use it. The recommended browser is Firefox. To use Breedbase, you or your project will usually need to have your own Breedbase instance. Information how you can obtain your own instance can be obtained from the Breedbase project (https://breedbase.org/).

Once an instance is set up, the site needs to be configured and some metadata uploaded before you can design, run and analyze trials:

- define user accounts and their access privileges.
- add a trait ontology with the traits that you require.
- add the locations that you use in your breeding program
- add the foundational germplasm and the respective pedigree data
- "historical" trial data can be uploaded as needed. Usually more recent trials are prioritized over older trials

In this chapter, we will cover how you can manage the user accounts and some basic website features such as lists and how to navigate the menus. The other topics are covered in subsequent chapters. Note that Breedbase instances are highly customizable, and that not every instance of Breedbase will have the same options in the same location. Refer to the site specific documentation, if any, for site specific information.

## 1.1 Creating a User Account

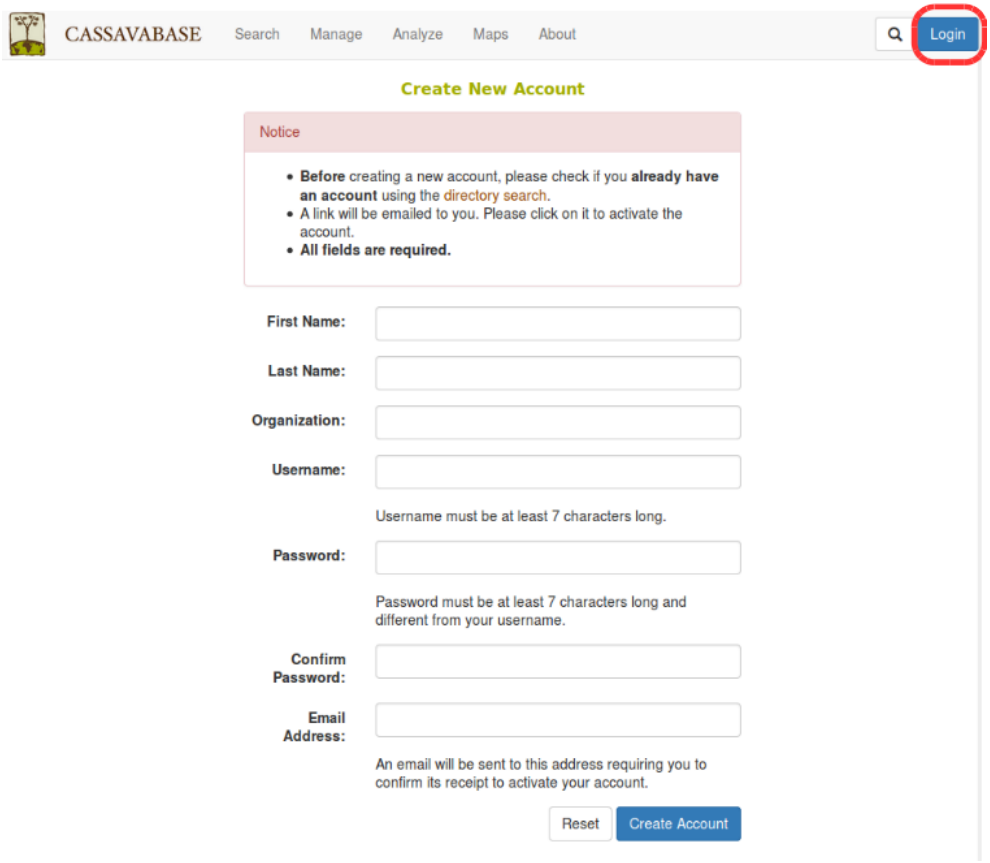
### 1.1.1 Verifying first that you do not already have an account

Before creating an account, please verify first that you don't already have an account. You can use "Search" menu to check if you already registered as a user.

In the "Search" menu, selecting the "People" tab and search your name. If nothing is found, proceed with the instructions below. Otherwise, continue by clicking the "Login" button. If you have forgotten your password, you can retrieve it by clicking the "Forgot your password?" link on the login page.

### 1.1.2 Creating a user account

On the right of the toolbar, click on "Login", which will take you to the login dialog. On the login dialog, click on the link "sign up for an account." It will take you to the page below:




Filling in all of the information, then click "Create Account."

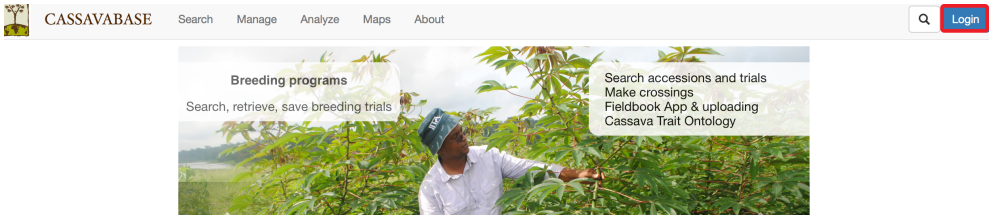
After you submit the information, an email will be sent to the provided email address. Check your email and click on the link to activate your account.

## 1.2 Managing your Account

### 1.2.1 Login

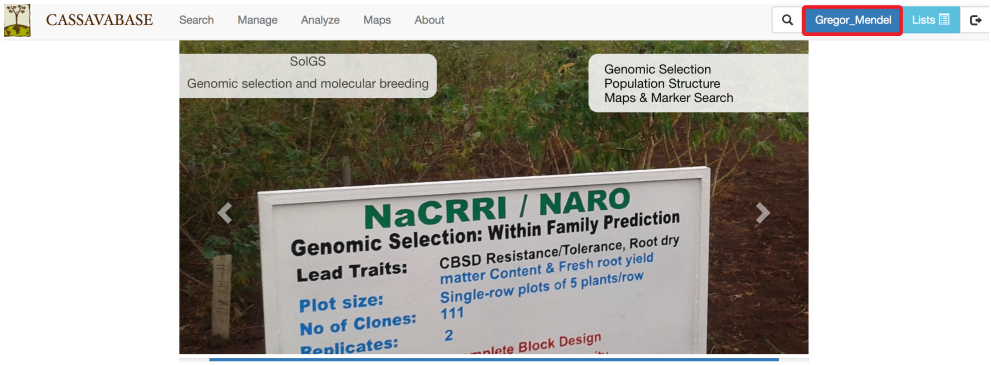
To login, click the "Login" link in the toolbar on any page and enter your username and password.

If you have forgotten your password, you can retrieve it by clicking the "Forgot your password?" link on the login page.



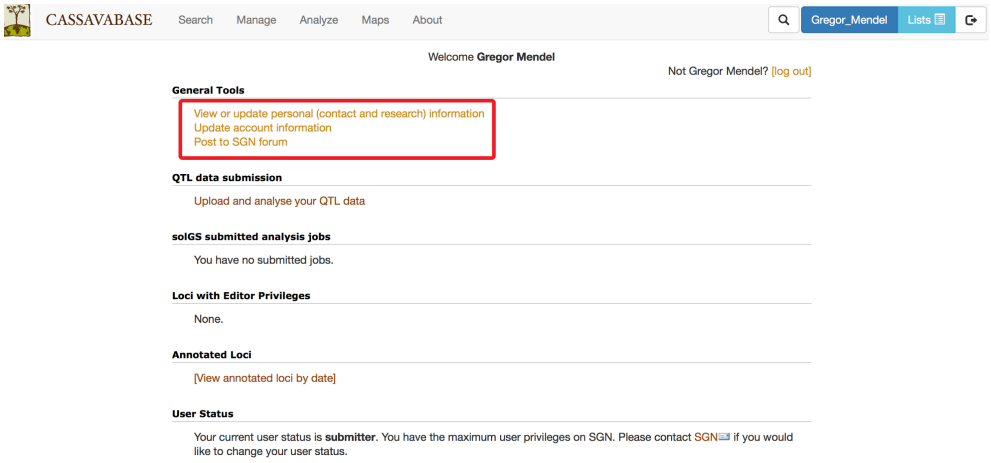

### 1.2.2 Editing Account Settings

Account settings can be edited by clicking on the "my profile" link displayed as your user name, on the right of the toolbar. You must be logged in order to access and change account settings.




You can add personal information to your account using the "View or update personal information" link.

To change your password, username, or your contact email, click on "Update account information" link. You must provide your old password before you can make any changes.




### 1.2.3 Changing Your Account Status: From "User" to "Submitter"

After you create an account, your account has a "user" status. This account has limited privileges.

Accounts with "user" status are able to:

- Change personal information
- Post comments on pages
- Post to the forum

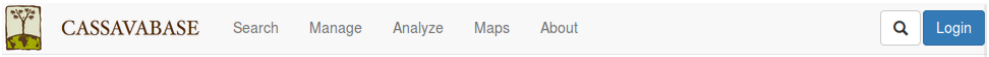
To upgrade your account status to "submitter," contact the database curators using the "contact" link provided at the footer of each page. Submitter accounts can add data, such as new plots, accessions, phenotypic data and images.

### 1.2.4 Submitting Feedback on an SGN Database

We appreciate your feedback! Feel free to submit any questions or suggestions by using the "Feedback" link provided at the footer of each page.

## 1.3 Menu Layout

SGN Database websites have a toolbar on the top of each page with a number of menus for convenient access of major functions. The menus, as pictured below, are "search," "manage," "analyze," and "maps." The toolbar also provides a quick search, a "log in" button, and a "new user" button. The menus can be customized for each instance, so they may not appear exactly as shown on your instance; for example, most instances do not have a map menu.




### 1.3.1 Menu Options

#### Search

In the Search menu, the options are:

| Tab | Description |
|---|---|
| Wizard | Search different accessions and plots by location, year, trial, and trait data. Can also be used to create lists of different types. |
| Accession and plots | Search accessions and plots using a variety of criteria |
| Trials | Search trials by name, description, breeding program, year, location, and trial type. |
| Markers | Search different markers |
| Images | Search images contained in the SGN database |
| People | Search database users |

#### Manage

In the Manage menu, the options are:

| Tab | Description |
|---|---|
| Breeding Programs | View, add and delete breeding programs |

| Tab | Description |
|---|---|
| Locations | View, add and delete locations |
| Accessions | Manage and search different accessions |
| Seedlots | Manage and search different seedlots |
| Crosses | Create new crosses in the database |
| Field Trials | Manage field trials. Create trials using different field layouts. |
| Genotyping Plates | Manage genotyping plates. Create 96 or 384 well plates. |
| Phenotyping | Upload phenotyping files from the Tablet Field Book application |
| Field Book App | Manage the field book app data (download files to tablet) |
| Barcodes | Refers to the old barcode system, mainly historical |
| Download | Download information in the database based on lists |

**Analyze**

**Clicking on the "Analyze" link will give a full menu of all analysis functions**
In the Analyze menu, the options are:

| Tab | Description |
|---|---|
| **Breeder Tools** | |
| Breeder Home | Access breeding functionalities. Lists important and helpful links. |
| Barcode Tools | Manage, create, and download barcodes. Also access barcode tools. |
| Genomic Selection | Can search for traits, start building a GS model, and predict values based on genotypes |
| **Sequence Analysis** | |
| BLAST | Sequence homology search |
| **Other** | |
| Ontology Browser | Browse all recorded ontologies |

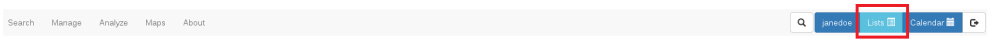
## 1.4 Working with Lists

Lists are collections of identifiers that are stored in the database. Lists can be composed of accessions, plots, traits, locations, and trials. A given list can only contain one type of items, for example, all items have to be of type location in a list of locations. Lists are attached to the individual user's account, and can only be created and seen by the user while logged in, however, lists can be made public for all other users to see. SGN databases make heavy use of lists in a number of tools on the website. For example, trials are created using lists of accessions.

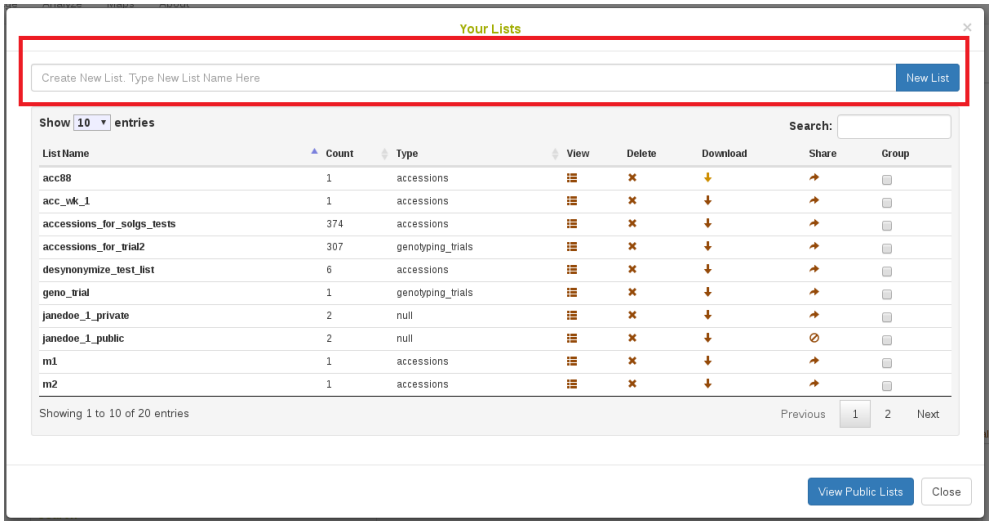
### 1.4.1 Creating lists

Lists can be generated in various ways:

One way to create a list is by clicking on the "Lists" link located on the toolbar.

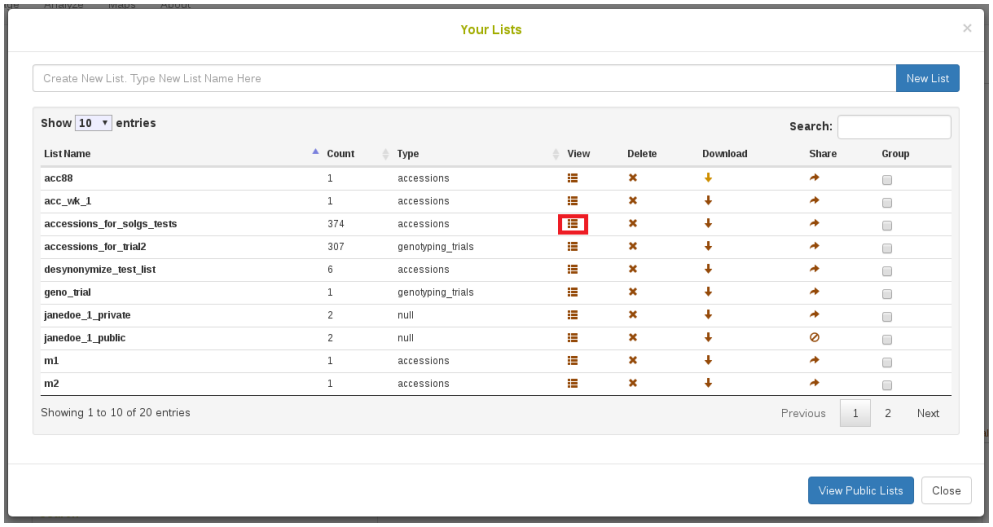



To create a new list, enter the name of your new list and then click on the "New List" button. The name of the list can be anything, but should be unique and should be something to help you easily identify the list.

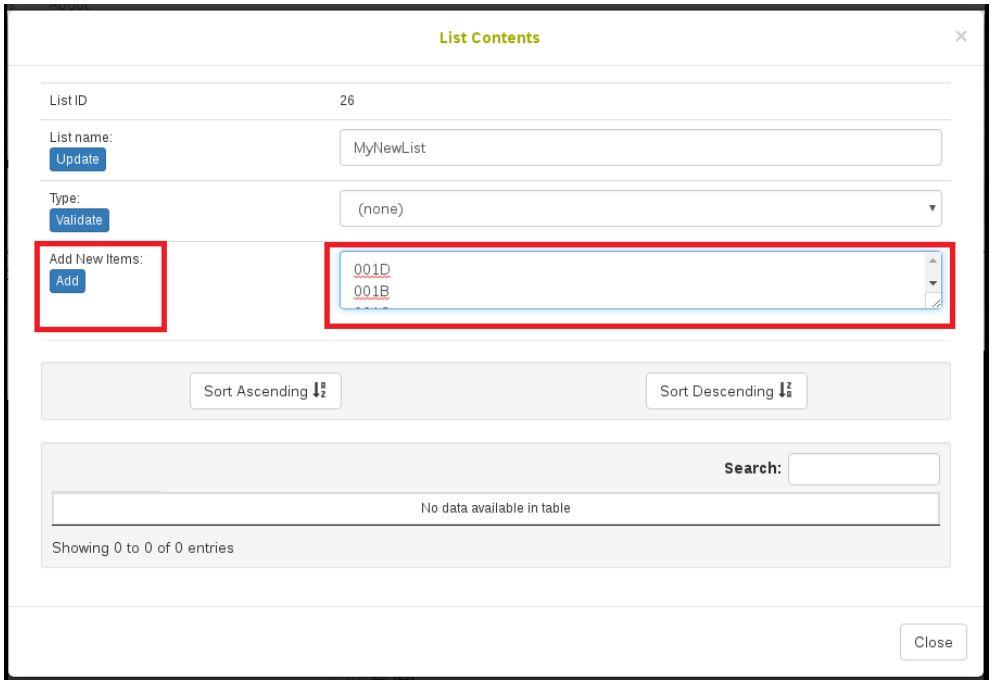



If the list already exists, it will appear on the "Your Lists" dialog. To add items to your list, click on the "View" icon to open the "List Contents"

section.

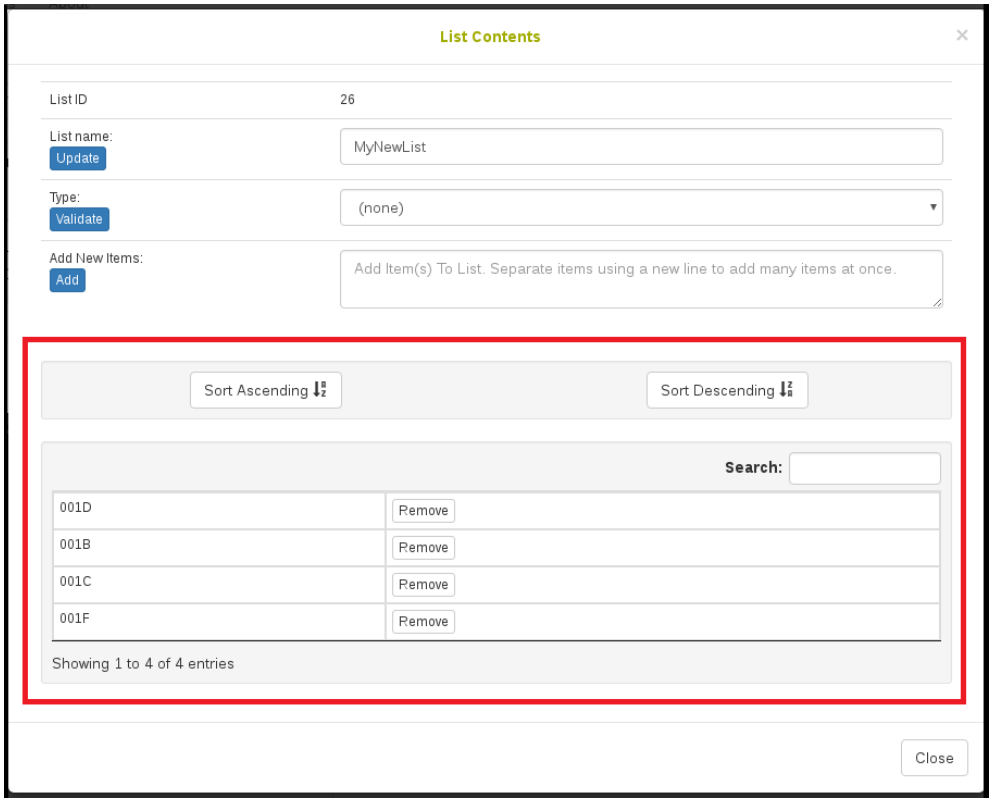



On the "List Contents" page, enter items that you want to add to the list, then click on "Add" button.

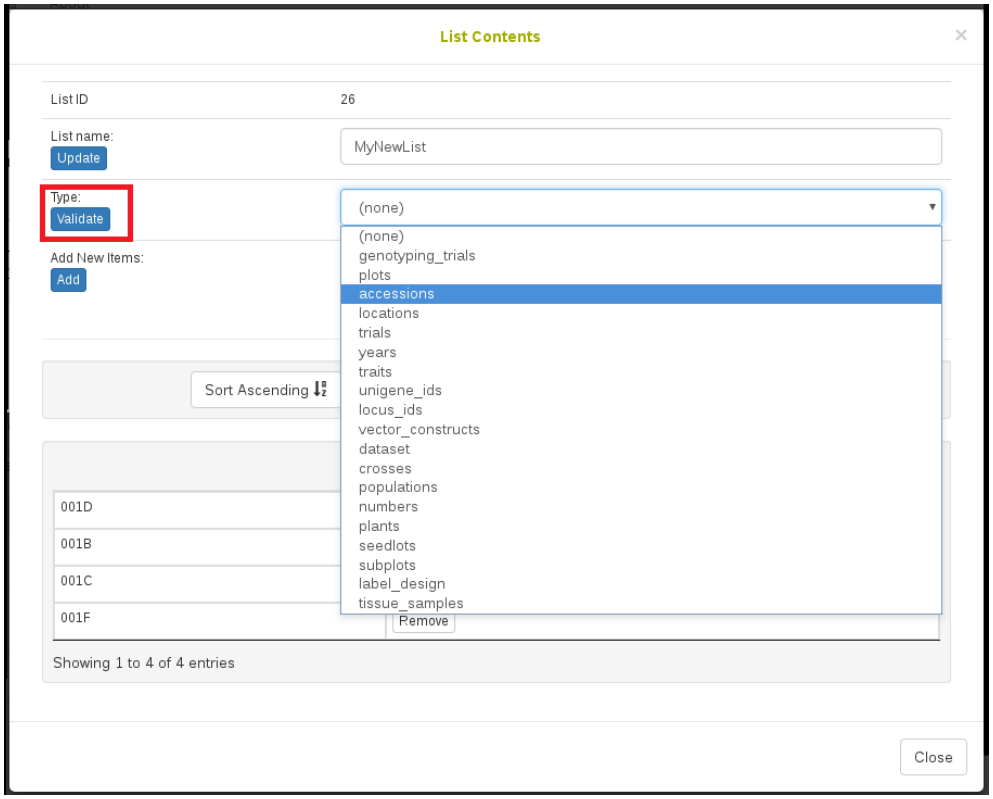



The page will be updated and will display your items in a table at the bottom

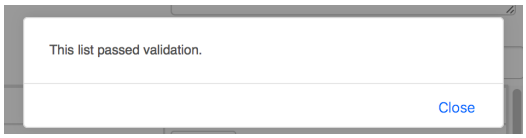
of the dialog It is possible to sort the list if you need.




Select the type of items in your list. To verify that the items that you added to your list are already stored in the database and that you selected a correct type for the items, click on the “Validate” button.




If those items are already in the database, a message will indicate that "This list passed validation"

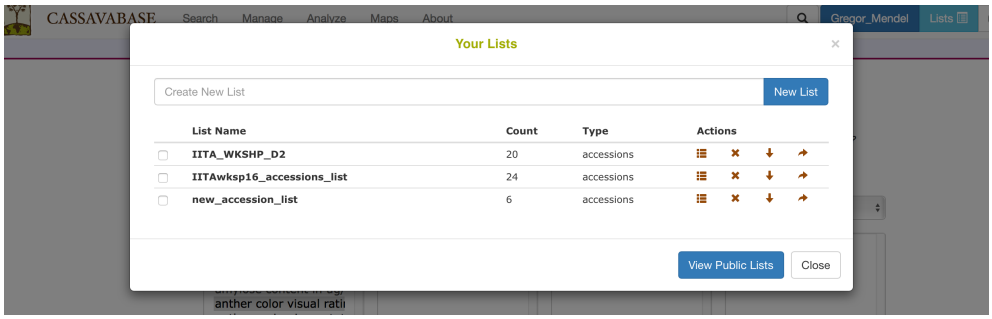



Note that a list cannot contain duplicate elements. If a duplicate item is entered, the list manager will inform the user that the element is already in the list and will not add it again.

Another easy way to create a list is to use the 2.1, which can be accessed from the Search menu.

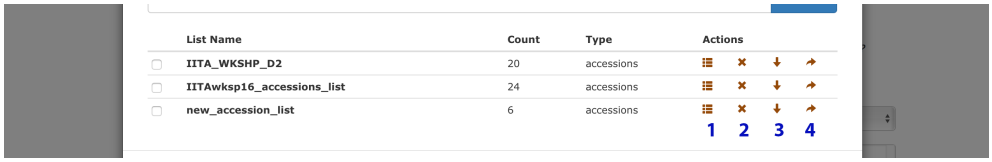
### 1.4.2 Viewing and editing lists

Lists can be viewed and edited using the "Lists" link on the toolbar. Clicking on the link will open a dialog that displays all of your lists, as well as an option to create new lists.




This page shows all lists that have been created, including those created by using the Search Wizard. You can view and edit your lists by using "Actions" buttons.

1. Clicking on the "view" icon will open a new window called "List Contents" that allows you to change the list name, the type of the list, add new items, or delete existing items.
2. Clicking on the "delete" icon will delete your list. **Caution: this action cannot be undone**.
3. Clicking on the "download" icon will download the contents of your list to your computer.
4. Clicking on the "make public" icon will make your list available for other users to view and use your list.




## 1.5 User Permissions

Breedbase accounts are assigned one or more of four different roles to determine the level of access they have within the database. The possible roles

are **User**, **Submitter**, **Sequencer**, and **Curator**. Each role grants specific permissions, and careful management of them helps prevent data from being altered or deleted in error.

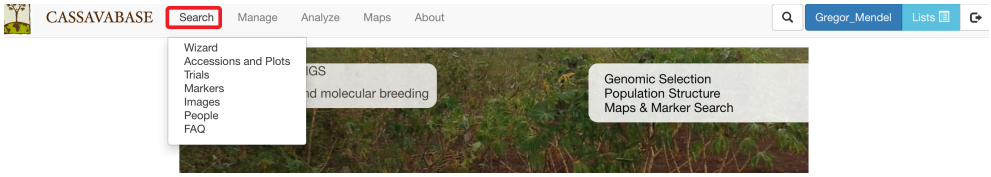
Accounts are also assigned Breeding Program role(s) to grant access to the specfic breeding program(s) they work with.

- The **User** role gives an account permission to view and download data throughout the database.
- The **Submitter** role gives an account permission to design field experiments and to upload and edit data using the tools in the "Manage" section. In order to submit and manage breeding data within a given breeding program, a submitter also must have a matching Breeding Program role.
- The **Sequencer** role gives an account permission to design genotyping experiments and submit plates to a genotyping service.
- The **Curator** role gives an account permission to do all of the above, as well as to delete data within the database. The Curator role also enables the addition or deletion of roles for all database accounts in the 'Manage User Roles' tool.

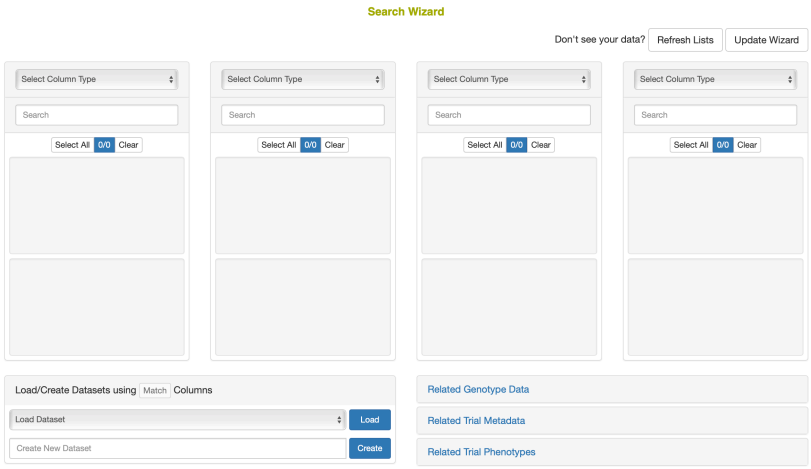
# Chapter 2

# Searching the Database

You can search for information on the database by using the following search options: Wizard, which uses combined criteria specified by users; Accessions and Plots; Trials; Markers; Images; People; FAQ.



## 2.1 The Search Wizard

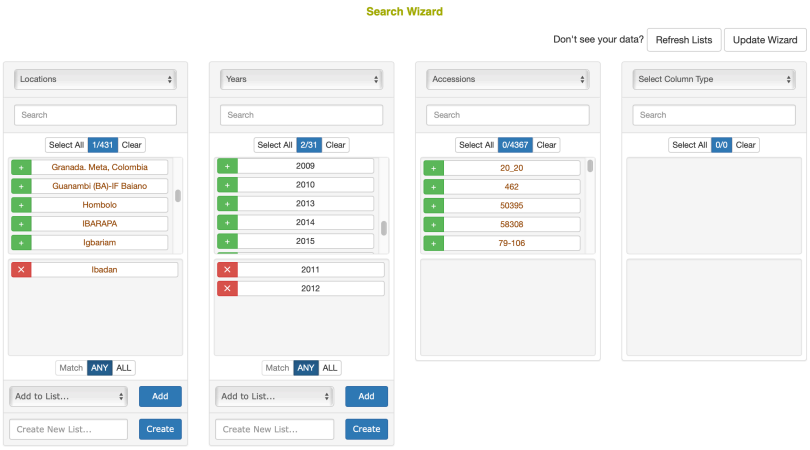



### 2.1.1 How the Search Wizard Works

The search wizard presents a number of select boxes, which are initially empty. You start searching by picking a category of data from the dropdown above the left-most select box.

Once a category has been picked, the database will retrieve all the options within this category and display them within the first select box. You then select one or more options from the first select box, which activates the second dropdown.

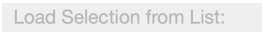
You can then select a category from the second dropdown, and repeat this same search process through all four dropdowns and select boxes.




- In the example above, the "locations" category was chosen in the first dropdown. The first select box then displayed all the possible locations in the database. The option Ibadan was selected.
- This activated the second dropdown. The category "years" was chosen in the second dropdown. The second select box then displayed all the years that are linked in the database to the location Ibadan. From that list, the options 2011 and 2012 were selected.
- This activated the third dropdown. A final category, "accessions", was chosen in the third dropdown. The third select box was then populated with the 3847 accessions in the database that are linked with the location Ibadan in the years 2011 or 2012.

In addition to the basic search operations demonstrated above, users can take advantage of two more features:

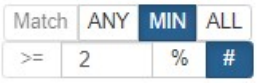
**Load Selection from List**




- Instead of picking a category in the first dropdown, users can instead populate the first selectbox from a list by scrolling down in the first dropdown to the "Load Selection from List" subheading and selecting a list. This is useful for starting queries with a list of plots, as this category is not among the options in the first dropdown.

**ANY/MIN/ALL** Toggle

- By default, the search wizard combines options within a category using an OR query. In the example above, in the third panel the wizard retrieved accessions associated with the location 'Ibadan' in **ANY** of the years "2011 **OR** 2012"
- If the user clicked the toggle below the second select box to change it to **ALL** before choosing accessions in the third dropdown, the wizard would instead retrieve accessions associated with the location 'Ibadan' in the years "2011 **AND** 2012". This will be a smaller set of accessions, because any accessions used only in 2011, or only in 2012 will be excluded.
- A more advanced search could use the **MIN** toggle option. This allows the user to make a query in between an ANY or ALL query, where a minimum number of matches from the selected column will be used as a filter for the next column. The minimum can be provided as either a percentage (%) or an actual count of items (#). In the example above, if the years 2011, 2012, and 2013 were selected in the second column, the user could enter '2' in as the minimum and select '#' as the minimum match type. This would select accessions in the third column that were used in 2 or more of the selected years.



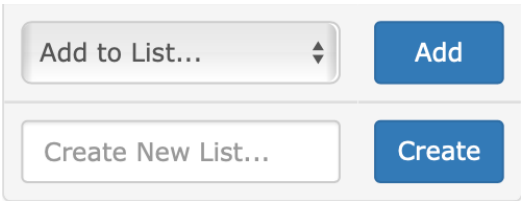

### 2.1.2 How to use retrieved data

#### Getting more Info

Any option in the wizard select boxes (except for years) can be clicked to open a page with more details. The new page is opened in a new tab.

#### Saving to a list

You can store the highlighted items in any selected box to lists. This is done using the inputs and buttons directly below the select box. **Don't forget, you must be logged in to work with lists!**




- To **add items to an existing list**, first pick an existing list using the "Add to List..." dropdown on the left. Then click the "Add" button. A popup window will confirm the action, and display the number of items added to your existing list.

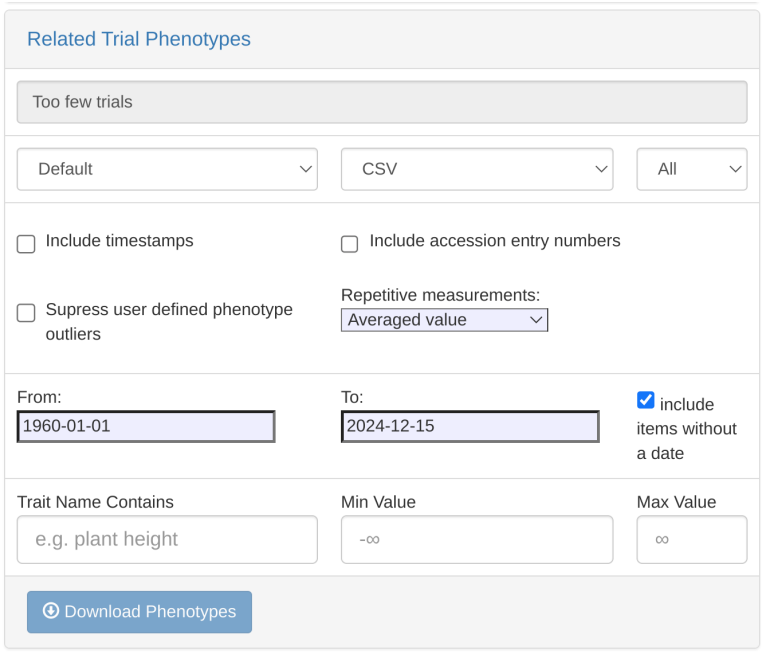
- To **store items to a new list**, first type a new list name in the "Create New List..." text input on the left. Then click on the "Create" button. A popup window will confirm the action, and display the number of items added to your new list.

### Downloading Data

You can download trial metadata, phenotypes and genotypes associated with the highlighted items in the wizard select boxes. This is done using the buttons in the download section at the bottom of the page. **Don't forget, you must be logged in to download data!**

**Metadata** Trial metadata can be downloaded by selecting a subset of trials from the database or based on your search categories. To download, click on "Related Trial Metadata", a dialog will appear. Select download format and click the "Metadata" button to complete your download.

**Phenotypes** The phenotypes download is quite flexible, and can download a subset of all the trial data in the database based on whichever categories and options you currently have selected. Simply click on the "Related Trial Phenotypes" link, review the options, changing or adding any additional parameters you like, then click 'Download Phenotypes'.



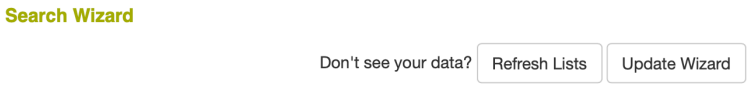

**Genotypes** The genotype download is more stringent. It requires a minimum of one accession and one genotyping protocol to be selected in the wizard select boxes. The text box in the download section of the page will help track what has been selected. Once clicked, the "Download Genotypes" button will download a genotype file for the selected accessions.

### Saving the wizard selections

As discussed above, the selections of the individual select boxes in the wizard can be saved separately to a list. The lists can be used as inputs in other tools on the site. However, sometimes creating a selection is quite time consuming and restoring the selections from four different lists would be cumbersome too. Therefore, the selections can be saved together in a dataset, and named for later retrieval. This is done in the section "Load/Create Datasets" that is below the first two wizard select boxes. To select an existing dataset, one uses the "Load Dataset" dropdown. A particular dataset can be chosen, and the "Load" button can be clicked to retrieve and display the dataset in the wizard. To create a new dataset using items that are selected in the wizard, one can enter the name of the new dataset in the "Create New Dataset" text box. Once the dataset has been given a name, clicking the "Create" button will save the new dataset.

## 2.1.3 Updating the Wizard

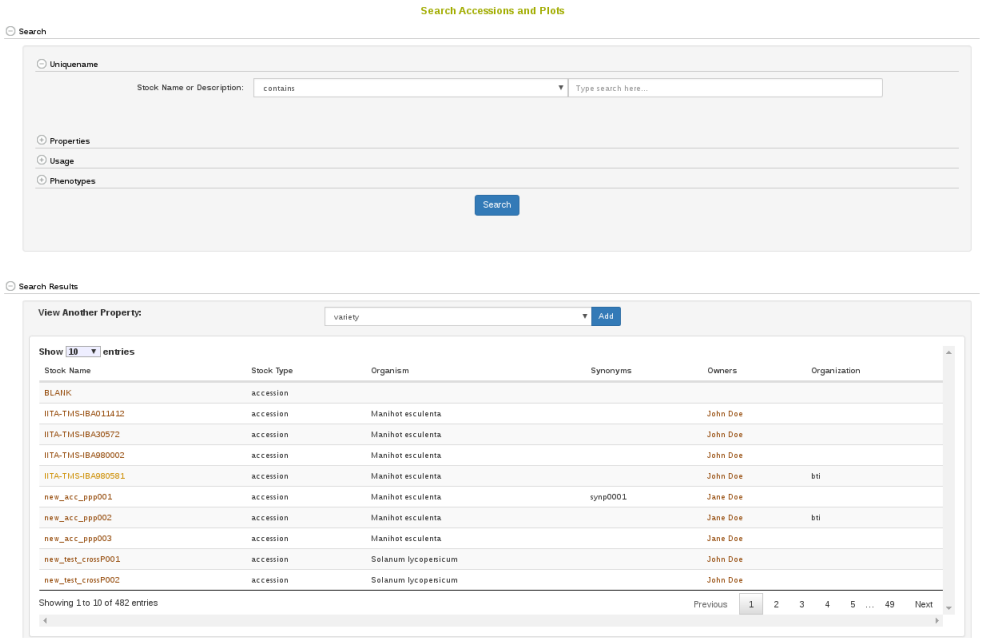
The search wizard uses a copy of the database, or a cache, to return results quickly. If data appears to be missing, it usually means that the cache needs to be updated. Users with submitter privileges or above can do this using the 'Update Wizard' button. One can also use the 'Refresh Lists' button to update the available lists.



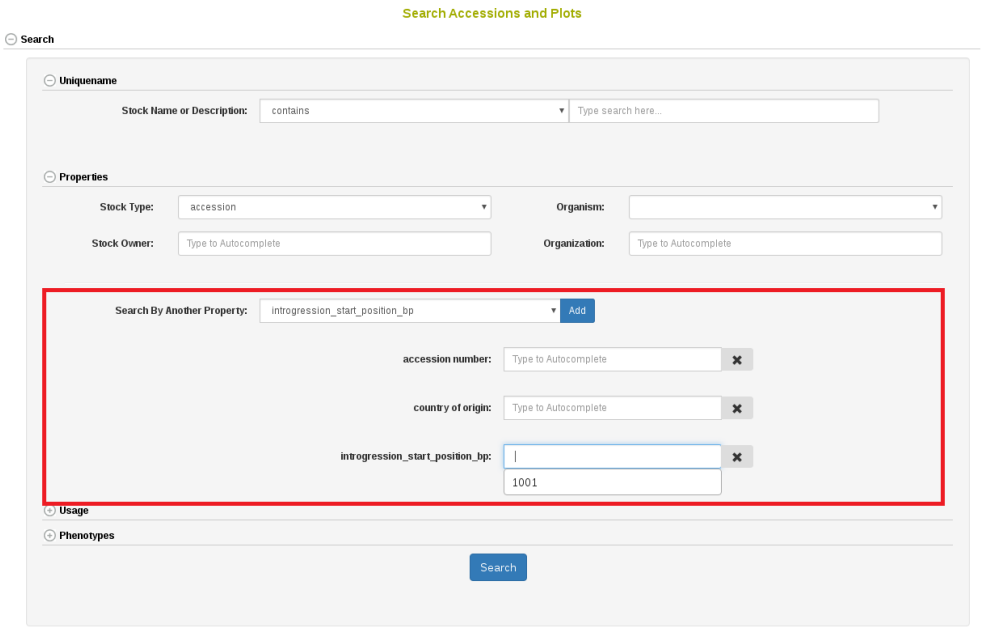

This will take just a few seconds in small databases, but may take a few hours to complete in larger databases.

## 2.2 Accessions and Plot Search

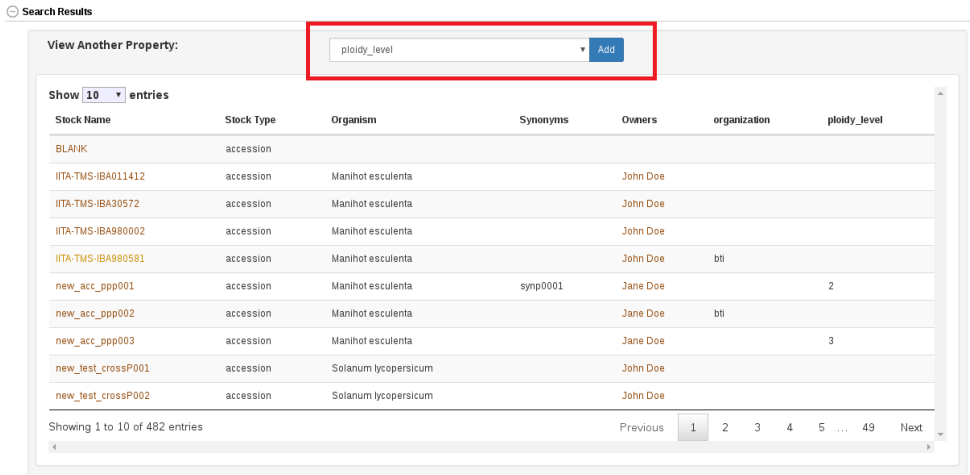
Accessions and their related materials (cross, plant, plot, population, tissue_sample, training population) can be searched by using "Search Accessions and Plots" page. On this page, "accession" is the default stock type; however, you can change stock type by selecting an option from the dropdown list. From this page you can construct detailed queries for stock types. For example, by using the "Usage" section, the "Properties" section, and the "Phenotypes" section you could search for accessions which were diploids used in a specific year and location and were also phenotyped for height. You can also search for accessions based on genetic properties, such as the location of an introgression on a specific chromosome.



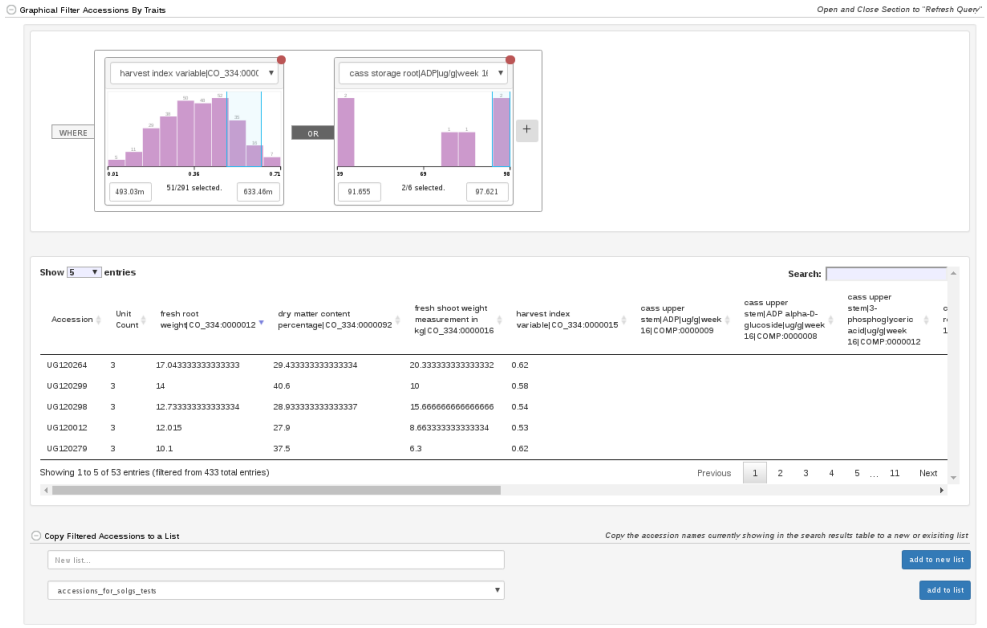

It is possible to query over any of the available properties, such as "ploidy_level", "country of origin", "introgression_chromosome", etc.




In the search result table it is possible to select any of the available properties to view.



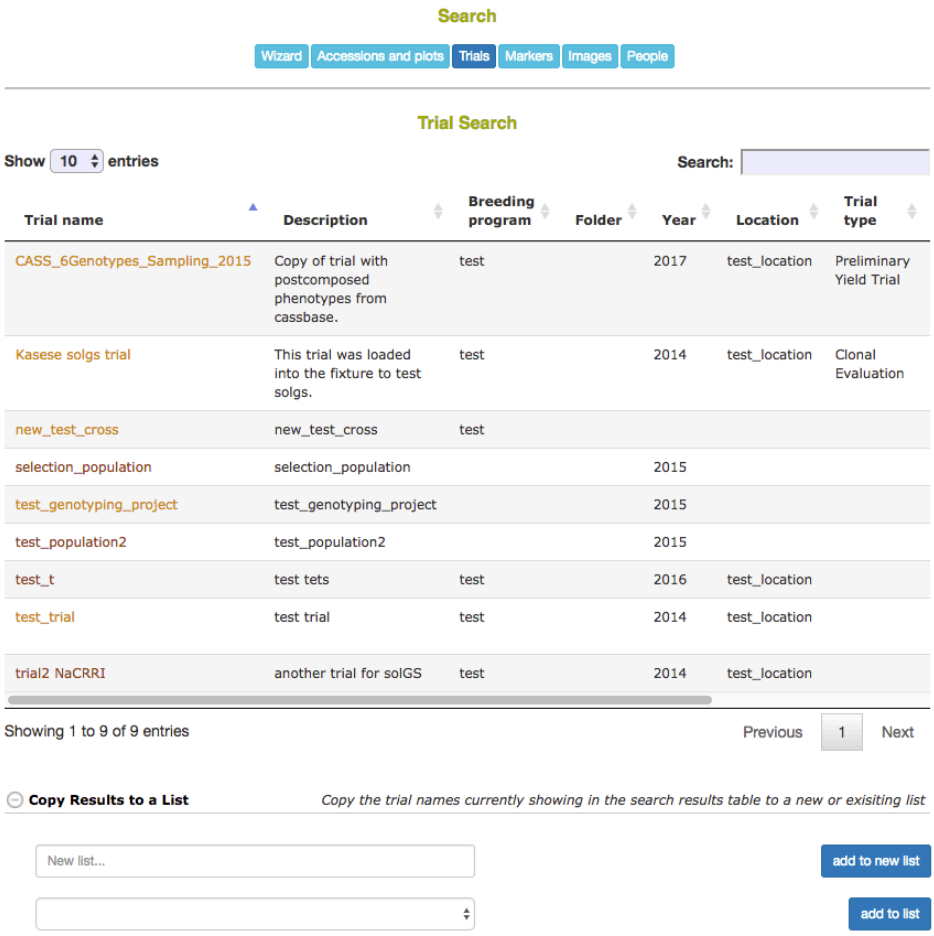

At the bottom of the accession search there is a phenotype graphical filtering tool. Here you can filter down accessions based on combinations of trait performance. The filtered down accessions are then able to be saved to a list.




For information on adding Accessions please see the Managing Accessions help. For information on how field trial plots, plants, tissue samples, and subplots are added to the database, please see the Managing Field Trials help.

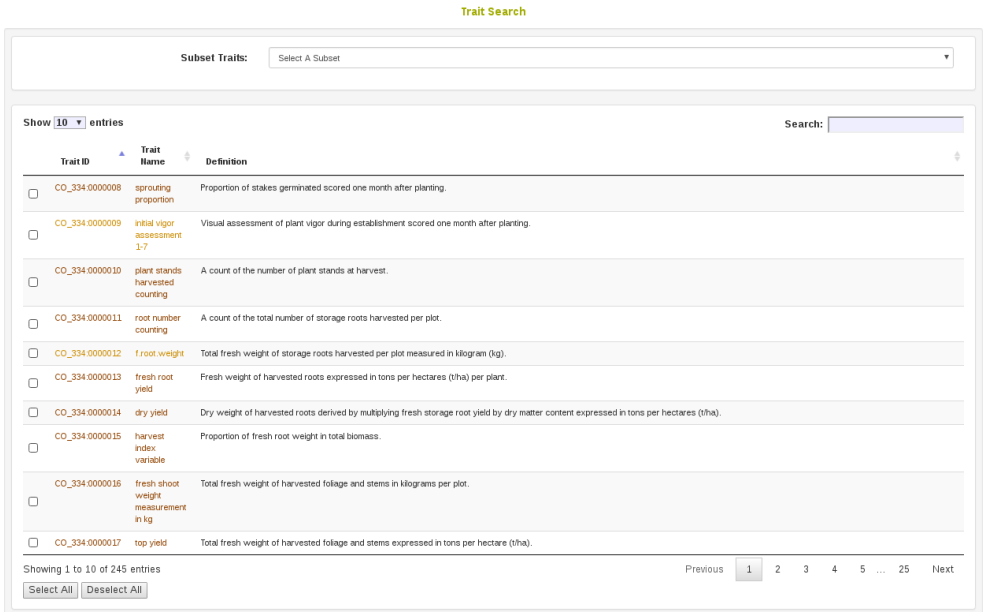
## 2.3 Trials Search

Trials on the database can be searched based on trial name, description, breeding program, year, location, trial type, design, planting date, and harvest date.

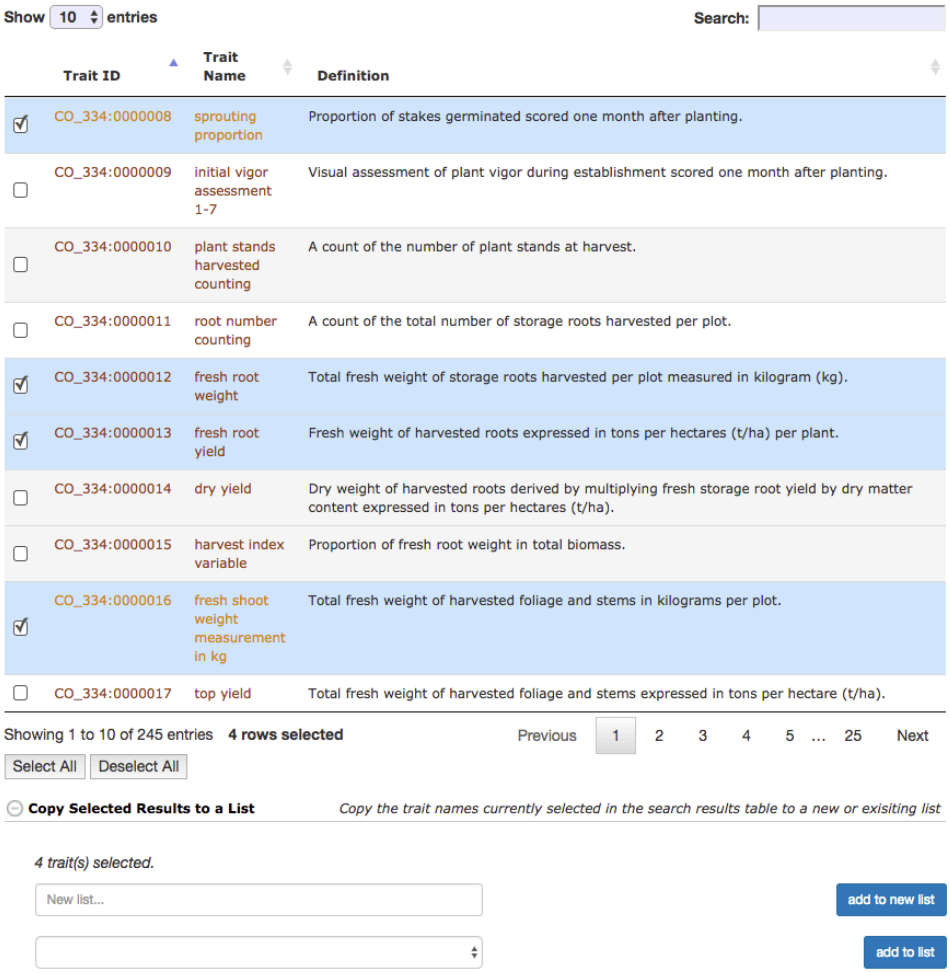



## 2.4 Trait Search

On the Trait Search page (menu item `Search > Traits`), traits in the database can be searched by ID, name, or descripiton. Optionally, a starting list of traits can be selected to filter down results.



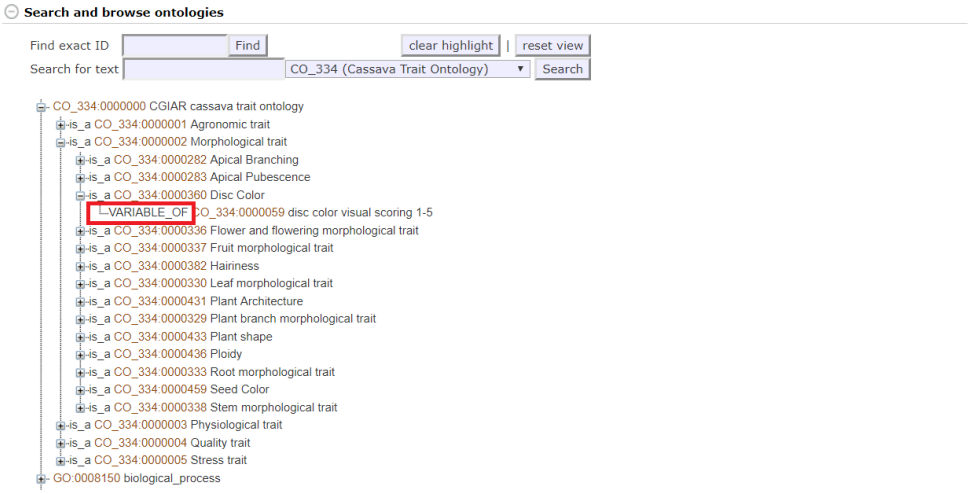

Selecting traits in the results of the search allows one to add the selected results to a trait list, or create a new trait list from the select results.




## 2.5 Ontology Browser

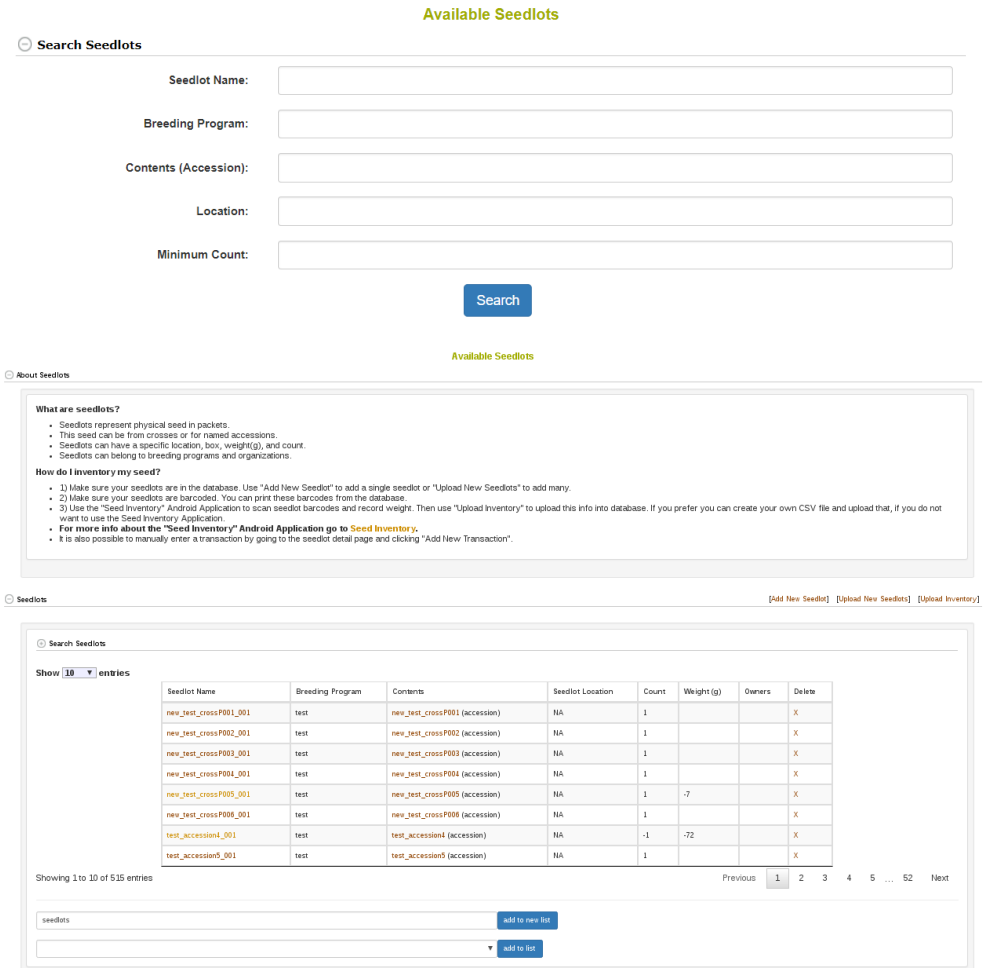
A more advanced tool for searching for Traits is the ontology browser, available by clicking on Analyze and Ontology Browser. From here you can search ontologies and see the various classifications of terms in a tree display.

The terms which appear in the Trait Search in 2.4 are only variable terms. The ontology browser shows these variables as different from their grouping terms by indicating VARIABLE_OF like in the following screenshot.



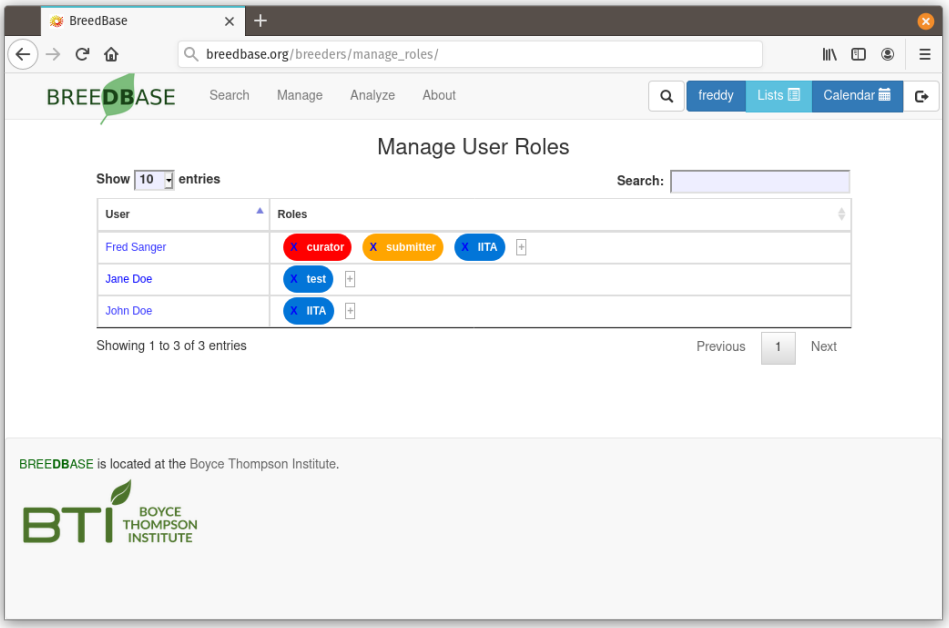

## 2.6 Search Seedlots

Seedlots are different from Accessions in that they represent the physical seed being evaluated in an experiment. Seedlots have things like physical storage locations and seed quantities, which accessions do not. To search for available seedlots you go to Manage and then click Seed Lots. By clicking Search Seedlots, you can specify query information. The results from your search will be in the table below the search form.


Available Seedlots
Search Seedlots
Seedlot Name:
Breeding Program:
Contents (Accession):
Location:
Minimum Count:
Search
Available Seedlots
About Seedlots
What are seedlots?
Seedlots represent physical seed in packets.
This seed can be from crosses or for named accessions.
Seedlots can have a specific location, box, weight(g), and count.
Seedlots can belong to breeding programs and organizations.
How do I inventory my seed?
1) Make sure your seedlots are in the database. Use "Add New Seedlot" to add a single seedlot or "Upload New Seedlots" to add many.
2) Make sure your seedlots are barcoded. You can print these barcodes from the database.
3) Use the "Seed Inventory" Android Application to scan seedlot barcodes and record weight. Then use "Upload Inventory" to upload this info into database. If you prefer you can create your own CSV file and upload that, if you do not want to use the Seed Inventory Application.
For more info about the "Seed Inventory" Android Application go to Seed Inventory.
It is also possible to manually enter a transaction by going to the seedlot detail page and clicking "Add New Transaction".
Seedlots
[Add New Seedlot] [Upload New Seedlots] [Upload Inventory]
Search Seedlots
Show 10 entries
Seedlot Name
Breeding Program
Contents
Seedlot Location
Count
Weight (g)
Owners
Delete
new_test_crossP001_001 test new_test_crossP001 (accession) NA 1 X
new_test_crossP002_001 test new_test_crossP002 (accession) NA 1 X
new_test_crossP003_001 test new_test_crossP003 (accession) NA 1 X
new_test_crossP004_001 test new_test_crossP004 (accession) NA 1 X
new_test_crossP005_001 test new_test_crossP005 (accession) NA 1 -7 X
new_test_crossP006_001 test new_test_crossP006 (accession) NA 1 X
test_accession4_001 test test_accession4 (accession) NA -1 -72 X
test_accession5_001 test test_accession5 (accession) NA 1 X
Showing 1 to 10 of 515 entries
Previous 1 2 3 4 5 ... 52 Next
seedlots
add to new list
add to list

# Chapter 3

# Managing User Roles




## 3.1 What are User Roles?

Every user account in Breedbase has one or more associated "roles" that determine the authorizations (what the user is allowed to do) in the database.

There are three fundamental roles, “curator”, “submitter”, and “user”, which determine basic read/write levels. The “curator” status can read and write everything in the database. The “submitter” status can add information and edit or delete previously submitted information. The “user” type can only read data. Additional roles represent the breeding programs, and are sometimes used to fine-tune write and edit capabilities, as it necessary for multiple users in a breeding program to edit each other’s data.

## 3.2 The Manage User Roles page

In the “Manage” menu, select the item “User Roles”. This will show the current users in the database with their associated roles. If you are logged in as a curator, the table will show system roles as well as breeding program roles; if you are logged in as a submitter or user, it will show breeding program membership.

If logged in as a “curator”, the roles can be added or deleted.

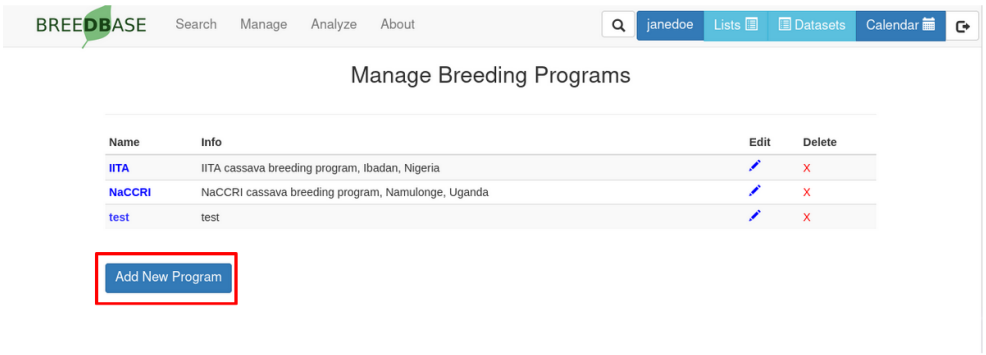
- To delete a role, click on the X in the role name. A confirm dialog will be displayed to prevent accidental deletion.
- To add a role, click on the plus sign next to the roles. A dialog will pop up with a list of roles. Select the desired role and click “Submit”.
- The new role should be displayed next to the user immediately.
- Role deletions and additions will be effective immediately.

It is recommended that few users be given the “curator” privileges to avoid confusion over data ownership and accidental data overwriting and deletion.

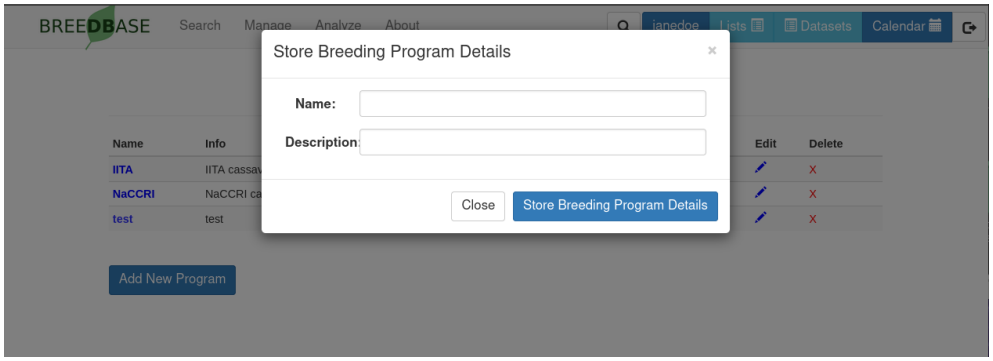
# Chapter 4

# Managing Breeding Programs

New breeding programs can be added by using "Add New Program" button on the "Manage Breeding Programs" page.



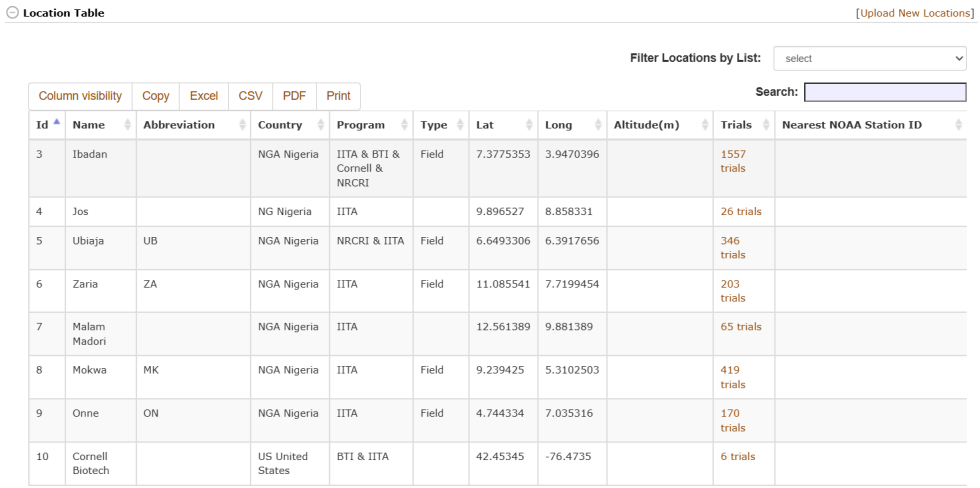

Clicking on the "Add New Program" button will generate a blank form for you to fill out the name and description of the breeding program that you want to add. After completing the form, click on "Add Breeding Program" button to finish the process.


BREEDBASE
Search
Manage
Analyze
About
janedoe
Lists
Datasets
Calendar
Store Breeding Program Details
Name:
Description:
Close
Store Breeding Program Details
Name
Info
Edit
Delete
IITA
NaCCRI
test
Add New Program

# Chapter 5

# Managing Locations

Field locations can be managed using the "**Manage Locations**" page. On this page, locations in the database are organized based on their breeding programs. Each location has a link to trials conducted in that location. To add a new location, click on the "Upload New Locations" button that links to the "Upload Locations" form.



Location Table [Upload New Locations]

Filter Locations by List: select

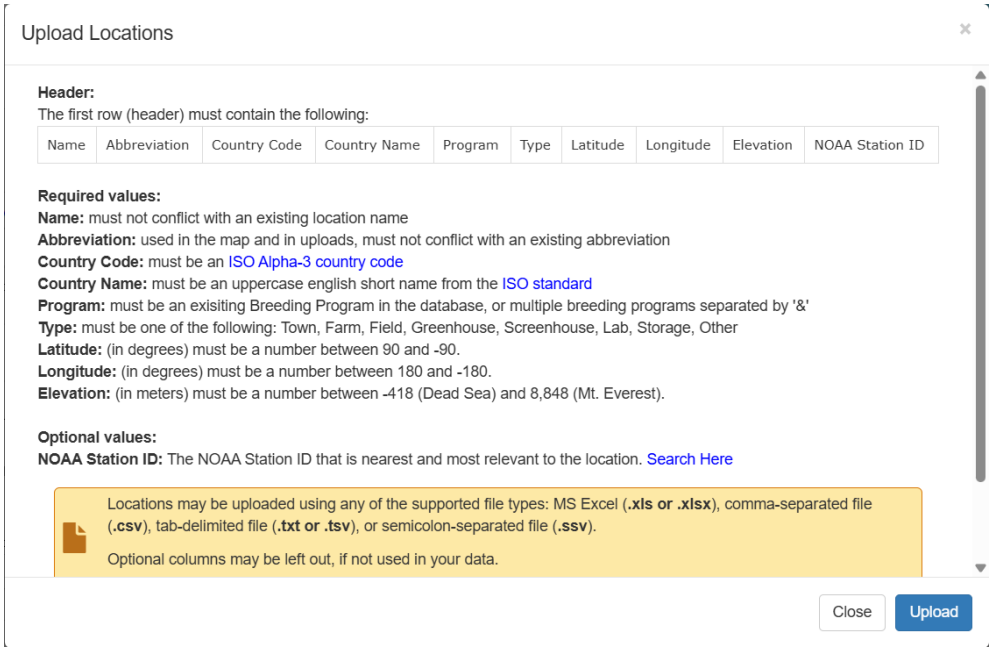
Column visibility | Copy | Excel | CSV | PDF | Print

Search:

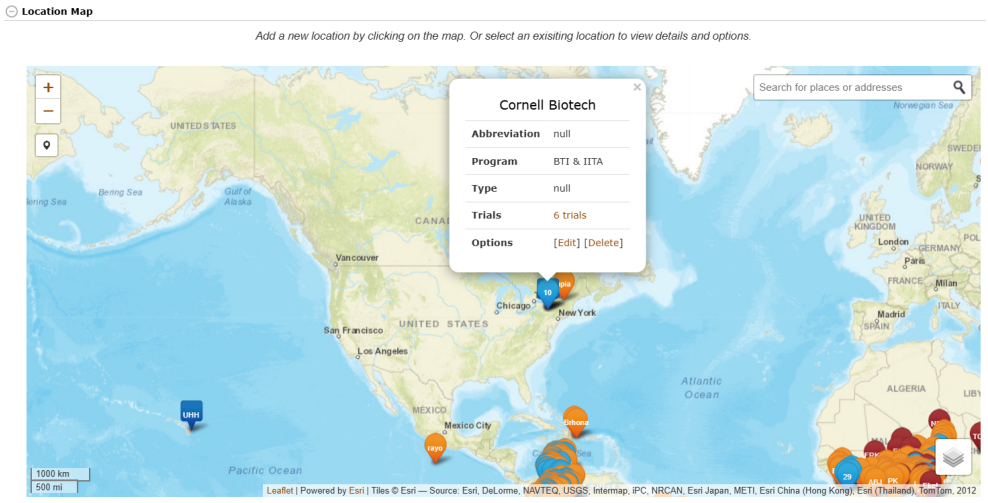
| Id | Name | Abbreviation | Country | Program | Type | Lat | Long | Altitude(m) | Trials | Nearest NOAA Station ID |
|---|---|---|---|---|---|---|---|---|---|---|
| 3 | Ibadan | | NGA Nigeria | IITA & BTI & Cornell & NRCRI | Field | 7.3775353 | 3.9470396 | | 1557 trials | |
| 4 | Jos | | NG Nigeria | IITA | | 9.896527 | 8.858331 | | 26 trials | |
| 5 | Ubiaja | UB | NGA Nigeria | NRCRI & IITA | Field | 6.6493306 | 6.3917656 | | 346 trials | |
| 6 | Zaria | ZA | NGA Nigeria | IITA | Field | 11.085541 | 7.7199454 | | 203 trials | |
| 7 | Malam Madori | | NGA Nigeria | IITA | | 12.561389 | 9.881389 | | 65 trials | |
| 8 | Mokwa | MK | NGA Nigeria | IITA | Field | 9.239425 | 5.3102503 | | 419 trials | |
| 9 | Onne | ON | NGA Nigeria | IITA | Field | 4.744334 | 7.035316 | | 170 trials | |
| 10 | Cornell Biotech | | US United States | BTI & IITA | | 42.45345 | -76.4735 | | 6 trials | |

The "Upload Locations" describes how to build a spreadsheet with location data for upload. Name, abbreviation, country code, country name, program, type, latitute, longitude, and elevation are all required. The NOAA station ID is optional. Link a spreadhsheet to the form and click "Upload" to add

those locations to the database.



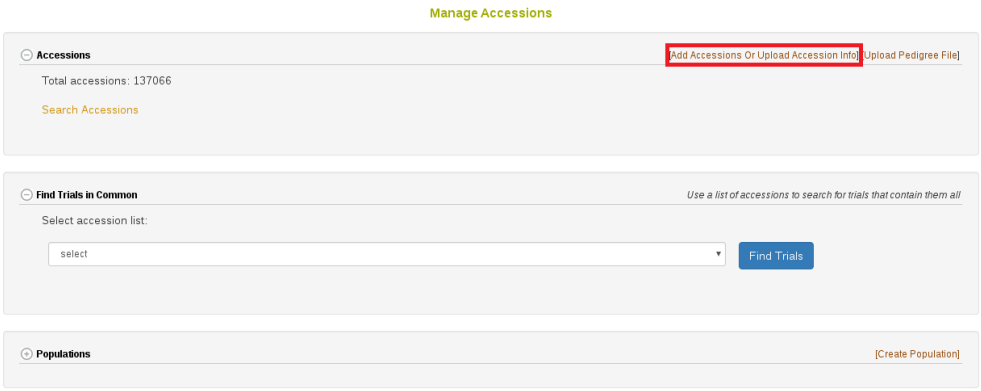

Alternatively, locations can be viewed and added via the map. Hover over an icon on the map to see the location details and trials linked to that location. Click on the map to open the new location dialog. Fill in the same information that would be used in the spreadsheet upload to add a new location.



# Chapter 6

# Managing Accessions

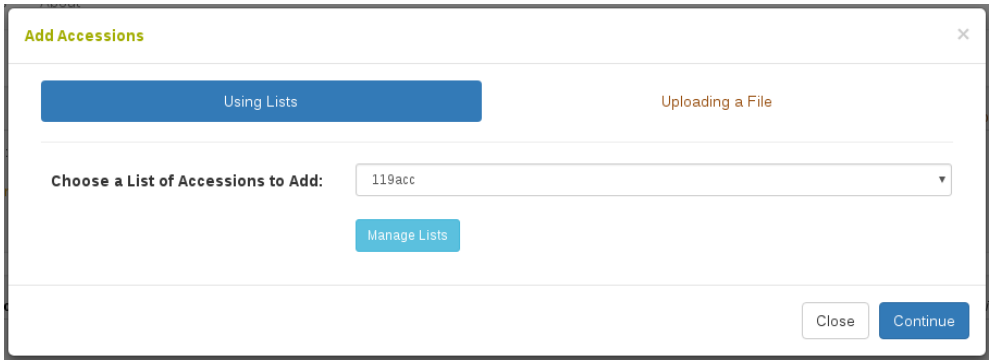
The "Manage Accession" page provides links for adding new accessions. New accessions can be added to the database by either using a List or by uploading an Excel file (either XLS or XLSX format). Both options are explained in more detail below. To begin, click on the "Add Accessions or Upload Accession Info" link.




This will open a dialog allowing you to select either "Using Lists" or "Uploading a File".

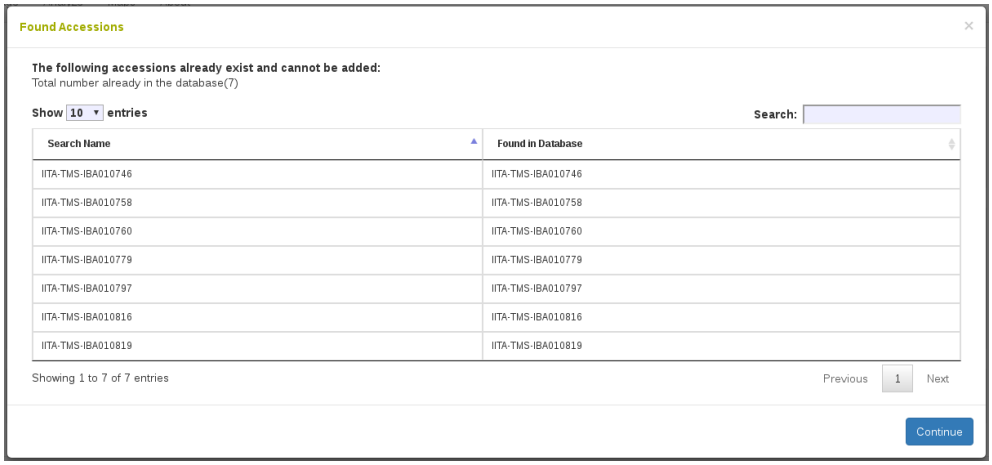
## 6.1 Add Accessions Using A List

First we will show how to add accessions "Using Lists".

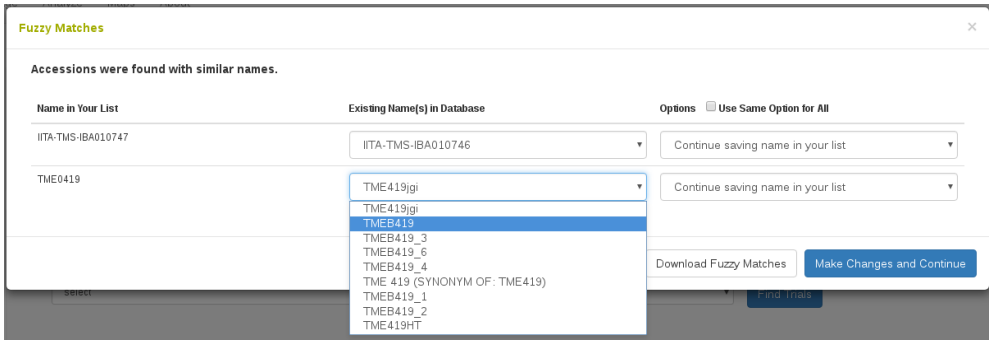



Here you select an accession list which you have previously made (see List Manager chapter). If you need to create or edit your list you can do so now by clicking "Manage Lists". After selecting your list, click "Continue".

The contents of the list will be checked against the database, and elements that are already present will be flagged. A dialog will appear that will show the accessions which already exist in the database.



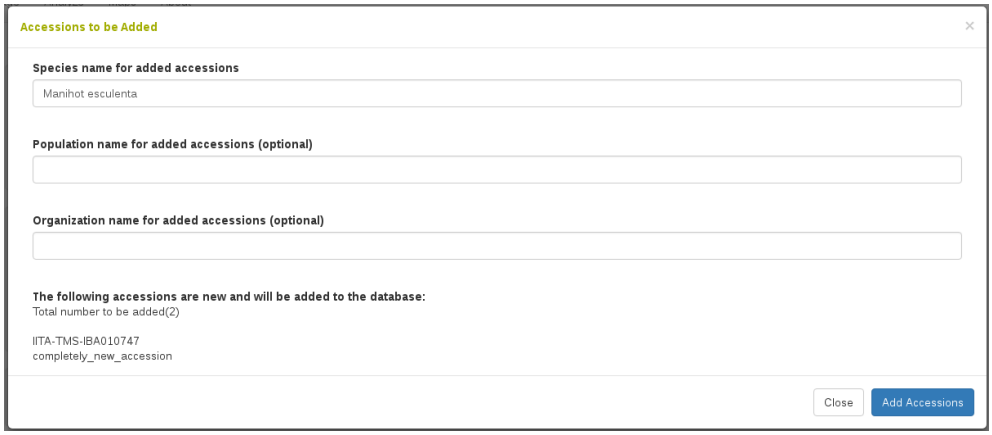

After clicking on the "Continue" button, the next dialog will appear with accessions that have very similar names as the accession that you are adding. In the example below, there are two accession with very similar names to accessions already in the database. 'TME0419' is very similar to 'TME419', and probably represent the same line, so it would be a mistake to add this the database again. Duplicate lines in the database should be avoided, as they cause problems when evaluating lines; data is divided up among several duplicates, making it harder to get the full picture about an accession.



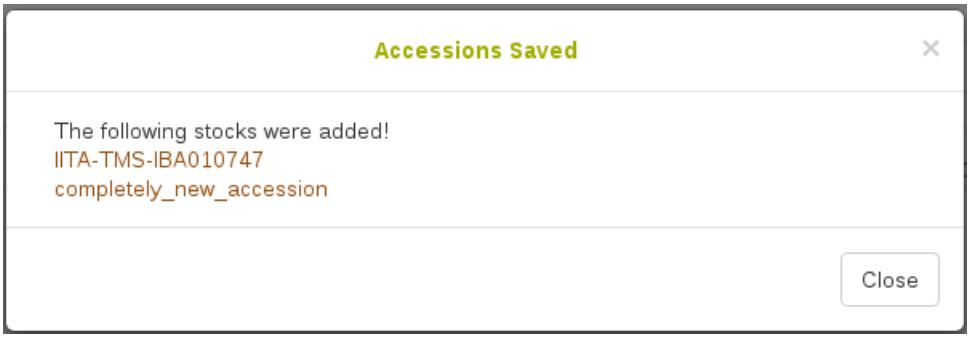

To avoid situations in adding a mistaken duplicate accession, the database gives you options for moving forward with these very similar looking accession names. You can either "continue saving the name in your list", "replace name in your list with selected existing name", "remove name in your list and ignore", or "add name in your list as a synonym to selected existing name".

Clicking "Download Fuzzy Matches" will return a tabular result of the "fuzzy" accession name results shown. Click "Make changes and continue" to move on.

The final dialog shows the accessions that will be added. Here you need to assign the species of these accessions. You can optionally group the accessions into a population and/or add an organization for the accessions.



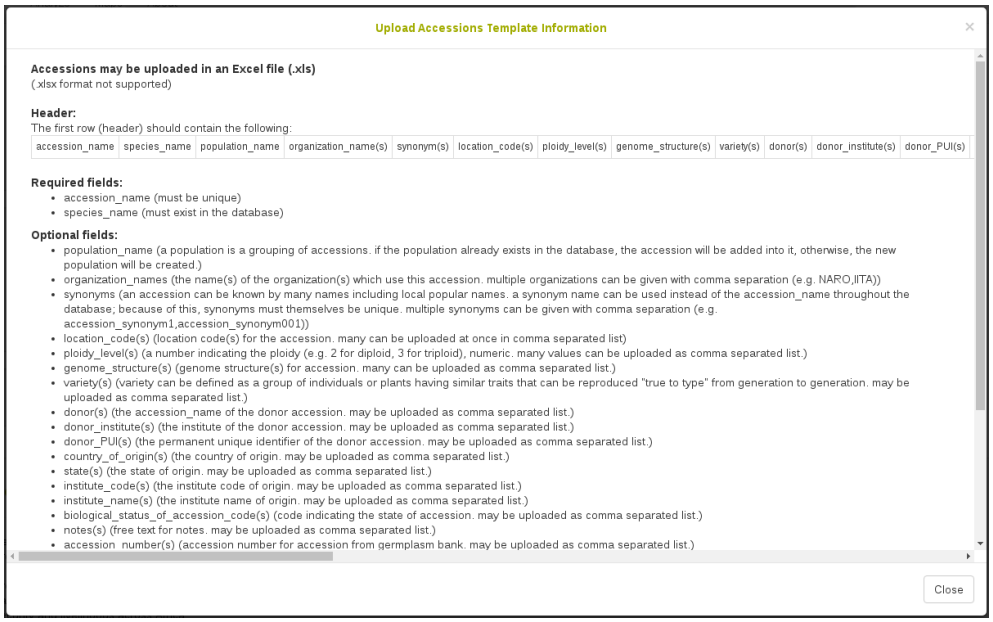

Once you click "Add Accessions", the new accessions will be created in the database and you will see the following confirmation dialog, which includes links to the newly created accessions.




## 6.2 Uploading Accessions and Accession's Info From A File

Uploading accessions using a file is very similar to using a list, but enables you to add a variety of attributes, such as synonyms or ploidy levels, to the accessions in bulk.

Clicking on "Spreadsheet format" will show the required structure of the spreadsheet. The file must be XLS or XLSX format and can contain a number of header columns as attributes. It is important that you use exactly the same header column names as listed here. In columns that indicate that many attribute values can be passed at once using (s), such as synonym(s), you can pass a comma separated list of values, such as 'synonym1,synonym2'.




Once you have selected your XLS or XLSX file for upload, click "Continue".

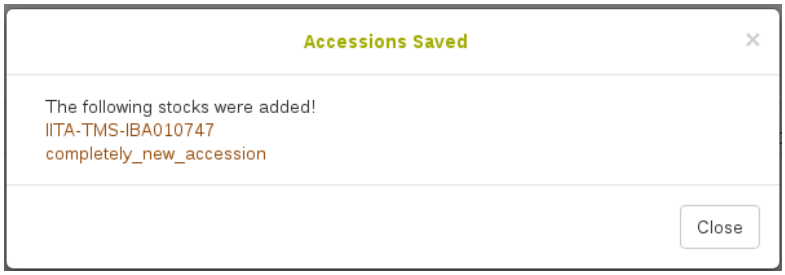
The following process is the same way as with lists:

The first dialog which can appear will show accession names which are already in the database.

Click "Continue" and the next dialog that can appear will show "fuzzy" matches for the accession names you are trying to upload. Here you can choose to prevent adding accession names which look very similar to each other as wrongly duplicated accessions.

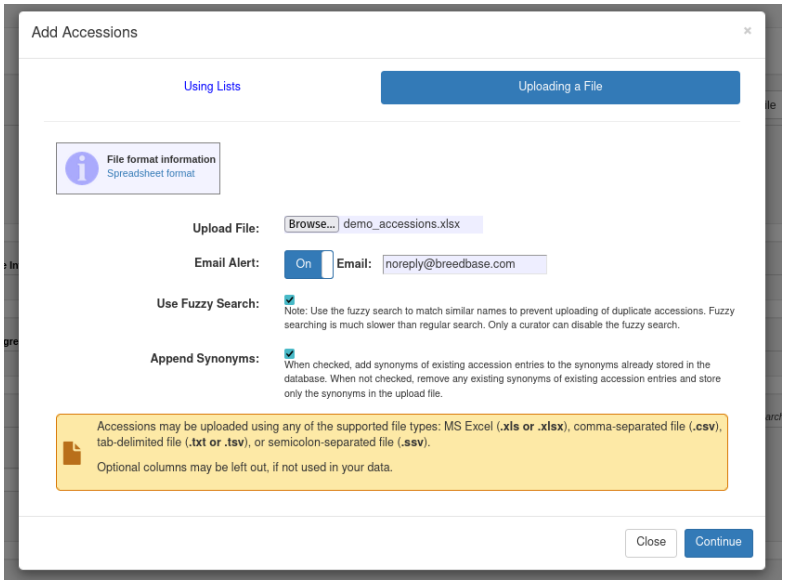
Click "Continue" and the final dialog that will appear will show the information to be added into the database. Here it is divided into accession names that are new and accession names that already exist in the database; however, for the accession names that already exist it will show additional attributes that originated from your file that will be added to these accessions.

Once you click "Add Accessions", the new accessions and information will be created in the database and you will see the following confirmation dialog, which includes links to the created and updated accessions.



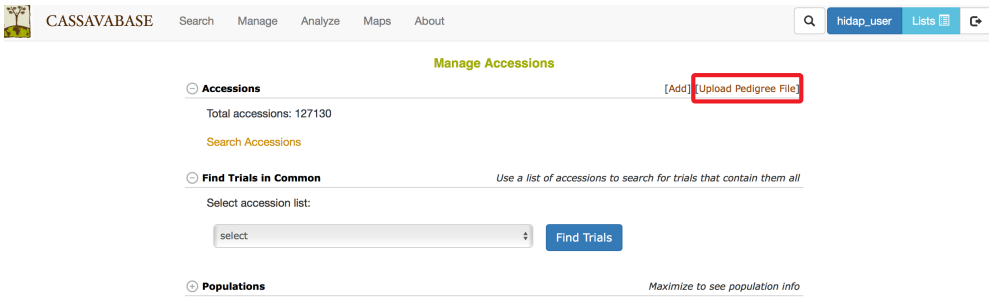

## 6.3 Email alert for accession upload

When uploading a large number of accessions from a file, uploads can take a while, as the system needs to perform a series of checks on each entry. You have the option to receive an email notification about the status and results of your upload by clicking the "Email Alert" checkbox. By default, the system will use the email address associated with your account, but you have the option of entering a different email address if you prefer. After submitting, the upload process runs in the background, allowing you to continue using the interface without interruptions. Once the process completes, you will receive an email with the upload results, including any warnings or errors that may have occurred during the upload.



## 6.4 Add Parentage (Pedigree) Information to Accessions

Pedigree data can be uploaded from your computer by clicking on "Upload Pedigree File"

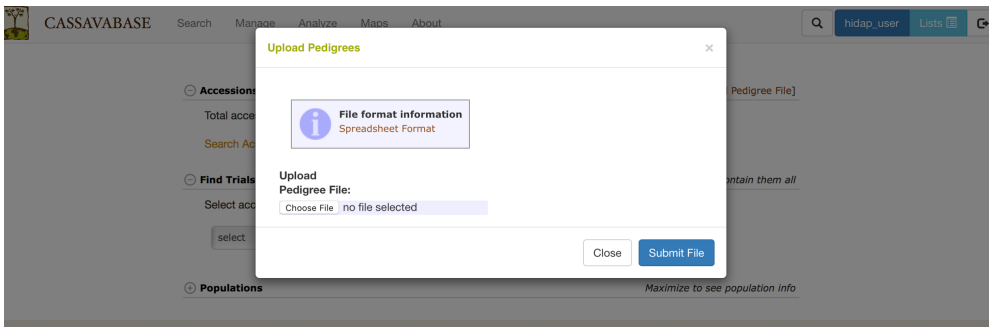


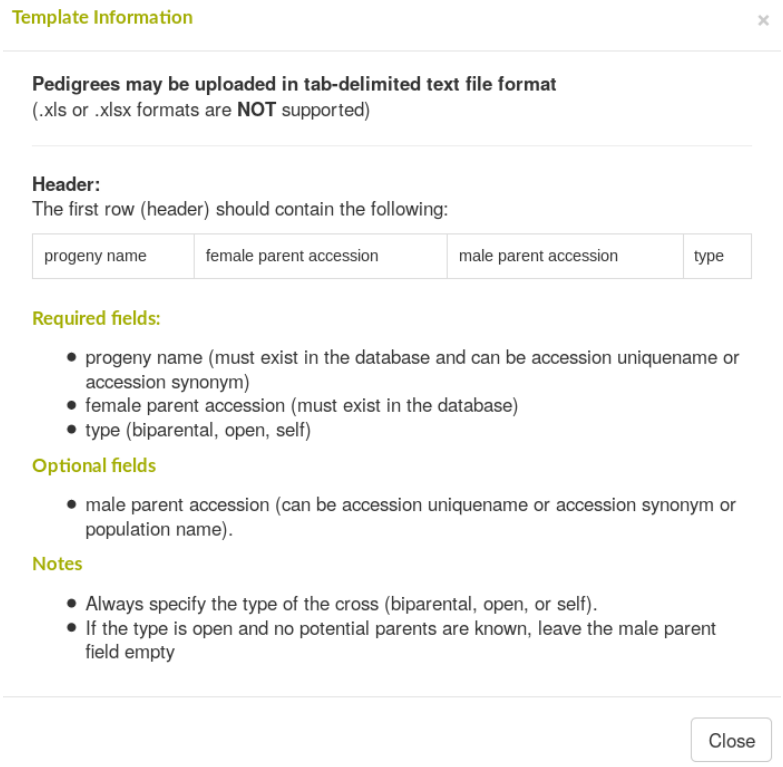

You can find detailed information on how to prepare pedigree file by clicking on "File format information". The file format can be tab or comma delimited text file, or Excel files (.xls or .xlsx).

The currently supported format has four columns:

progeny name female parent accession male parent accession type

Type can be biparental, self, backcross, sib, polycross, reselected, or open. In the case of the open type, the male parent accession field can remain blank. For all other types, both columns should be filled, even if they contain the same information as another column (such as self).






## 6.5 Working with grafts

Grafts are plants that are composed of a rootstock and a scion, which are genetically different and fused together, usually at the stem level.

To work with grafts, the grafts interface needs to be activated by a system administrator. Please contact your Breedbase provider. Briefly, a configuration parameter needs to be added to the sgn_local.conf file, show_grafting_interface. It should be set to 1 in sgn_local.conf, the default is 0 in sgn.conf.

Grafts to be created need to be specified using an Excel file (xlsx format) with two columns. The first column should have the header "scion accession" and should list accession names that will be scions. The second column should have the header "rootstock accession" and should list accession names that will be rootstocks.

In the database, the graft accessions will be created as single accessions. The graft accession will have two relationships, one to the scion accession (scion_of relationship) and one to the rootstock (rootstock_of relationship). These relationships are displayed on the pedigree viewer. The graft accession name is created from the scion accession name and the rootstock accession name, separated by the graft separator character. By default, the graft separator character is the plus sign '+'. The graft separator character can be changed in the sgn_local.conf file, using the parameter graft_separator_string. The graft separator string should not occur in any other accession names that are not grafts.

When the grafting interface is activated, a new button will be shown on the manage accessions page, called "Upload Grafts".

Clicking the button brings up the upload grafts dialog.

Select the Excel file containing the grafting information. The system will validate the file, for example, check whether the accessions are in the database, and if the headers are correct.

The validation result will be presented, and if problems are found, they will be listed. In addition, if there are problems, the Upload button will be grayed out and upload will not be possible. Conversely, if there are no problems, the Upload button will be activated and can be clicked to store the data.

If the upload completes, a completion message is displayed with a summary what was uploaded.

Grafted accessions can be used like any other accession, for example, they can be used on field layouts. If you create a list of graft accessions, use the list type 'accessions'.

Note that you shouldn't create new grafts based on other grafts. The scion accession and the rootstock accession have to be different, otherwise they will not be created.

## 6.6 Bulk renaming of accessions

Accessions can be renamed in bulk using the rename accessions feature. To rename accessions, prepare a tab delimited file with two columns: the first

column should have the header “old name” and contain the accession names that need to be changed. The second column should have the header “new name” and contain the names that the accessions in column 1 should be renamed to.

The accession renaming feature is available from the Manage->Accessions page. Click on the “Rename Accessions” button. The first step is the upload of the file with a verification step. The verification step checks whether all the accession names in column 1 exist in the database, and whether all the accession names given in column 2 do NOT exist in the database. Only if both conditions are met, will the “rename” button become active, otherwise an error message is displayed listing the offending accession names.

Optionally, the old name can be automatically added as a synonym to the renamed accession, using the checkbox on the submit form. This option is clicked by default. Unclick the checkbox to NOT save any old names as synonyms.

Note that accession renaming should not be undertaken lightly. This feature is intended for special use cases, such as where accessions are created in a nursery with a name that is different from the accession name in the downstream breeding program.

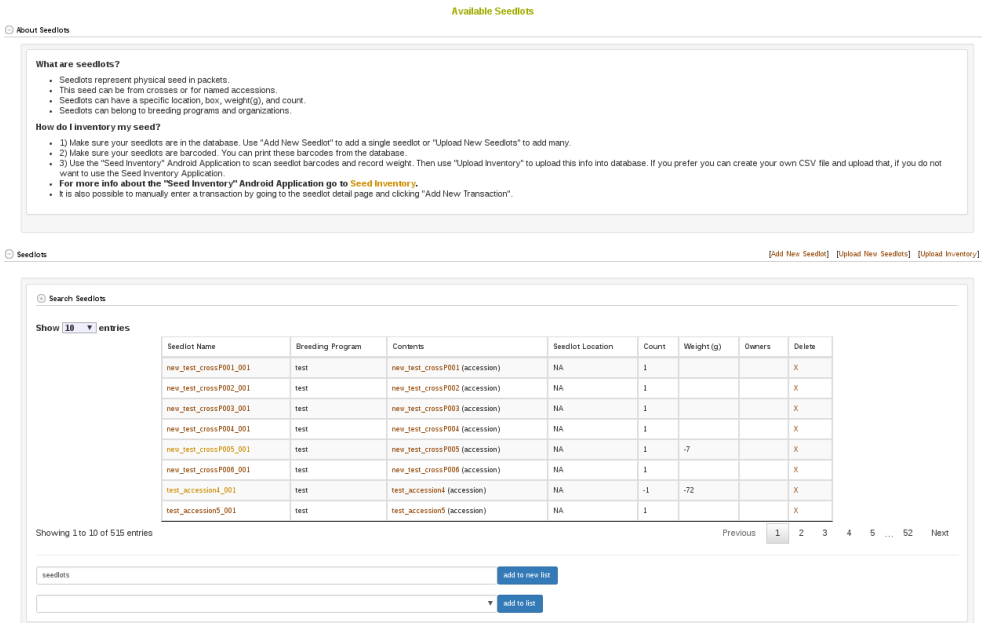
It can also be used to rename accessions in bulk that have spelling mistakes and other issues. Please note however, that the tool does not make any attempt to change the names of associated elements, such a plots, that may have been constructed using accession names.

Because of the many implications of accession renaming, the feature is limited to accounts with the curator role.

# Chapter 7

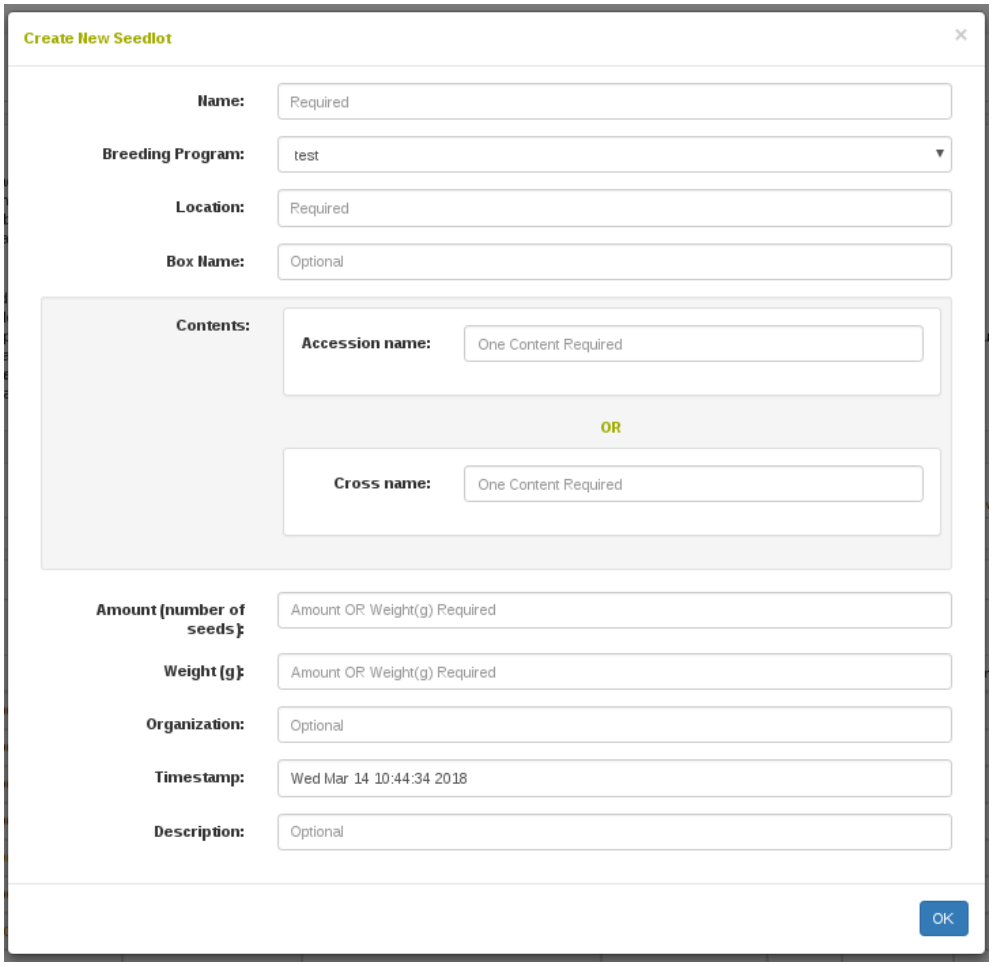
# Managing Seed Lots

Seedlots are different from Accessions in that they represent the physical seed being evaluated in an experiment. Seedlots have things like physical storage locations and seed quantities, which accessions do not. The seed in seedlots can be from crosses or can be named accessions. Seedlots from crosses would represent seed harvested. Click Manage and then Seed Lots to begin.



## 7.1 Add New Seedlot(s)

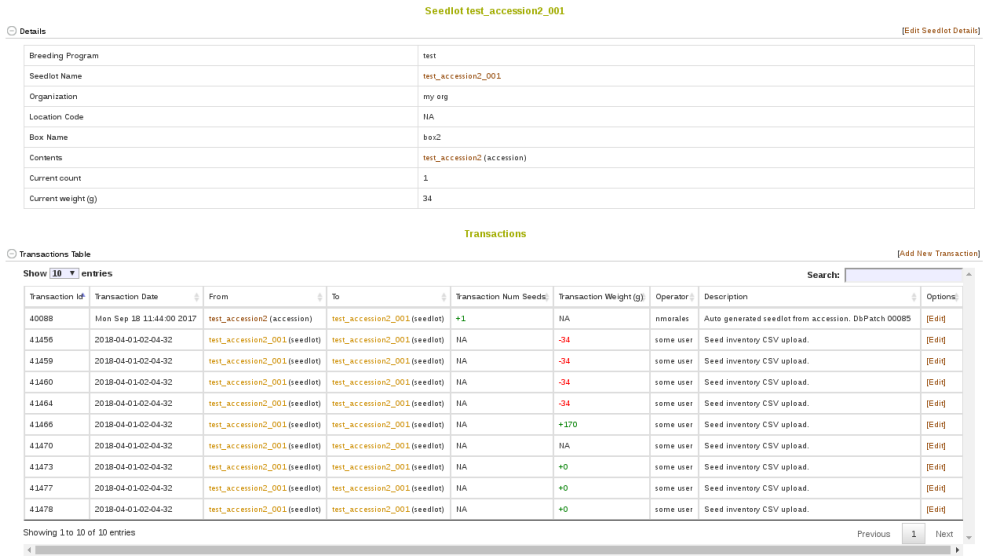
To add a single new seedlot, click on "Add Seedlot". This will bring up the following dialog where you enter information about where the seedlot exists, what accession or cross is contained in it, and how many seeds there are. A seedlot must contain either an accession or a cross, and not both. A seedlot must have a weight in grams or a seed count or both of these.




In the case where you have many seedlots to add to the database, you can upload an excel XLS or XLSX file instead. Click "Upload Seedlots" to see the following dialog.

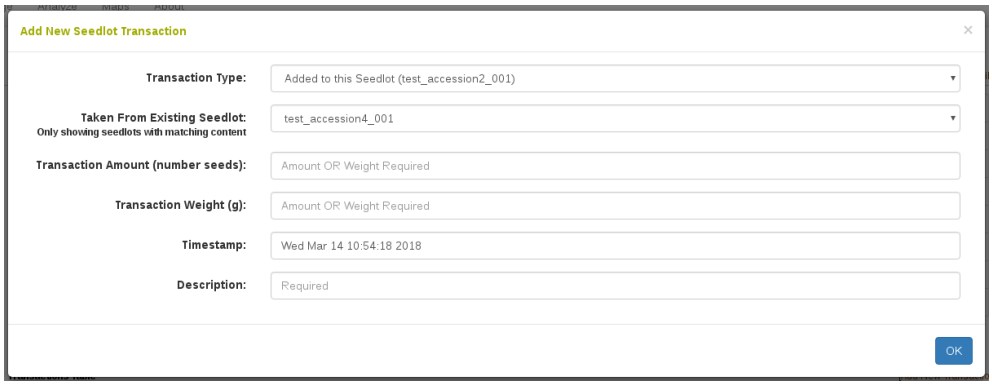
## 7.2 Seedlot Transactions

Seedlots are capable of tracking where seeds came from, such as from crosses, and to where seeds go, such as to plots in the field. If you navigate to a seedlot detail page you will see the following.




On this page you see and can edit information regarding a single seedlot,

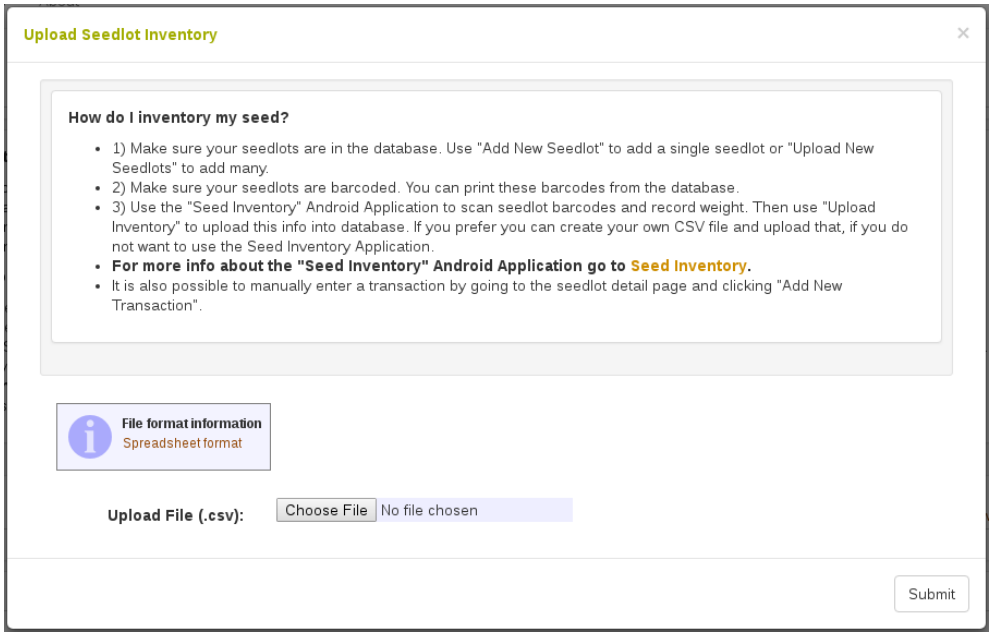
such as its name and location. You will also see a table indicating all t he transactions that a seedlot has been involved in, such as if it was planted in a plot in the field. Transactions to field plots are created when adding or uploading a new trial or from a trial's detail page. Clicking on "Add New Transaction" let you add a transaction from between this seedlot and another seedlot. This kind of transaction is useful for representing if you have distributed seed to different locations.



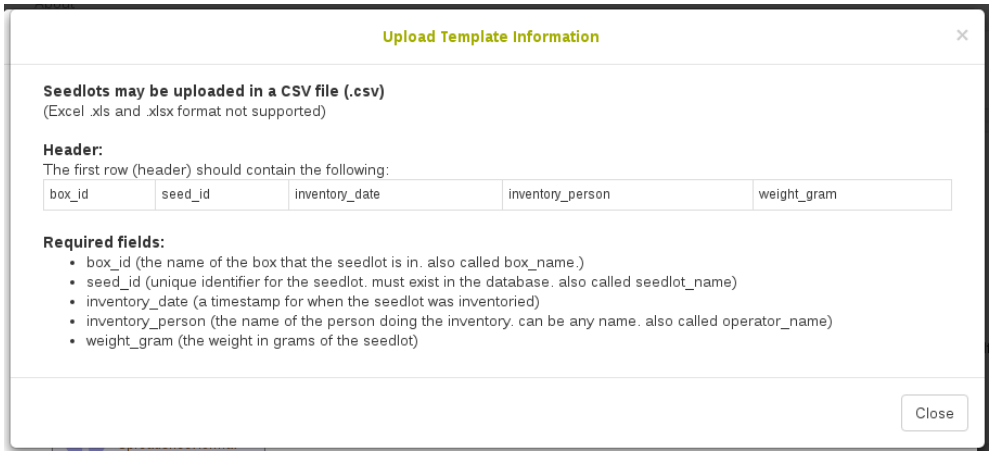

## 7.3 Seed Inventory

To inventory your seed: 1) Make sure your seedlots are in the database. Use "Add New Seedlot" to add a single seedlot or "Upload New Seedlots" to add many. 2) Make sure your seedlots are barcoded. You can print these barcodes from the database. 3) Use the "Inventory" Android Application to scan seedlot barcodes and record weight. Then use "Upload Inventory" to upload this info into database. If you prefer you can create your own CSV file and upload that, if you do not want to use the Inventory Application. For more info about the "Inventory" Android Application go to Inventory.

Clicking the "Upload Inventory" button will bring the following dialog:



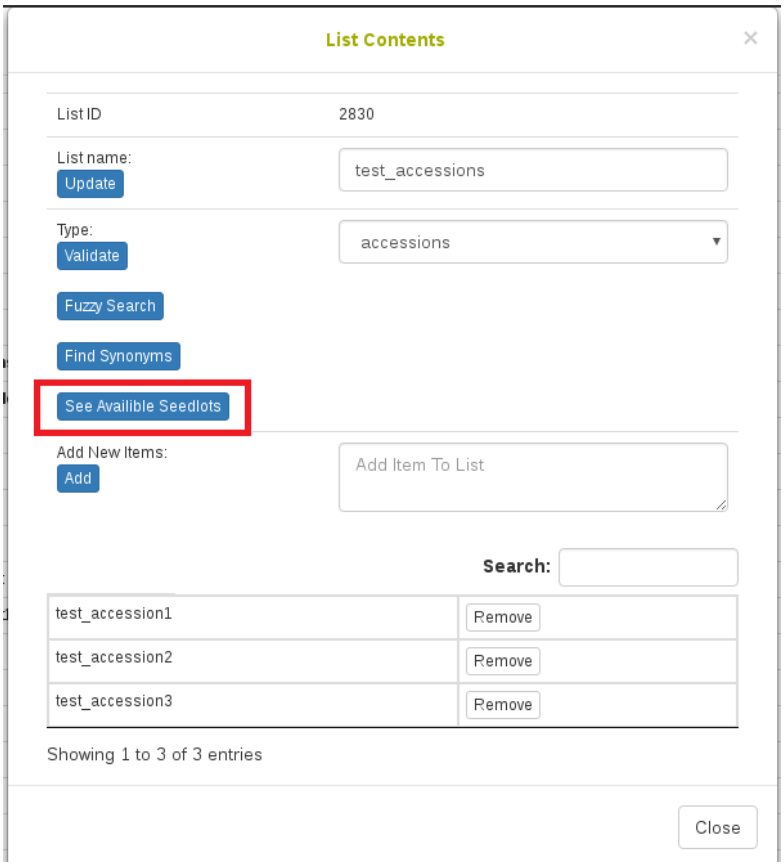

The CSV file that should contain your inventory should meet these Template requirements. The Seed Inventory Android Application exports this exact file.



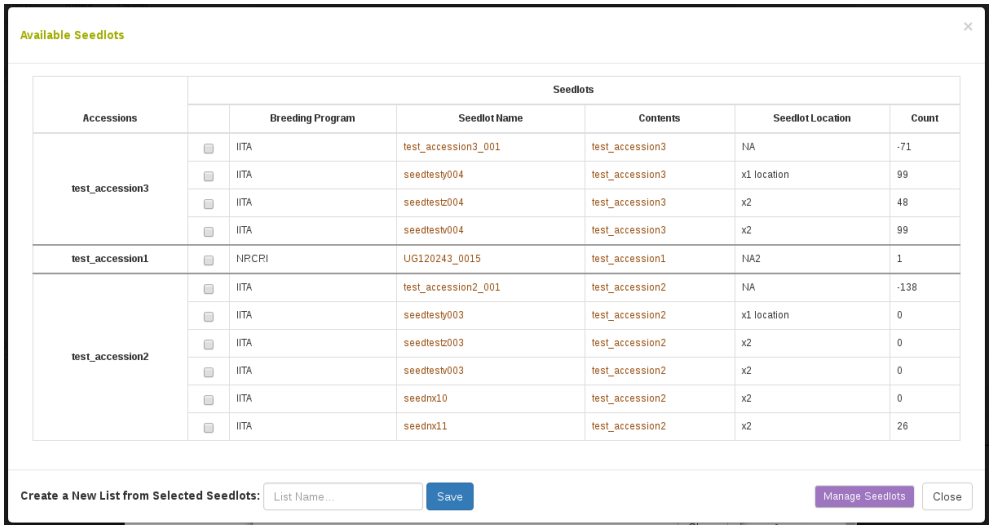

## 7.4 Find Seedlots For a List of Accessions

A convenient tool for searching available seedlots for a list of accessions is available in the list tool. First open up your list of accessions. For help

opening a list of accessions please see the List section help. There is a button called "See Available Seedlots".

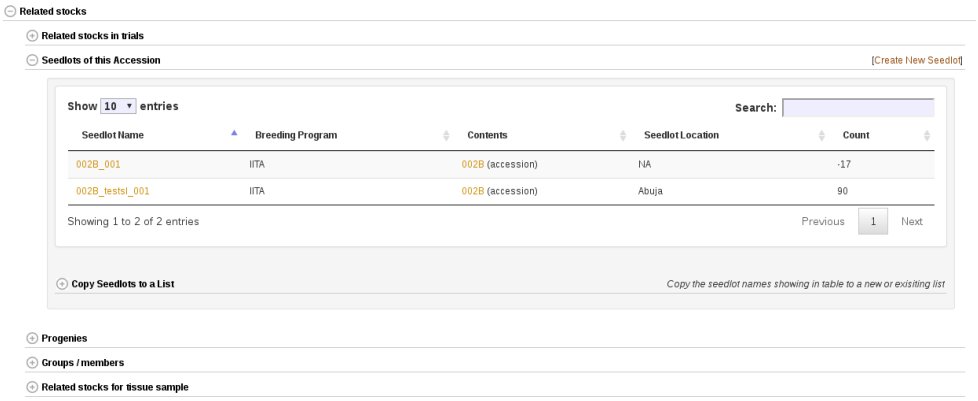



Once you click this, you will see the following table in a dialog. From here you can create a list of seedlots using the checkboxes and the input at the bottom.




## 7.5 Create a seedlot for an Accession or Cross

Complementary to what we saw above for creating seedlots from the "Manage Seedlots" page, it is possible to create a new seedlot from an accession's detail page or from the cross detail page. On the accession detail page, this is visible in the "Related Stocks" section as seen below. The cross detail page has an identical section. Notice the link for creating a new seedlot, which streamlines adding the seedlot.



## 7.6 Add quality data to a seedlot

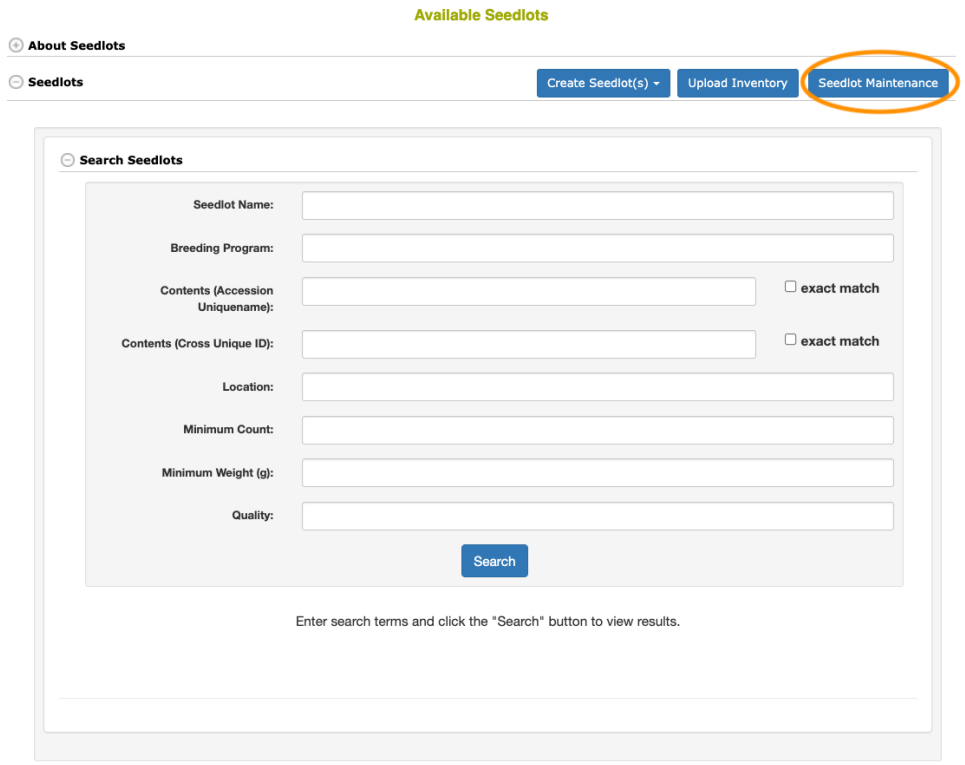
Quality information can be added to a seedlot in the quality field. This is also available as a column in the file upload format. It is recommended to use a controlled vocabulary, defined by the user, for the quality field. For example, good quality seed should be labelled “ok”, whereas other quality descriptors could be “moldy”, “insect damage”, or “low sprouting”, etc.

## 7.7 Seedlot Maintenance Events

For some crops, such as sugar kelp, a “seedlot” requires routine maintenance for the successful long-term storage of the seedlot. (For example, a Seedlot Maintenance Event for sugar kelp would be the routine change of the water that gametophytes are kept it). Breedbase can now store a record of these Seedlot Maintenance Events associated directly with existing Seedlots. Maintenance Events can be uploaded using a simple Excel template or recorded directly on the website.

### 7.7.1 Setup

Each Breedbase instance needs to be configured to support the storage of Seedlot Maintenance Events since each crop will have their own distinct set of maintenance events for their seedlots. To check if your Breedbase instance supports this feature, go to the Manage menu and select the Seed Lots page. Make sure you are logged in and look for the **Seedlot Maintenance** button near the top, next to the **Create Seedlot(s)** and **Upload Inventory** buttons. If you don’t see this button, contact the developer(s) supporting your Breedbase instance and ask if they can setup this feature.




*The location of the Seedlot Maintenance button on the Manage > Seed Lots page*

### 7.7.2 Adding Events

Seedlot Maintenance Events can be added using two methods: 1) Uploading an Excel template or 2) Recording events directly on the website

#### Uploading Events with Excel Template

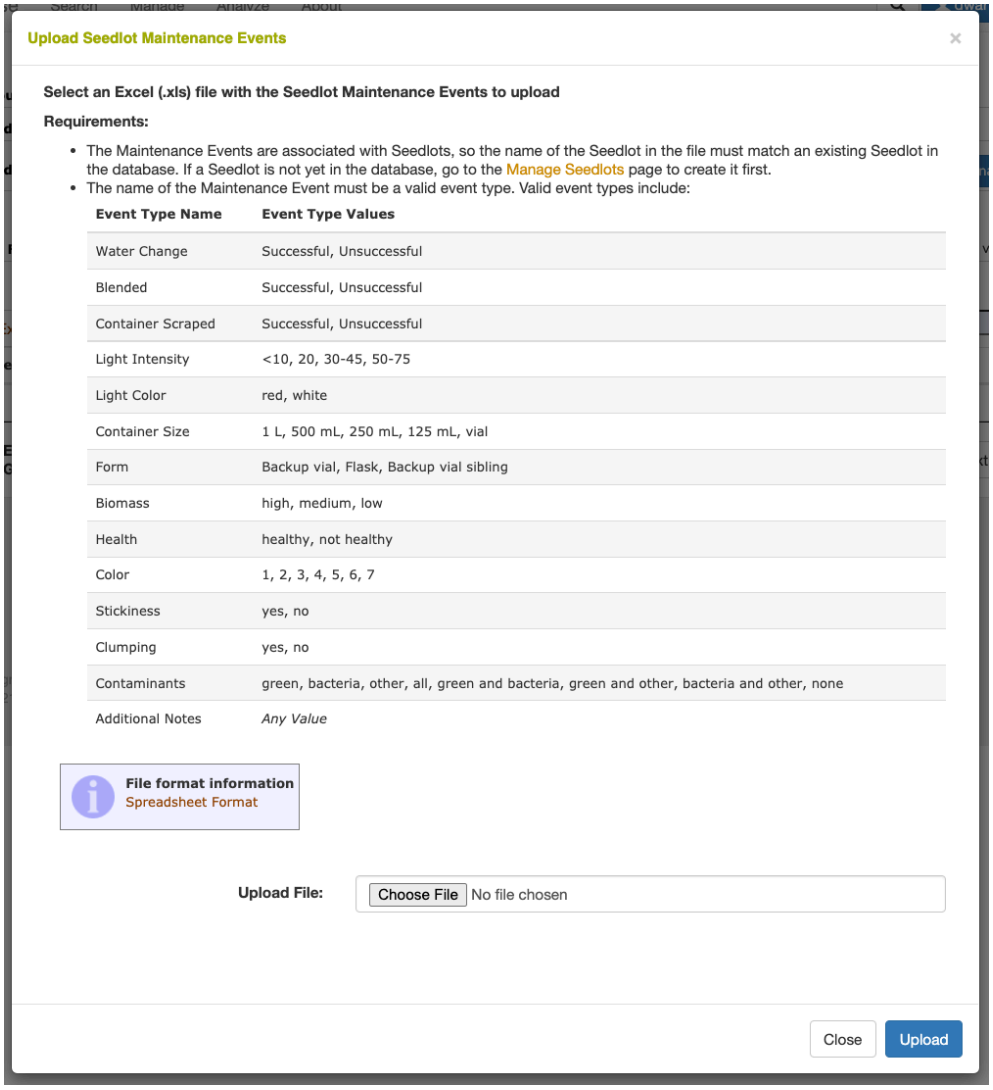
To bulk-upload a file of Seedlot Maintenance Events, first create an Excel (.xls or .xlsx) file with the following headers:

- **seedlot** - the name of the Seedlot to associate the event with (must exactly match an existing Seedlot in the database)
- **type** - the name of the Seedlot Maintenance Event type (these vary between Breedbase instances, a list of supported event types is displayed

on the upload page)
- **value** - the value of the Seedlot Maintenance Event (these may be different for each event type and vary between Breedbase instances, a list of supported event values is displayed on the upload page)
- **notes** - optional, additional notes/comments about the event
- **operator** - the username of the Breedbase user that recorded the event
- **timestamp** - the date/time the event was recorded, in 'YYYY-MM-DD HH:MM:SS' format

Once you have an Excel file with the events filled out, follow these steps to upload the events to the database:

1. Make sure you are logged in to your Breedbase instance
2. Go to the Manage > Seed Lots page
3. Select the **Seedlot Maintenance** button
4. Select the **Upload Maintenance** button
5. Choose your Excel (.xls or .xlsx) file to upload
6. Select the **Upload** button



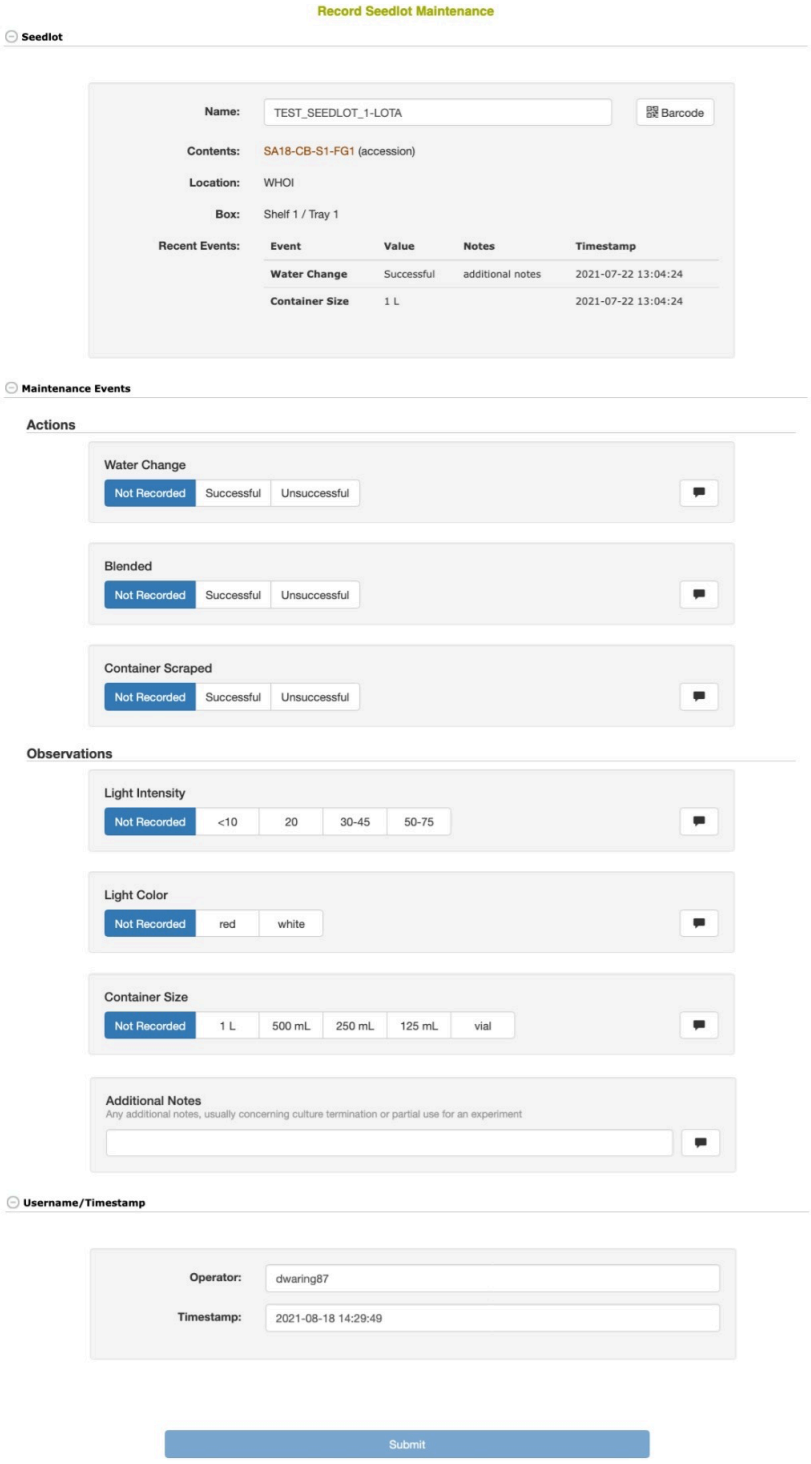

*The Seedlot Maintenance upload dialog, showing the supported event types and values (for sugar kelp)*

### Recording Events on Website

To add individual Seedlot Maintenance Events to the database in real time, as they're being recorded, use the **Record Maintenance** page. Follow these steps to record Seedlot Maintenance Events:

1. Make sure you are logged in to your Breedbase instance

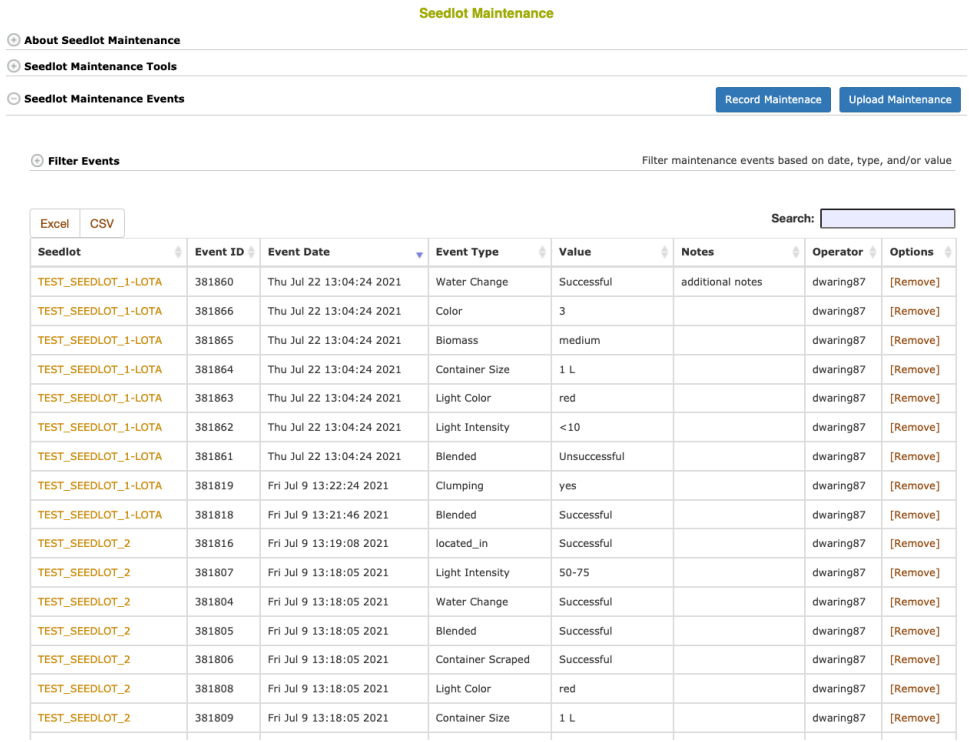
2. Go to the Manage > Seed Lots page
3. Select the **Seedlot Maintenance** button
4. Select the **Record Maintenance** button
5. Enter the **Seedlot Name** or scan a barcode that has the Seedlot Name encoded. Once entered, the box at the top of the page will display basic information about the Seedlot as well its recently recorded events.
6. Select or Enter the values of individual events
7. Optionally, notes button next to each event to add additional notes/comments about that specific event
8. Make sure the operator/username and timestamp are correct
9. Select the **Submit** button to add the recorded events to the database. NOTE: any events that remain selected as "Not Recorded" will not be submitted to the database.


Record Seedlot Maintenance
Seedlot
Name:
TEST_SEEDLOT_1-LOTA
Barcode
Contents:
SA18-CB-S1-FG1 (accession)
Location:
WHOI
Box:
Shelf 1 / Tray 1
Recent Events:
Event
Value
Notes
Timestamp
Water Change
Successful
additional notes
2021-07-22 13:04:24
Container Size
1 L
2021-07-22 13:04:24
Maintenance Events
Actions
Water Change
Not Recorded
Successful
Unsuccessful
Blended
Not Recorded
Successful
Unsuccessful
Container Scraped
Not Recorded
Successful
Unsuccessful
Observations
Light Intensity
Not Recorded
<10
20
30-45
50-75
Light Color
Not Recorded
red
white
Container Size
Not Recorded
1 L
500 mL
250 mL
125 mL
vial
Additional Notes
Any additional notes, usually concerning culture termination or partial use for an experiment
Username/Timestamp
Operator:
dwaring87
Timestamp:
2021-08-18 14:29:49
Submit

*The Seedlot Maintenance record page, as configured for sugar kelp*

### 7.7.3 Displaying Events

Recently recorded Seedlot Maintenance Events are displayed in a table from the main Seedlot Maintenance page, as well as the detail page for individual Seedlots.



Seedlot Maintenance

About Seedlot Maintenance

Seedlot Maintenance Tools

Seedlot Maintenance Events — Record Maintenace — Upload Maintenance

Filter Events — Filter maintenance events based on date, type, and/or value

Excel CSV — Search:

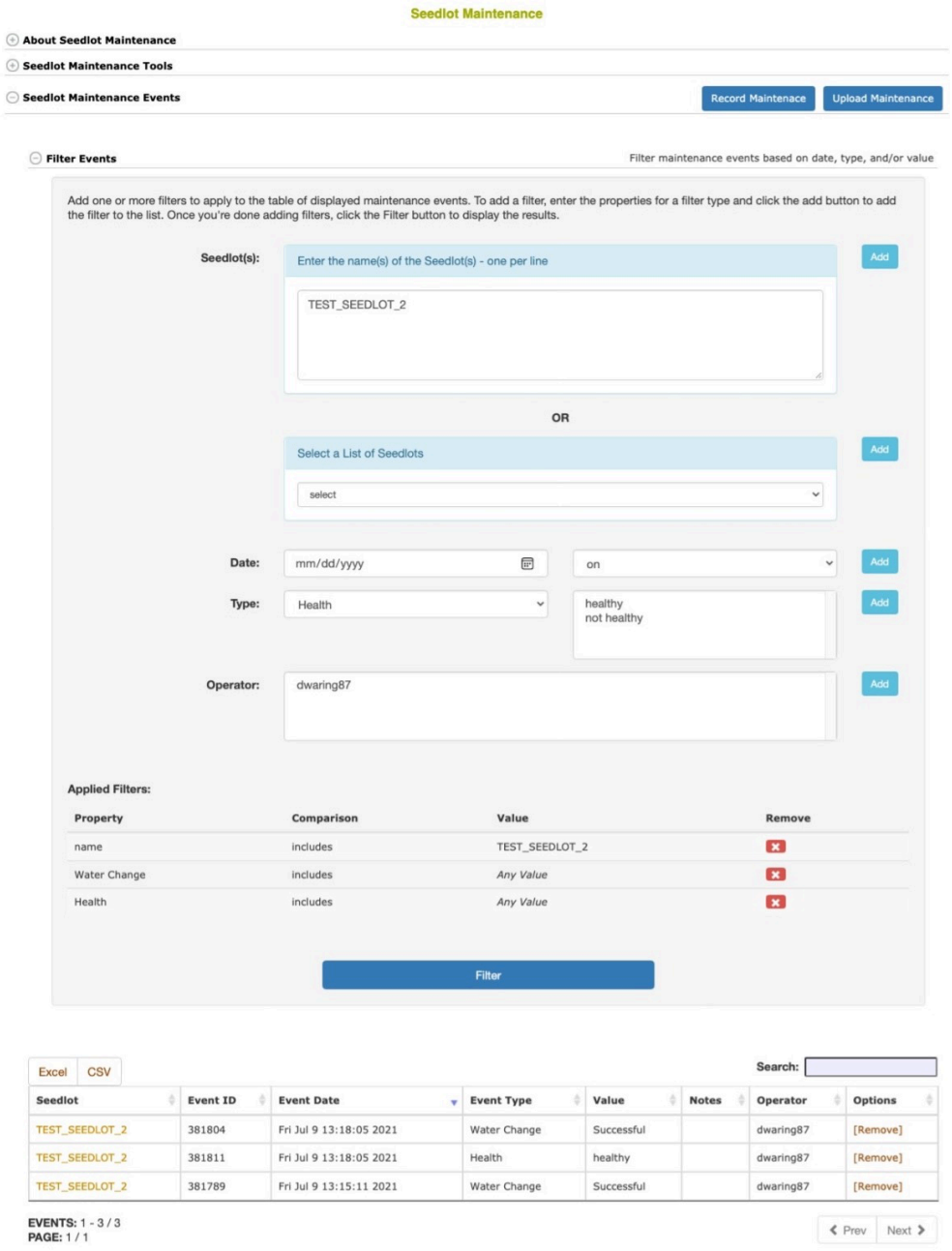
| Seedlot | Event ID | Event Date | Event Type | Value | Notes | Operator | Options |
|---|---|---|---|---|---|---|---|
| TEST_SEEDLOT_1-LOTA | 381860 | Thu Jul 22 13:04:24 2021 | Water Change | Successful | additional notes | dwaring87 | [Remove] |
| TEST_SEEDLOT_1-LOTA | 381866 | Thu Jul 22 13:04:24 2021 | Color | 3 | | dwaring87 | [Remove] |
| TEST_SEEDLOT_1-LOTA | 381865 | Thu Jul 22 13:04:24 2021 | Biomass | medium | | dwaring87 | [Remove] |
| TEST_SEEDLOT_1-LOTA | 381864 | Thu Jul 22 13:04:24 2021 | Container Size | 1 L | | dwaring87 | [Remove] |
| TEST_SEEDLOT_1-LOTA | 381863 | Thu Jul 22 13:04:24 2021 | Light Color | red | | dwaring87 | [Remove] |
| TEST_SEEDLOT_1-LOTA | 381862 | Thu Jul 22 13:04:24 2021 | Light Intensity | <10 | | dwaring87 | [Remove] |
| TEST_SEEDLOT_1-LOTA | 381861 | Thu Jul 22 13:04:24 2021 | Blended | Unsuccessful | | dwaring87 | [Remove] |
| TEST_SEEDLOT_1-LOTA | 381819 | Fri Jul 9 13:22:24 2021 | Clumping | yes | | dwaring87 | [Remove] |
| TEST_SEEDLOT_1-LOTA | 381818 | Fri Jul 9 13:21:46 2021 | Blended | Successful | | dwaring87 | [Remove] |
| TEST_SEEDLOT_2 | 381816 | Fri Jul 9 13:19:08 2021 | located_in | Successful | | dwaring87 | [Remove] |
| TEST_SEEDLOT_2 | 381807 | Fri Jul 9 13:18:05 2021 | Light Intensity | 50-75 | | dwaring87 | [Remove] |
| TEST_SEEDLOT_2 | 381804 | Fri Jul 9 13:18:05 2021 | Water Change | Successful | | dwaring87 | [Remove] |
| TEST_SEEDLOT_2 | 381805 | Fri Jul 9 13:18:05 2021 | Blended | Successful | | dwaring87 | [Remove] |
| TEST_SEEDLOT_2 | 381806 | Fri Jul 9 13:18:05 2021 | Container Scraped | Successful | | dwaring87 | [Remove] |
| TEST_SEEDLOT_2 | 381808 | Fri Jul 9 13:18:05 2021 | Light Color | red | | dwaring87 | [Remove] |
| TEST_SEEDLOT_2 | 381809 | Fri Jul 9 13:18:05 2021 | Container Size | 1 L | | dwaring87 | [Remove] |

*Unfiltered table of recent Seedlot Maintenance events*

The events displayed in these tables are sorted by timestamp, with the most recently recorded events displayed first. The displayed events can be filtered using any number of supported filter criteria, such as: - seedlot names (as entered on the page or using an existing seedlot list), - dates (on, on or before, before, on or after, and/or after the entered dates) - event types - event type values - operator/username

Select the properties of the filter(s) you want to apply, then select the **Add** button next to the button to add the filter to the list of applied filters. Once

you're done adding filters, select the **Filter** button to search the database for the filtered events.




*A filtered table of Seedlot Maintenance events*

The filtered events can be downloaded directly from the table using the **Excel** or **CSV** buttons at the top of the table. Or Seedlot Maintenance Events can be bulk-downloaded (this includes all events for a Seedlot) using a list of Seedlots from the main downloads page (see below).

### 7.7.4 Downloading Events

To bulk-download all events for a specific subset of Seedlots:

1. Create a list containing the Seelots you are interested in.
2. Go to the **Download Using Lists** page (Manage > Download)
3. Find the **Download Seedlot Maintenance Events** section
4. Select your list of Seedlots
5. Select the **Download** button to generate the download file

The downloaded file will follow the same format as the upload template and will contain all recorded Seedlot Maintenance Events for each Seedlot in the list.

## 7.8 Deleting Seedlots

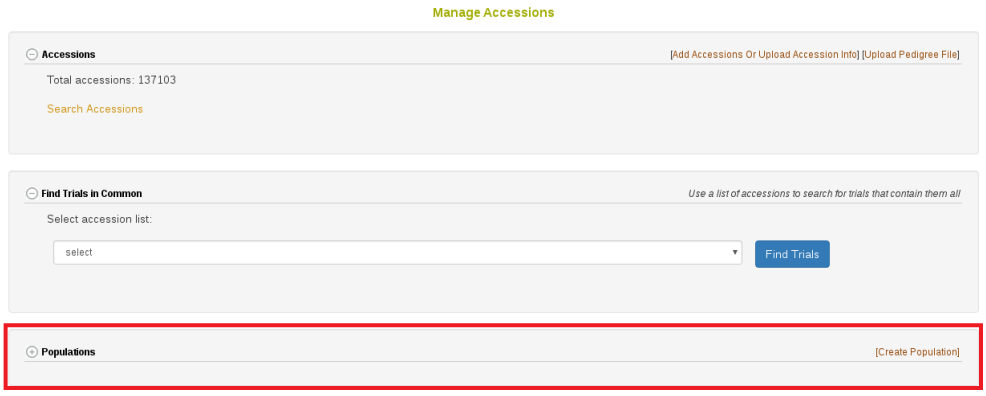
Seedlots can be deleted on the Manage Seedlots page (/breeders/seedlots) by search the seedlot and then clicking the X to delete one seedlot at a time. To delete a seedlot, the logged in user needs the required delete privileges on the seedlot. The seedlot also should not have any transactions associated with it (except for the initial transaction).

To delete seedlots in bulk, generate a list of type seedlot, for example, using the wizard. Open the section “Delete seedlots using a list” on the Manage Seedlots page and select the list. Seedlot deletion using a list is only available to user with curator status.

# Chapter 8

# Managing Populations

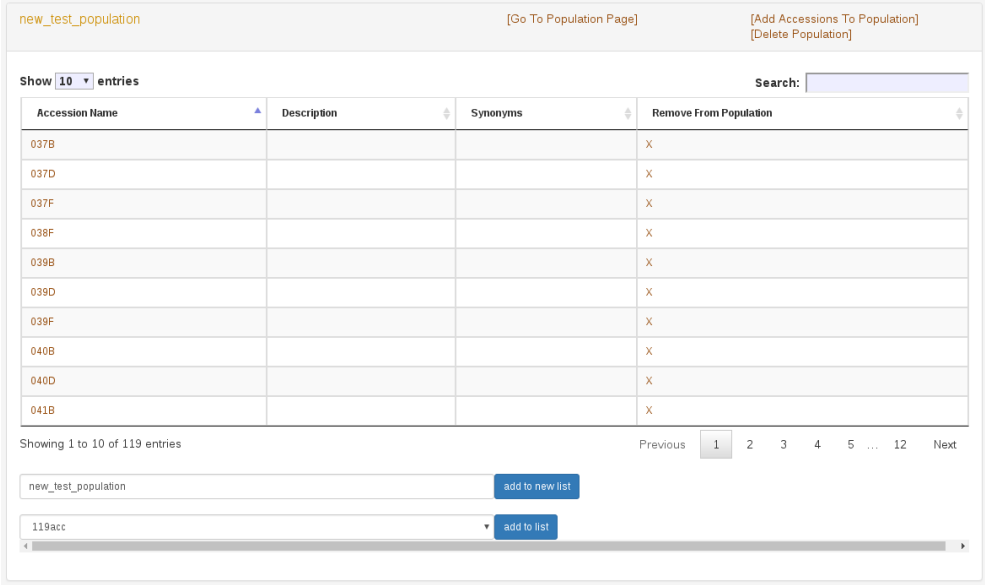
Populations are modeled as groups of accessions. This grouping can be useful in downstream analyses. To manage these populations go to Manage Accessions and scroll tp the bottom.




To add a new population click “Create Population”. The following dialog will appear where you choose a list of accessions and give a name to the new population. Please note it is also possible to create a population when you are uploading new accessions into the database.

Click on the plus (+) button next to Populations to see all the available populations. Click on a population name to see the accessions in the population.

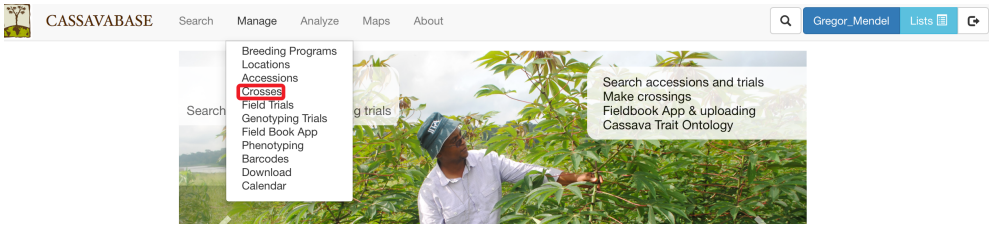
From here you can delete accessions from a population as well as add new accessions to the population.



# Chapter 9

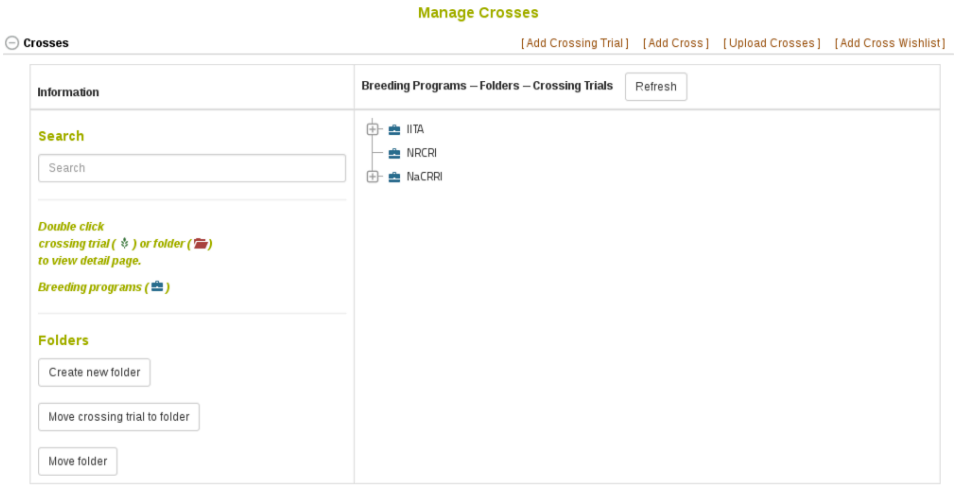
# Managing Crosses

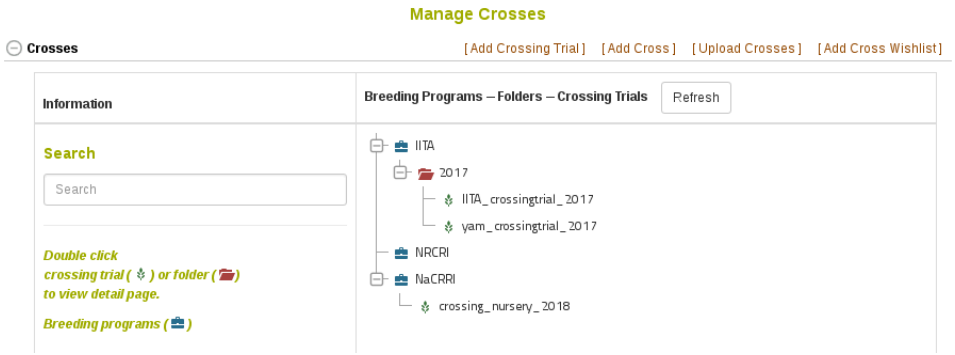
Information for crosses can be managed using the "Crosses" option in the Manage menu.




## 9.1 Crossing Experiment

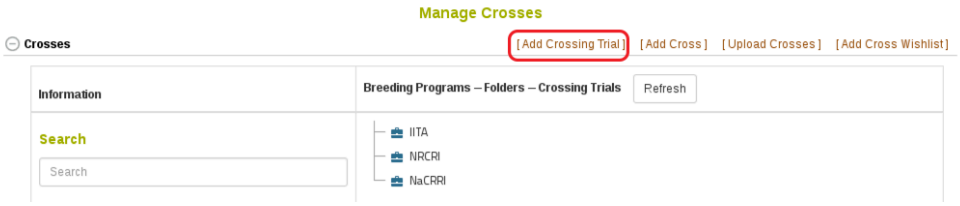
Different crosses in the same trial/nursery/project are grouped via "**crossing experiment**". Crossing experiments are organized based on their breeding programs. To find a crossing experiment, you can either type the crossing experiment name in the "Search" box, or look for the crossing experiment directly in its breeding program by clicking on the "+" icon. In each breeding program, crossing experiments can be placed directly in the breeding program, or organized in folders. The "**Folders**" section allows you to place crossing experiments in folders, move a crossing experiment in a folder to another folder, or rearrange your folders within a breeding program.




### 9.1.1 Add New Crossing Experiment

To add a new crossing experiment, click on "Add Crossing Experiment" link.




Required Information:

- "**Crossing Experiment Name**": enter a name for the crossing experiment. The crossing experiment name must not already exist in the database.

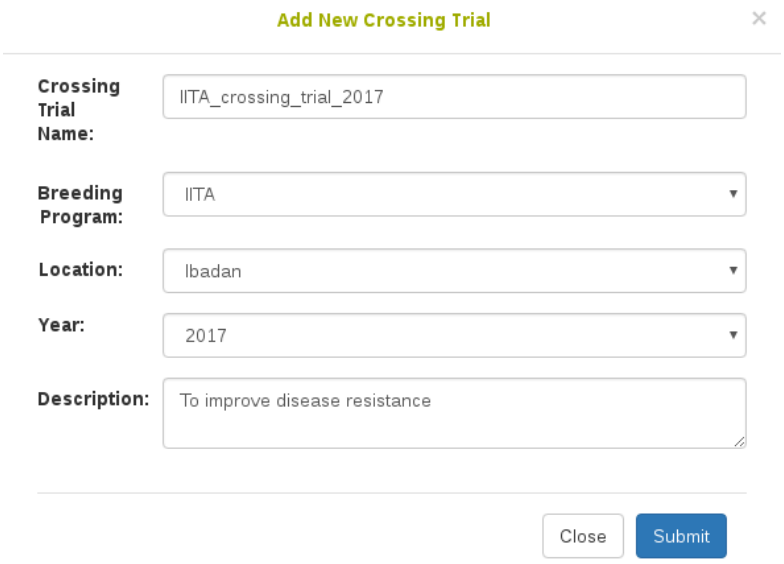
• "**Breeding program**": select a breeding program that is available in the database. New breeding programs can be added on the "Breeding program" page, accessible from the "Manage" menu. *Breeding Program Page*

• "**Location**": select a location for the crossing experiment. New locations can be entered on the "**Locations**" page, accessible from the "**Manage**" menu. *Location Page*

• "**Year**": select a year.

• "**Description**": enter a description for the crossing experiment.

After filling in the information, click "**Submit**" to generate the crossing experiment.

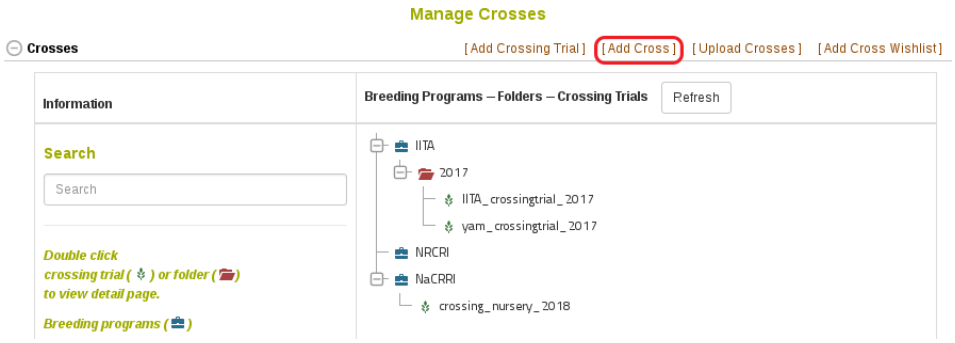



# 9.2 Cross

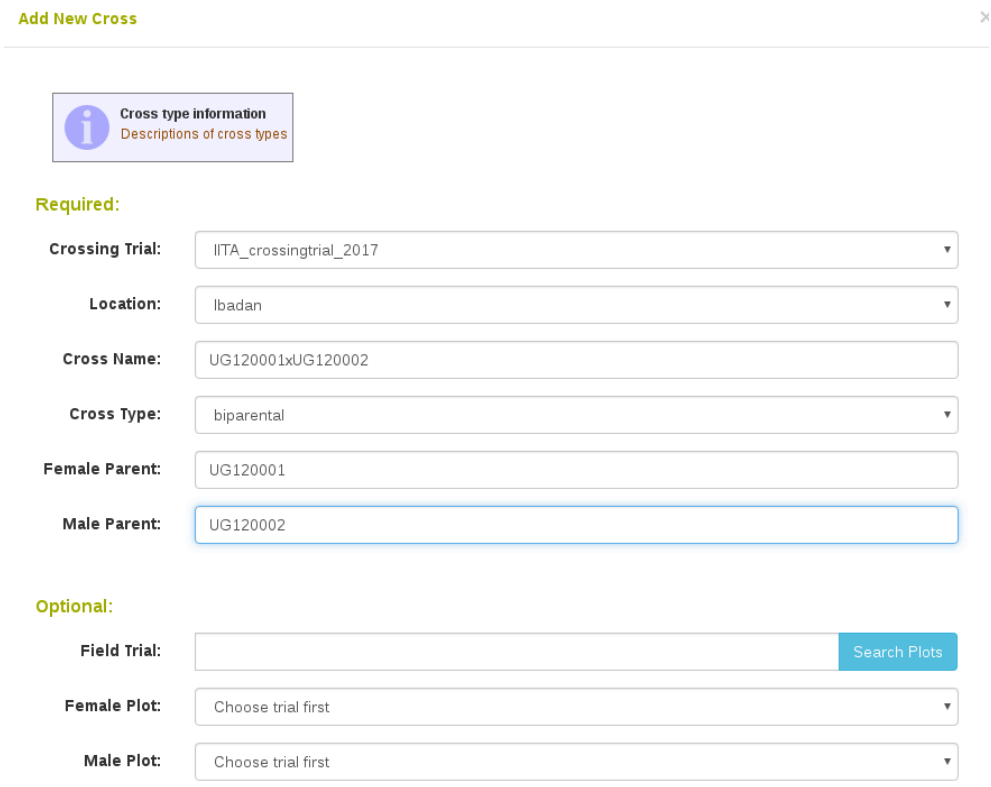
## 9.2.1 Add New Crosses

### Add a cross by using the "Add New Cross" dialog

To add a single new cross, click on "Add Cross" link.




Enter cross information in the popup dialog.

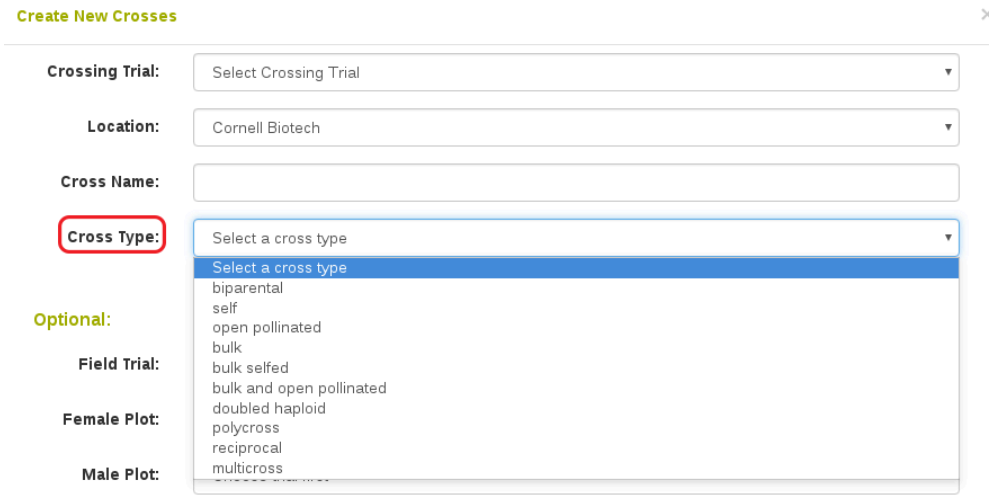



Required Information:

- "**Crossing experiment**": select a crossing experiment available in the database.
- "**Location**": select a location available in the database.

• "**Cross name**": enter a name for the cross. The cross name must not already exist in the database.

• "**Cross type**": the options for cross types are: biparental, self, open pollinated, bulk, bulk selfed, bulk and open pollinated, double haploid, polycross, reciprocal and multicross.



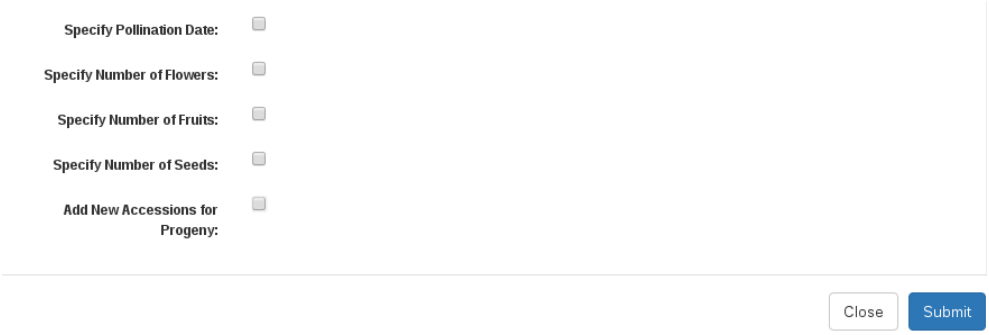

• The "**Female Parent**" and "**Male Parent**" field are auto-complete fields for accessions that are already in the database. The parents specified will be entered in the pedigree of the new accessions generated by this cross.

Optional Information:

• "**Female Plot and/or Male Plot**": In addition to the accession names, specific plots used in the cross can also be added to the database. To retrieve plot names associated with each female/male accession, enter your trial name, then click "**Search Plots**". Plot names of each parental accession in that field trial will be shown in the drop-down list, you can then select the plot used in the cross.

Additional crossing experimental information such as pollination date, number of flowers, number of fruits, number of seeds can be specified during adding new cross. Alternatively, this information can be updated or edited directly on the "**Cross Details**" page.

If you know the number of accessions that are generated from the cross, they can be instantiated immediately in the database by clicking the "**Add accessions for progeny**" checkbox and specifying the number.

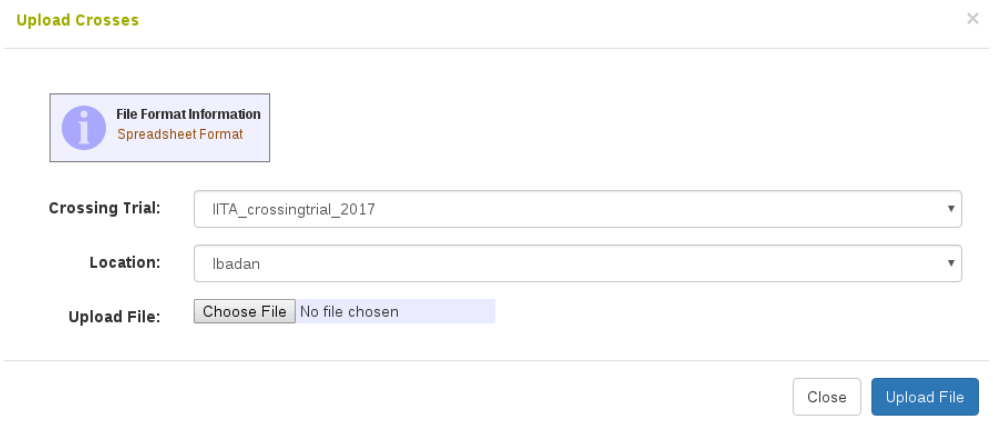



Click "Submit" to generate the cross.

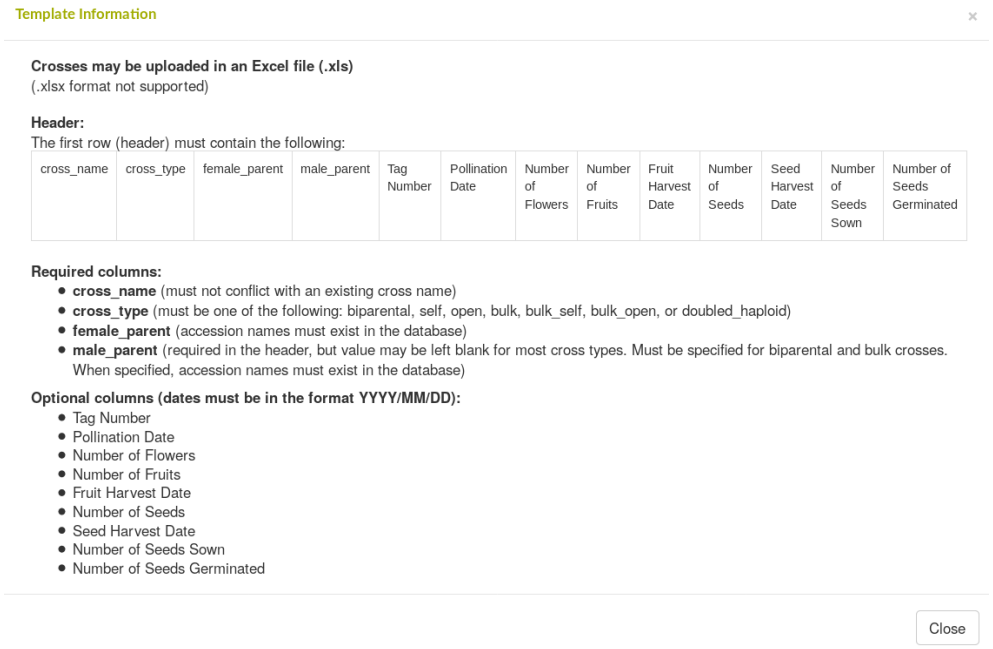
### Upload New Crosses

To upload new crosses from an Excel file (.xls or .xlsx), click on "Upload Crosses" link.

Select a crossing experiment and a location available in the database from drop-down lists and choose a file that you want to upload, then click "**Upload File**".




Please check spreadsheet format carefully. The file must be an Excel file (.xls or .xlsx).

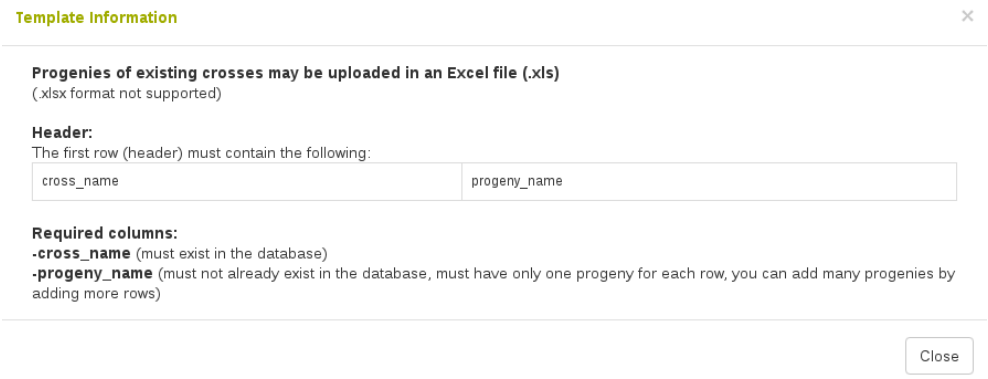


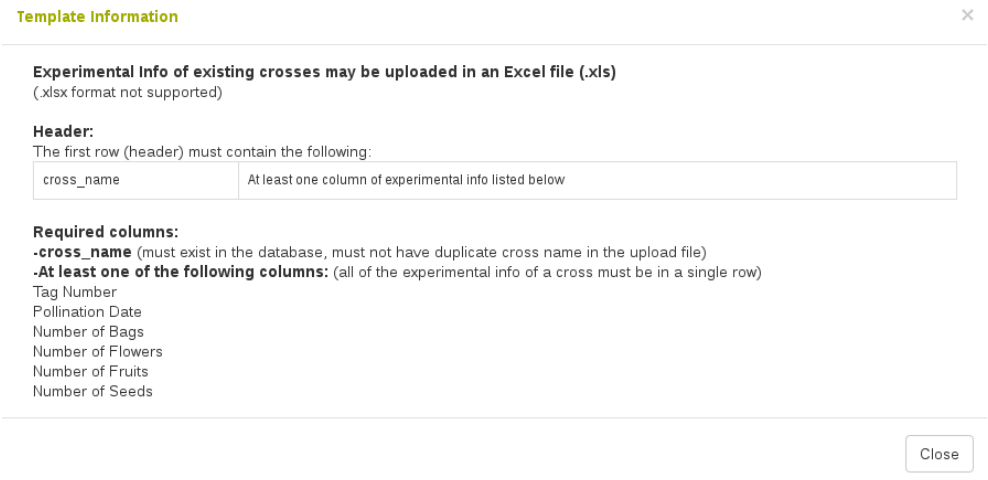

### 9.2.2 Update Crosses by Uploading

To upload progenies and/or experimental info of crosses already in the database, go to "**Manage-Upload**" page.

In the "**Crosses**" section, there are links for uploading progenies and experimental info.

Please check spreadsheet format in each link carefully. The file must be an Excel file (.xls or .xlsx).



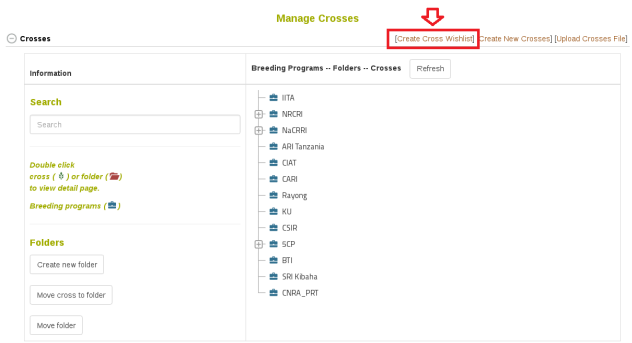

Note: crossing experimental information is customized based on the need for

each crop. As a result, column headers for experimental info in your database may be different from the information shown in this manual.

## 9.3 Cross Wishlist

An Android ODK application is being developed to record cross information on a mobile device in the field. To link this mobile application with the database, the Cross Wishlist can be used to create a plan for which crosses to perform.

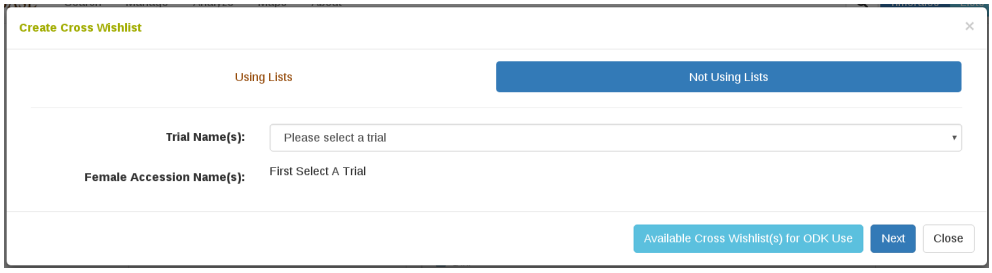
This tool is available on the Manage Cross page. It is currently only available on certain databases, so when you click this link you may see an alert mentioning that the cross wishlist is not available on your database.



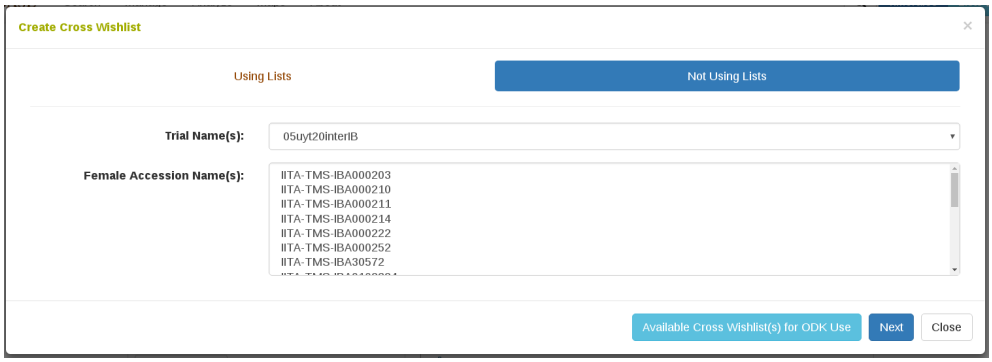

### 9.3.1 Create a Cross Wishlist

#### Step 1. Select the accessions to be crossed in your trial

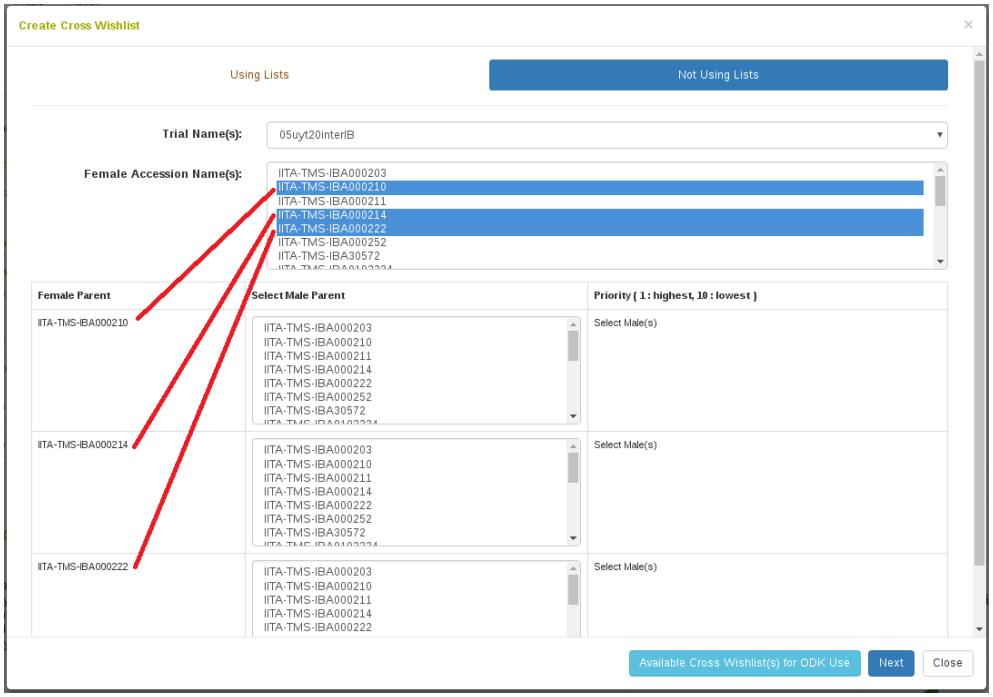
There are two interfaces for this step, either "Not Using Lists" or "Using Lists". Depending on if you already have a list of female and male accessions to use, you can decide on which interface to use. The end result of using either interface is the same.



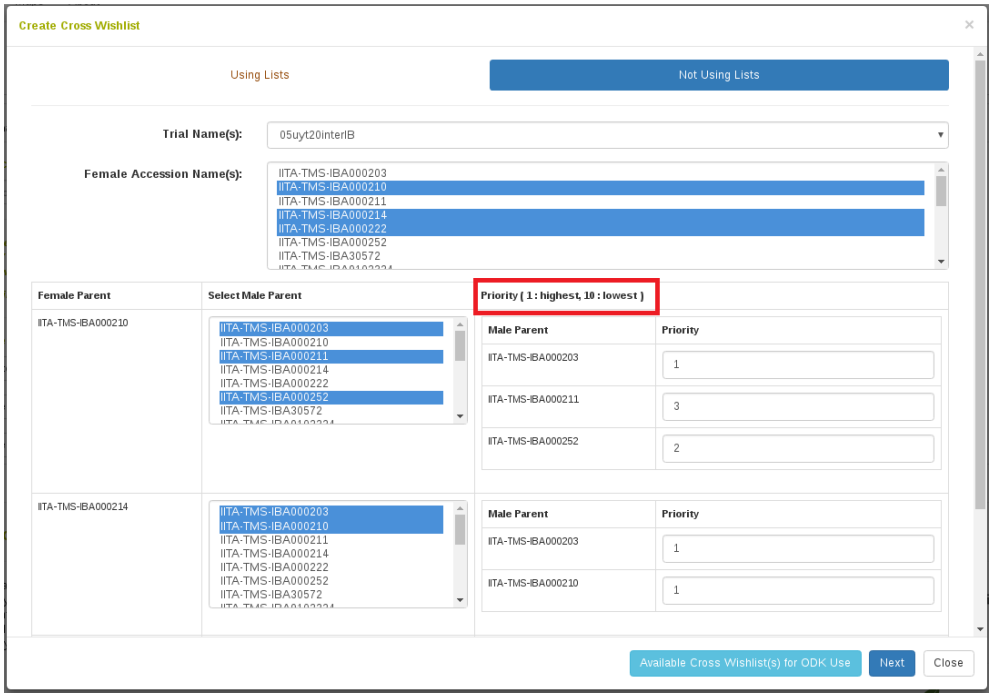

We will start by showing "Not Using Lists". First select the trial in which the crosses are to be performed. This will populate a select box with all the accessions used in that trial. From here, one or many accessions can be selected as the female accession.



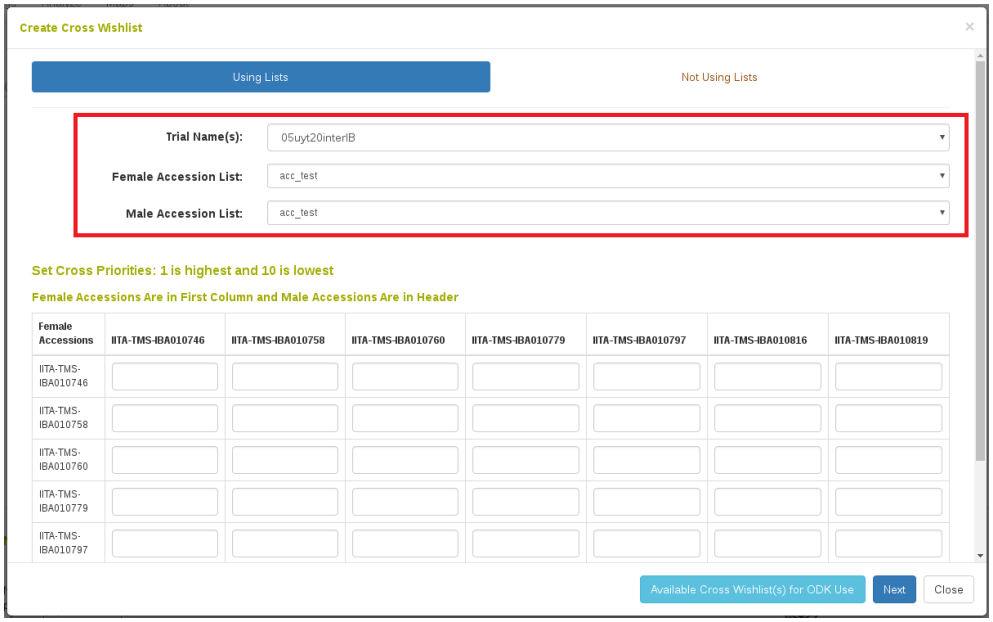

Once the female accessions are selected, a table is populated. Each row in this table begins with the female accession that was selected, followed by a select box with all the accessions used in the trial. From here, one or many accessions can be selected as the male to use in the cross.




Once the male accessions are selected to cross with each female accession, a table indicating priorities appears. Priority is meant to indicate an order in which to attempt the cross; first the highest priority male will be considered, but if this cross is not possible then subsequent males will be considered. An equal priority can be given and this will not indicate a specific order to follow.

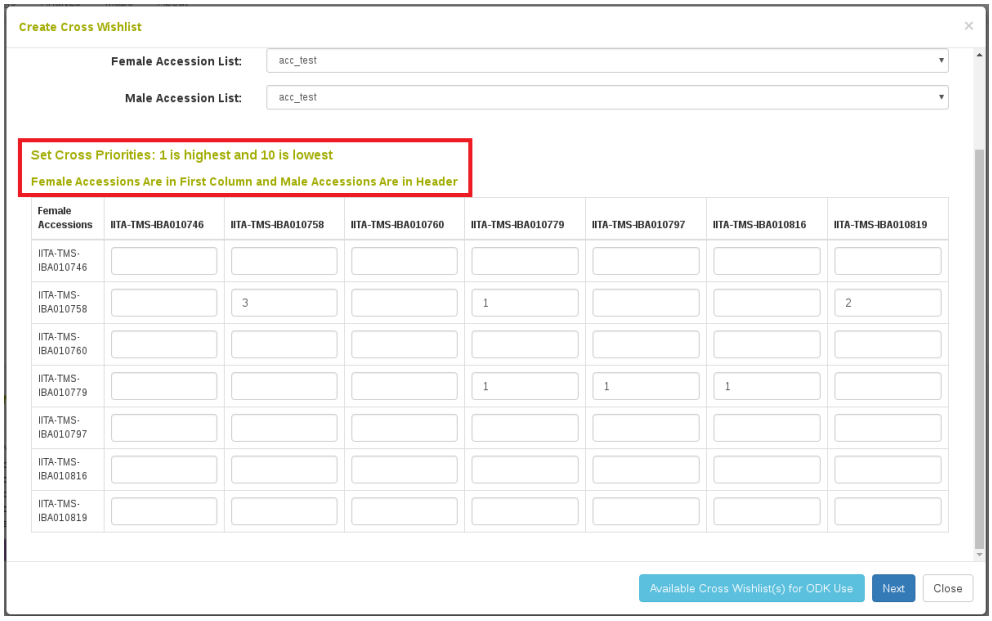



Alternatively, we could have used the "Using List" interface instead. Here we select the trial in which the crosses will be performed and we provide a list of accessions to consider for the females and the males to be crossed.

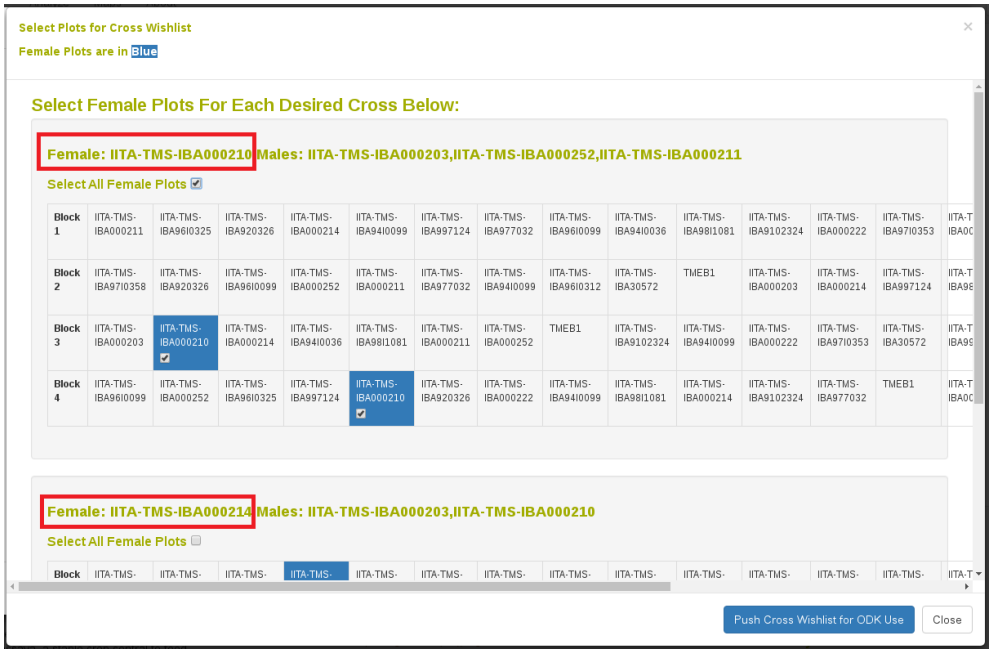



**Step 2. Select the female plots to be considered in the crosses**

After selecting your lists, the table below is populated. The first column has all the female accessions specified and the header row has all the male accessions specified. The males to consider crossing with each female are indicated with priority.




After female and male accessions are selected to cross, either by the "Nor Using List" or "Using List" interface, click Next. The next dialog will allow selection of specific female plots to use for the cross. Sections for each female accession selected will appear with the field layout displayed. Selecting all plots in which the female is present indicates that the cross should be performed on all plots where that female accession is present.

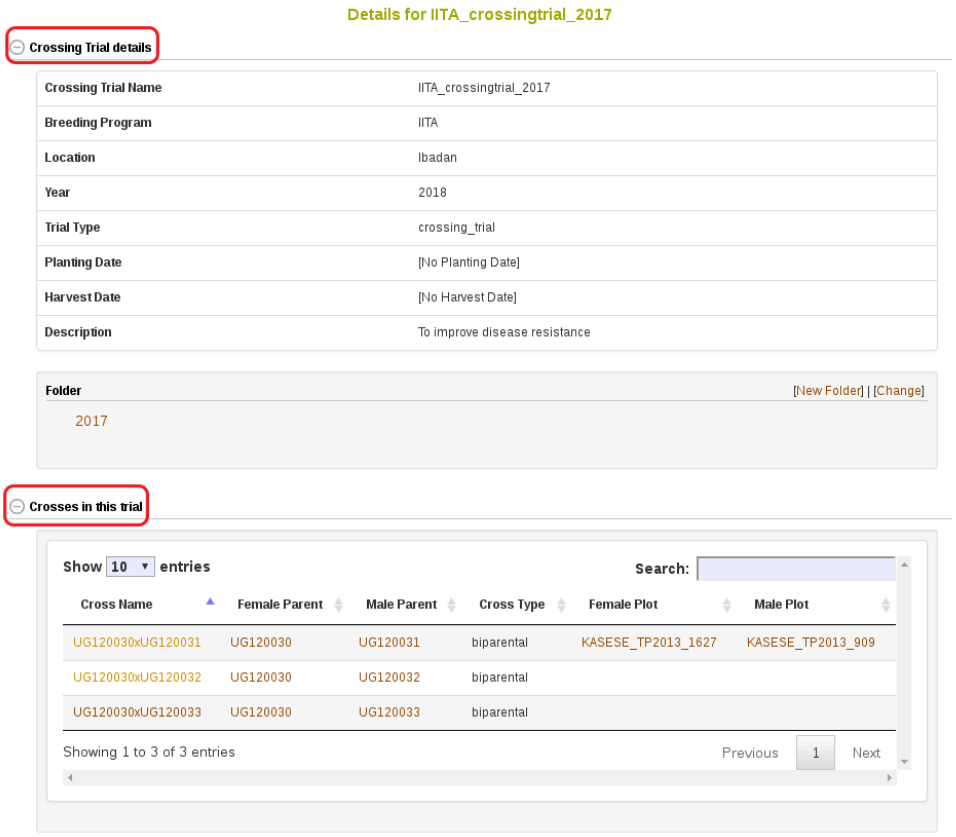


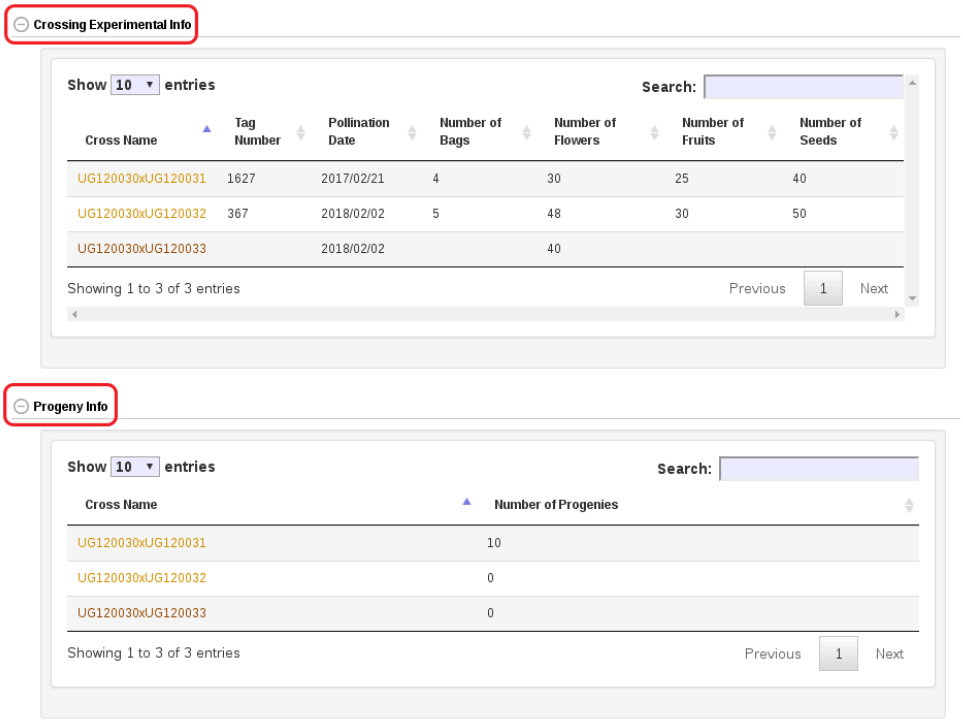

**Step 3. Transfer the cross wishlist to your mobile crossing application**

Clicking "Push Cross Wishlst for ODK Use" will send the cross wishlist plan to the ONA server for use by the mobile ODK application. Crosses can then be performed and recorded in the field using the mobile application. Afterwards, the crosses are sent back to our database and stored.

## 9.4 Crossing Experiment Detail Page

Information for crosses in the same crossing experiment is compiled in the crossing experiment detail page.

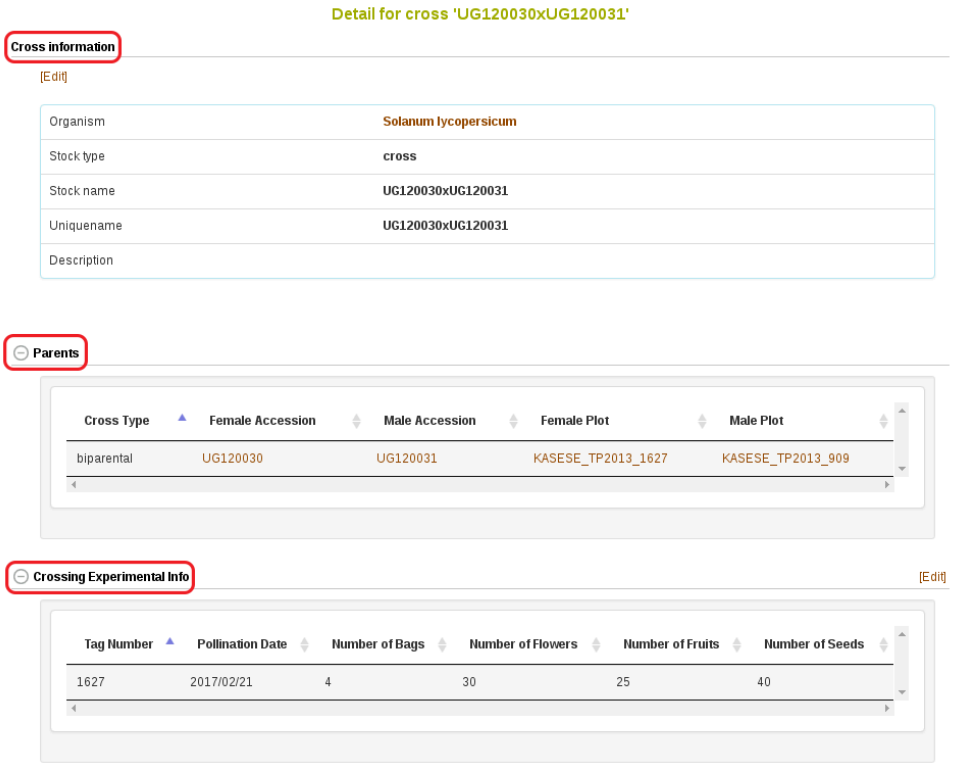

Details for IITA_crossingtrial_2017
Crossing Trial details
Crossing Trial Name IITA_crossingtrial_2017
Breeding Program IITA
Location Ibadan
Year 2018
Trial Type crossing_trial
Planting Date [No Planting Date]
Harvest Date [No Harvest Date]
Description To improve disease resistance
Folder [New Folder] | [Change]
2017
Crosses in this trial
Show 10 entries
Search:
Cross Name Female Parent Male Parent Cross Type Female Plot Male Plot
UG120030xUG120031 UG120030 UG120031 biparental KASESE_TP2013_1627 KASESE_TP2013_909
UG120030xUG120032 UG120030 UG120032 biparental
UG120030xUG120033 UG120030 UG120033 biparental
Showing 1 to 3 of 3 entries
Previous 1 Next

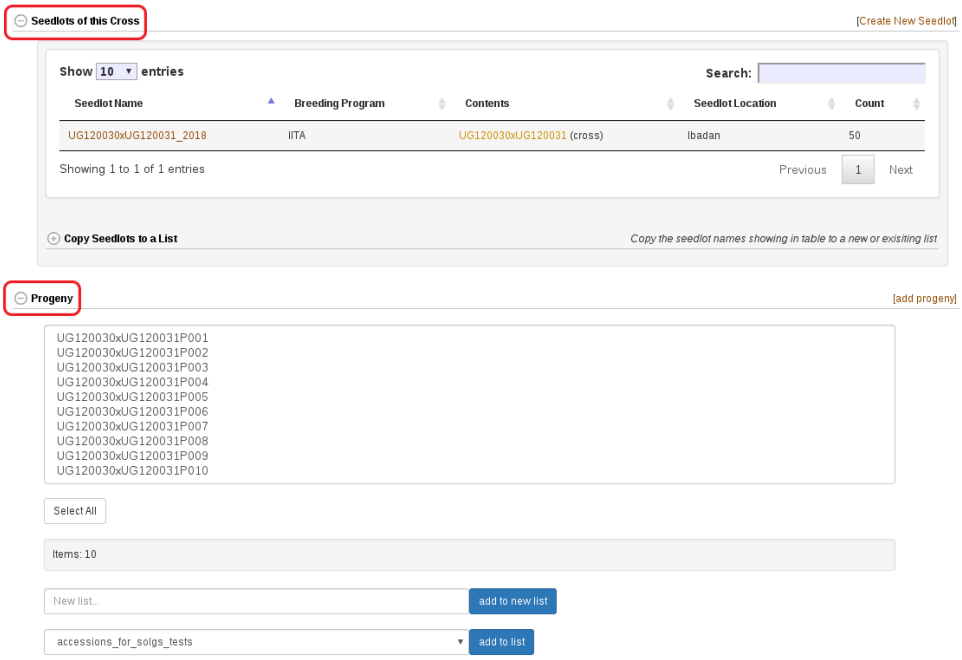



Each cross name, female parent, male parent, female plot and male plot has a link to its own detail page, which contains information specific to each one. Note: crossing experimental information is customized based on the need for each crop. As a result, the details of the information in your database may be different from the information shown in this manual.

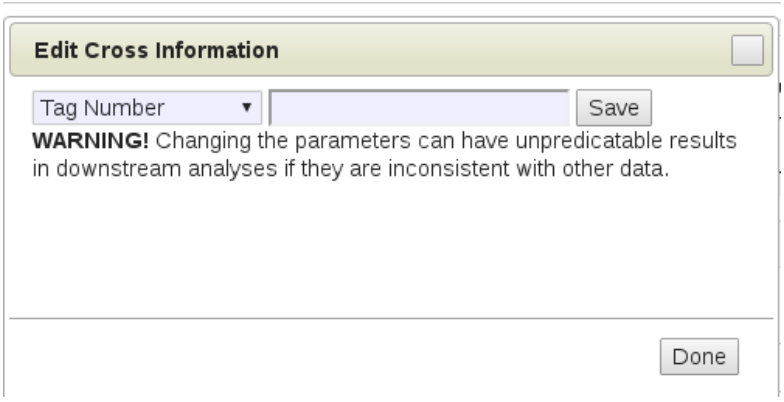
## 9.5 Cross Detail Page

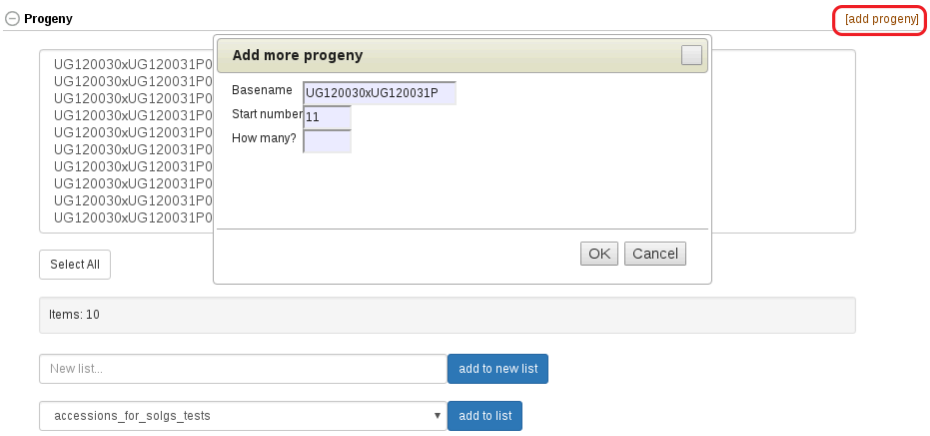
Information of each cross can also be viewed in its detail page.


Detail for cross 'UG120030xUG120031'
Cross information
[Edit]
Organism
Solanum lycopersicum
Stock type
cross
Stock name
UG120030xUG120031
Uniquename
UG120030xUG120031
Description
Parents
Cross Type
Female Accession
Male Accession
Female Plot
Male Plot
biparental
UG120030
UG120031
KASESE_TP2013_1627
KASESE_TP2013_909
Crossing Experimental Info
[Edit]
Tag Number
Pollination Date
Number of Bags
Number of Flowers
Number of Fruits
Number of Seeds
1627
2017/02/21
4
30
25
40




This page allows you to update or edit crossing experimental information and add progenies related to that cross. Note: crossing experimental information is customized based on the need for each crop. As a result, the details of the information in your database may be different from the information shown in this manual.

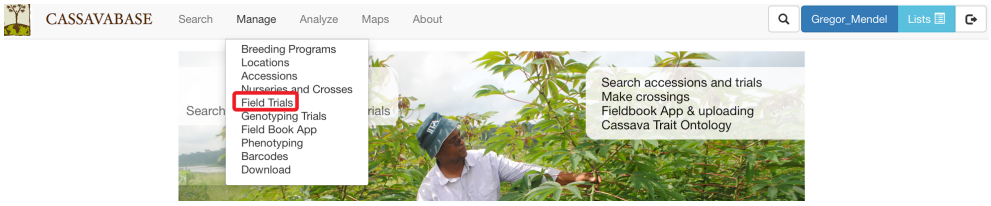



Progeny
[add progeny]
UG120030xUG120031P0
UG120030xUG120031P0
UG120030xUG120031P0
UG120030xUG120031P0
UG120030xUG120031P0
UG120030xUG120031P0
UG120030xUG120031P0
UG120030xUG120031P0
UG120030xUG120031P0
UG120030xUG120031P0
Add more progeny
Basename
UG120030xUG120031P
Start number
11
How many?
OK
Cancel
Select All
Items: 10
New list...
add to new list
accessions_for_solgs_tests
add to list

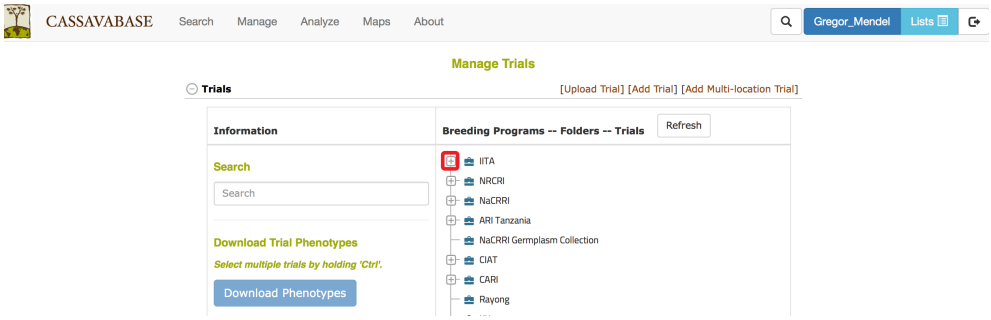
# Chapter 10

# Managing Field Trials

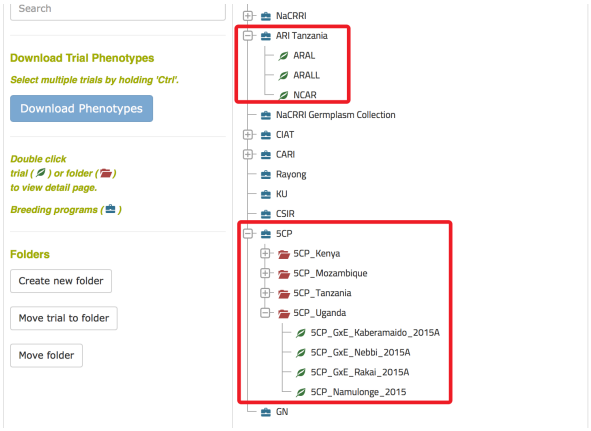
To view trial details on the database, click on the “Field Trials” link under the “manage” menu on the toolbar.




Clicking on the “Field Trials” link will bring you to the “Manage Trials” page. On this page, trials are organized according to their breeding programs. To access trial details, click on the + icon next to your breeding program.

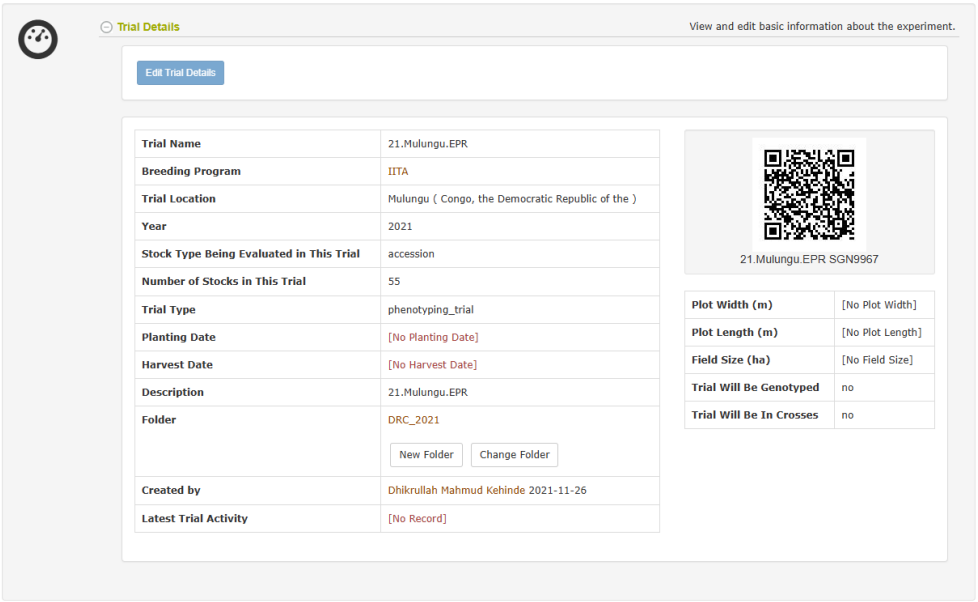



Trials can be placed directly in their breeding program. Alternatively, they can be organized by using folders within each breeding program. Clicking on trial name will take you directly to the trial details page.



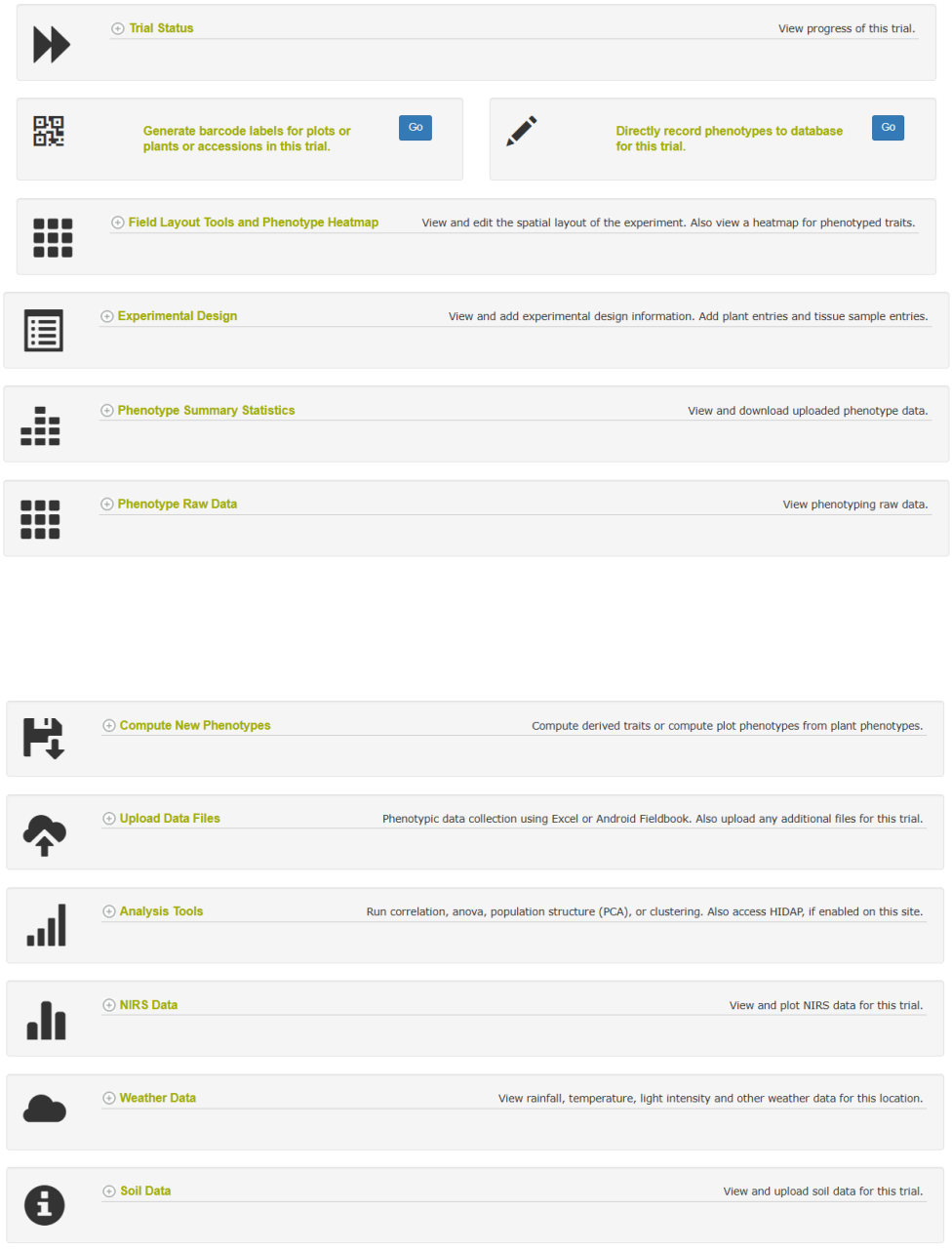

## 10.1 Trial Detail Page

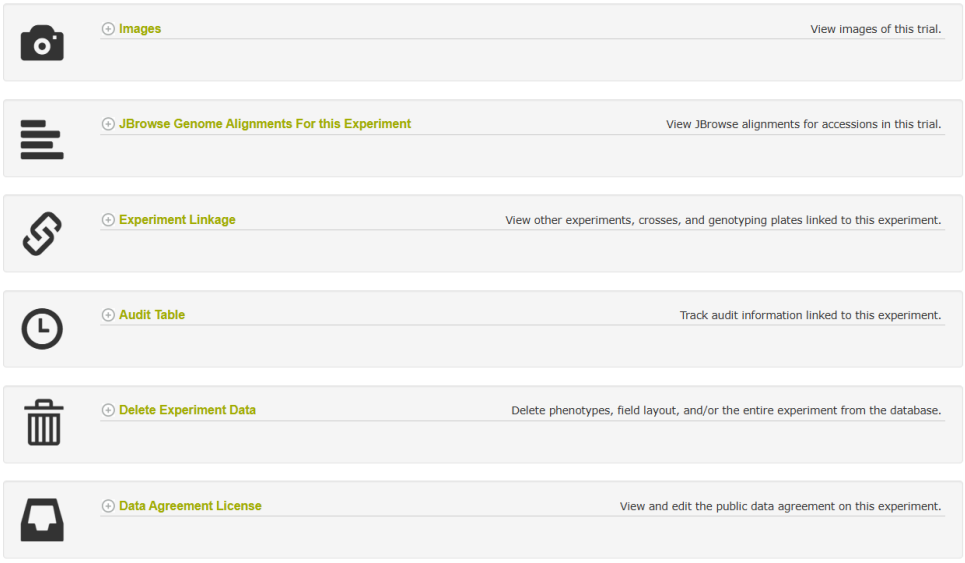
The trial detail page displays important information about individual trials including breeding program, location, year, description of the trial, design, and any files associated with that trial.



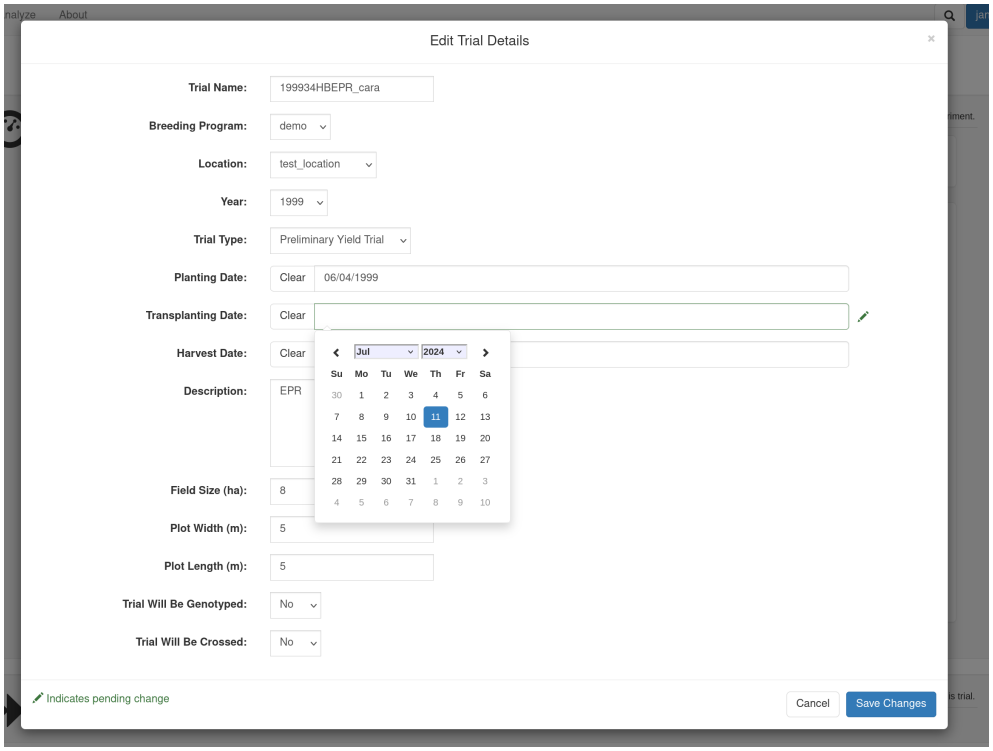

Below the trial details you will find various menus for accessing and modifying trial data. There are sections for printing labels for your plots or plants, recording phenotypes, viewing your trial layout or design, viewing

phenotypes for this trial, or conducting analyses.






The "transplanting date" field feature will only be displayed if it has a value. To add a transplanting date after creating a trial, change the show_transplanting_date parameter from 0 to 1 in the SGN config file. As a result, you will be able to add a date under the transplanting date field by clicking the "Edit Trial Details" on the trial detail page.




## 10.2 Adding Trials

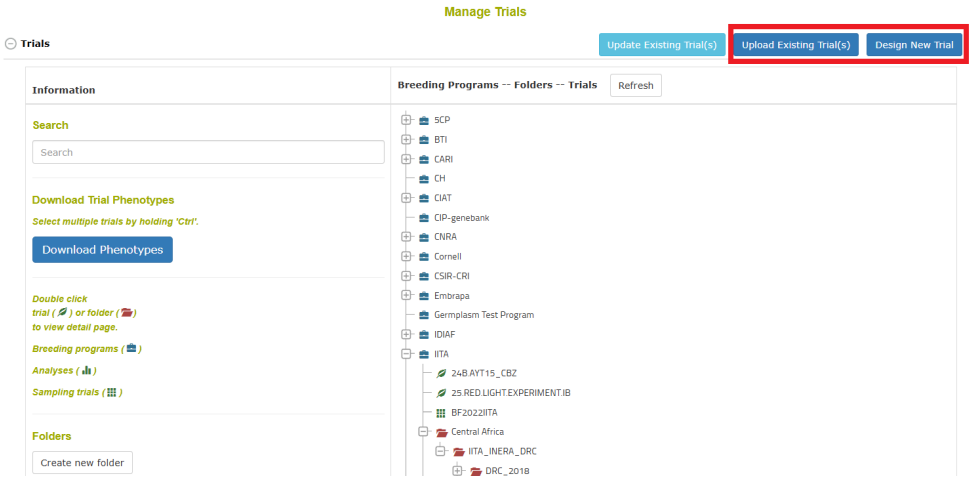
Only users with the account status of "submitter" may create trials. To learn how to change your account status from "user" to "submitter" visit section 1.2.

### 10.2.1 Prerequisites

- To add a trial, all of your accessions should already exist in the database before you begin to design a trial. If you have accessions that are not in the database, see the instructions in *Managing Accessions*.
- The breeding program and location for your trial should also exist in the database. If you need to add breeding program and/or location to the database, see the instructions in *Managing Breeding Programs* and *Managing Locations* respectively.

On the "Manage Trials" page, there are two methods to create trials: by

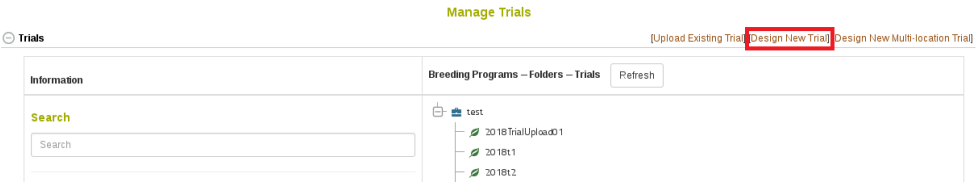
selecting "Upload Existing Trial(s)" to create a trial or trials from a spreadsheet; or by selecting "Design New Trial" and entering the trial data by hand.



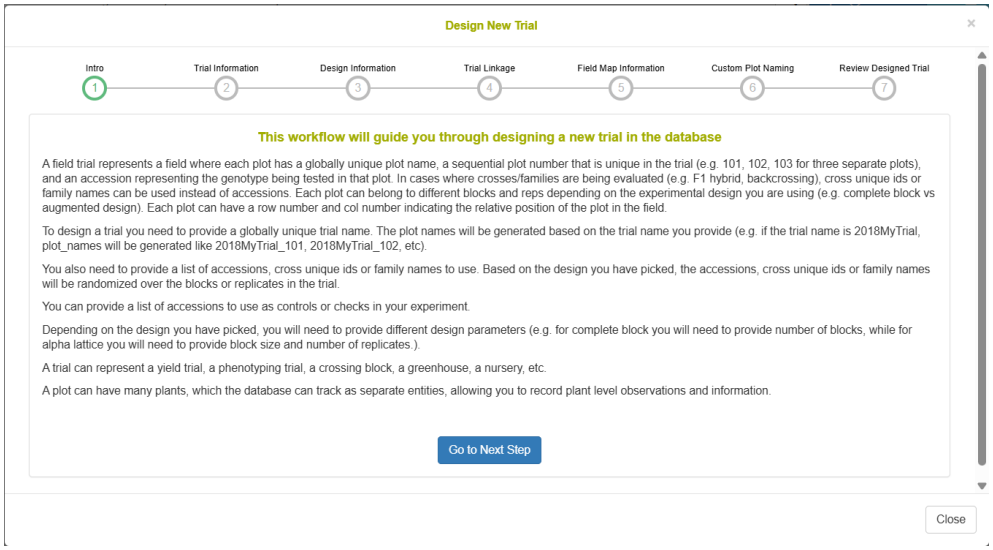

## 10.2.2 Adding a trial by using "Design New Trial" form

### Step 1. Begin the "Design New Trial" workflow

Click on "Design New Trial" to begin.



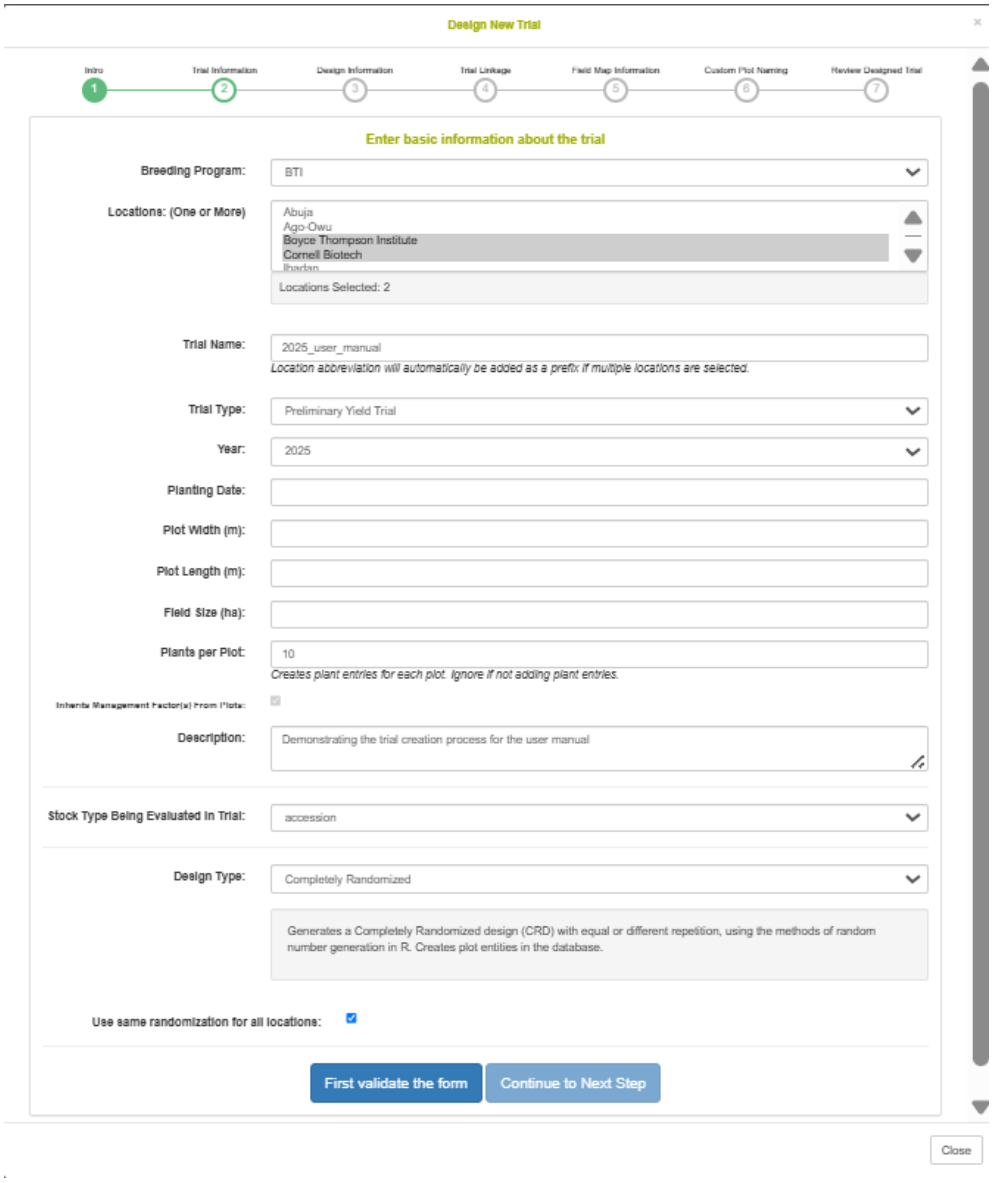

The first step in this workflow is an introduction that looks like this:




Here it gives information about what is required for a trial, including that to create a new trial, you need to create a list of the accessions that you would like to use in the trial. Lists can be viewed, created, and modified with the "lists" tool at the upper right of the screen. For more information on lists, click here.

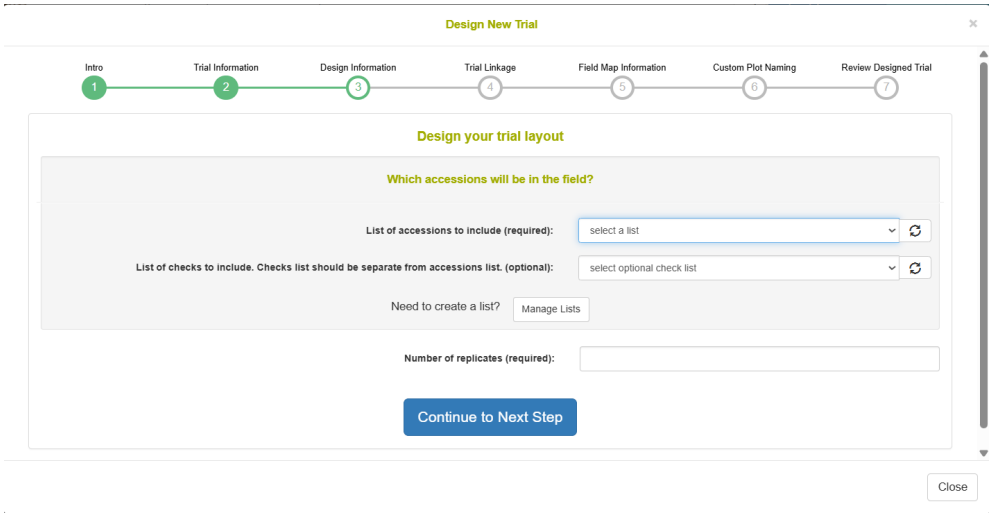
### Step 2. Enter Trial Information

On this screen you need to enter basic information about the trial, such as breeding program and location(s). You must also select a design type, such as Complete Block Design. The design is important because it influences how your genotypes are distributed and randomized over the trial. You must first click validate before proceeding to the next step.



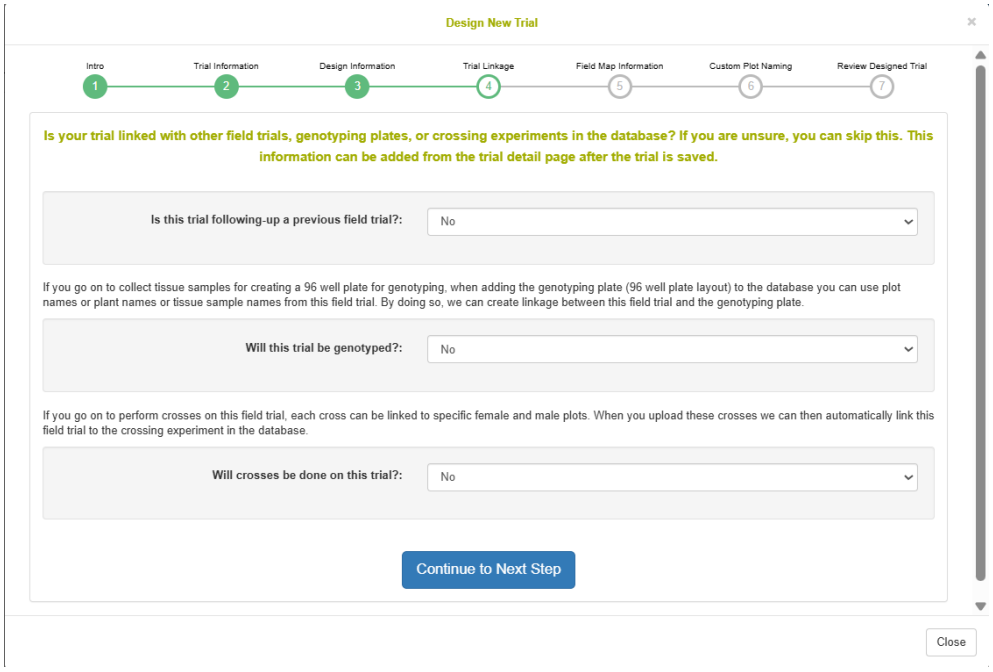

### Step 3. Enter Design Information

On this screen you need to specify a list of accessions to use in the experiment. This list must be a valid list of accessions. You must also specify all required design information, such as number of replicates. In this case, the number of blocks must be given.



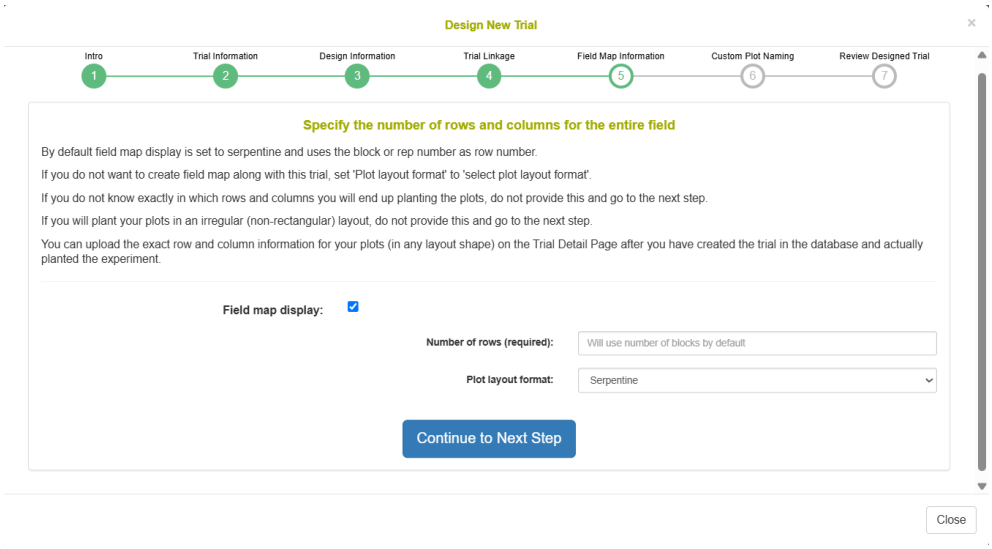

### Step 4. Enter Trial Linkage Information (Optional)

This next sections allows you to associate this new trial with other field trials, crossing experiments, or genotyping plates already present in the database. This is optional, and can be completed at a later date from the trial detail page.



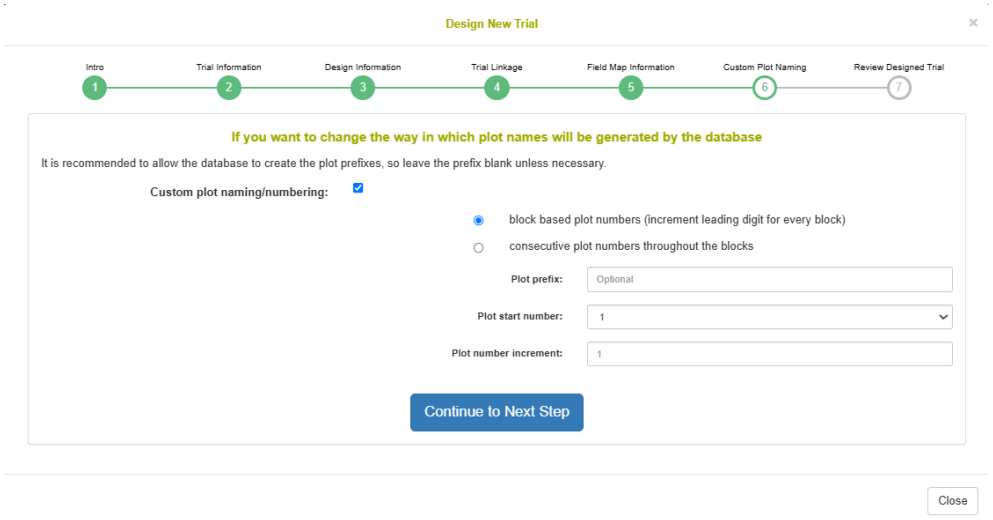
### Step 5. Enter Field Map Information (Optional)

On this screen you can specify how the row and column numbers will be generated for the plots in the trial. The row and column number represent a relative position of the plot in the field. If you are not exactly sure of how you will plant the plots in the field or you have an irregular (non-rectangular) layout, you can skip this step for now. This information can be added on the Trial Detail Page once the trial is saved in the database in order to reflect exactly how the plots were planted in the field.




### Step 6. Custom Plot Naming (Optional)

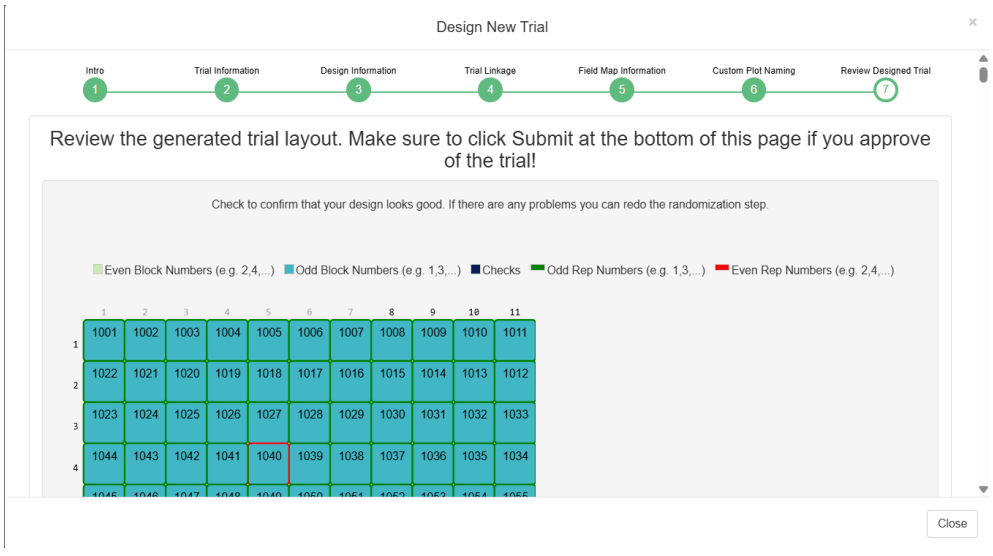
On this screen it is possible to change the format in which plot names will be generated for your trial. It is recommended to skip this step and just use the format generated by the database by default.




### Step 7. Review Designed Trial

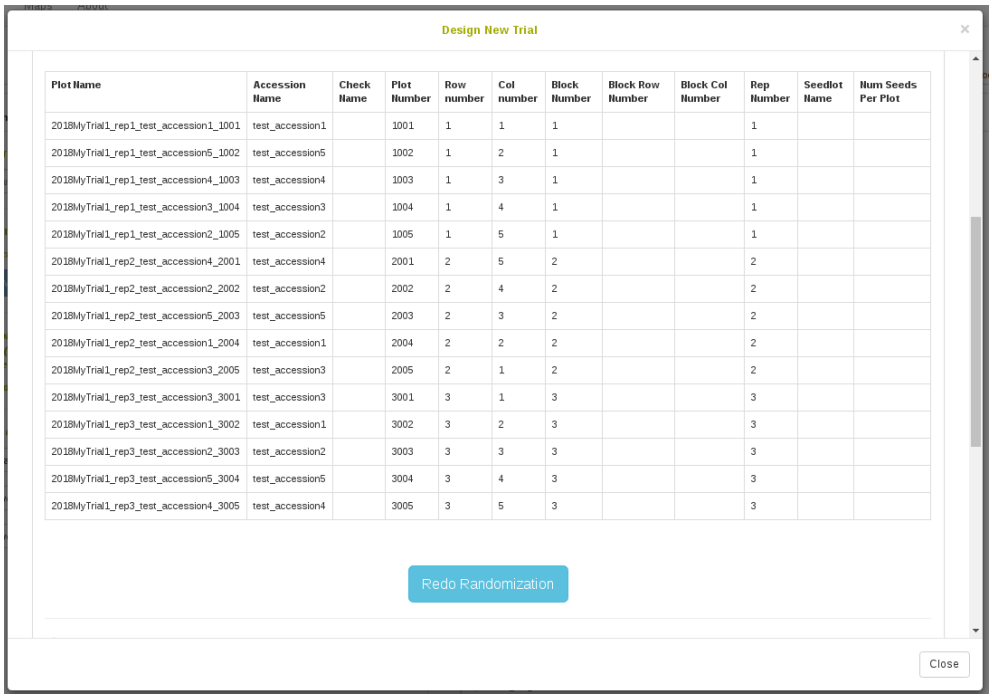
On this screen you can review the trial that the database has generated.

You will see a graphical representation of the trial. The numbers on the squares represent the plot_number of each plot and on mouse hover you can see further information about the plot.

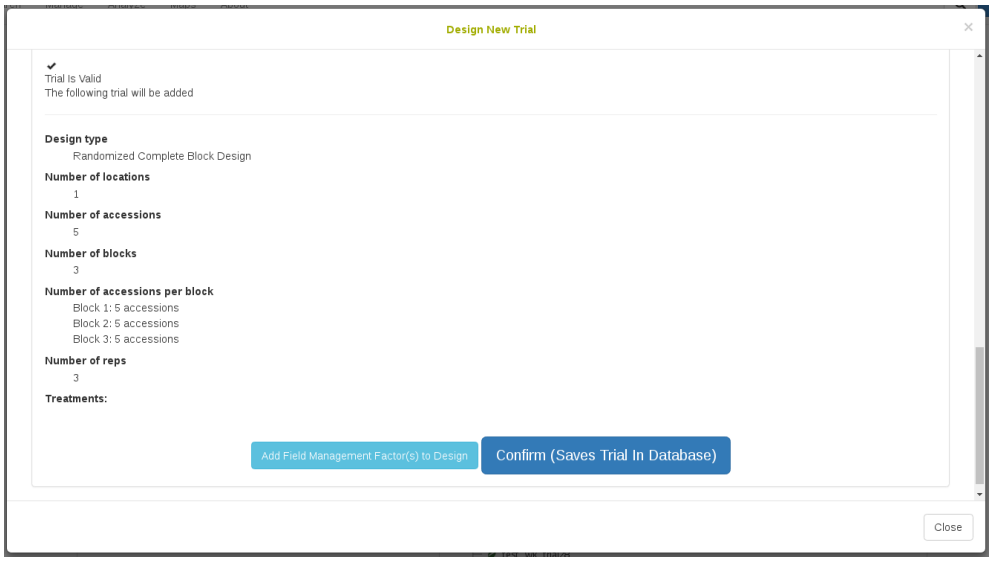



You will also see a table representation of all the plots and their information.

If you want to redo the randomization, you can click the "Redo Randomization" button.

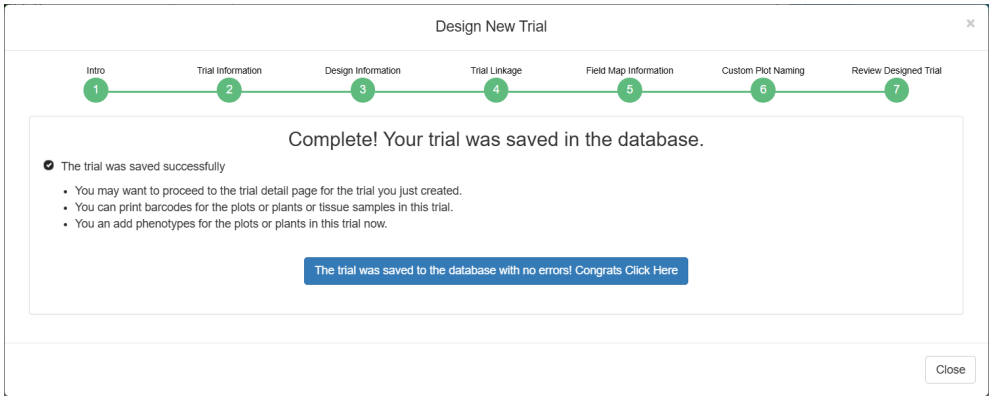



At the bottom there is a brief summary of the trial followed by two buttons.



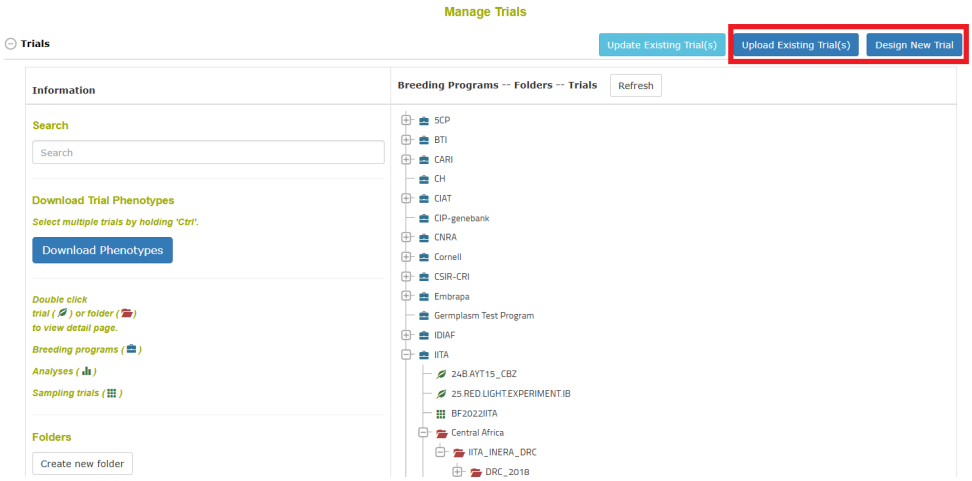
### Step 7. Saving new trial in the database

Once you are done reviewing the trial you can click "Confirm" to save the generated trial into the database. Once the trial has saved you will see the final completion screen:




## 10.2.3 Adding a trial from an uploaded file

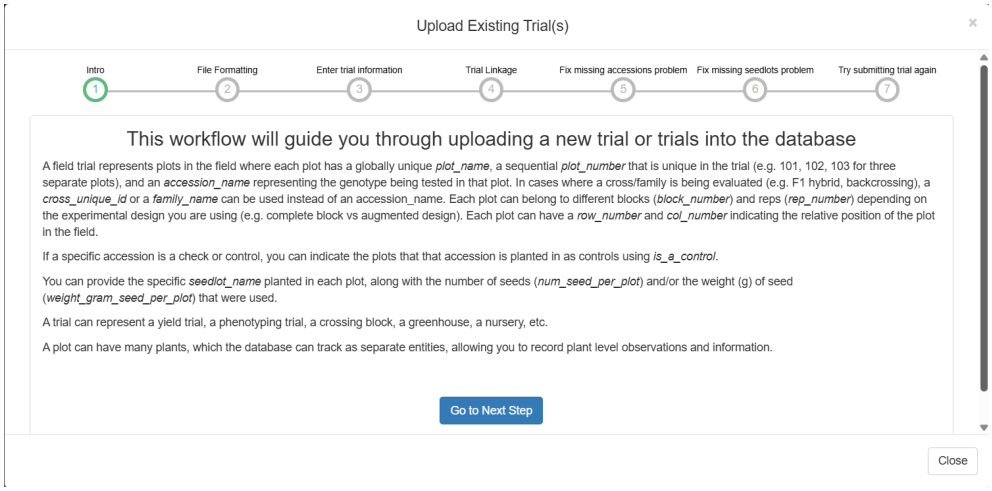
If you already have trial design layout in a spreadsheet, you can add your trial into the database by using the "Upload Existing Trial(s)" button on the *Manage Trials* page.



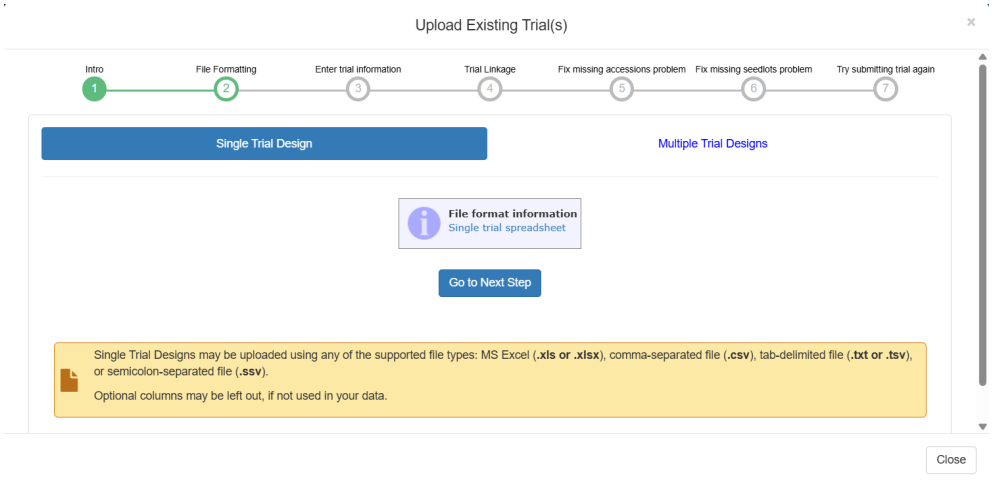

Enter the information in the workflow to upload a trial from a spreadsheet.

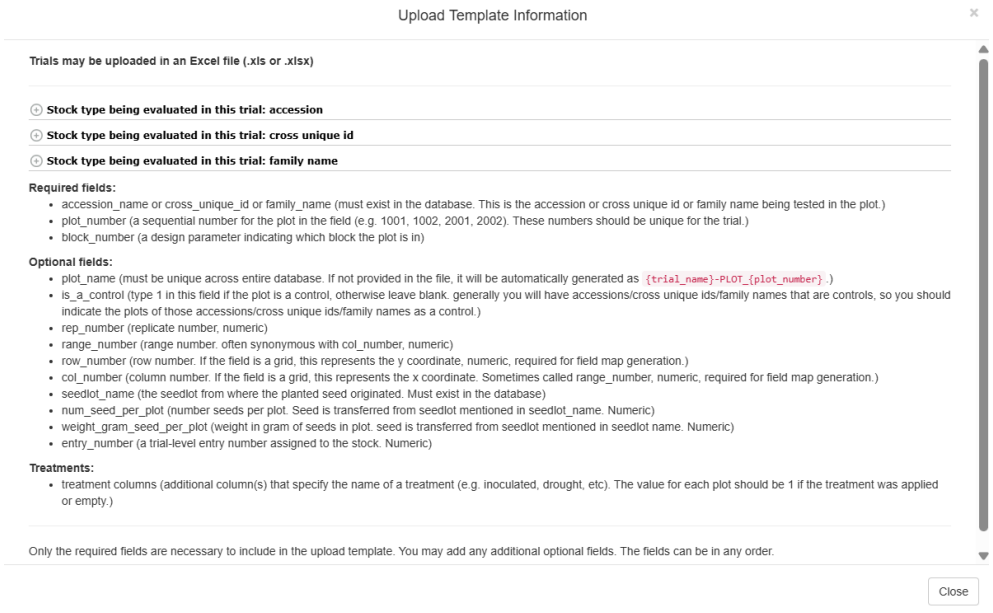
**Step 1:**

The first step is to understand what the format of the trial upload is. It is important to understand that the field layout represents plots in the experiment. Each plot has a globally unique plot_name, a sequential plot_number that is unique in the trial (but not globally unique. e.g. 101, 102, 103 for three separate plots), an accession_name representing what genotype is planted in that plot, and a block_number representing design replication. Each plot can be thought of as having a row_number and a column_number representing the relative position of the plot in a grid (e.g. the top left plot is row 1 column 1 following by row 1 column 2). Each plot can be planted with an amount of seed from a seedlot, where the seedlot_name represents the specific seed packet that was used, and num_seed_per_plot and weight_gram_seed_per_plot represent amount that were transferred from the seedlot_name to the plot_name. Treatments can be applied onto plots using additional column names in your file, where a 1 represents if the treatment was applied to the plot and an empty cell means it was not applied.




The following page will allow you to pick a file for upload, including uploading multiple trials at once. On this page you can also inspect the file requirements for both single trial and multi-trial uploads.




## Minimum File requirements

- All accession names in the file must exist in the database. See adding accessions for more information.
- The uploaded file can be excel (.XLXS or .XLS), comma-separated values (.CSV), tab-separated values (.TSV) or semicolon-separated values (.SSV).

- The first row (header) must contain the column names: plot_name accession_name plot_number block_number is_a_control rep_number range_number row_number col_number seedlot_name num_seed_per_plot weight_gram_seed_per_plot entry_number
- Only accession_name, plot_number, and block_number are required.

Minimal Example:

| plot_name | accession_name | plot_number | block_number | is_a_control | rep_number | range_number | row_number | col_number | seedlot_name | num_seed_per_plot | weight_gram_seed_per_plot | entry_number |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2018plot1 | accession1 | 101 | 1 | | 1 | | | | | | | |
| 2018plot2 | accession2 | 201 | 2 | | | | | | | | | |
| 2018plot3 | accession1 | 102 | 1 | | 2 | | | | | | | |
| 2018plot4 | accession2 | 202 | 2 | | 1 | | | | | | | |

When uploading multiple trials, the requirements are the same, but with additional headers to differentiate trials in the same file: trial_name breeding_program location year transplanting_date design_type description trial_type trial_stock_type plot_width plot_length field_size planting_date harvest_date. Additionally, when uploading multiple trials, you may choose to get an email when the upload is complete (see section 10.2.4 )

### File validation

- In case of errors in the uploaded file such as missing or invalid data, a window will appear listing the specific errors in the file that must be corrected before a successful upload.

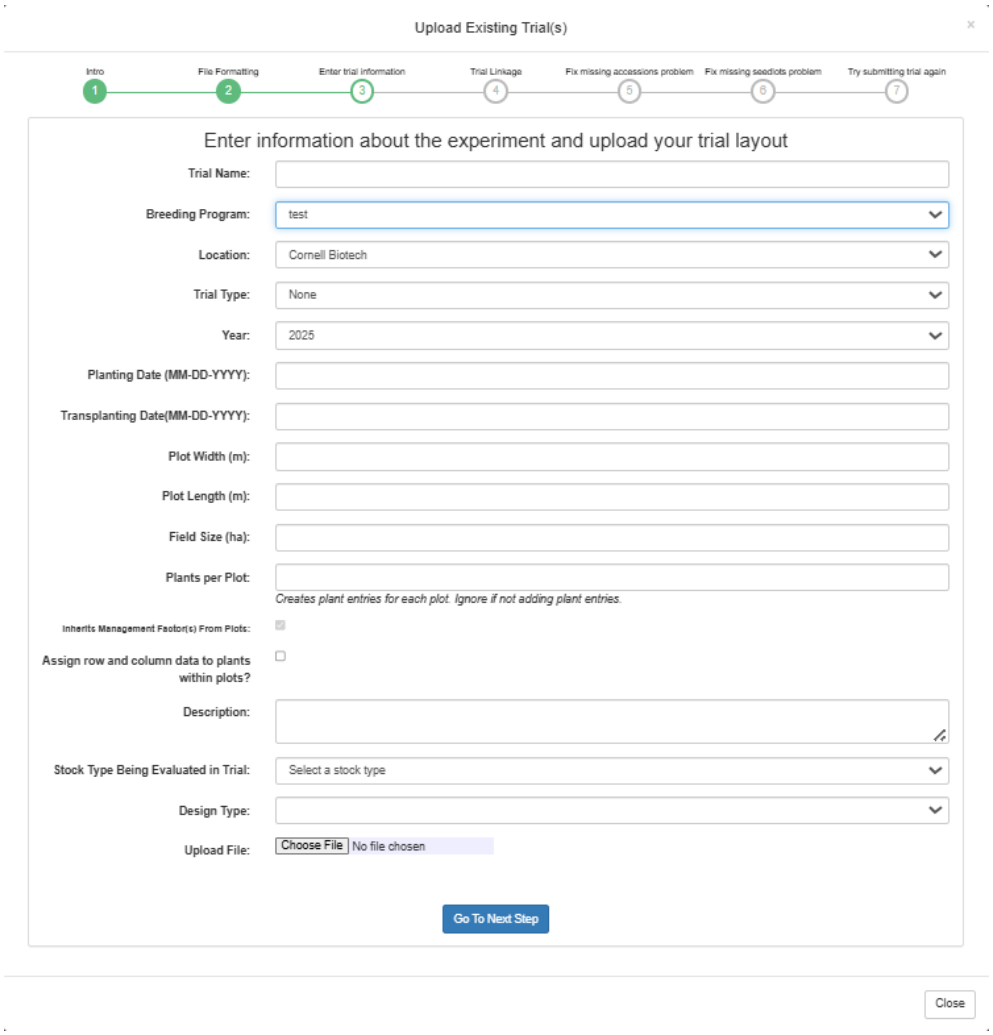
### Uploading a trial with Treatments

- You can upload a trial with treatments by adding additional column(s). The column header will be the treatment e.g. fertilizer, watering regime, inoculation, etc. and the values in these columns will be either 1 or empty, indicating that the treatment was applied to the plot or not.

### Step 2:

Once you feel that your experiment field layout is in the right format, click on to the Next Step. You will see the following form which must be filled in

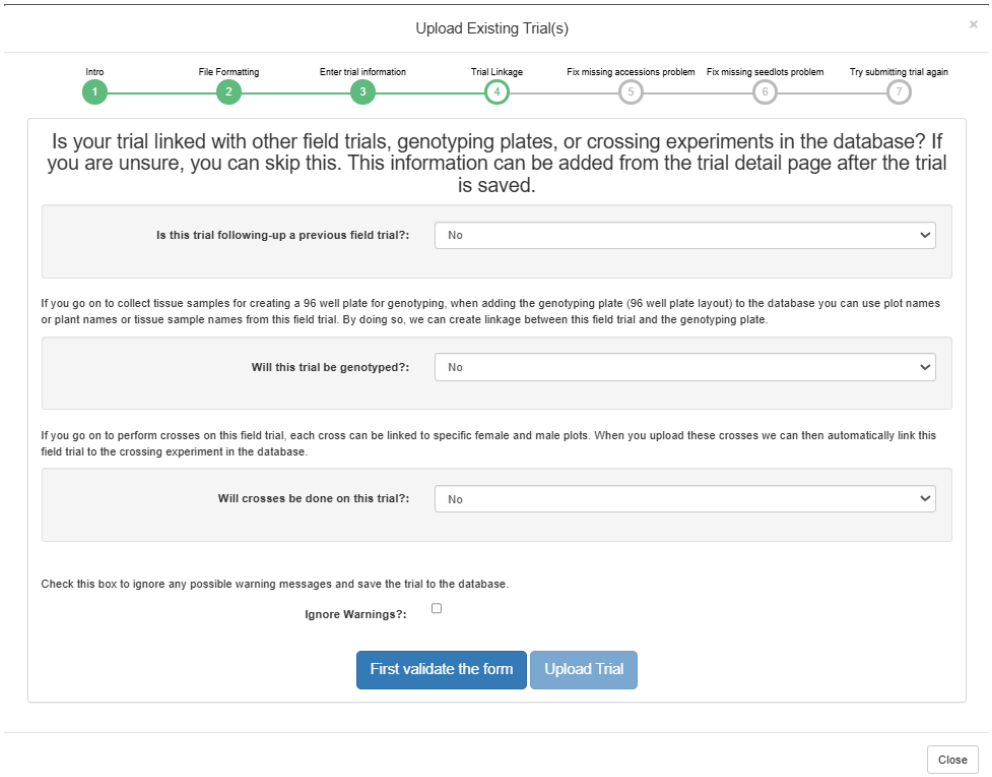
completely:




The trial name must be globally unique in the database. Please try to follow standard naming conventions for your group.

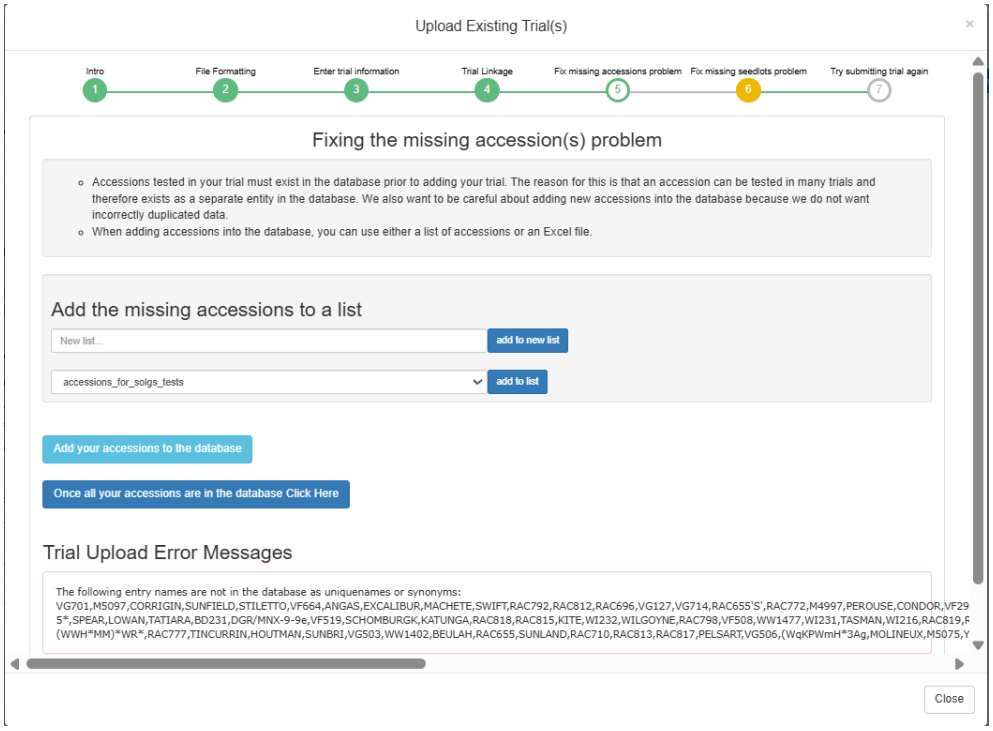
### Step 3:

Go to the next page where you can link this trial to other projects. Validate the form, and then you can click “Upload Trial”.




**Step 4:**

In the case where you have uploaded an experiment using accession_names that are not already present in the database, you will be taken to this screen. If the accession_names in your file are all already in the database, this step will be skipped. The reason it is necessary for your accessions to be in the database before you can add a trial using them is that a single accession can be used among many trials and therefore must exist as a separate entity in the database; because of this it is also very important to be careful about adding wrongly duplicated accession_names into the database. From this screen it is possible to make a new list with the missing accession_names and then click "Add Accessions to the database" to immediately resolve the issue. Once all your accessions are in the database, click to move to the Next Step.



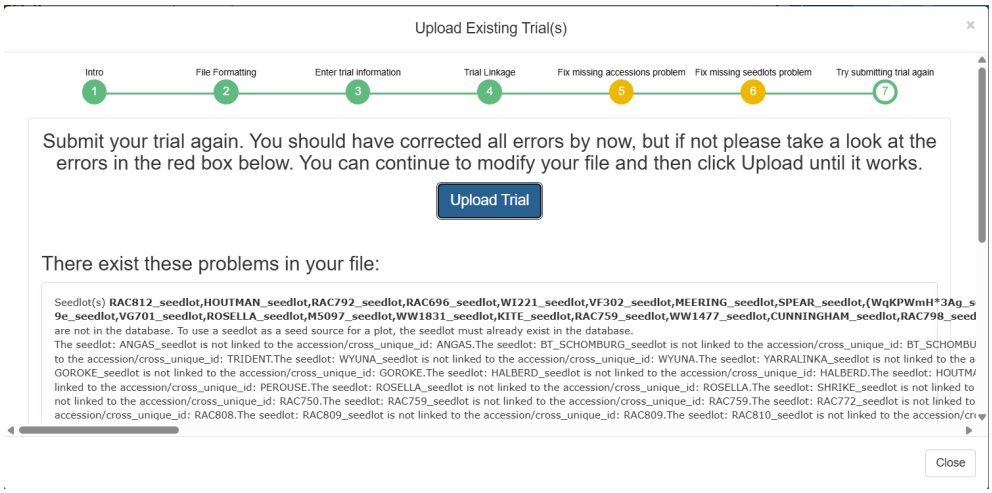

**Step 5:**

In the case where you have uploaded an experiment using seedlot_names that are not already present in the database, you will be taken to the next screen. If the seedlots in your file are all already in the database, this step will be skipped. The reason it is necessary for your seedlots to be in the database before you can add a trial using them is that a single seedlot can be used among many trials and therefore must exist as a separate entity in the database. Once all your seedlots are in the database, click to move to the next step.

**Step 6:**

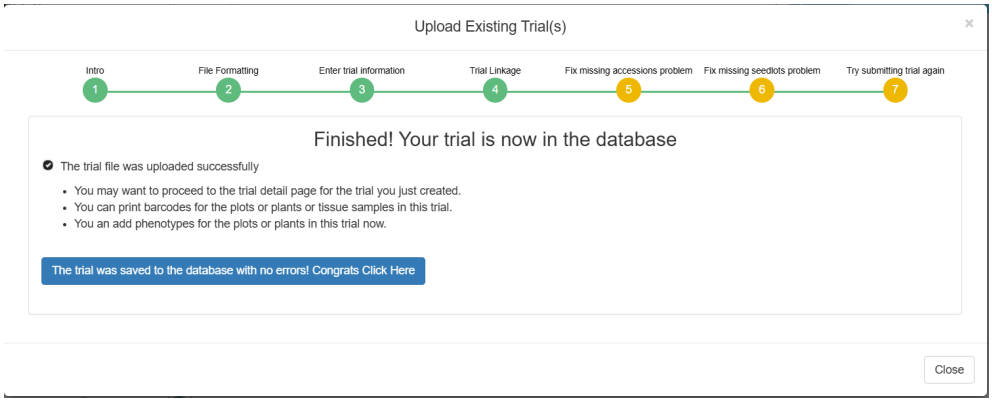
If there are any other errors with your file, such as if the plot_names are not globally unique in the database or your plot_numbers are not unique in your trial or row_number is not an integer or any other error, you will see the errors listed in the red box. It is up to you to correct these errors in your file. Simply open up the file you selected earlier in Excel and correct

the issues and then save the file. Then you can click "Submit Trial" and it will resubmit it for you. You can continue to edit your file here and submit as many times as you need until it is accepted.




### Completion screen

Whether you were lucky enough to submit your trial successfully on Step 2 or if you tried many times on Step 5, once your trial has been saved in the database you will see the following screen:




## 10.2.4 Multi-location trials

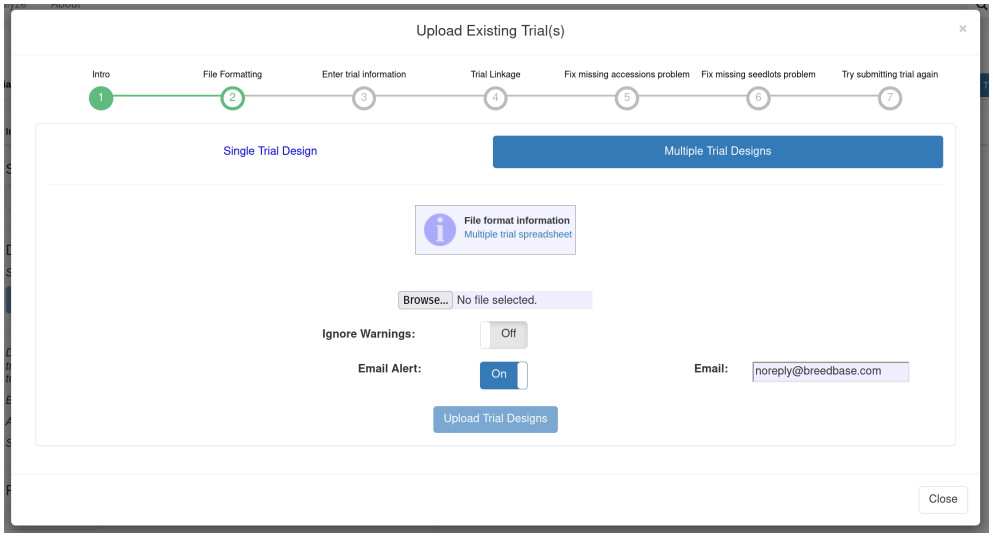
To add multi-location trials, simply select the multiple locations while using the 'Add Trial' form.

This will create a separate trial for each selected location, but they will share the same design and will be grouped in a single folder.

By default each trial design will have a fresh randomization, but if desired you may check the "Use same randomization for all locations" option.

### Email alert for multiple trial design upload

When uploading multiple trials from a file, you have the option to receive email notifications by clicking the "Email Alert" checkbox. By default, the system will use the email address associated with your account, but you have the option to enter a different email address if you prefer. After submitting, the upload process runs in the background, allowing you to continue using the interface without interruptions. Once the process completes, you will receive an email with the upload results.

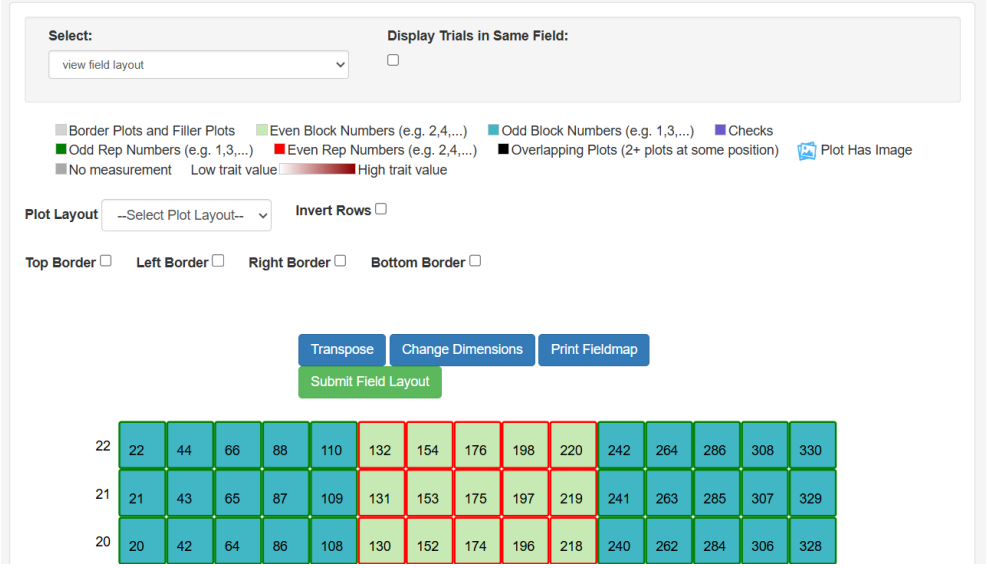



## 10.2.5 Viewing Plot Layout and Trait HeatMap

### 10.2.5.1 Viewing plot layout

In the "Field Layout Tools and Phenotype Heatmap" section of a Trial Detail page, the trial physical layout is displayed by default. The relative position of the plots will be displayed based on the row and column positions given to the plots during the trial creation or upload steps. The plots are color-coded

based on the plot's rep and block numbers and whether or not it is used as a check. Hover the mouse over the plot to see details about a specific plot.




#### 10.2.5.2 Viewing plot layout for multiple trials

If there is more than one trial grown in the same physical field, the trial layouts of all of the trials can be shown together if the trials share these properties:

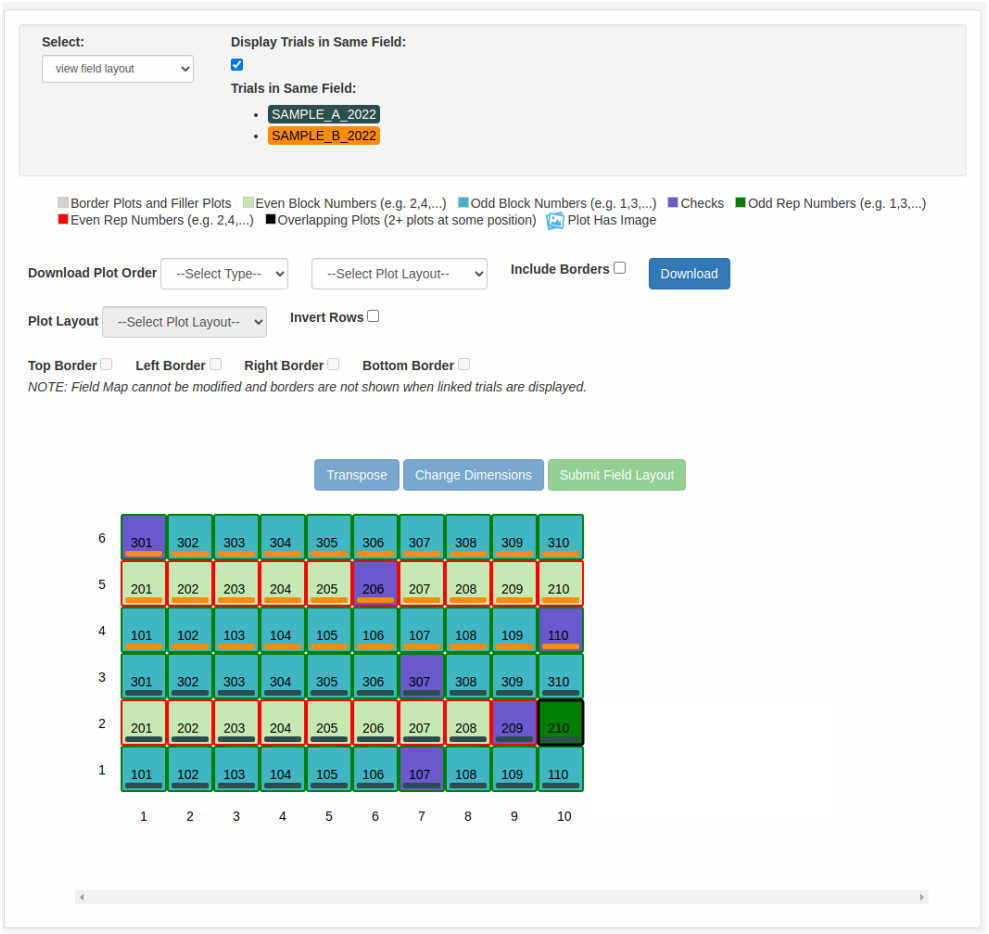
Each trial has the same year

Each trial has the same location

The location type of the trials' location is set to Field

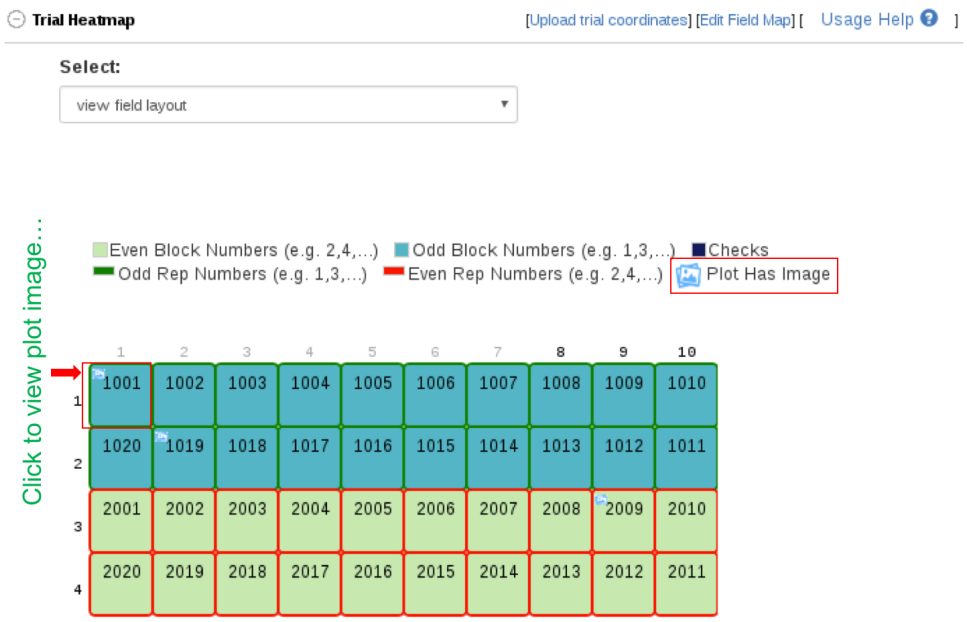
The row and column positions of all of the plots (across the related trials) don't overlap. For example, trial #1 starts at row 1 and trial #2 starts at row 10.

When these conditions are met and you check the "Select Trials in Same Field" checkbox, the plots from all of the related trials will be displayed on the same field layout. The plots will be color-coded by trial. The planting order and harvest order downloads will include the plots from all of the displayed trials in the order in which the plots occur in the field.



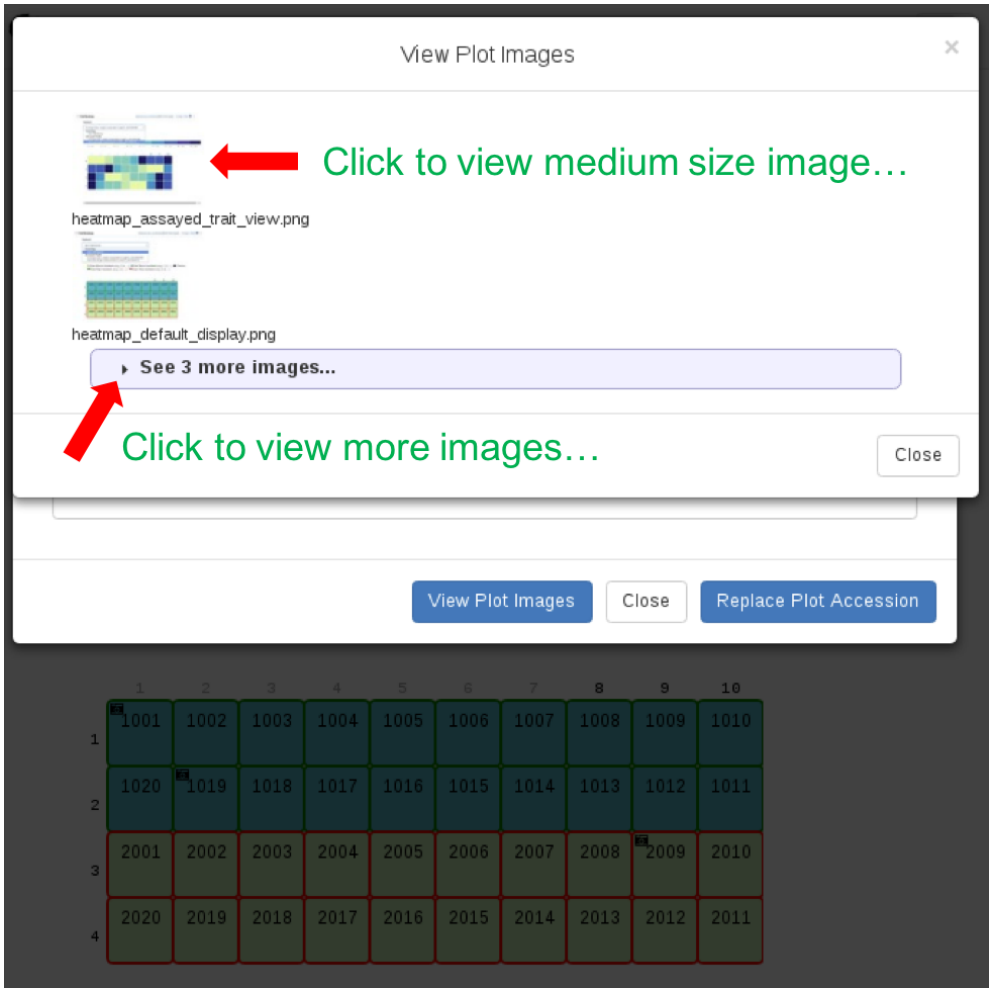

### Tracking plot images on fieldMap

Plot images can be seen on fieldMap if a plot is associated to any image.



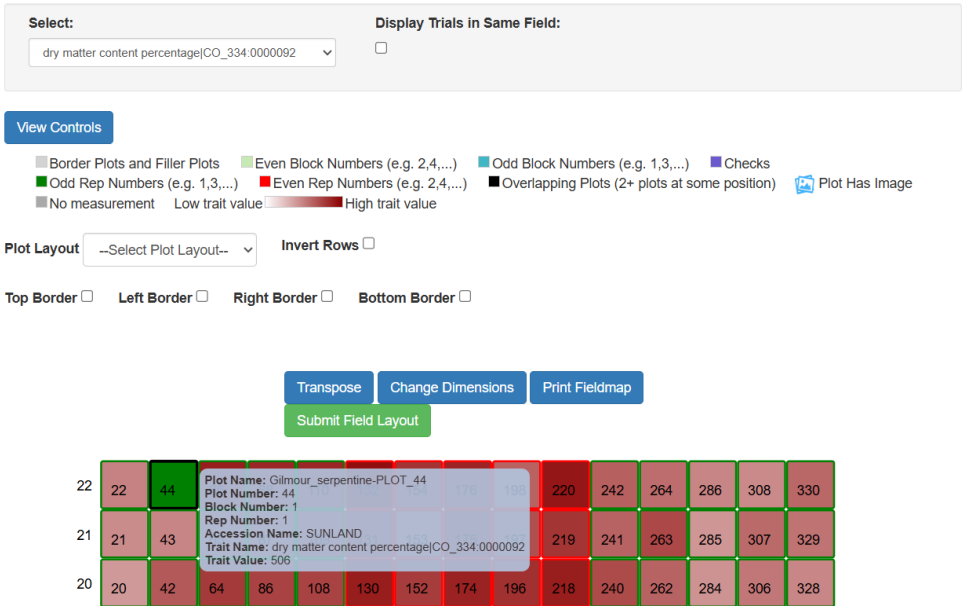

To view plot image(s), click on a plot, a dialog will appear. In this dialog you will see a detailed overview of the plot, including what stock(s) are contained within it and their layout in the plot, if applicable. At the bottom of the dialog, you will see options to change the plot name or accession as well as view plot images.

On the resulting dialog, click on "View Plot Images." To see more images if a plot has more that 2 images, click on See more images... Medium size of an image can be viewed by clicking on an image.



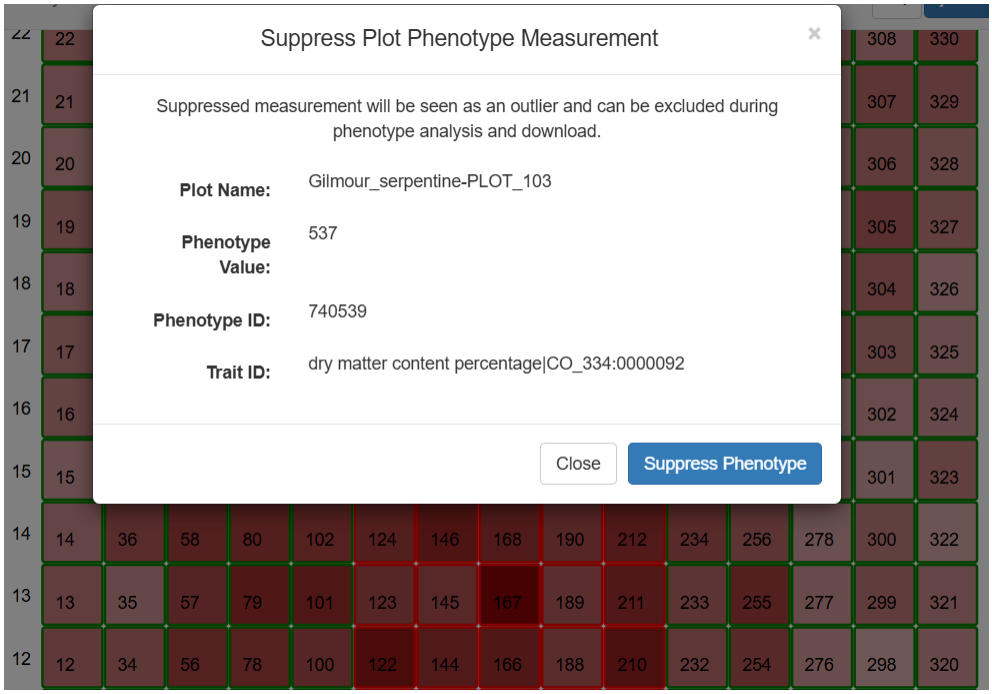

### Viewing assayed trait heatmap

The phenotype heatmap can be viewed by selecting a specific assayed trait from the selectbox drop-down. Mousing over the plots highlights the plot in green and also displays the plot's field information including the selected trait's phenotype value.



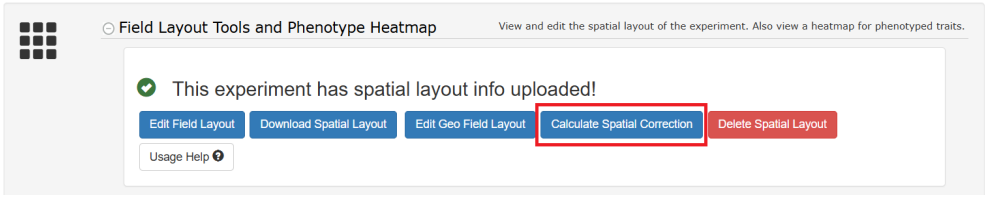

### Suppressing Plot Phenotype

Clicking on a plot on the heatmap would display a dialog that has a button for suppressing a plot phenotype value for a given trait. A suppressed plot value can be excluded during trial analysis and phenotype download.




#### 10.2.5.3 Correcting spatial autocorrelation

For trials with spatial layout information and stored phenotypes, you can check and correct for spatial autocorrelation by clicking the "Calculate Spatial Correction" button above the field map. Doing so will open the spatial corrections dialog. For an in-depth review, visit the chapter on *data analysis tools*.

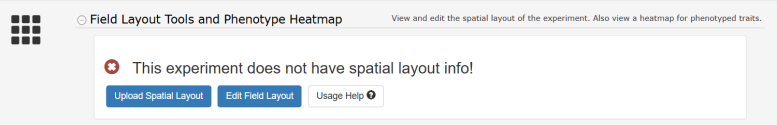



### 10.2.6 Adding additional information in the *Trial Detail* page

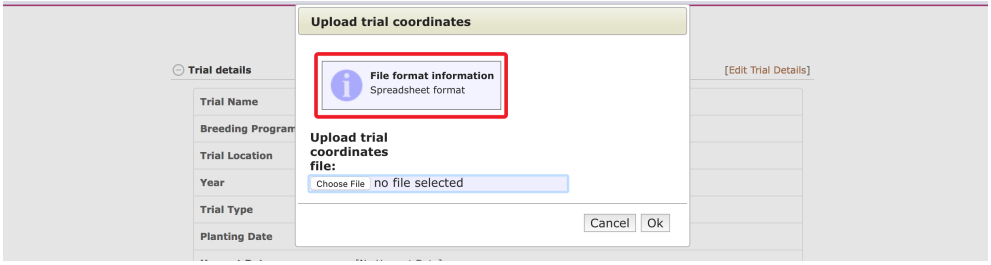
After you added a new trial to the database, you can edit trial details or add more information for that trial through the *Trial Detail* page.

### Uploading Physical Trial Layout

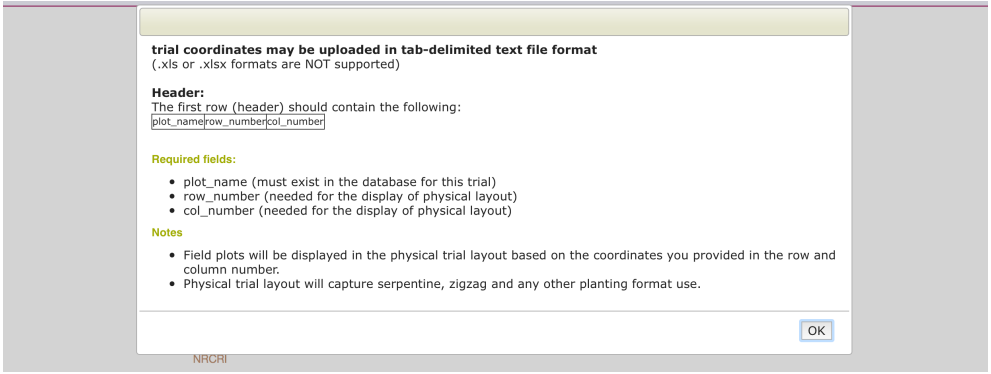
You can upload physical trial layout by clicking on the "Upload Spatial Layout" button on the *Trial Detail* page.




Please check file format carefully. You can find file format information by clicking on the "Spreadsheet format" on the "Upload Spatial Layout" window.




Spreadsheet format:



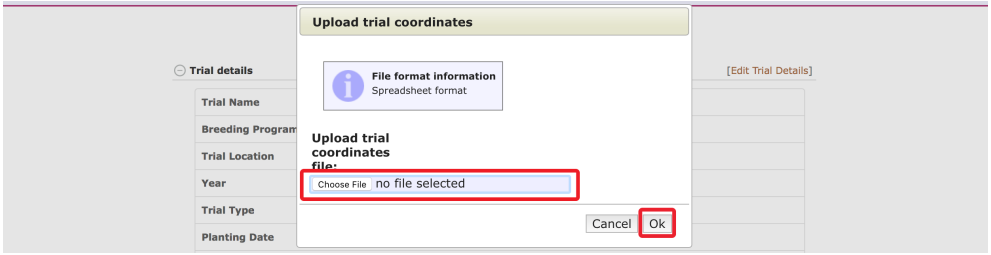

### Physical Trial Layout File requirements

- All plot names in the file must exist in the database.
- The uploaded file should be tab delimited (txt).
- The first row (header) must contain the column names

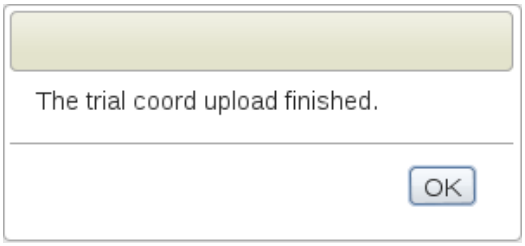
Example:

| plot_name | row_number | col_number |
|---|---|---|
| plot1 | 1 | 1 |
| plot2 | 1 | 2 |
| plot3 | 1 | 3 |

Select the trial layout coordinates file that you want to upload for this trial, then click "OK" button to upload the file.

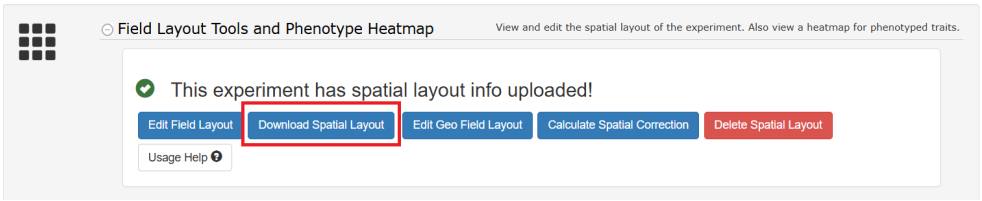



The following message is displayed after the coordinates are uploaded.



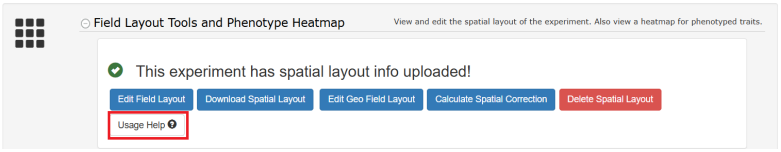

### Downloading Field Map Spreadsheet

Field map spreadsheet can be downloaded if the trial has field coordinate (row and column numbers) uploaded for it plots. To download, click on the Download FieldMap Layout link on the Trial Heatmap section.



## Editing Physical Trial Layout

The "Usage Help" button contains information on how to edit physical trial layout.




How to Use and Edit Field Map

### Background:

Field map is a tool that enable users to view the physical layout of plots in a trial. Maps can be generated on the fly while adding a trial or uploading a trial. Field map coordinates can also be uploaded independently after trials have been added or uploaded. It is a very intuitive, flexible and user friendly tool for manipulation/making changes to field trial layouts before phenotypes are uploaded.

### Editing Options:

**Replace Plot Accession**

A plot accession can be replaced by an accession within or outside of the trial. To do this, **click on the plot** and **provide the name of the new accession** (must already exist in the database).

**Replace Trial Accession**

An accession used in a trial can be replaced by a new accession or another accession from the trial. When this replace option is used, it replaces every instances (plots and plants) of that accession in the trial. To do this, **click on the Edit Field Map link** by the top right of the physical trial layout section; **click on Replace Accession button**; **select accession** to replace from the trial and **provide a new accession** (must already exit in the database); **click on Replace Trial Accession button** to complete the operation.

**Change dimensions**

Click on the change dimensions button to alter the dimensions of the field map, and the respective row and column attributes of the plots in your trial. Clicking on this button will open a dialogue that prompts you to enter an accession name if the change you are making will result in dimensions that are larger than the amount of plots in your trial. The result is that the map will fill in the extra space with filler plots, and the filler plots will have the accession entered in input in the dialogue.

**Invert rows**

Check invert rows to observe your trial from the top down instead of the bottom up. This change is only present on your visualization, and the trial data will not be changed.

**Plot Layout**

You can select how you want your map to be ordered. The options include serpentine and zigzag.

**Download Harvesting Order**

This button allows you to download an excel file which has an incrementally ordered list of the harvesting order for your trial based on the current layout of the field map. You can play around with the four options given to experiment with the different traversal patterns available. The harvesting order file will omit border plots, but it will include filler plots.

**Download Planting Order**

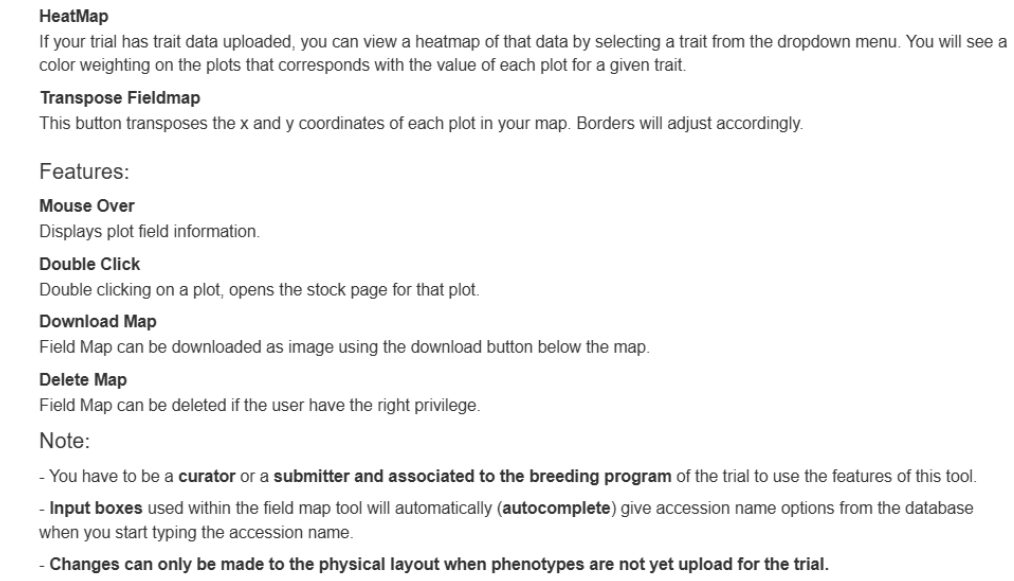
This button allows you to download an excel file which has an incrementally ordered list of the planting order for your trial. It is very similar to the harvesting order, but since you will need to plant the borders too, it includes the border plots in the traversal of your map.

**Add Borders**

These checkboxes allow you to add borders around your field map on any sides you would like. These plots are purely visual and have no real data associated with them. They will not be included in the database, but you can see them in your harvesting order if they are present.

**Submit Field Layout**

This button allows you to update the databases with the changes you have made to your trial. You can continuously change your trial data, just remember to submit the changes when you are satisfied so that it is saved.



**HeatMap**
If your trial has trait data uploaded, you can view a heatmap of that data by selecting a trait from the dropdown menu. You will see a color weighting on the plots that corresponds with the value of each plot for a given trait.

**Transpose Fieldmap**
This button transposes the x and y coordinates of each plot in your map. Borders will adjust accordingly.

Features:

**Mouse Over**
Displays plot field information.

**Double Click**
Double clicking on a plot, opens the stock page for that plot.

**Download Map**
Field Map can be downloaded as image using the download button below the map.

**Delete Map**
Field Map can be deleted if the user have the right privilege.

Note:

- You have to be a **curator** or a **submitter and associated to the breeding program** of the trial to use the features of this tool.
- **Input boxes** used within the field map tool will automatically (**autocomplete**) give accession name options from the database when you start typing the accession name.
- **Changes can only be made to the physical layout when phenotypes are not yet upload for the trial.**

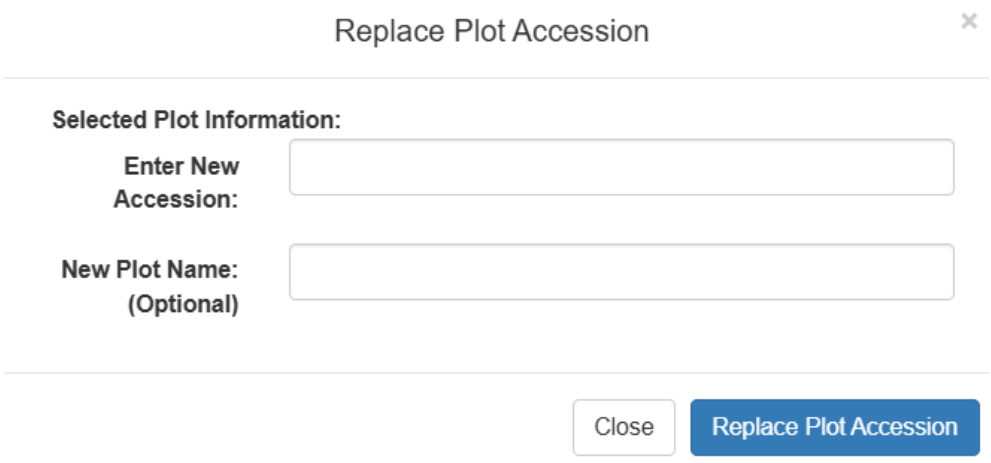
There are two different options for editing trial layout:

- Replacing plot accession by clicking on the plot in the layout.

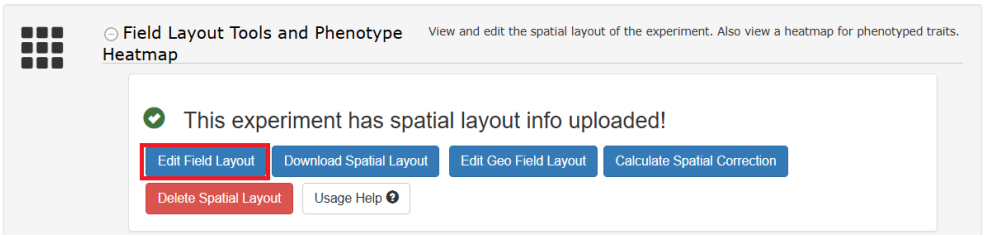
- Replacing trial accession by using the “Edit Field Layout” button.

To edit a specific plot, click on that plot. Enter a new accession on the “Edit Plot Info” form, then click the “Replace Plot Accession” button.

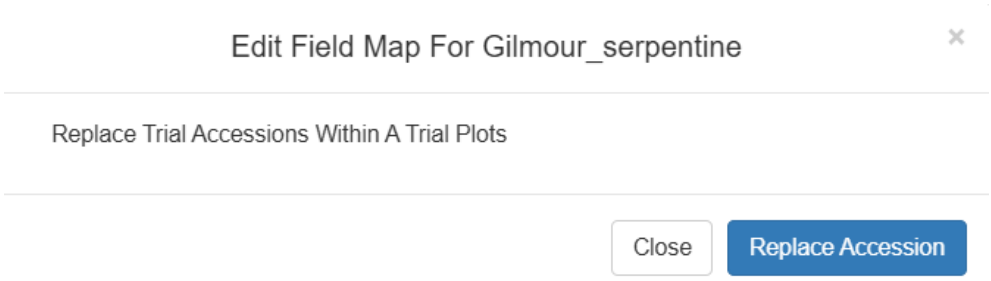
Plot Details
Plot Name:
Gilmour_serpentine-PLOT_22
Plot Number:
22
Plot Database ID:
41972
Accession:
CORRIGIN
Plot Structure:
▶ attributes
▶ has
3
Gilmour_serpentine-PLOT_22_plant_7
Gilmour_serpentine-PLOT_22_plant_8
Gilmour_serpentine-PLOT_22_plant_9
2
Gilmour_serpentine-PLOT_22_plant_4
Gilmour_serpentine-PLOT_22_plant_5
Gilmour_serpentine-PLOT_22_plant_6
View Plot Images
Close
Edit Plot Info



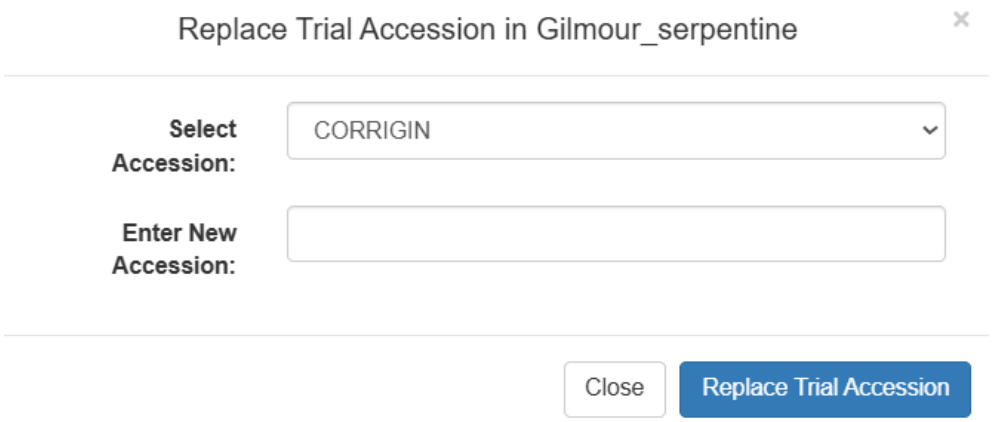

To replace an accession (in every plot/plant of that accession), click on the "Edit Field Layout" button.




On the "Edit Field Layout" window, click the "Replace Accession" button.

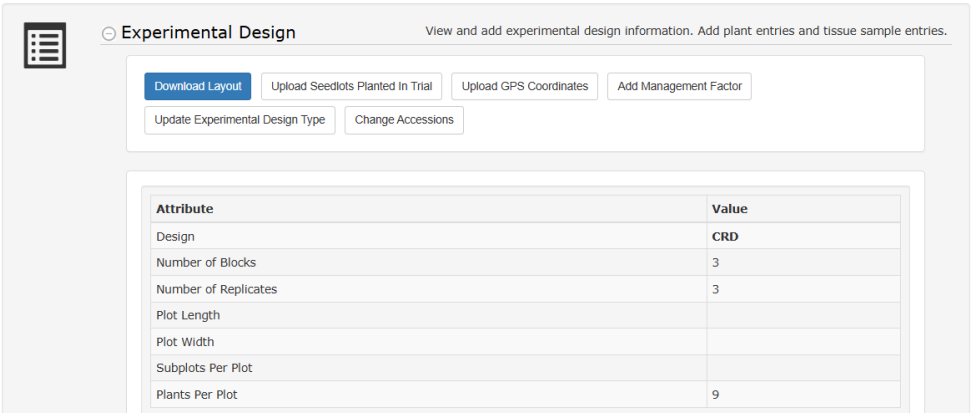



Select any accession that you want to replace, enter your new accession, then click the "Replace Trial Accession" button.

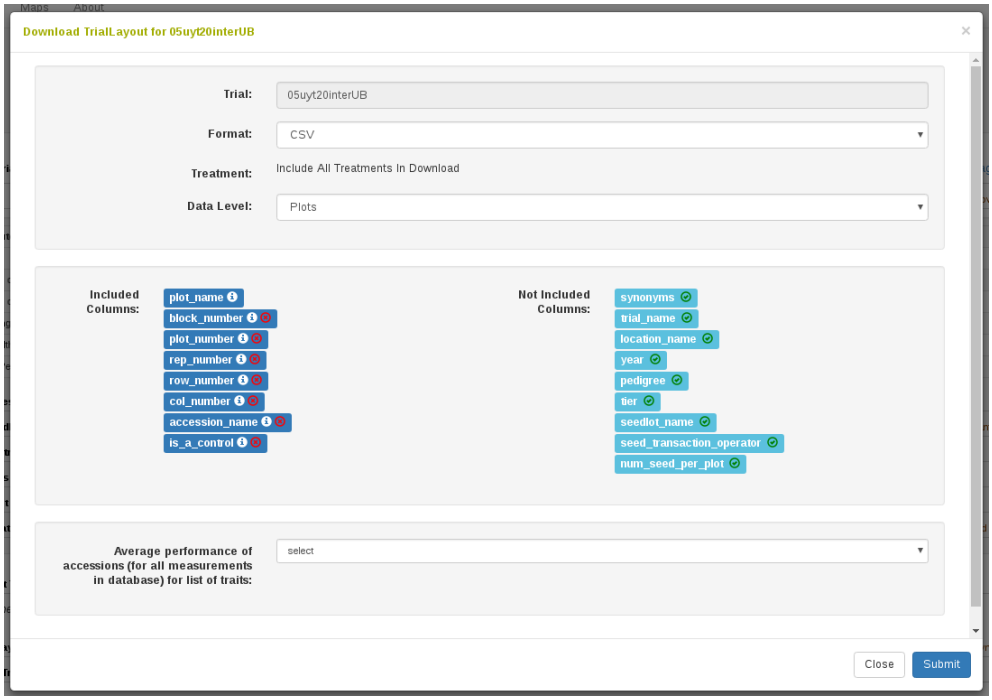



### 10.2.7 Downloading the Trial Layout from the *Trial Detail* page

Click on "Download Layout" on the *Trial Detail* page under "Experimental Design".

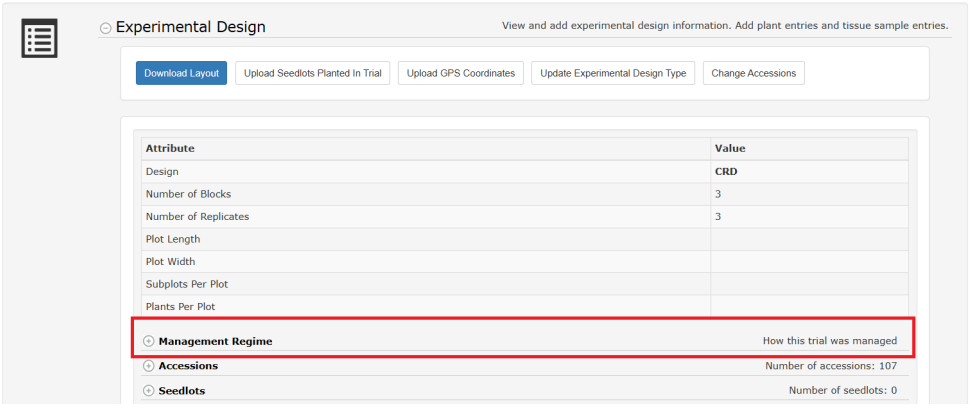



The trial layout includes all information regarding the observation units in the experiment. The observation units can be plots, plants, or subplots. The trial layout can include trial design information such as the block_number and rep_number. It can also include physical map information such as the row_number and col_number, if that information is available for the trial. The trial layout also includes information regarding treatments that have been applied in the field. Optionally, the layout can give information regarding accession's global performance for a list of traits.



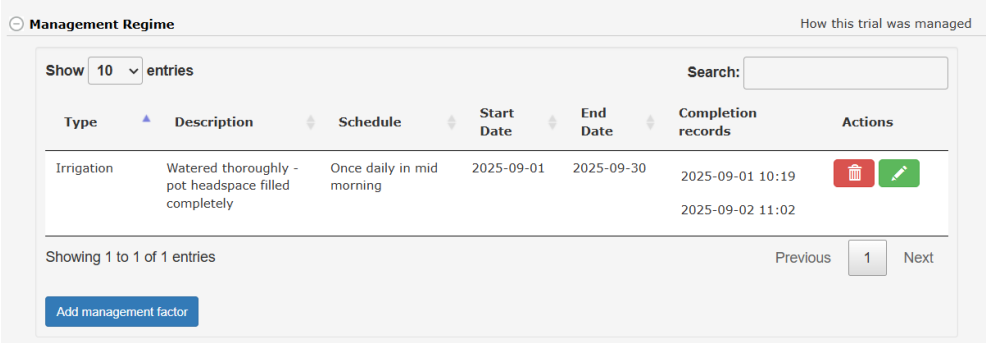

### 10.2.8 Adding Management Regimes

In addition to trial designs and trial layouts, you can specify the specific ways a trial was managed using the *Management Regime* section of the *Experimental Design* tab of the trial detail page.

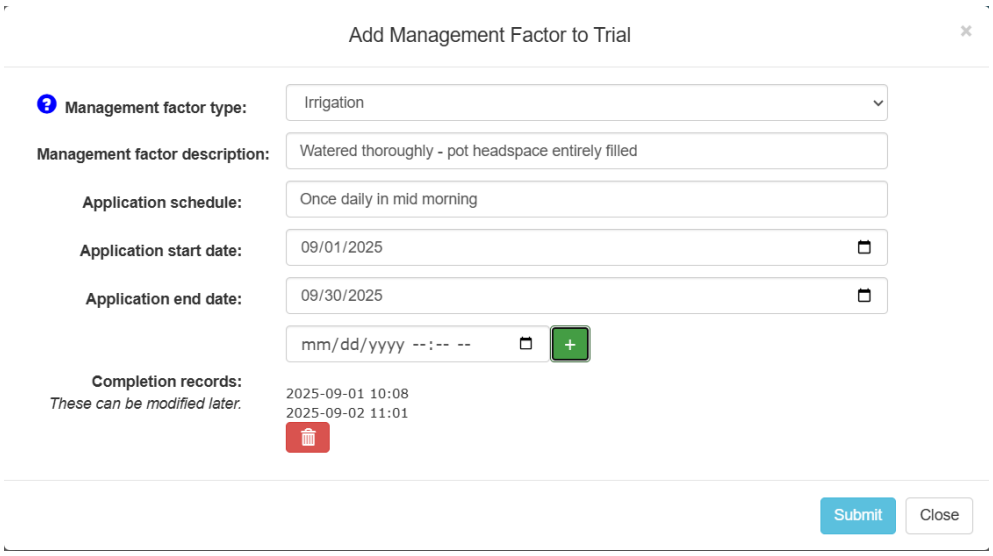



Here, you can view information like fertilization, irrigation, and weed management regimes. The total set of allowed management factor types is a matter

of server configuration - contact your server administrator with questions.



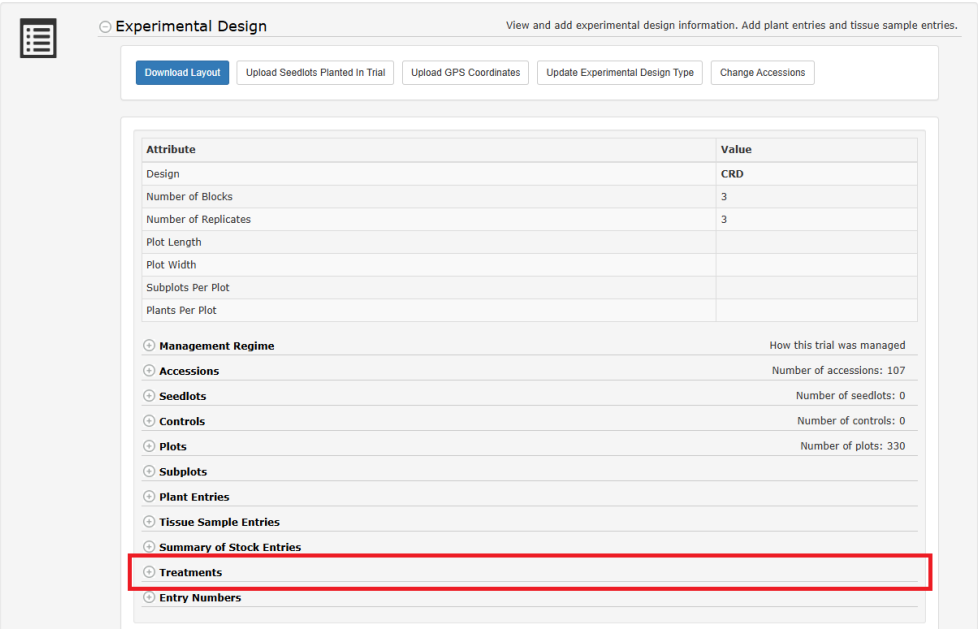

To add a new management factor, click on the 'Add management factor' button in the management regime section. Most of the data is optional, although the type and description must be filled. You may specify the start date, end date, and schedule of the treatment, and you may even record specific instances of the factor being applied if you choose. Click 'Submit' to add the factor.




Once added, a management factor can be edited by clicking the green pen icon next to it. The same dialog as before will open pre-populated with that management factor's data, and you can make changes and submit them. Click on the red trash can to delete the management factor from the trial.

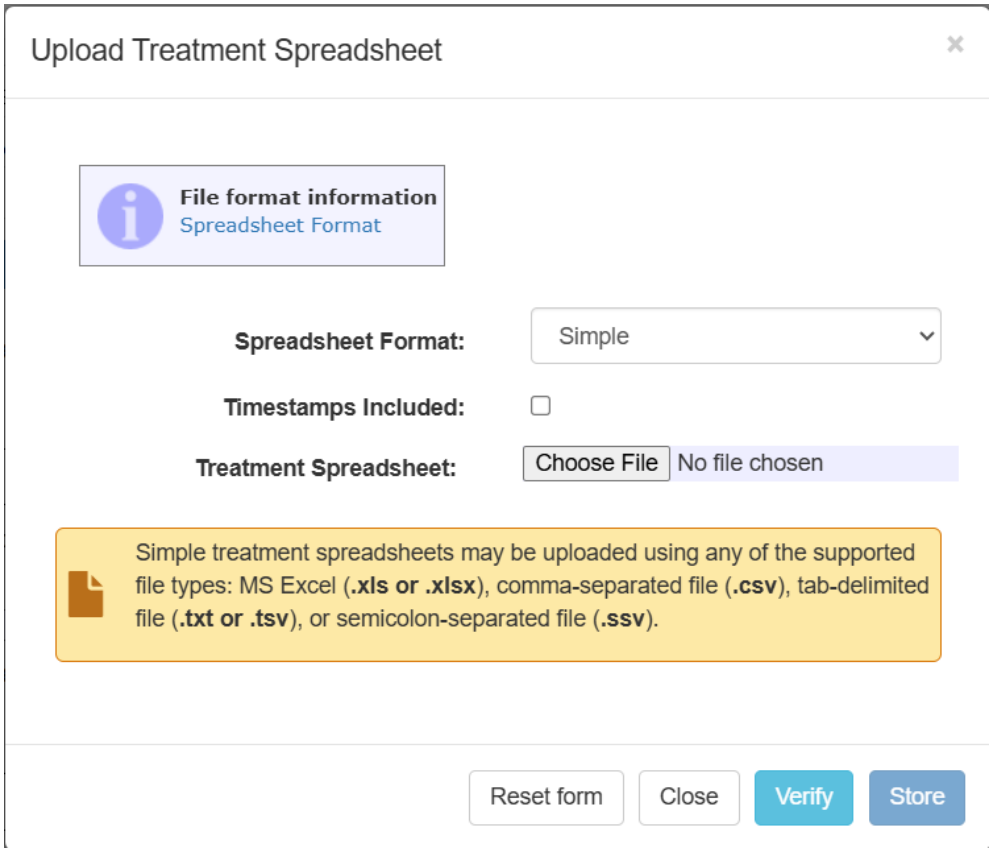
### 10.2.9 Adding Experimental Treatments

In contrast to management regimes, which describe how a field trial as a whole was managed, you may wish to store information regarding treatments applied differentially to the stocks of a trial. Treatments may be added from the *Treatments* section of the *Experimental design* tab.




Opening this tab gives a brief overview of the total number of stocks to which a treatment is applied and a table that allows you to search treatment values across different stock levels.

Adding treatments is very similar to uploading phenotypic data: click on the 'Add Treatments' button to view the treatment upload dialog:




Just as with phenotype uploads, you can supply a spreadsheet specifying the stocks and the treatments with optional timestamps. To ensure the treatment column headers are correct, consult the treatment search page or the ontology browser.

Unlike phenotypic data, however, treatments are automatically applied to the child stocks of any stock receiving a treatment. That means if a plot receives a treatment, all the subplots, plants, and tissue samples in that plot receive the same treatment.

### 10.2.10 Adding Plant Entries To Your Trial

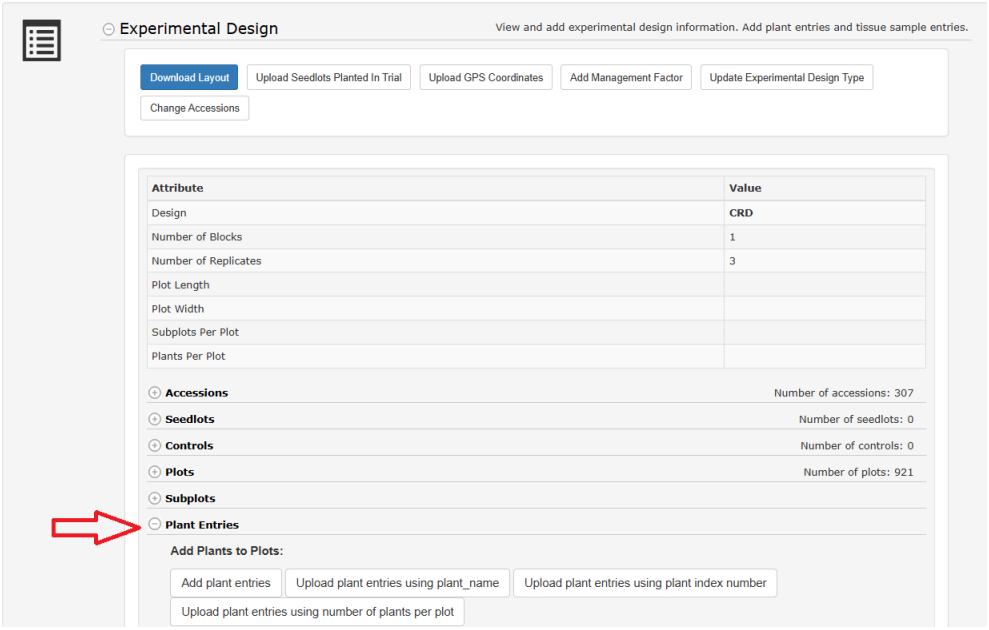
After you added a new trial to the database you can choose to add plant entries to your trial. Adding plant entries enables plant level phenotyping. It is generally better to enter data at the plant level into the database because it is always possible to calculate plot level phenotypes from the individual

plant data.

Plant entries can be added to your trial in four ways: (1) Automatically generated by the database. The only input required is the number of plants per plot. (2) Uploaded in an XLS or XLSX file. This allows you to specifically name your plant entries. (3) Uploaded using plant index number instead of name (the name is automatically generated) (4) Uploaded using the number of plants per plot. This allows different numbers of plant per plot, as opposed to option 1.

Additionally, if a trial has had subplots added, each of these options can be used to add plants to subplots. Note that once subplots are added, plants cannot be added directly to plots, and once plants have been added, subplots cannot be created.

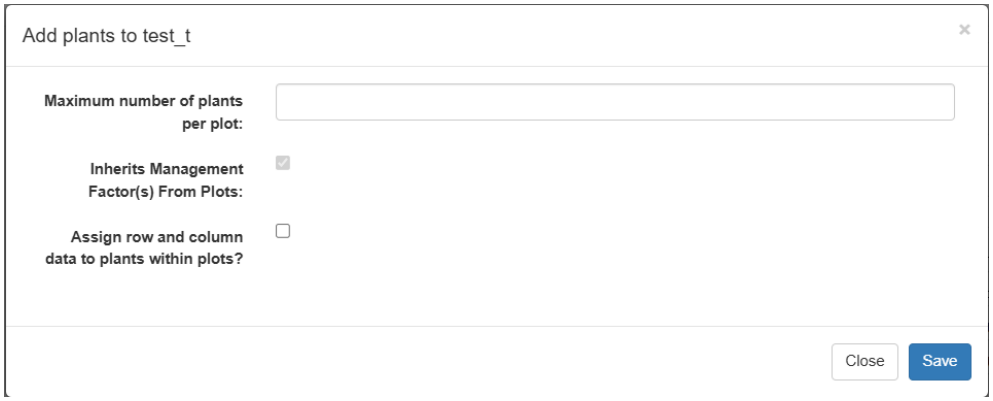
These options are available in the "Plant Entries" section on the *Trial Detail* page under "Experimental Design," as shown below.




### Automatically Generate Plant Entries

Clicking on "Add plant entries" opens the following dialog box. The only input required is the number of plants per plot. This will create plant entries

that are named as a concatenation of the plot_name and the plant's index number e.g. plot_name_plant_1. You may optionally add row and column data to give plants a spatial layout within each plot.



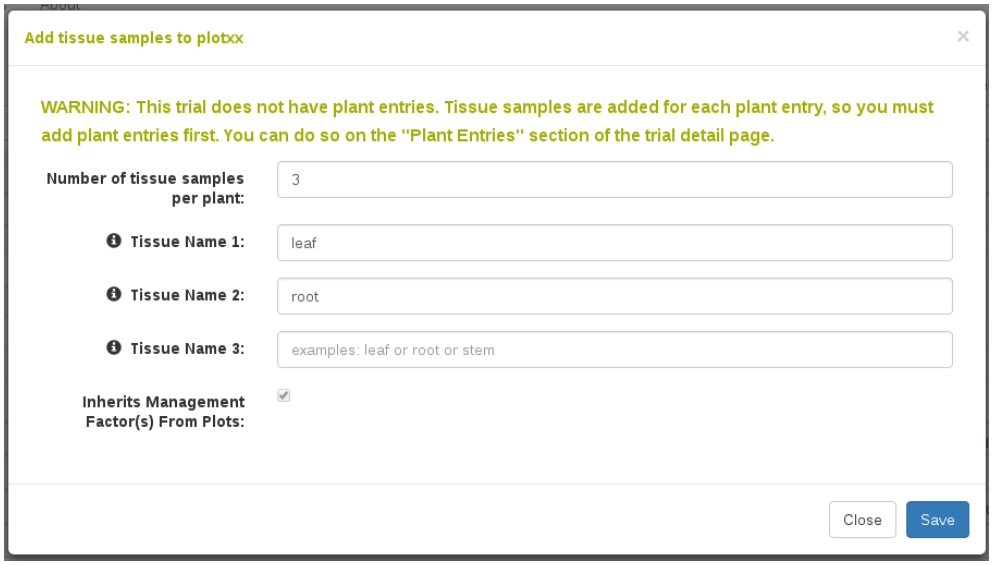

### Upload Plant Entries

Alternatively, you can choose to upload an XLS or XLSX file that contains the names of the plant entries, the plant index numbers, or the number of plants per plot. Each option comes with a dialog that specifies the file formats.

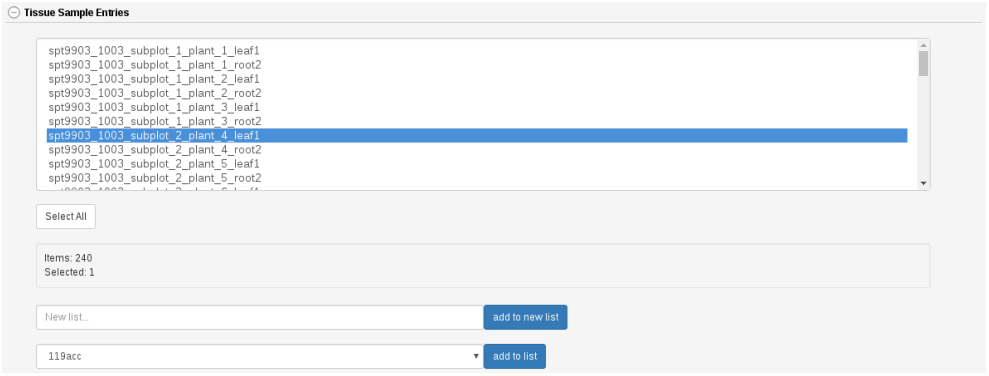
## 10.2.11 Adding Tissue Sample Entries To Your Trial

Some trials require tissue samples to be collected from plants in a field trial. The database will generate these tissue sample identifiers for you and will maintain all relationships with the plant, plot, accession, etc. To begin, go to the Design section of a trial's detail page and open the "tissue sample entries" section. Please note that tissue samples are directly related to plants, therefore your trial requires plants before you can add tissue samples.

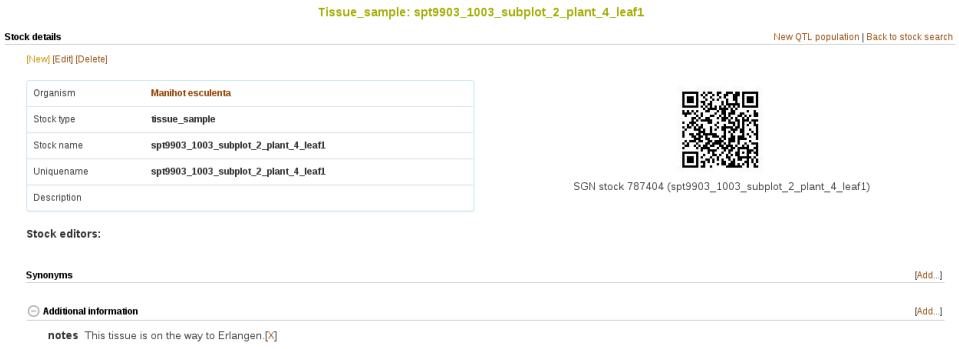
When you click on "Add tissue sample entries" you will see a dialog where you specify the number of tissue samples you require per plant. Once you have specified how many tissues samples, you can give specific words to distinguish samples, such as "root" or "stem", as seen below.



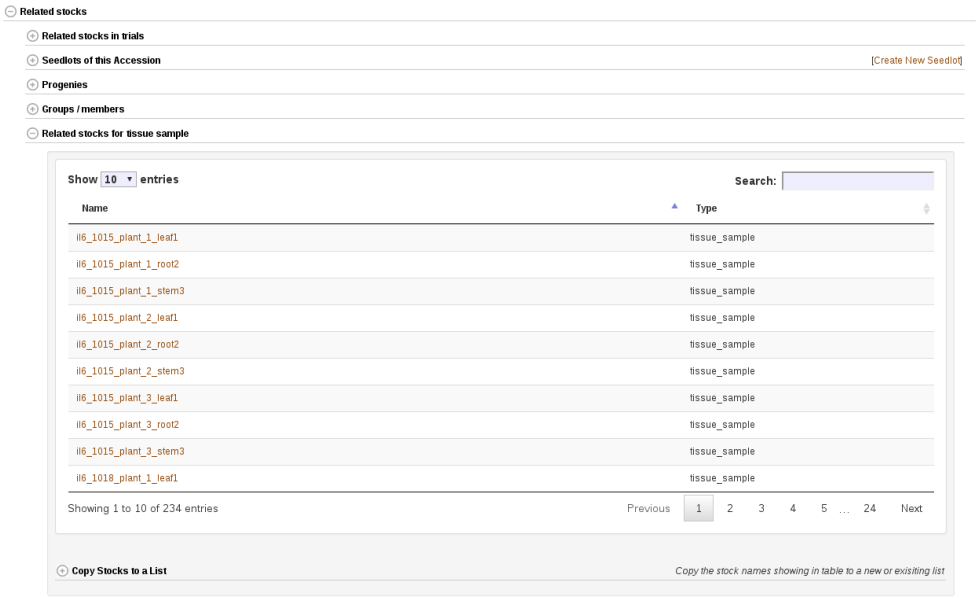

Once you have added tissue sample entries they will appear in the design section of the trial as seen below.




Each tissue sample has a detail page where you can add information about the sample, such as if it is in transit or in storage somewhere.




The related stocks section near the bottom of this detail page displays the relationships between all stocks, including tissue samples.

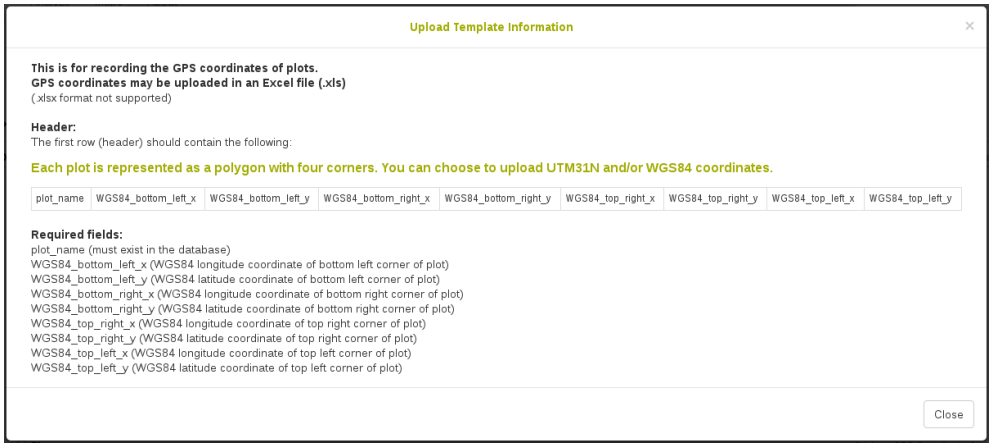



### 10.2.12 Uploading GPS Coordinates For Plots

You can upload GPS coordinates for the plots in your trial. There is a link on the Trial Detail Page as shown below.

Clicking on this link will bring up the following dialog.

Here you can upload an XLS or XLSX file. To see information on the format of the file that should be uploaded, click on "Spreadsheet format". This will bring up the following dialog.

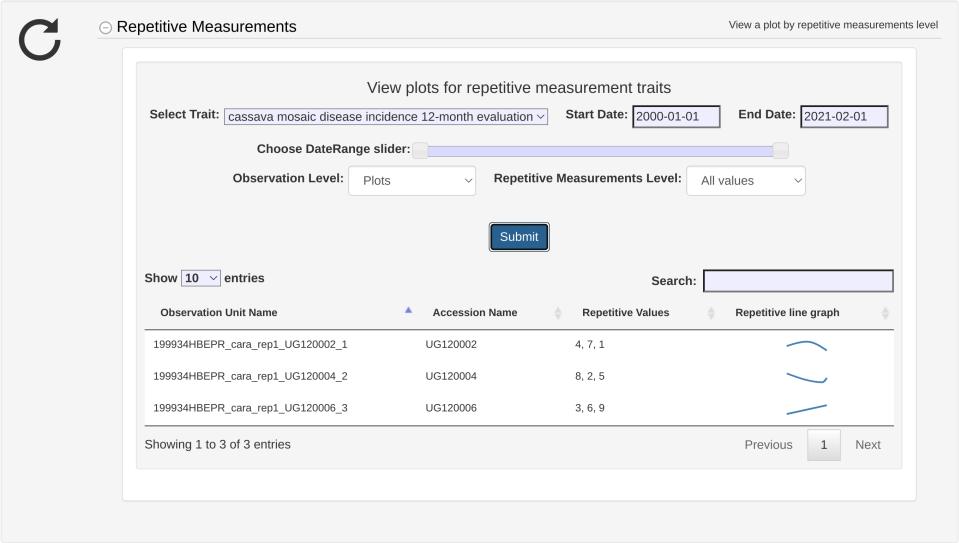



This dialog tells you that the file must be XLS or XLSX and must contain: plot_name WGS84_bottom_left_x WGS84_bottom_left_y WGS84_bottom_right_x WGS84_bottom_right_y WGS84_top_right_x WGS84_top_right_y WGS84_top_left_x WGS84_top_left_y The GPS coordinates should be WGS84 format and specify a four-pointed polygon around the plot.

### 10.2.13 Repetitive Measurements Section

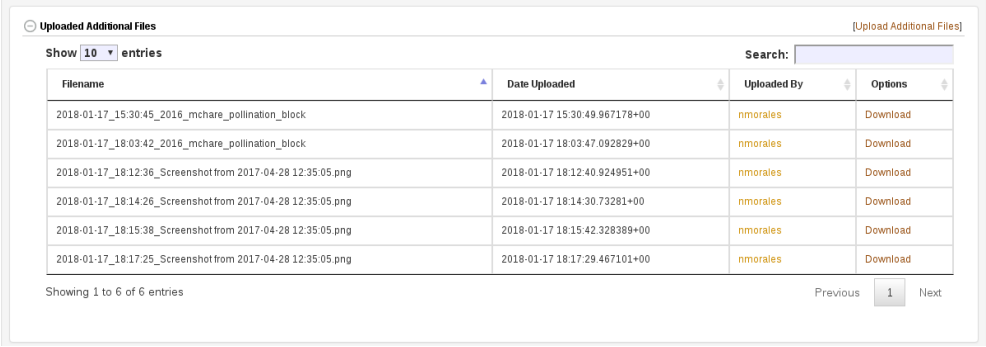
If a trial includes repetitive traits or time-series values, you can effectively view and analyze these values through the Repetitive Measurements Section. Start by selecting the desired trait from the trait drop-down menu. Next,

define the date range by either using the date-range picker or an interactive slider, which allows you to dynamically adjust the period you wish to examine. Once the date range is set, determine how to handle the repetitive measurements by choosing from various options such as First Value, Last Value, Averaged Value, Sum Values, or All Values. Choosing the "All Values" option enables an additional feature that visualizes the trend of the values over time, helping you identify patterns and trends within the data.

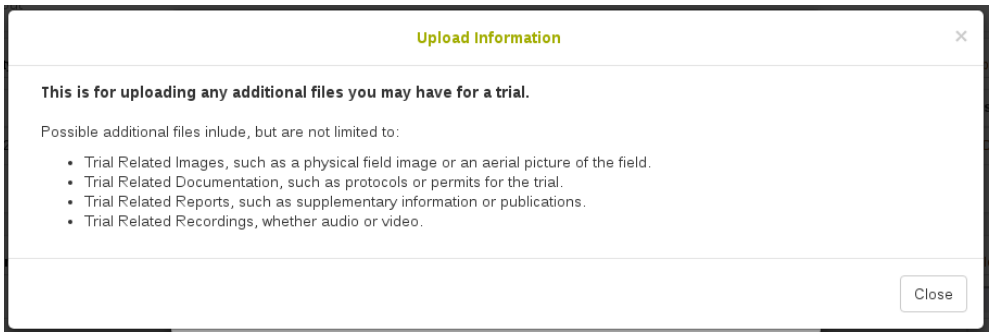



### 10.2.14 Uploading Additional Files To Trial

It may be of interest to you to upload additional documents, images, or recordings to your trial. To do this, scroll down to the "Uploaded Additional File" section on the trial detail page. From here you can view and download any of these additional files.

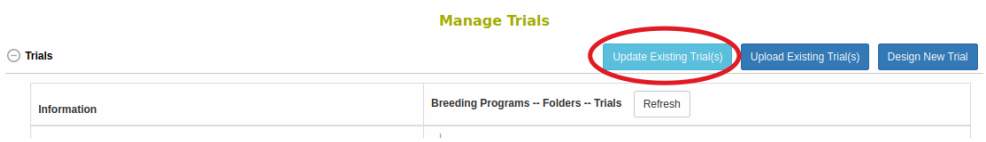



To upload an additional file, click on the "Upload Additional Files" link. A dialog will appear where you simply select your desired file. For information, you can click "Upload information" to see the following message.




## 10.3 Updating Trial Data

To updated the trial-level metadata (such as the planting date, design type, description, etc) of one or more existing trials, click the "Update Existing Trial(s)" button from the Manage > Field Trials page. This upload can also be used to rename trials or move trials to a different breeding program. In order to update a trial, you must be a curator or a submitter (that is associated with the breeding program of the trials).




Here you can upload a file that contains the new metadata for the existing

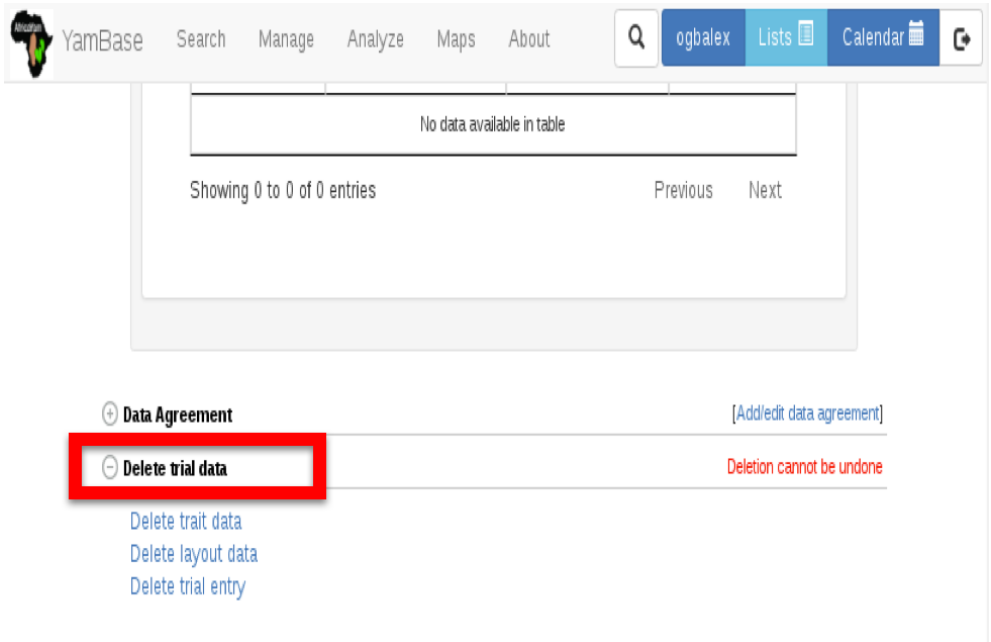
trials in the database. The first column is labeled 'trial_name' and includes the name of the existing trial. Additional columns can be included for the metadata you want to update. Any columns not included in the file or values left blank will leave the existing metadata unchanged. The columns that can be included are:

- new_trial_name: A new name for the trial, must not already exist in the database
- breeding_program: The name of breeding program that managed the trial, must exist in the database.
- location: The name or abbreviation of the location where the trial was held, must exist in the database.
- year: The year the trial was held.
- transplanting_date: The transplanting_date of the trial was conducted. Date in YYYY-MM-DD format or 'remove' to remove the date
- planting_date: Date of Planting in YYYY-MM-DD format or 'remove' to remove the date
- harvest_date: Date of Harvest in YYYY-MM-DD format or 'remove' to remove the date
- design_type: The shorthand for the design type, must exist in the database. Possible values include CRD: Completely Randomized Design, RCBD: Randomized Complete Block Design, RRC: Resolvable Row-Column, DRRC: Doubly-Resolvable Row-Column, ARC: Augmented Row-Column, Alpha: Alpha Lattice Design, Lattice: Lattice Design, Augmented: Augmented Design, MAD: Modified Augmented Design, greenhouse: undesigned Nursery/Greenhouse, splitplot: Split Plot, p-rep: Partially Replicated, Westcott: Westcott Design
- description: Additional text with any other relevant information about the trial.
- trial_type: The name of the trial type, must exist in the database. Possible values include Seedling Nursery, phenotyping_trial, Advanced Yield Trial, Preliminary Yield Trial, Uniform Yield Trial, Variety Release Trial, Clonal Evaluation, genetic_gain_trial, storage_trial, heterosis_trial, health_status_trial, grafting_trial, Screen House, Seed Multiplication, crossing_block_trial, Specialty Trial
- plot_width: plot width in meters
- plot_length: plot length in meters

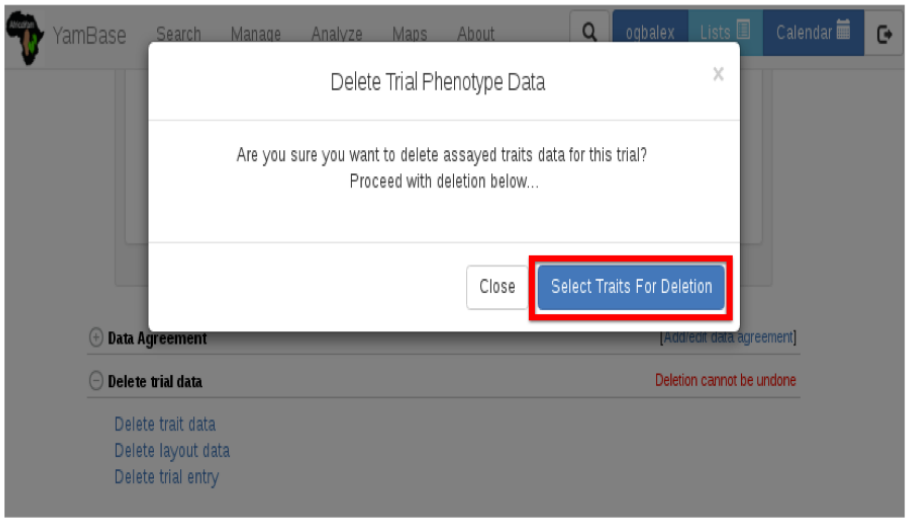
- field_size: field size in hectares

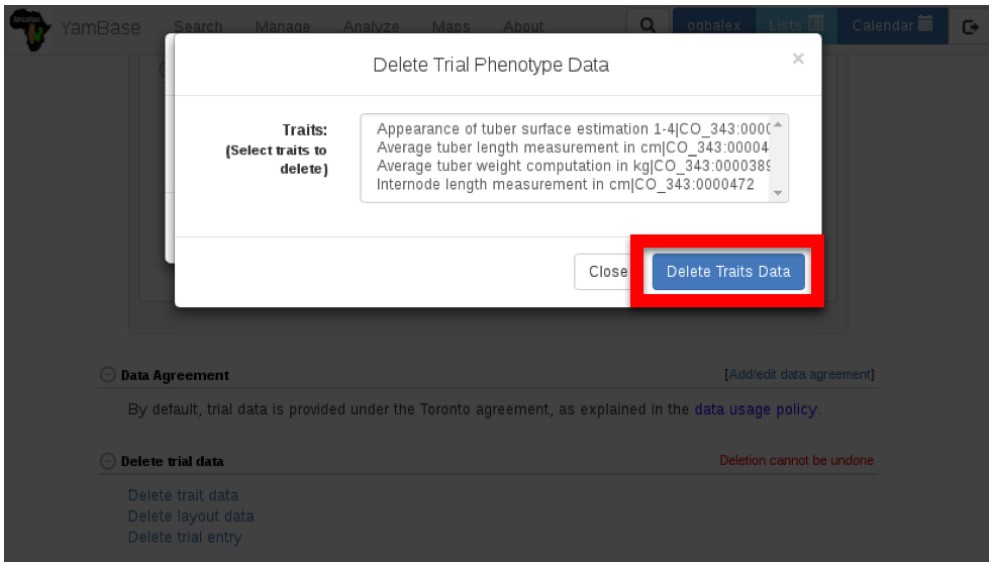
## 10.4 Deleting Trial Data

To delete a trial data, click on the "Delete trial data" section. There are links to delete traits, layout and trial entry data.




To delete assayed trait data, click on "Delete trait data" link. On the appeared dialog, confirm deletion by clicking on the "Select Traits For Deletion" button, then select one or more traits to delete from the trial.




To delete trial layout data, click on the "Delete layout data" link. Confirm deletion on the appeared dialog.

To Delete trial entry, click on "Delete trial entry" link. Confirm deletion on the appeared dialog.

## 10.5 Intercropping Support

Breedbase has started to add support for intercropping trials, allowing for more than one crop to be grown in a trial and more than one accession to be associated with a plot. Recent changes also allow for more than one trait ontology to be stored in the database, keeping the trait terms organized separately for each crop.

### 10.5.1 Trial Creation

Currently, intercropping is only supported when uploading the trial design using a file (it is not yet supported in the interactive trial design tool).

The trial upload template (both the single- and multiple-trial design formats) has an `accession_name` column, which is used to specify the name of the primary accession for each plot (each plot must have a primary accession). The upload templates now support an optional `intercrop_accession_name` column, which can be used to add one or more additional accessions to a plot. You can include more than one intercropping accession by separating their names with a comma or leave the column empty when there is no intercropped accession in the plot.

| trial_name | plot_name | plot_number | accession_name | intercrop_accession_name |
|---|---|---|---|---|
| B4I_2025_NY | B4I_2025_NY-PLOT_5001 | 5001 | IL22-5365 | NDP160034Y |
| B4I_2025_NY | B4I_2025_NY-PLOT_5002 | 5002 | IL22-5365 | NDP250123Y,NDP250155Y |
| B4I_2025_NY | B4I_2025_NY-PLOT_5003 | 5003 | IL21-4364 | NDP250178Y |
| B4I_2025_NY | B4I_2025_NY-PLOT_5004 | 5004 | IL21-4020 | |
| B4I_2025_NY | B4I_2025_NY-PLOT_5005 | 5005 | IL23-1619 | NDP150221G |

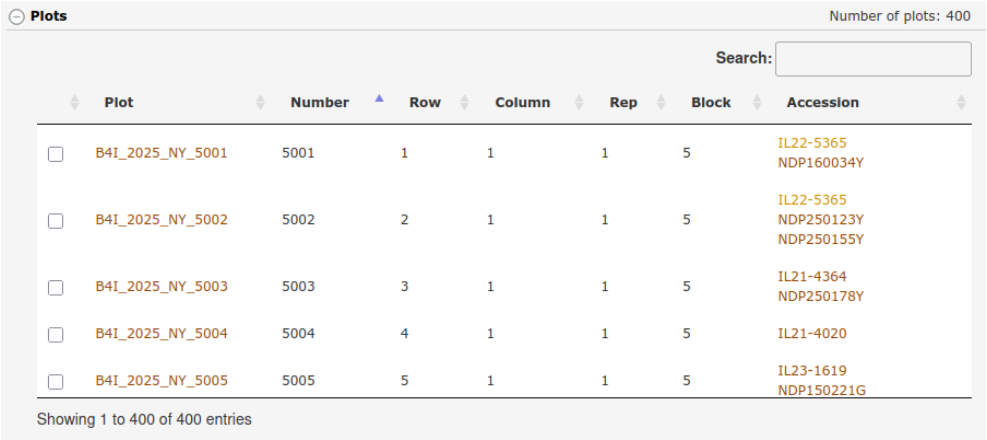
**An example multi-trial upload template that includes multiple accessions per plot**

*NOTE: some required columns are left out for clarity*

### 10.5.2 Trial Details

The intercropped accessions are displayed as metadata in various parts of the trial detail page.

The *Experimental Design* section displays the intercropped accessions in the Accessions table as well as the Plots table.



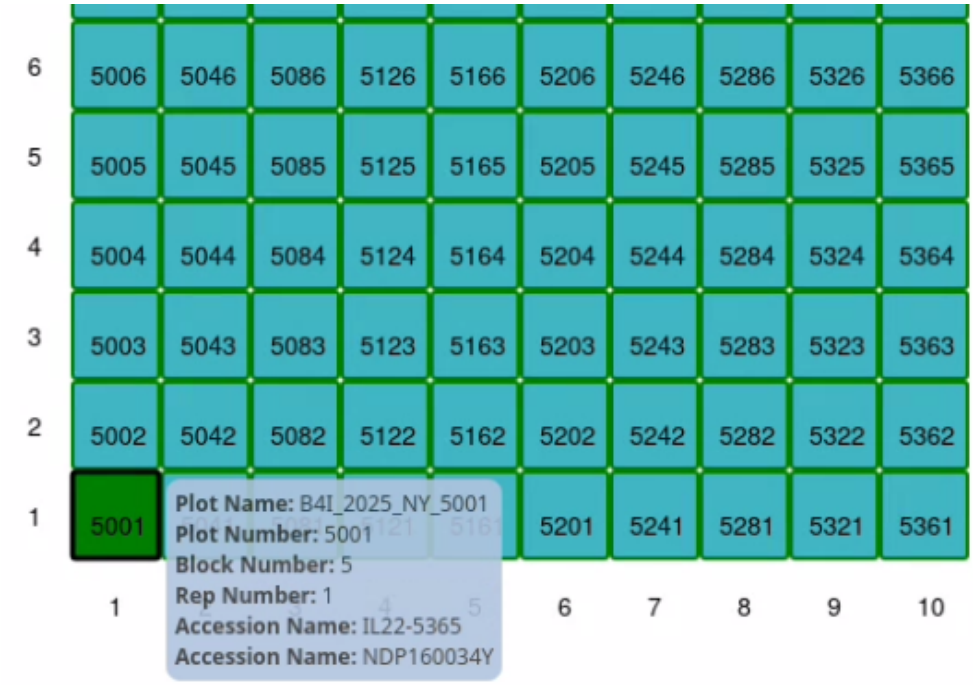
Plots — Number of plots: 400

Search:

| Plot | Number | Row | Column | Rep | Block | Accession |
|---|---|---|---|---|---|---|
| B4I_2025_NY_5001 | 5001 | 1 | 1 | 1 | 5 | IL22-5365<br>NDP160034Y |
| B4I_2025_NY_5002 | 5002 | 2 | 1 | 1 | 5 | IL22-5365<br>NDP250123Y<br>NDP250155Y |
| B4I_2025_NY_5003 | 5003 | 3 | 1 | 1 | 5 | IL21-4364<br>NDP250178Y |
| B4I_2025_NY_5004 | 5004 | 4 | 1 | 1 | 5 | IL21-4020 |
| B4I_2025_NY_5005 | 5005 | 5 | 1 | 1 | 5 | IL23-1619<br>NDP150221G |

Showing 1 to 400 of 400 entries

The interactive field map tooltip (displayed when you hover the mouse pointer over a plot) will display all of the Accessions associated with the plot. The spatial layout download will list all of the Accessions in the plot as well.




### 10.5.3 Phenotype Downloads

All of the phenotype trait data downloads (from the trial detail page, search wizard, etc) still include the `germplasmDbId` and `germplasmName` columns, which include the the database id and name of the primary accession. The `intercropGermplasmDbId` and `intercropGermplasmName` columns will include the id(s) and name(s) of any intercropped germplasm added to the plots. If there are no intercropped germplasm for a particular plot, the columns will be empty. If there are more than one intercropped germplasm added to a plot, the IDs and names will be separated by a comma.

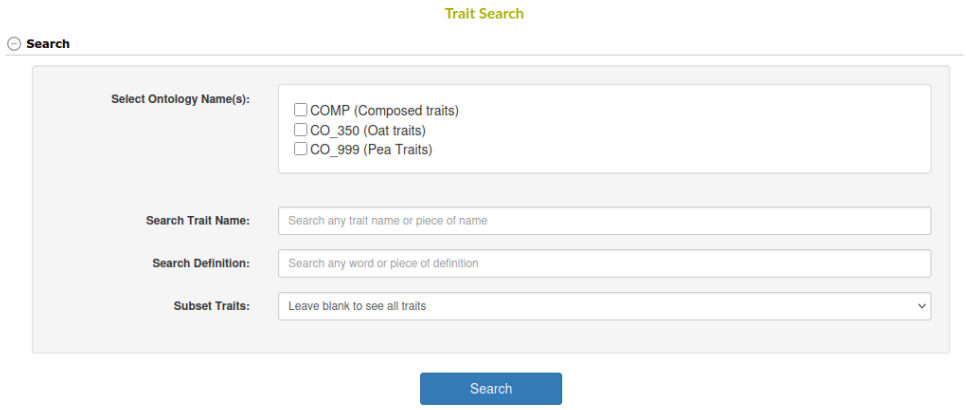
| | A | B | C | D | E | F | G | H | I |
|---|---|---|---|---|---|---|---|---|---|
| 1 | studyName | germplasmDbId | germplasmName | observationUnitName | plotNumber | intercropGermplasmDbId | intercropGermplasmName | Lodging severity - 0-9 Rating\|CO_350:0005009 | Pea plant height - cm\|CO_999:0000010 |
| 2 | B4I_2025_NY | 544279 | IL22-5365 | B4I_2025_NY_5001 | 5001 | 704516 | NDP160034Y | 7 | 75 |
| 3 | B4I_2025_NY | 544279 | IL22-5365 | B4I_2025_NY_5002 | 5002 | 704599,704806 | NDP250123Y,NDP250155Y | 8 | 78 |
| 4 | B4I_2025_NY | 530654 | IL21-4364 | B4I_2025_NY_5003 | 5003 | 704820 | NDP250178Y | 9 | 82 |
| 5 | B4I_2025_NY | 530784 | IL21-4020 | B4I_2025_NY_5004 | 5004 | | | 2 | 79 |
| 6 | B4I_2025_NY | 704687 | IL23-1619 | B4I_2025_NY_5005 | 5005 | 704746 | NDP150221G | 6 | 86 |

**An example phenotype download that includes multiple accessions per plot**

*NOTE: some columns are left out for clarity*

### 10.5.4 Trait Ontologies

A separate trait ontology for each crop can be loaded into the database. This allows for the clear dilineation and separation of trait definitions between different crops.

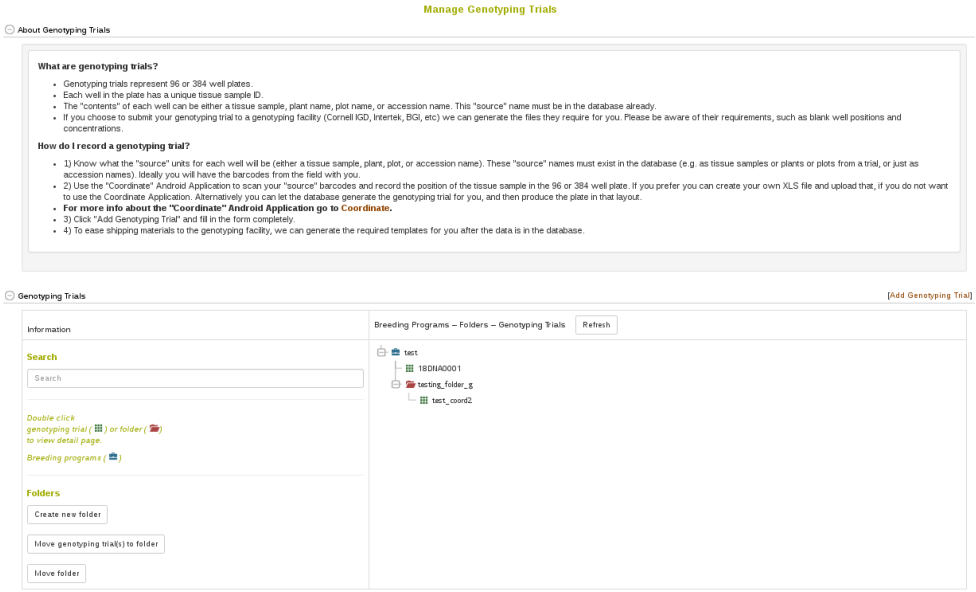


# Chapter 11

# Managing Genotyping Plates

Genotyping Plates represent the content of a genotyping plate sent to a genotyping facility (e.g. samples in specific wells). To streamline this process, it is possible to upload this information or let the database create a plate for you. Once the genotyping plate is saved in the database it is then possible to export the information directly to genotyping facilities that are BrAPI compliant. The genotyping facility can then provide status information to us via BrAPI.

To begin go to Manage->Genotyping Plates.

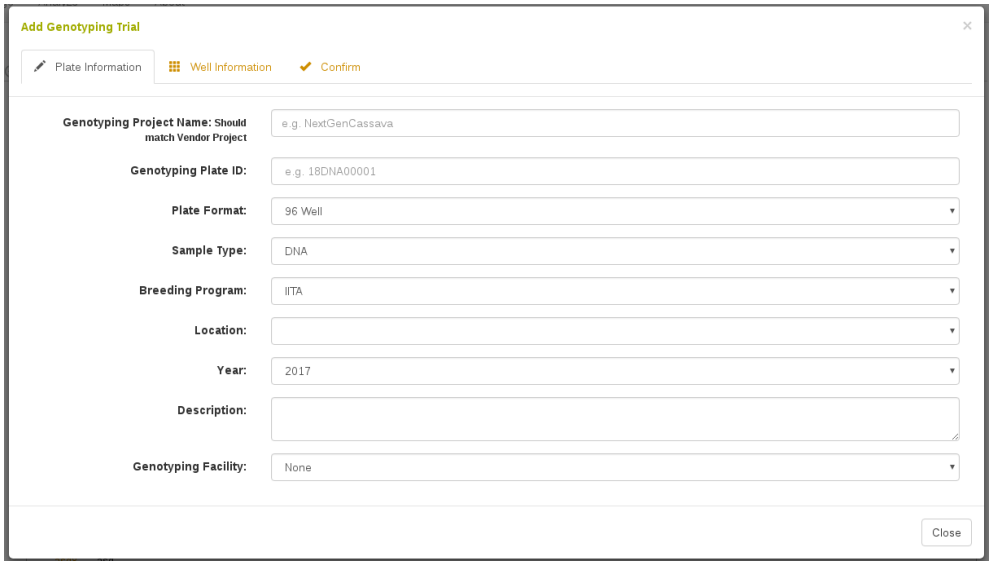



Here the genotyping plates are divided by Breeding Program. These sections can be expanded by clicking on one.

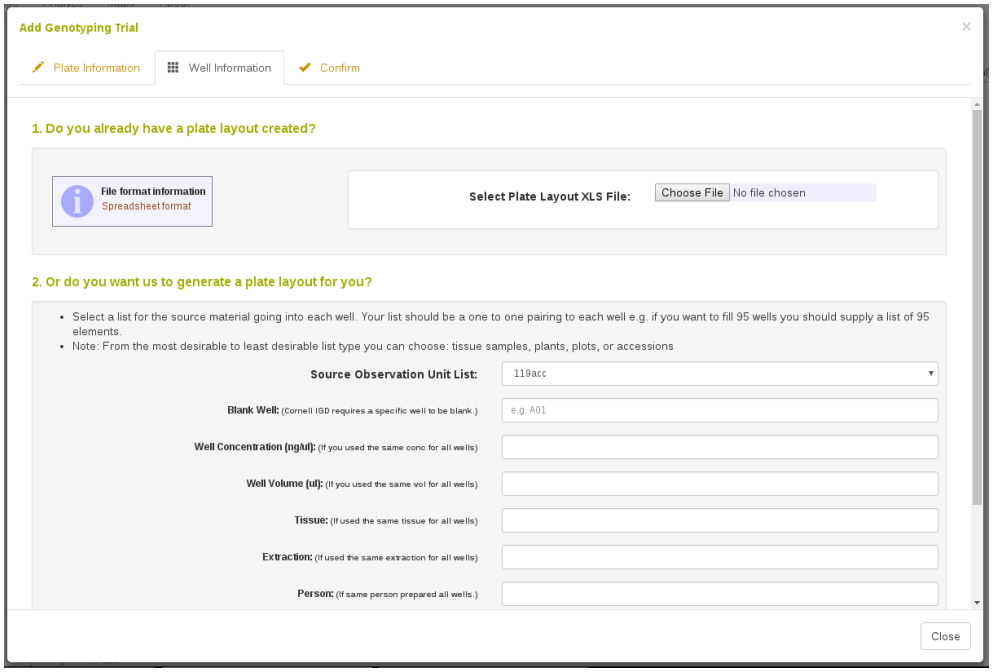
## 11.1 Adding a New Genotyping Plate

To begin, click on "Add Genotyping Plate". Notice that this form is split into three sections: "Plate Information", "Well Information", and "Confirm". The first section is for defining information about the genotyping plate, such as a Plate identifier, plate format (96 well), etc. The second section is for defining the samples in the wells, such as sample names, sample concentrations, well position, etc. The final section is for Submitting the info.

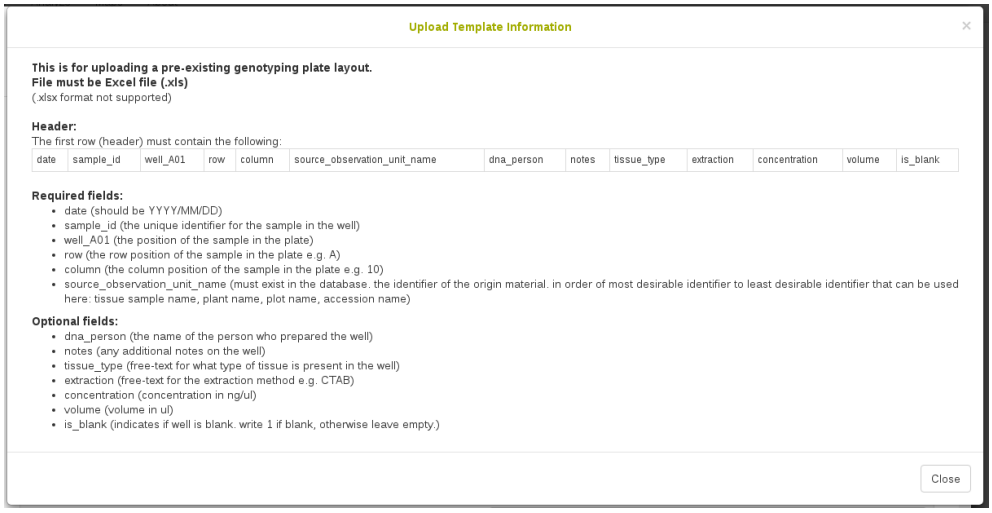
All fields in the Plate Information section are required.




In the Well Information section you can choose to either 1) Upload an XLS or XLSX spreadsheet with your sample layout or 2) let the database create the sample layout.



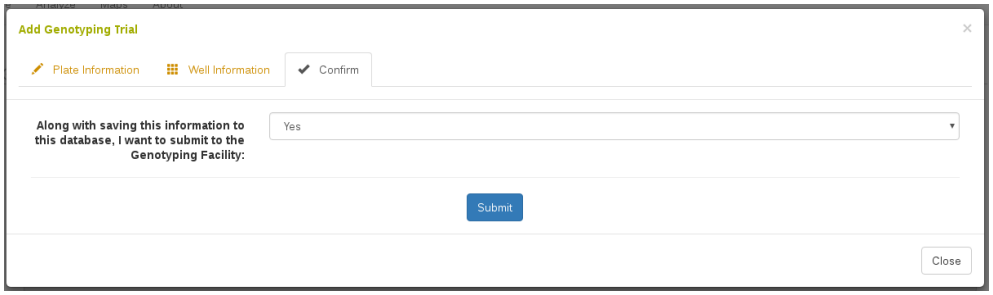

If you choose to upload an XLS or XLSX spreadsheet, the Spreadsheet Template info requires the following:




In either case, the sample identifier is generally a concatenation of Plate name and well position, e.g. MyGenotypingTrial1_A01. In either case, you need to provide a "source_observation_unit_name" for each sample. This

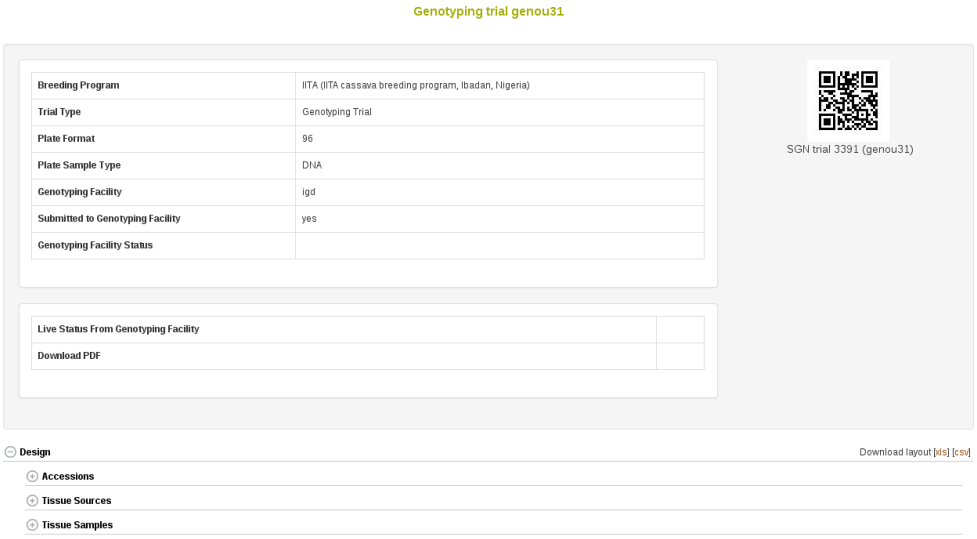
can be a tissue sample name, a plant name, a plot name, or an accession name; however, in any case, the identifier must already exist in the database. This allows us to link the sample in the well to specific field trial plots, or, plants, or tissue_samples. If you only know which accession is in the well, you can use the accession name.

In the final Confirm section you can decide whether to submit this information to the genotyping facility you selected. This requires that the genotyping facility is BrAPI compliant to work.

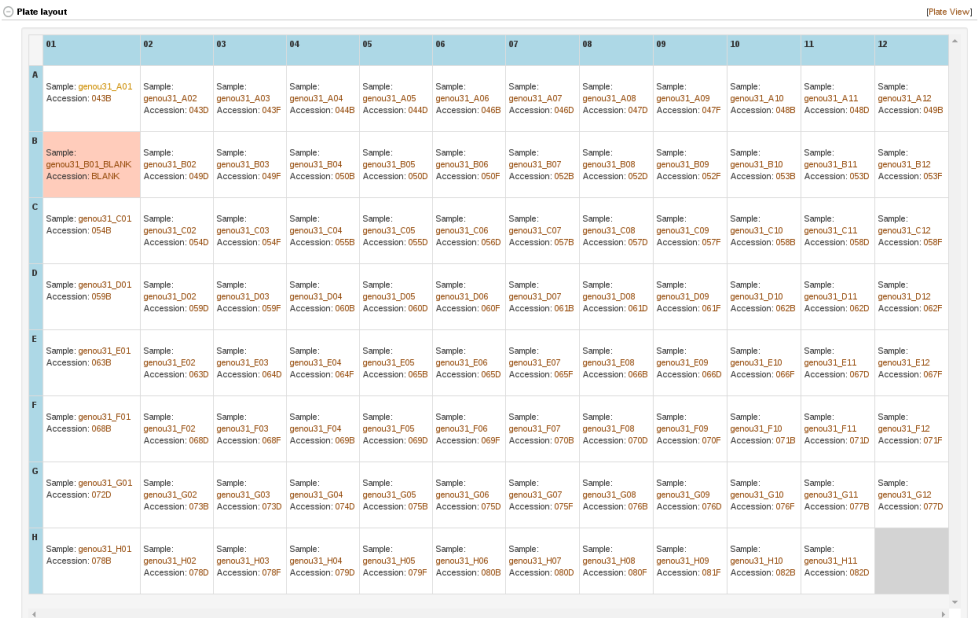



## 11.2 Genotyping Plate Detail Page

If you open a specific genotyping plate, it will take you to the detail page. Here you can see the Accessions used in the plate (if you created the trial and the source_observation_unit_names you used were plots, this will still work because we know the accession of the plot or plant or tissue sample).




Further down you can see a graphical representation of your plate with well positions. This can be 96 well or 384 well depending on your plate format.



# Chapter 12

# Using Field Book and BreedBase for Data Collection and Storage

## 12.1 Introduction

**Field Book** is an open-source Android application designed to streamline phenotypic data collection, replacing traditional paper field books and reducing transcription errors.

**BreedBase** supports integration with Field Book by enabling users to generate properly formatted field layout and trait files, import collected data, and optionally exchange data via **BrAPI**.

This section provides step-by-step guidance on preparing Field Book for data collection using trial data stored in BreedBase, including file generation, manual file import, and data export. It assumes that trial fields and trait defi-

nitions already exist in BreedBase and that users have access to an Android device.

*Field Book User Manual*

---

## 12.2 Installing Field Book

Field Book can be installed on an Android device via the Google Play Store or manually from GitHub. To ensure access to the latest features and fixes, we recommend checking for updates regularly.

### 12.2.1 Option 1: Downloading from the Google Play Store

Download the Field Book app directly from the Google Play Store or scan the QR code below.

**Walkthrough: Installation and Setup**

### 12.2.2 Option 2: Manual Installation from Github

**Walkthrough: Installing Field Book from GitHub**

### 12.2.3 Keeping Field Book Updated

Regular updates are recommended to access new features and bug fixes.

**Walkthrough: Updating the Field Book App**.

---

## 12.3 Preparing Field Layout Files in BreedBase

Field Book requires a **field layout file** containing a unique identifier for each plot or plant to be observed. Optionally, columns for row, column, pedigree, or previous phenotype data may be included for context.

BreedBase provides tools to generate layout files directly from trial metadata. These files can be exported and then uploaded to Field Book for data collection.

**Walkthrough: Creating a Field Layout File in BreedBase**

---

## 12.4 Importing Field Layout Files into Field Book

Field layout files can be imported into Field Book via cloud storage or directly from the device.

### 12.4.1 Option 1: Import from Cloud Storage

Requirements:

1. A generated field layout file.
2. Linked cloud service (e.g., Google Drive) on the device.
3. File uploaded to the cloud account.

**Walkthrough: Importing from Cloud Storage**

### 12.4.2 Option 2: Import from Local Storage Device

Requirements:

1. A generated field layout file.
2. File transferred to the Android device's local storage.

**Walkthrough: Importing from Local Device**

---

## 12.5 Adding Traits to Field Book Using BreedBase

For exported data to be compatible with BreedBase, trait names in Field Book must match the exact names stored in the database.

BreedBase supports generation of **trait files** containing properly formatted trait names, types, units, and metadata.

### 12.5.1 Creating Trait Files in BreedBase

**Walkthrough: Creating Trait Files in BreedBase**

### 12.5.2 Importing Trait Files to Field Book

**Walkthrough: Importing Trait Files**

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## 12.6 Collecting and Exporting Phenotype Data

This section describes the process for recording and exporting phenotype data using the Field Book application. It assumes that the required field layout and trait files have already been imported into the app.

Users will be guided through entering phenotypic observations in the field and exporting collected data in a format compatible with downstream processing and upload to **BreedBase**. This standardized workflow helps ensure data integrity, traceability, and seamless integration with breeding databases.

### 12.6.1 Collecting Phenotype Data in Field Book

Once field layout and trait files are loaded, data collection can begin. Field Book allows users to view each plot or plant entry, record trait values in

structured forms, and automatically timestamp and geotag observations (if device settings allow).

**Walkthrough: Collecting Data in Field Book**

### 12.6.2 Exporting Phenotype Data from Field Book and Importing to BreedBase

After data collection is complete, Field Book allows users to export data as a .csv file. This file can then be uploaded to BreedBase for storage and analysis.

**Walkthrough: Exporting Phenotype Data from Field Book**

Once exported, the .csv file can be imported into BreedBase through the trial's data upload interface. Proceed with uploading phenotype observations as described in the **Managing Phenotypic Data** section.

It is important that:

- Trait names match those defined in the database.
- Plot or plant identifiers align with the original layout.

---

## 12.7 Using BrAPI for Data Transfer Between Field Book and BreedBase

Field Book supports integration with **BreedBase** via the **BrAPI (Breeding API)** specification. BrAPI enables direct exchange of trial metadata and phenotype observations between Field Book and BreedBase without the need for manual file uploads.

This walkthrough covers:

- Connecting Field Book to a BrAPI-enabled BreedBase server
- Importing trial layout and traits from BreedBase to Field Book
- Exporting collected phenotype data from Field Book to BreedBase

**Prerequisites:**

- You must have a BreedBase user account with permissions to access trial and trait data.
- Your BreedBase instance must be configured with working BrAPI endpoints.
- The Field Book app must be installed on an Android device with internet access.

**Advantages of Using BrAPI:**

- Reduces manual errors from file handling.
- Supports real-time syncing of data.
- Enables remote team coordination and efficient workflows.

### 12.7.1 Connecting Field Book to a BrAPI Server

**Walkthrough: Connecting Field Book to a BreedBase BrAPI Server**

### 12.7.2 Importing Trials and Traits from BreedBase

**Walkthrough: Importing Trials from BreedBase via BrAPI**

**Walkthrough: Importing Traits from BreedBase via BrAPI**

### 12.7.3 Collecting Data

Proceed with entering phenotype observations as described in the **Collecting Phenotype Data in Field Book** section.

Ensure that:

- Trait values match expected formats (e.g., numeric, categorical).
- Observations are saved regularly during field use.

### 12.7.4 Exporting Phenotype Data to BreedBase via BrAPI

**Walkthrough: Sending Phenotype Data with BrAPI**

### 12.7.5 Verifying Data in BreedBase

After exporting data to BreedBase via BrAPI, it is important to verify that the export was successful directly in BreedBase.

1. Log in to your BreedBase instance.
2. Navigate to the Manage Trials or Trial Detail page for the uploaded study.
3. Confirm that:
    - Trait data appears in the Phenotype Data tab.
    - Data points align with expected plot/plant identifiers.
    - Trait units and scales match those defined in the database.

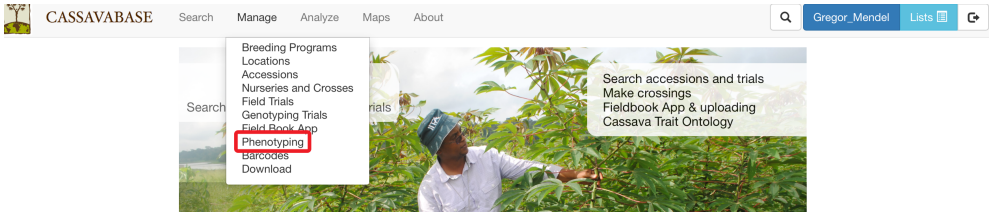
### 12.7.6 Troubleshooting

- Connection fails: Ensure you have the correct BrAPI server URL and valid credentials.
- Traits do not load: Confirm that the trait exists in BreedBase.
- Upload fails: Check for internet connectivity, correct user permissions, and trait formatting errors.

# Chapter 13

# Managing Phenotypic Data

To facilitate uploading process for phenotypic data, “Manage Phenotypic Data” page provides two options for uploading: Field Book Phenotype file in database format and phenotype file in Excel (.xls or .xlsx) file format. To access “Manage Phenotypic Data” page, clicking on “Phenotyping” in the “Manage” menu.



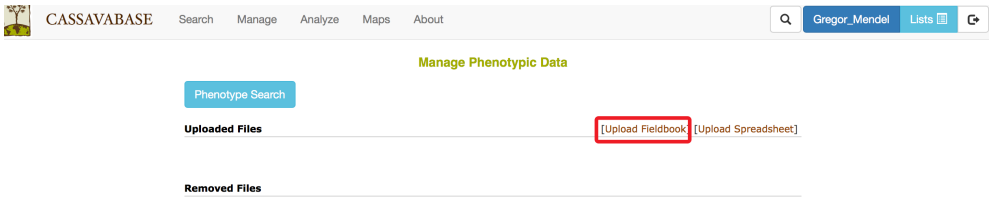

## 13.1 Uploading Fieldbook Phenotypes

### 13.1.1 Export Field Book Database File

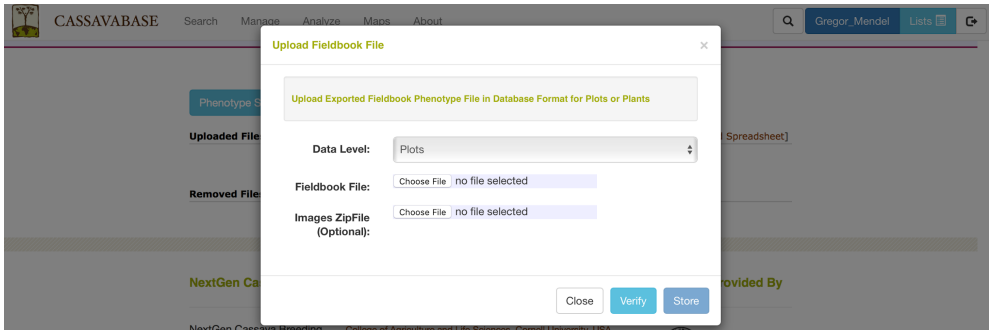
The database upload of Field Book phenotype data relies on the “Database” format from the Field Book. Please make sure to export the “Database” format from the Field Book if you intend to upload the data using the Field Book Upload we describe below. If you prefer to use the “Table” format that the Field Book exports, you can modify this format to work with the Speadsheet Upload we describe below.

### 13.1.2 Upload Field Book Database File

To upload a Field Book Phenotype file in a database format, click the "Upload Fieldbook" link

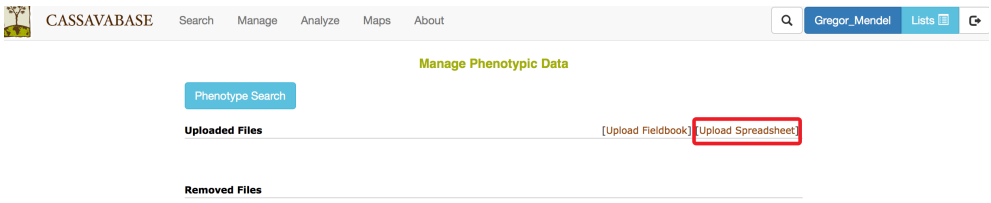



The "Upload Fieldbook" link on this page and "Upload" link on the "Field Book Tools" page open the same dialogue. Please follow instructions for uploading phenotypic files on the **??** page.

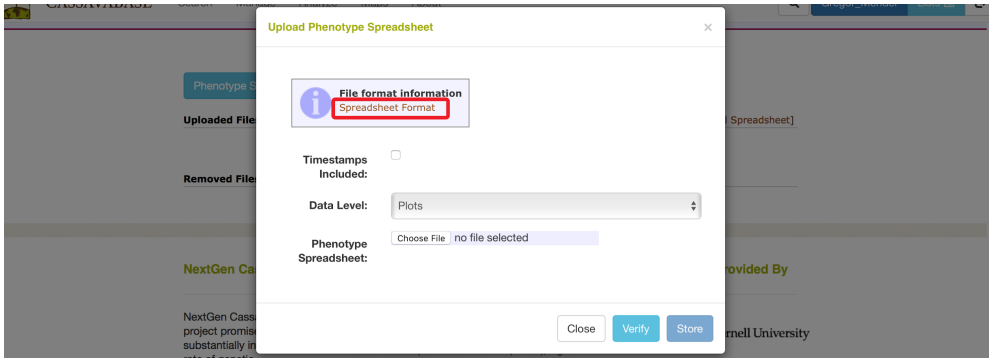



## 13.2 Uploading Spreadsheet Phenotypes

To upload a phenotype file in an Excel (.xls or .xlsx) file format, click the "Upload Spreadsheet" link.

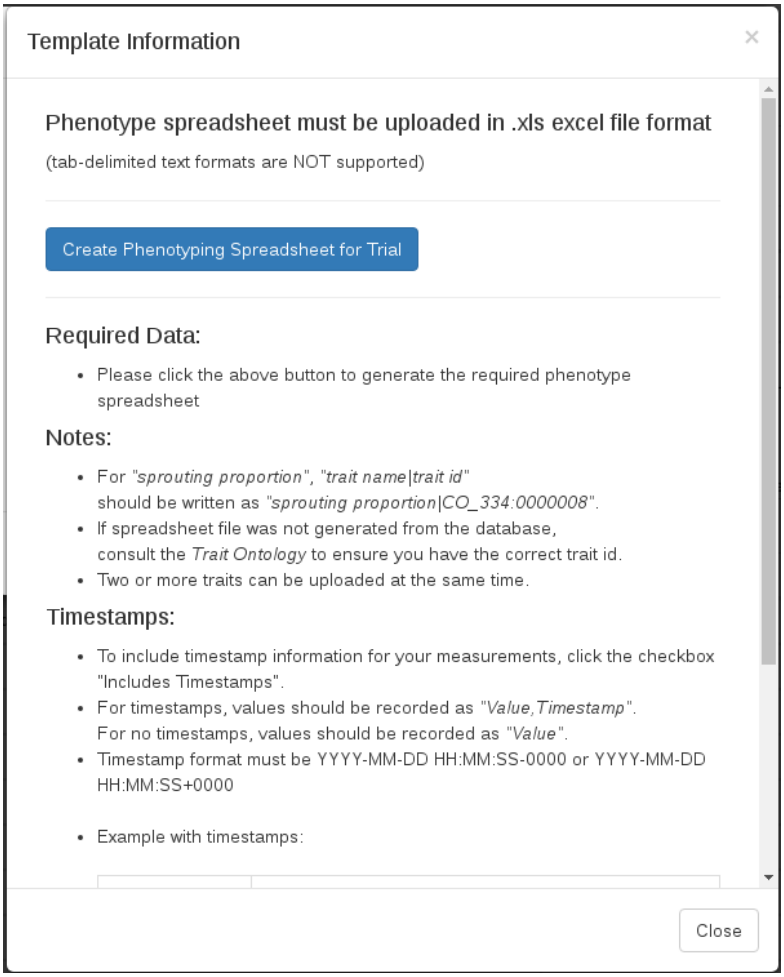



Please specify "Data Level" (Plots or Plants) and select the Excel file that you want to upload.

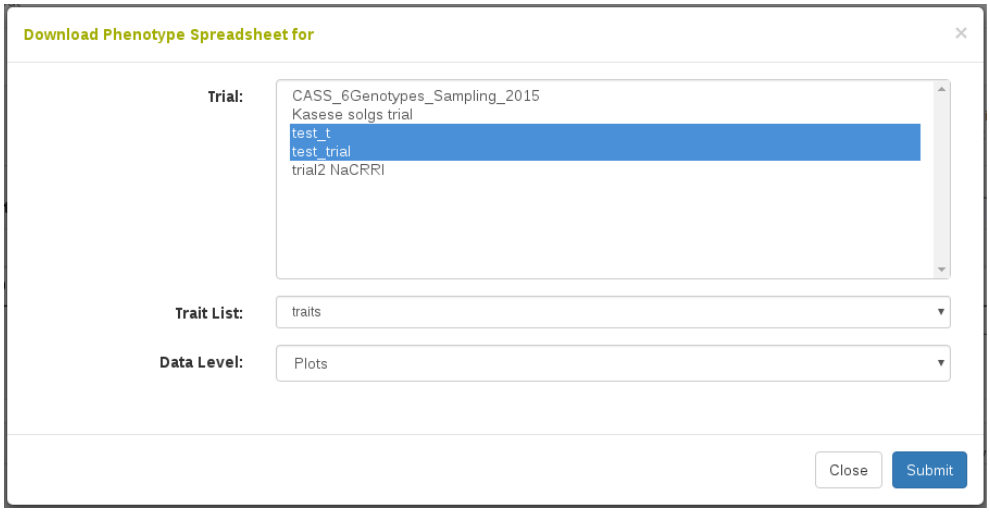



### 13.2.1 Generating Spreadsheet File

You can find more file format information by clicking on “Spreadsheet Format” link. Clicking on “Spreadsheet Format” will open the following dialog.

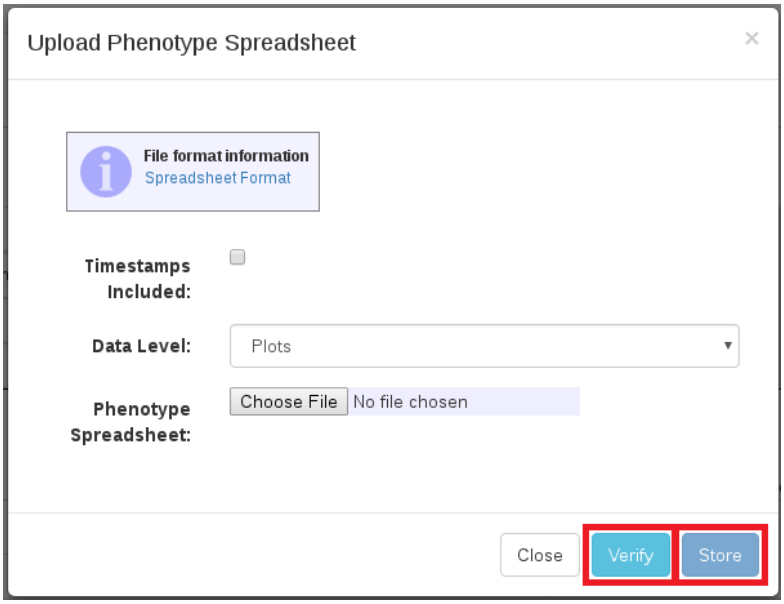



Clicking on "Create Phenotyping Spreadsheet" will bring up a dialog where you can indicate the trial(s) you are interested in and the trait list you are interested in. Clicking "Submit" will download the xlsx file onto your computer, where you can then fill in the phenotypes.




### 13.2.2 Uploading Spreadsheet File

To ensure that the file has a correct format for uploading, click on the "Verify" button. This will check the contents of the file and also perform quality checks on the values in the file. These checks include checking the trait definition for categorical values, minimum and maximum values, and data type checking. It will also check if there are already values uploaded for the given observation units and traits. If there are, there is an option to overwrite the existing values with the new values in your file. If the file is valid, only then can you click "Store" to store the information in the database.


Upload Phenotype Spreadsheet
File format information
Spreadsheet Format
Timestamps Included:
Data Level:
Plots
Phenotype Spreadsheet:
Choose File
No file chosen
Close
Verify
Store

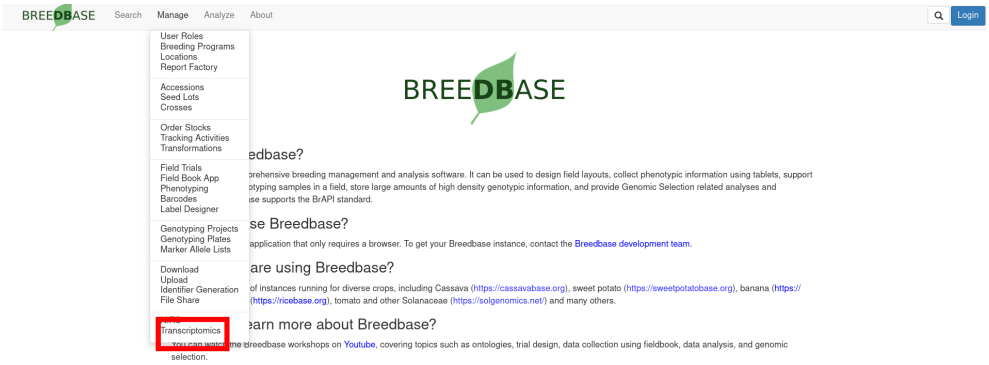
# Chapter 14

# Managing High Dimensional Phenotyping Data

## 14.1 Managing Transcriptomic Data

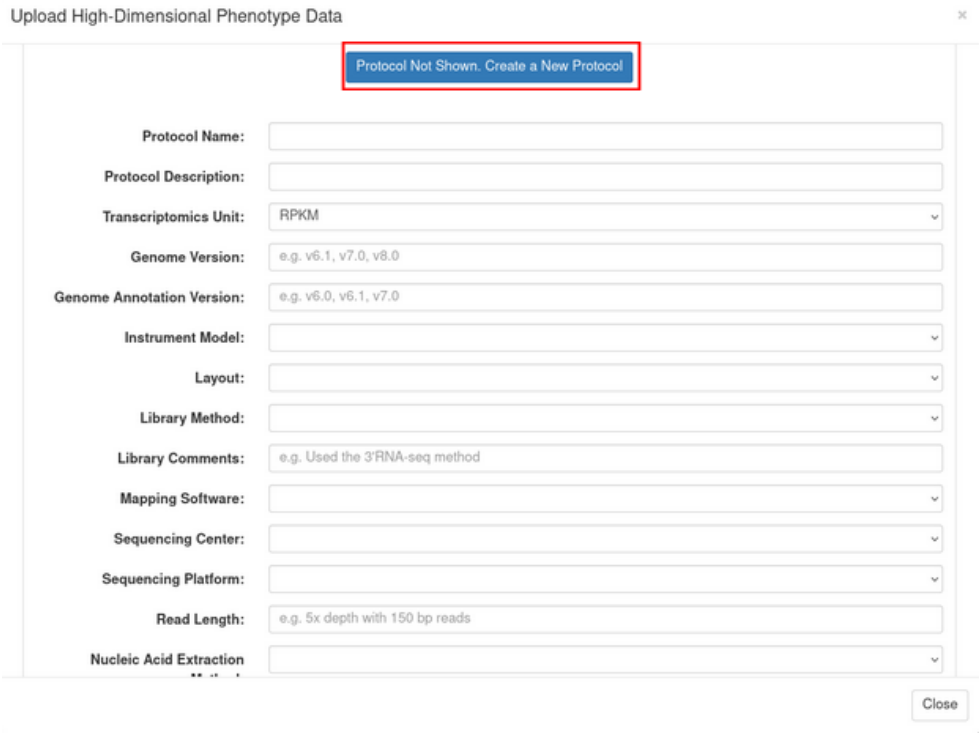
### 14.1.1 Uploading Transcriptomic Data

To upload transcriptomic data, go to the transcriptomics page by clicking "Transcriptomics" under the Manage tab.




Click the "Upload Transcriptomics Data" button to open the upload workflow dialog. On the second step "Samples", you will be prompted to make sure that your samples are already in the database. You can refer to the "Managing Tissue Samples" chapter for instructions on how to create a sam-

pling trial. After creating a sampling trial or confirming that your samples exist, move on to the "Protocol Info" step. Here you can select an existing protocol, or create a new protocol. To create a protocol, click the "Protocol Not Shown. Create a New Protocol" button. From here you can fill in information about your new protocol and when finished, click "Go to Next Step".




You will need to have two csv files when uploading transcriptomics data. The first is the data matrix file, which should have these headers:

Template Information

Upload Transcriptomics in a tabular spreadsheet (.csv) format with the following columns:

- **sample_name**: This should be a sample name that exists in the database. The sample can come from a sampling trial or a field trial or a genotyping plate.
- **device_id**: Optional identifier to differentiate device (e.g. serial number). Can be numeric or letter.
- **comments**: Optional comments for any information to include. Can include specific sample preparation for measurement.
- All other columns are **transcript names**. You can add as many columns as you want to upload, there isn't a limit.

| sample_name | device_id | comments | transcipt name columns |
|---|---|---|---|

The names in the sample_name column should match the sample names you created in your sampling trial file. The header marked "transcript_name_columns" should be replaced with the transcript names you want to upload, each in its own column in the header and the respective expression value below. Example:

| sample_name | device_id | comments | ZeamMp004 | ZeamMp005 | ZeamMp024 | ZeamMp032 |
|---|---|---|---|---|---|---|
| transcriptomics_test1 | | | 0.847898286111116 | 0 | 0 | 10.177049366007 |

The second file is the transcript details file, which should have these headers:

Upload details about the transcripts in a tabular spreadsheet (.csv) format with the following columns:

- transcript_name (required): This should be the same name as present in a column in your transcriptomics data matrix file.
- chromosome (required): The chromosome where the transcript exists.
- start_position (required): The start position in base pairs of where the transcript begins.
- end_position (required): The end position in base pairs of where the transcript ends.
- gene_description (optional): Descriptors for gene annotations.
- notes (optional): Any additional descrptions.

| transcript_name | chromosome | start_position | end_position | gene_description | notes |
|---|---|---|---|---|---|

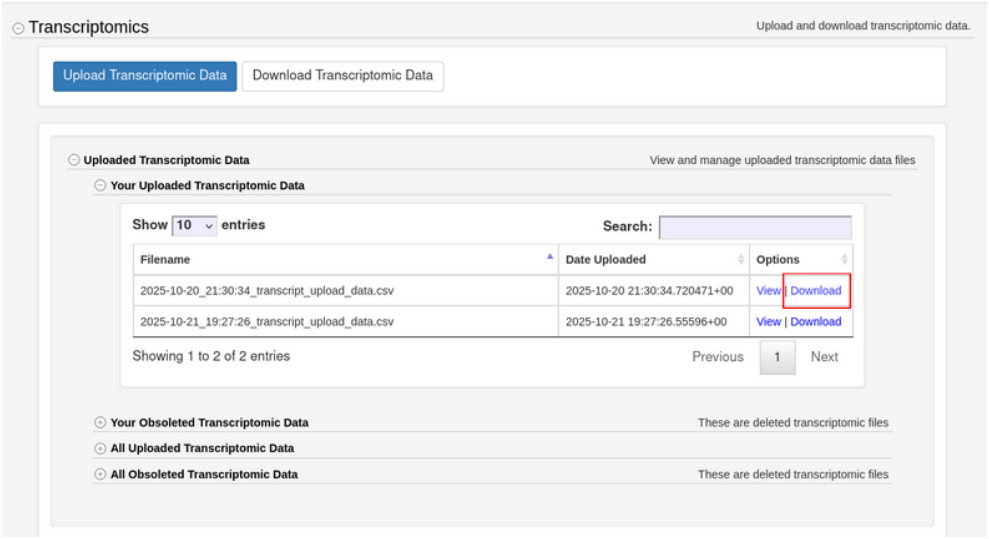
Each row under the transcript_name column should contain each transcript name from your data matrix file. For example:

| transcript_name | chromosome | start_position | end_position | gene_descrip | notes |
|---|---|---|---|---|---|
| ZeamMp004 | Mt | 8752 | 9915 | orf387 | |
| ZeamMp005 | Mt | 11111 | 13315 | orf734 | |
| ZeamMp024 | Mt | 161140 | 164661 | rps3 | |
| ZeamMp032 | Mt | 76132 | 84976 | nad4 | |

Once these two files are uploaded, click verify to check that they are formatted correctly and that the samples exist. If there are no errors, you can then click store to store your transcriptomics data.

### 14.1.2 Downloading Transcriptomics Data

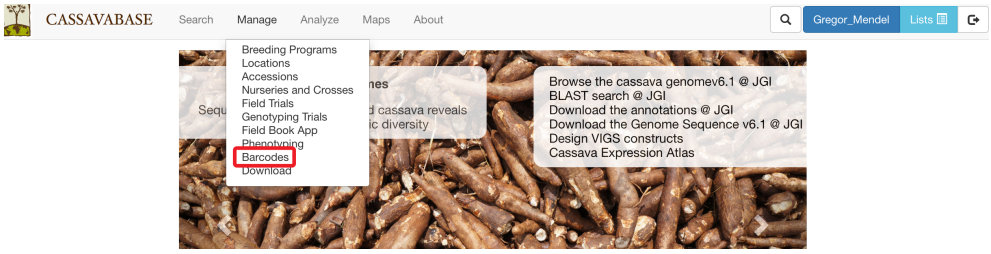
To download your data, click the "Uploaded Transcriptomics Data" dropdown menu on the Transcriptomics page. Then click the "Your Uploaded Transcriptomic Data" dropdown. From here you will be able to see the transcriptomics data that you have uploaded. Click the "Download" button next to the data that you wish to download.



# Chapter 15

# Managing Barcodes

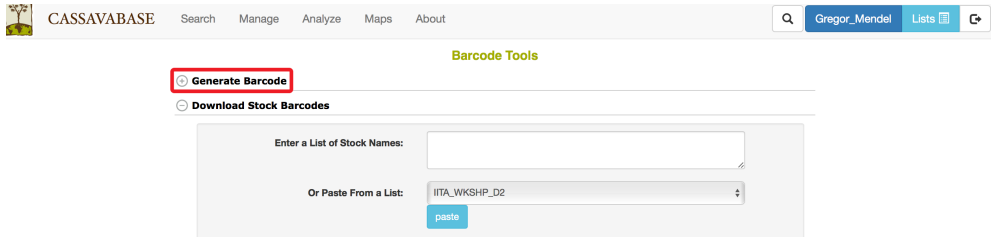
SGN databases provide tools for generating barcodes for stock identification. To access "Barcode Tools" page, clicking on "Barcodes" in the "Manage" menu.



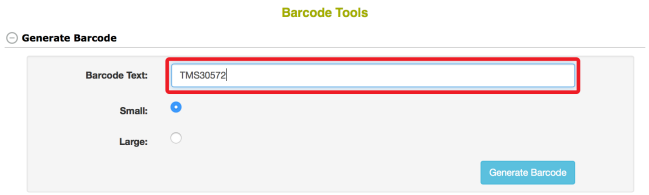

"Barcode Tools" page provides four options for generating barcodes:

- Single barcode
- Multiple barcodes
- Plot phenotyping barcodes
- Trial barcodes

To generate single barcode, clicking on "Generate Barcode" link on the "Barcode Tools" page.




In the "Generate Barcode" section, specify the name of the barcode, size of the barcode, then clicking on "Generate Barcode"



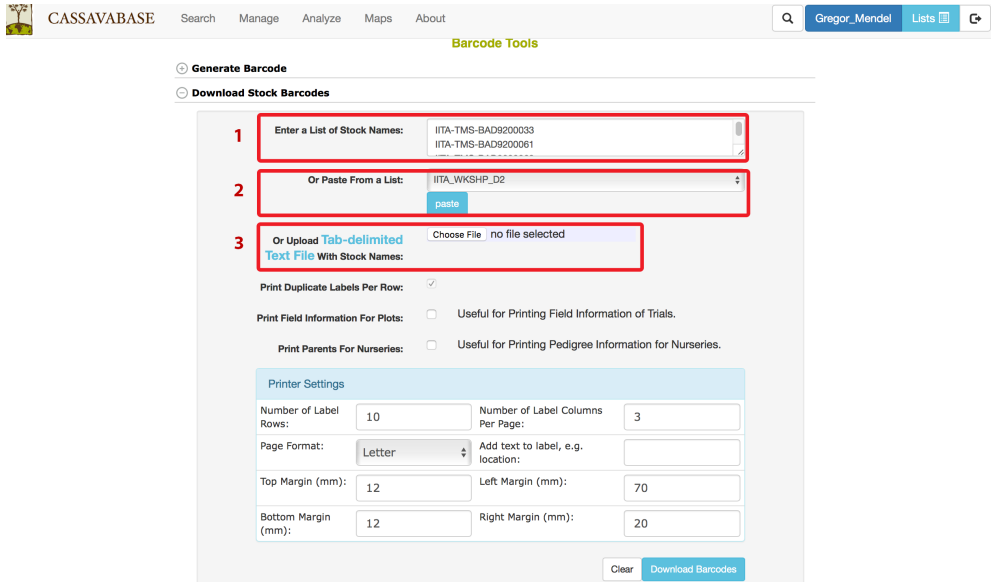

The database will generate a barcode for your stock. The barcode can be printed for your stock identification. It also appears on its corresponding stock page.

If you have a list of stocks that you want to generate barcodes, you can use "Download Stock Barcodes" section. You have three options for entering stock names:

1. Typing in stock names, or copy and paste from other file into the box (1)
2. Choosing a list of stocks from your "Lists" (2), and transferring the list into the box (1) by clicking on "paste" button.
3. Uploading a "Tab-delimited Text File" with stock names.
4. Select an optional printing format from the available formats.

You can select printer settings that you prefer in the "Printer Settings" sec-

tion. After you enter stock names and specify printer settings, clicking on "Download Barcodes" button at the bottom of the page.




If you have a list of plots that you want to generate phenotyping barcodes, you can use "Download Plot Phenotyping Barcodes" section. You have three options for entering plot names:

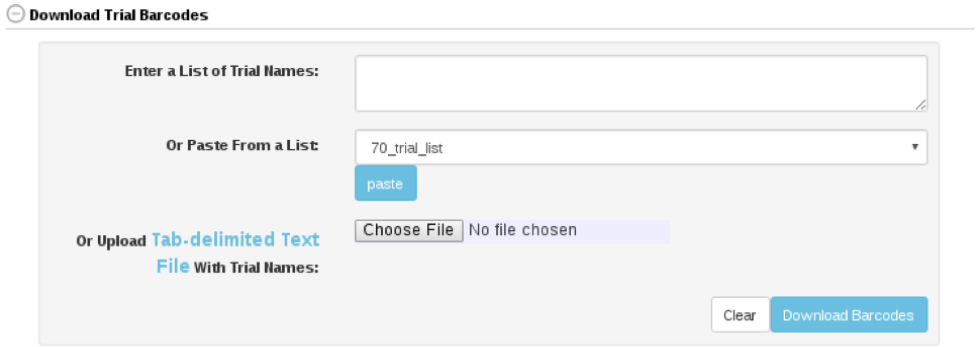
1. Typing in plot names, or copy and paste from other file into the box (1)

2. Choosing a list of plots from your "Lists" (2), and transferring the list into the box (1) by clicking on "paste" button.

3. Uploading a "Tab-delimited Text File" with plot names.

If you have a list of trials that you want to generate barcodes, you can use "Download Trial Barcodes" section. You have three options for entering trial names:

1. Typing in trial names, or copy and paste from other file into the box (1)
2. Choosing a list of trial from your "Lists" (2), and transferring the list into the box (1) by clicking on "paste" button.
3. Uploading a "Tab-delimited Text File" with trial names.



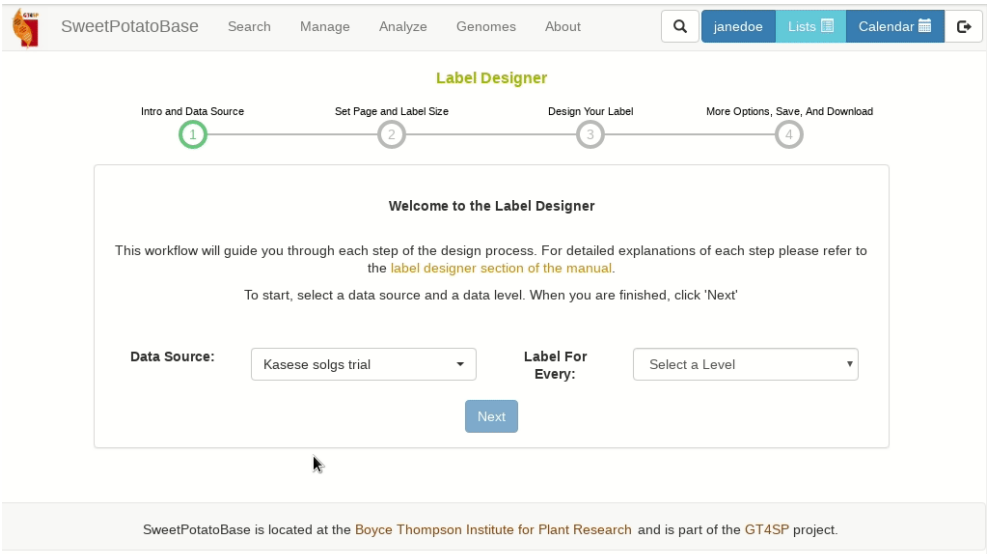
# Chapter 16

# Using the Label Designer

Breedbase provides an interactive design tool for creating custom labels. To access the label design tool, click on "Label Designer" in the "Manage" menu. The following sections explain your many options as you advance through each step of the design workflow.

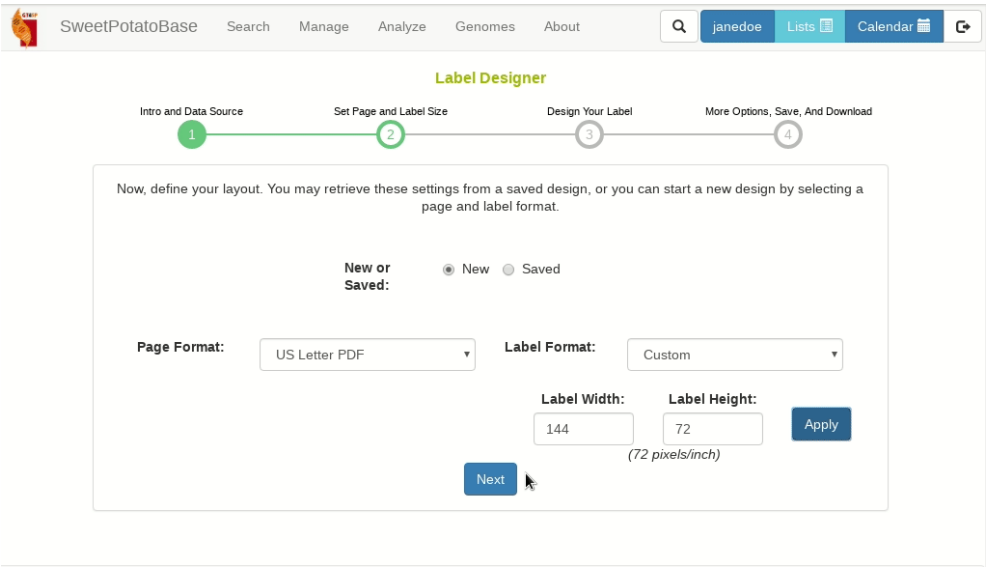
## 16.1 Select a Data Source

The first step is to select a data source. To do so, you may directly select a specific data source from the "Data Source" dropdown menu (either a field trial, genotyping trial, or crossing trial) to populate your labels with the trial design information. Alternately, from the same dropdown menu, you may select a list to populate your labels with the list contents. Prior to making a selection from the "Data Source" dropdown menu, you may also filter the data sources by using the "Datatype" dropdown menu. You must then choose a level (plot, plant, etc.) before proceeding, using the "Label for Every" dropdown menu. To generate plot-level labels for more than one trial at once, select a list of trials as the source and plot as the level. Once you have made your selections, click the 'Next' button to move to the next step in the workflow.



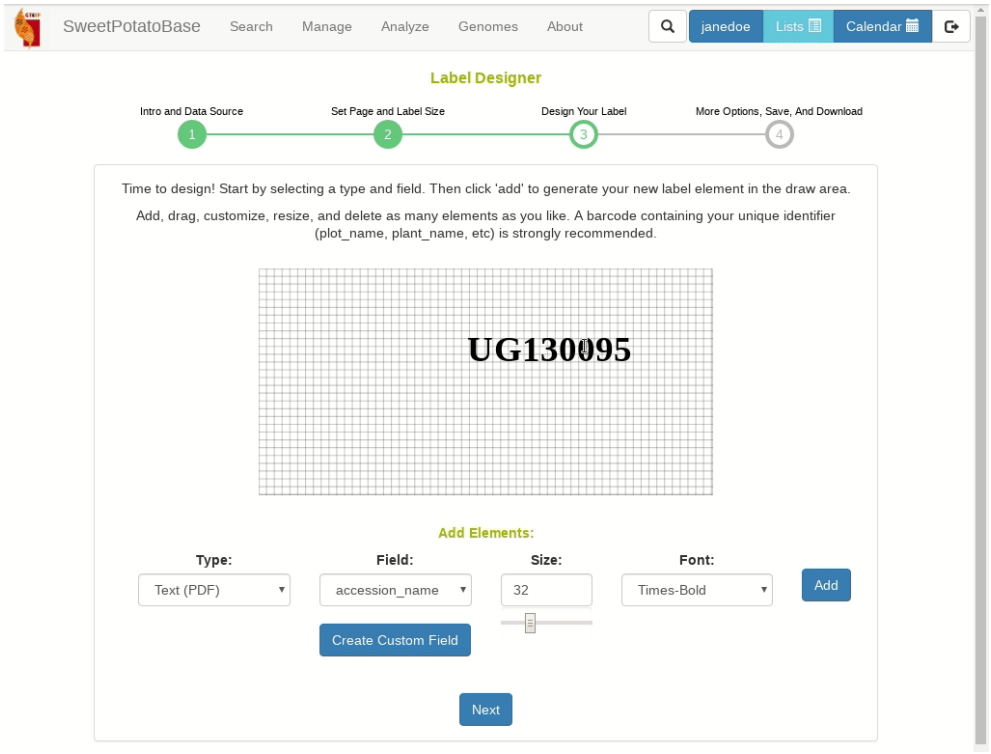

## 16.2 Set Page and Label Size

After selecting the data source, you must choose whether to create a new design or load a saved design. If you choose the "New" selector, you will be prompted to select a page size and label size. If you do not see your page size or label size as an option, then select "Custom" from the relevant dropdown menu and enter your desired dimensions in pixels (each pixel is 1/72nd of an inch). If you choose the "Saved" selector, you will be prompted to select a saved design. After selecting a saved design, you will be taken directly to the design step with the saved design elements pre-loaded.



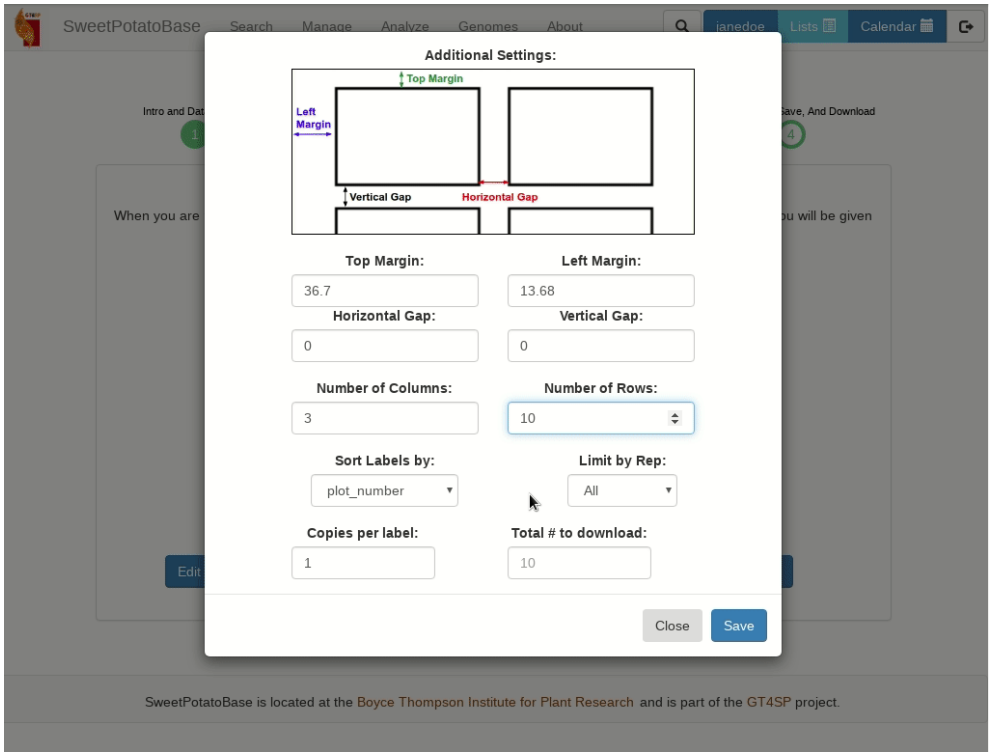

## 16.3 Design Your Label

Having set the page and label formats, you will be presented with a label canvas where you can begin adding elements to your label. Select a type, field, size, and font, then click "Add". You can add text to an existing field or create a completely custom field by clicking "Create Custom Field". Once added, you can drag and drop elements, or you can delete them by clicking on the red box in their upper left corners. Barcodes can also be resized by dragging the green box in their lower right corners. If you are creating labels for a trial, it is highly recommended that you include a barcode encoding your plot, plant, or tissue sample names. These are the unique identifiers that will need to be included with any phenotypic or genotypic measurements loaded into the database. When you are satisfied with your label design, click "Next".




## 16.4 Adjust Formatting, Save, and Download

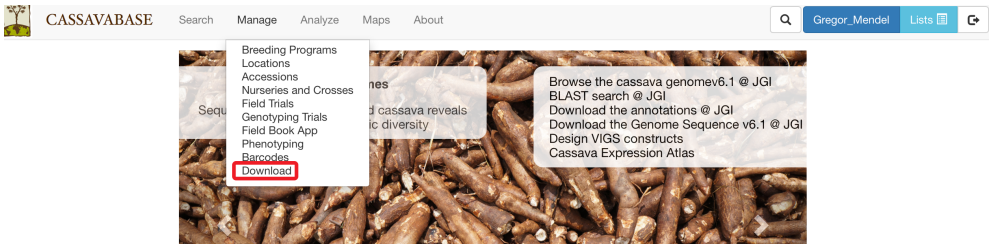
In the last step of the workflow, you can tweak your formatting and page layout, save your design, or download your labels. The "Additional Settings" dialog will allow you to adjust the page margins and margins between labels. The units are pixels (each pixel is 1/72nd of an inch). It is not recommended that you change these settings until you have already printed a test page. You can also set the number of copies per label, filter by replicate, or download only the first page for test purposes. To save your design, type a unique name in the text field and click "Save". This will save your design to your list manager, where you can make it public in order to share it with others. Finally, if you are ready, click download to generate and download your labels!


SweetPotatoBase
Search
Manage
Analyze
Genomes
About
janedoe
Lists
Calendar
Additional Settings:
Top Margin
Left Margin
Vertical Gap
Horizontal Gap
Top Margin:
36.7
Left Margin:
13.68
Horizontal Gap:
0
Vertical Gap:
0
Number of Columns:
3
Number of Rows:
10
Sort Labels by:
plot_number
Limit by Rep:
All
Copies per label:
1
Total # to download:
10
Close
Save
SweetPotatoBase is located at the Boyce Thompson Institute for Plant Research and is part of the GT4SP project.

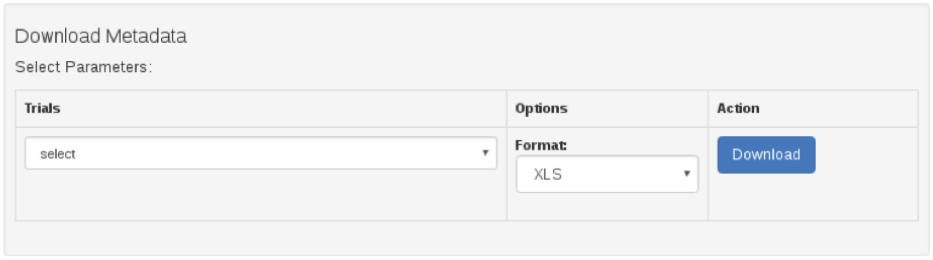
# Chapter 17

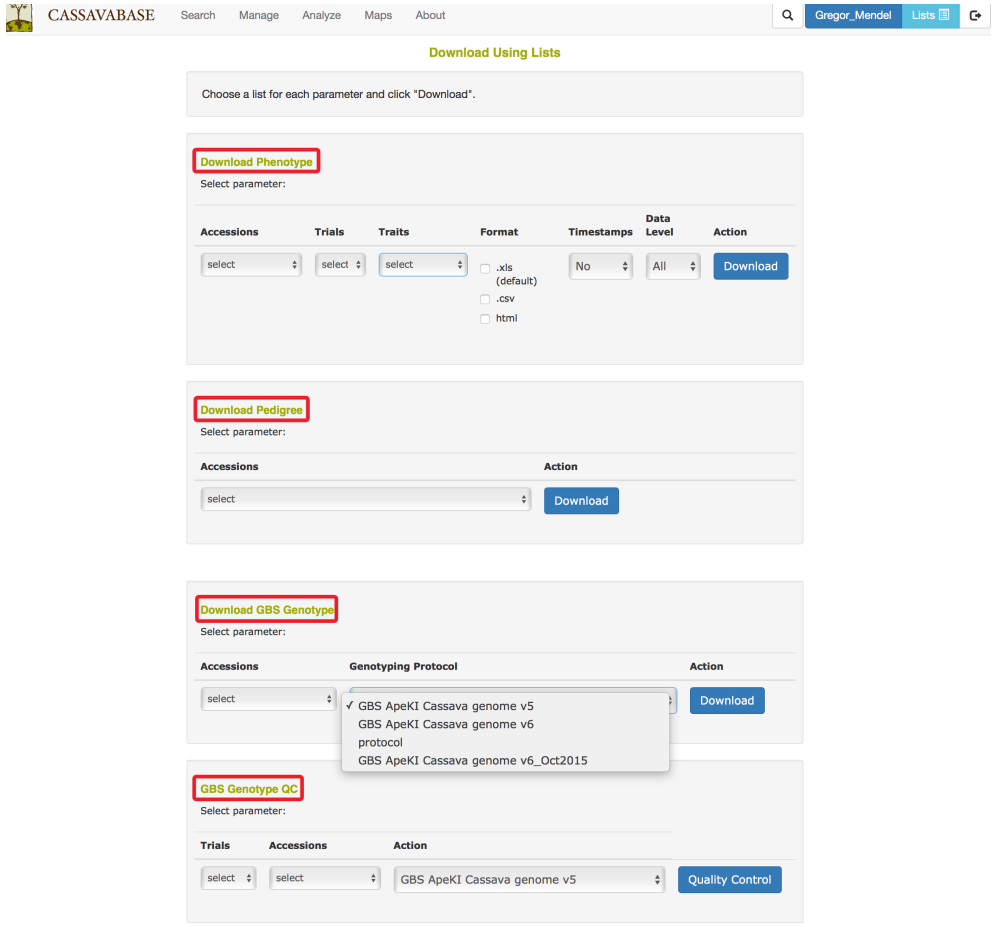
# Managing Downloads

You can download phenotype, trial meta-data, pedigree, GBS genotype and GBS genotype QC files from the database to your computer by using "Lists". To download, clicking on "Download" in the "Manage" menu.




For each category, you can select a list of accessions from your "Lists" to download their phenotypes, pedigree, GBS genotype, GBS genotype QC. In the case of downloading trial meta-data, you would provide a list of trials, while for downloading phenotype and GBS genotype QC, you can also use a list of trials or traits.




CASSAVABASE
Search
Manage
Analyze
Maps
About
Gregor_Mendel
Lists
Download Using Lists
Choose a list for each parameter and click "Download".
Download Phenotype
Select parameter:
Accessions
Trials
Traits
Format
Timestamps
Data Level
Action
select
select
select
.xls (default)
.csv
html
No
All
Download
Download Pedigree
Select parameter:
Accessions
Action
select
Download
Download GBS Genotype
Select parameter:
Accessions
Genotyping Protocol
Action
select
GBS ApeKI Cassava genome v5
GBS ApeKI Cassava genome v6
protocol
GBS ApeKI Cassava genome v6_Oct2015
Download
GBS Genotype QC
Select parameter:
Trials
Accessions
Action
select
select
GBS ApeKI Cassava genome v5
Quality Control

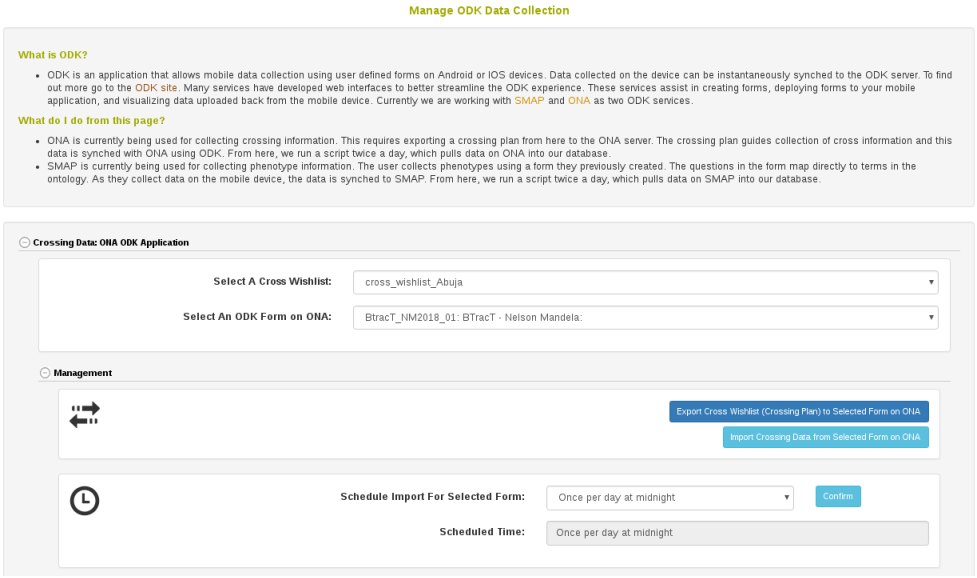
# Chapter 18

# Managing ODK Data Collection

To access this page go to Manage and then ODK Data Collection. ODK is used for remotely collecting data on Android and IOS devices. We currently are working to support two ODK service providers, namely ONA and SMAP. We are using ONA to collect crossing information, including all lab activities following seed production. We are using SMAP for phenotypic data collection.

# 18.1 ONA Crossing Information

## 18.1.1 Managing ONA Crossing Information




To begin collecting data using the ONA ODK form you must first have a crossing plan in the form of a Cross Wishlist. To do this from this page, click the "Export Cross Wishlist to ONA" button. Please refer to the "Create Cross Wihlist" help section for more information. It is possible to view the current available cross wishlists by clicking the "Export Cross Wishlist to ONA" button and then clicking "Available Cross Wishlists".

Once your cross wishlist is available, you can use your mobile ODK application to record crosses being done realtime. You can also record all laboratory activities following seed extraction up to greenhouse plantlet hardening.

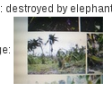
As you collect data using your mobile ODK application, your responses will be synchronized with our database. The "Schedule Import for Selected Form" section gives you options to perform the import daily or more frequently. It is also possible to initiate a data import from ONA at anytime by clicking "Import Crossing Data from Selected Form on ONA".

### 18.1.2 Reviewing Plant Status

The mobile ODK application has options to collect information about the status of plants in the field, such as if they are flowering. Images for each plant can also be recorded. The database will report this information here in a summary table that looks like the following. Notice that images are also transferred to the database.

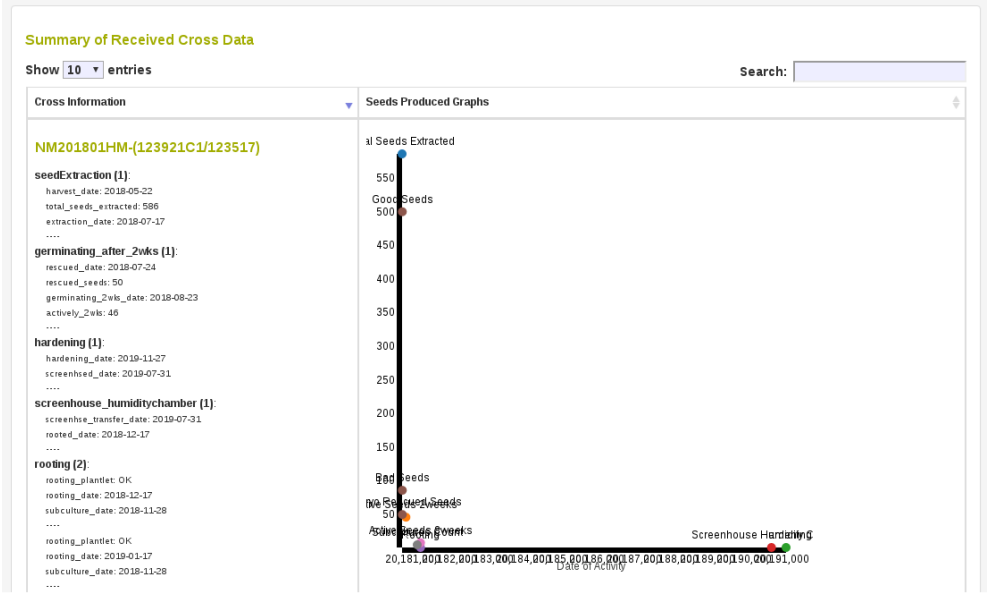
Summary of Received Plant Status

Show 10 entries Search:

| Plot Name | Date | Status |
|---|---|---|
| | null | **Status:**<br>Accession Name: null<br>Trial Name: null<br>User: null<br>Status Location:<br>Status: fallen<br>Note: null<br>Image: undefined |
| 16-Huti-white_r8c12_plot157 | 2018-02-12 | **Status:**<br>Accession Name: Huti-white<br>Trial Name: 2016 mchare polliantion block<br>User: HM<br>Status Location: in_field<br>Status: destroyed<br>Note: destroyed by elephants<br>Image:  |
| 16-ITC1460-Ijihulnkundu_r17c8_plot344 | 2018-01-18 | **Flowering:**<br>Accession Name: ITC1460-Ijihu Inkundu<br>Plant Sex: female |
| 16-ITC0712-AAcvRose_r1c1_plot1 | 2018-01-17 | **Flowering:**<br>Accession Name: ITC0712-AAcv Rose<br>Plant Sex: male |
| 16-ITC1468-Kahuti_r1c2_plot2 | 2018-01-17 | **Flowering:**<br>Accession Name: ITC1468-Kahuti<br>Plant Sex: female |

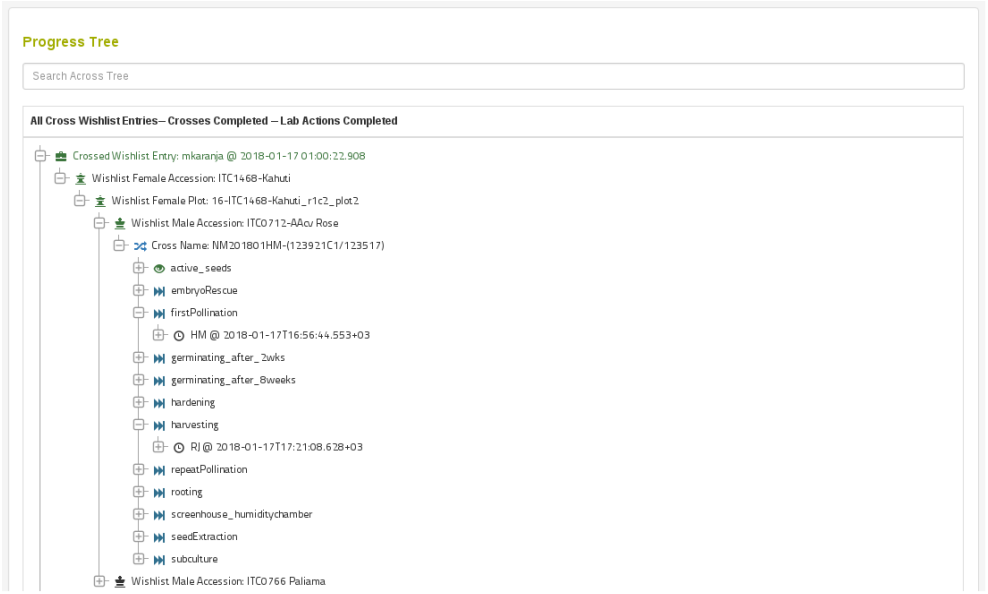
### 18.1.3 Graphical Summary For Performed Crosses

There is a section to summarize activities done for each cross. In this table each row represents a single cross performed. All the activities that have been performed will be shown here, such as “first pollination” and “embryo rescue”. The scatter plot shown tracks seed numbers generated on the Y axis and date of activity on the X axis.




## 18.1.4 Summary Information For Performed Crosses

There is a secondary section to summarize what has been done across the entire Cross Wishlist. This tree structure shows all activities performed for a cross and shows how these crosses relate to the Cross Wishlist.



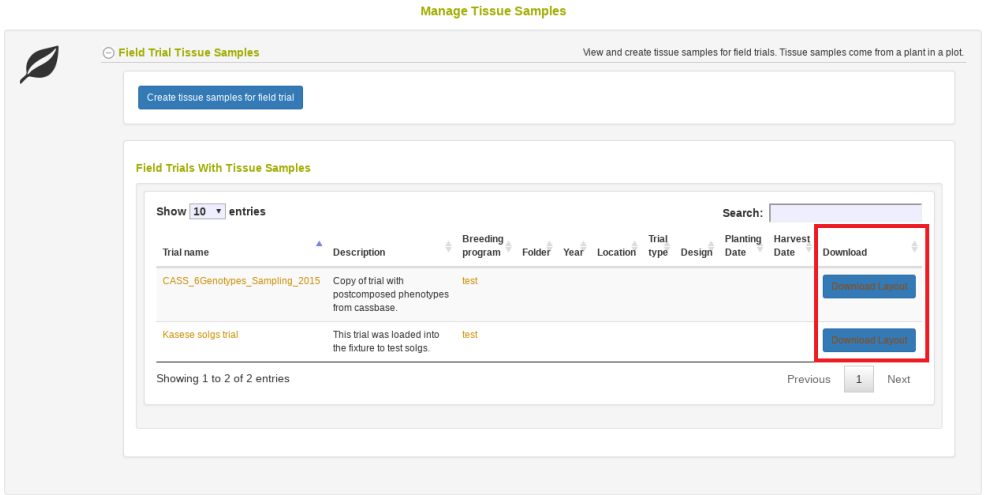
# Chapter 19

# Managing Tissue Samples

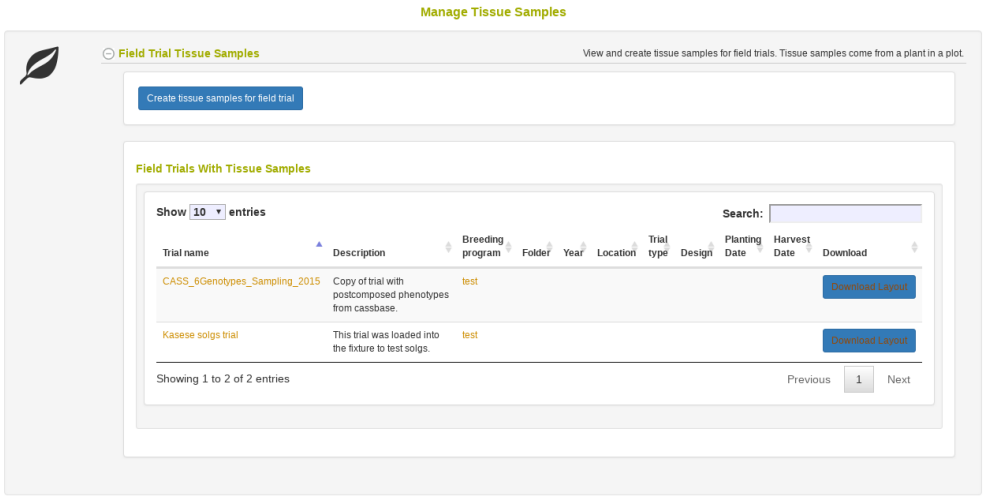
To access this page go to Manage and then Tissue Samples.

## 19.1 Tissue samples from field trials

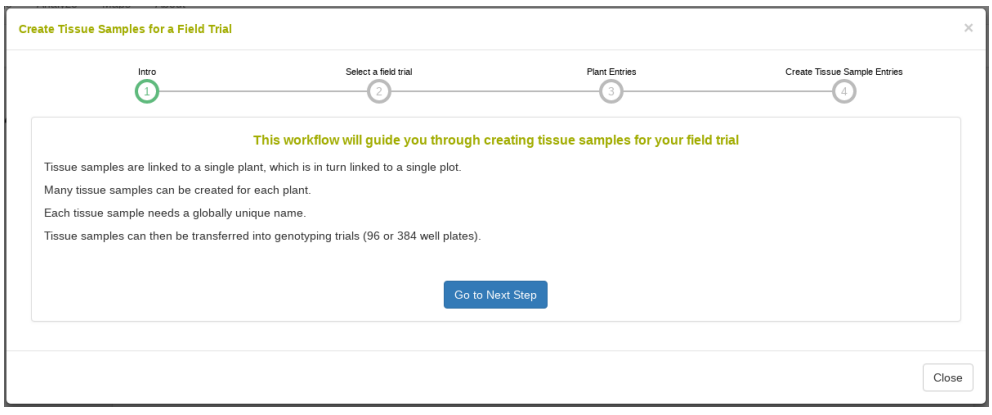
A field trial contains plots planted with a specific accession. Each plot can contain many plants, which in turn can contain many tissue samples. On the manage tissue sample page we can see the field trials that contain tissue samples already. We can choose to download the tissue sample layout as seen in the below picture.



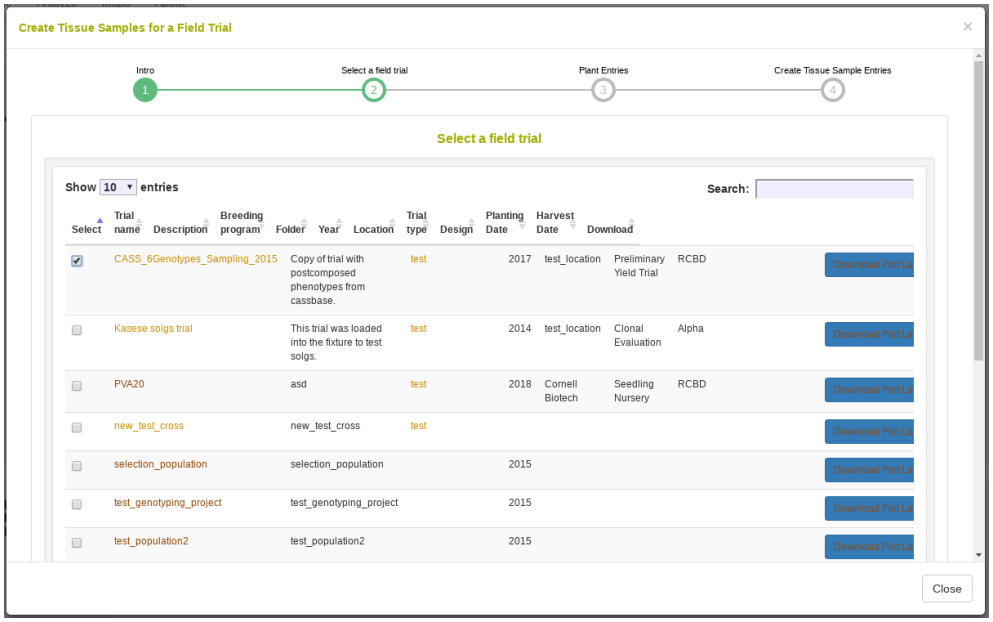
If the field trial you want to collect tissue samples from is not in the above table, you can click the button highlighted below.



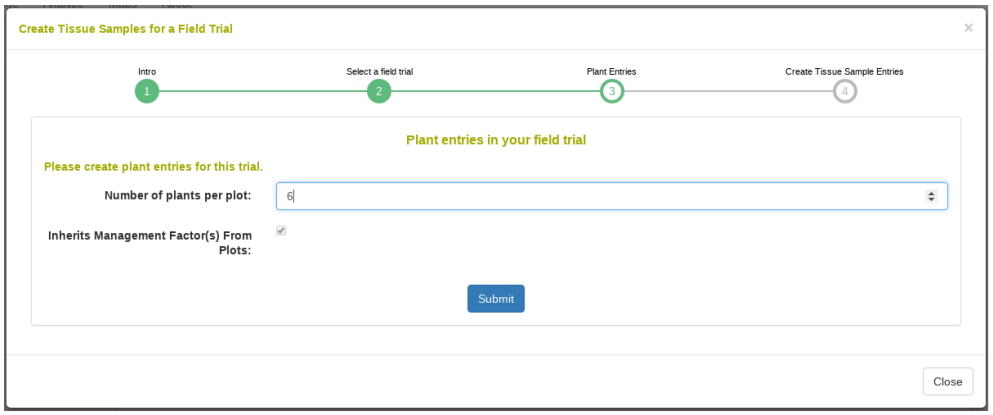

Once you have clicked this button, you will enter a workflow that begins with the following introduction.




Once you click next, you will need to select your trial.

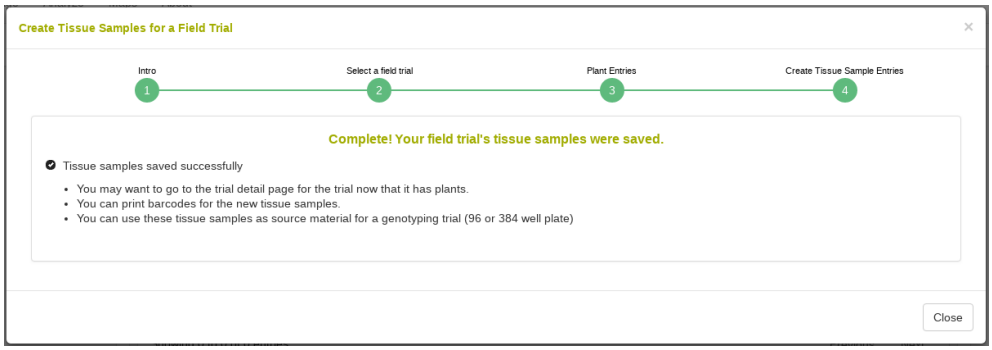



Next, if your trial currently only has plot entries saved, you will be asked to enter how many plants are in each plot.




Finally you will be asked how many tissue samples you want for each plant. You can specify a string to include in the tissue sample name, such as leaf or root.

Afterwards you should see the following success message, indicating that the tissue samples are saved.

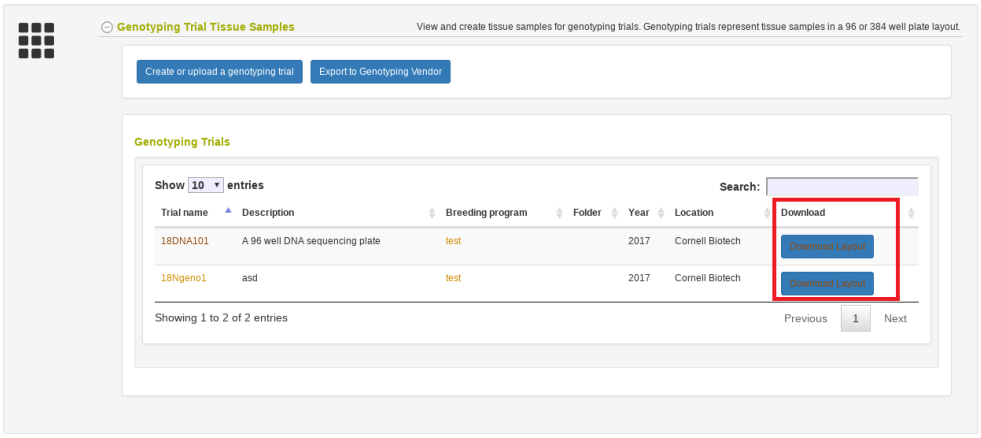



## 19.2 Genotyping Plate Tissue Samples (96 or 384 well plates)

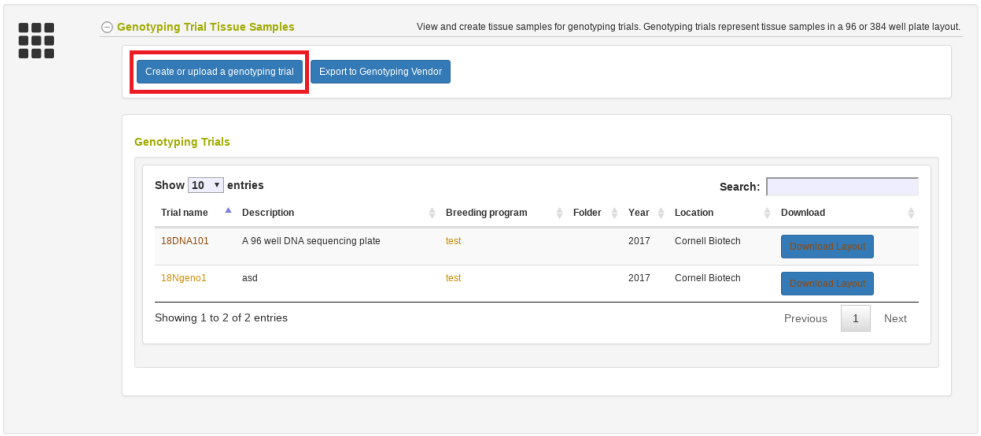
A genotyping plate represents a 96 or 384 well plate. You can use the Coordinate Android application to create your plate layout, or you can upload your own Excel plate layout, or you can use the database to generate a plate layout. Ideally, you will use tissue sample names originating from a field trial as the "source" for each well tissue sample, but you can also use plant names, plot names, or accession names.

From the manage tissue samples page, you can see the genotyping plates

saved in the database. You can also download the layouts as shown below.




If you need to create a new genotyping plate, you can click the button shown below. This will guide you through a workflow for uploading or creating the new plate layout.




Genotyping vendors require you to send a plate layout during submission. You can download the plate layout as shown above, or you can go to a genotyping plate detail page to download the Intertek formatted file.

In the future you will be able to directly export your genotyping plate plate layout to vendors.

# Chapter 20

# Managing Observation Variables

## 20.1 Managing Observation Variables with Traits, Methods, and Scales

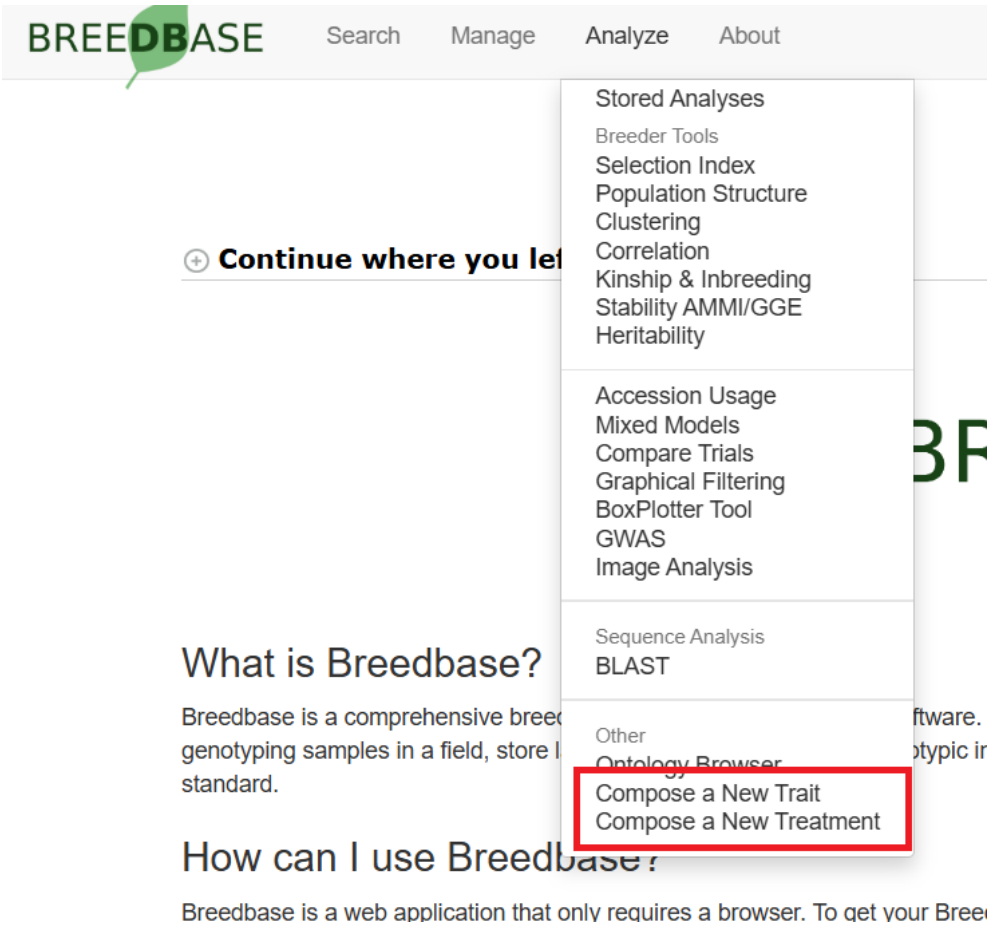
Though the information in this section pertains to traits, it also applies to experimental treatments. In general, anything that can be done with a trait can be done with a treatment.

Observation variables are the identifiers used when collecting phenotypic data. An observation variable is composed of a trait, a method, and a scale. The trait describes the attribute being measured e.g. 'Plant Height'. The method defines the protocol in which the trait was observed e.g. 'Using a one meter long measuring stick'. The scale defines the units or dimensions for which the measurement was taken e.g. 'Meters'.

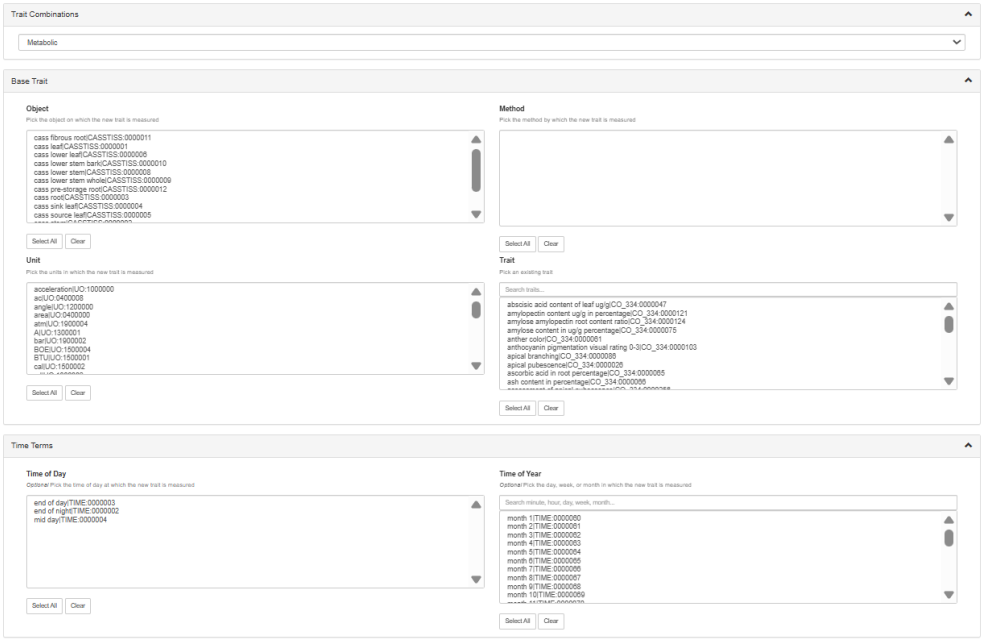
Generally, variables are defined in ontologies that are predefined. We often use ontologies from cropontology.org. In this case, you will not be able to define your own variables directly; instead, you will need to contact us and we will add the variable for you. We will discuss manual trait or treatment creation later in this section.

Before setting out to introduce new traits, visit the trait search page under the *Search* tab to make sure it isn't already present. Once a set of traits,

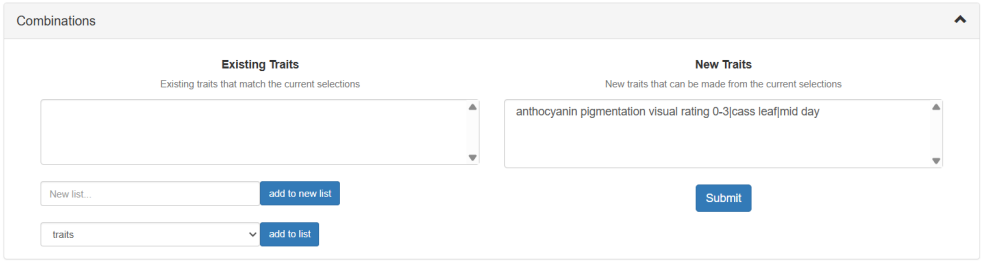
methods, units, scales, and other terms have been added to the database, they can be composed into an observation variable from the *Analyze* tab under *Compose a new trait*.




A base trait term can be combined with an object/plant part, collection method, measurement unit, and time terms to combine into a new observation variable. The types of combinations available to be made determine which of these can be combined. The types of composed traits available to be made is determined in your server configuration - contact a server administrator with questions.




To proceed, select one or more term from each box. At the bottom of the page, you will see existing composed traits that match your selections as well as the new composed traits that can be made. Select one or more of the possible new composed traits and click 'Submit' to create them.

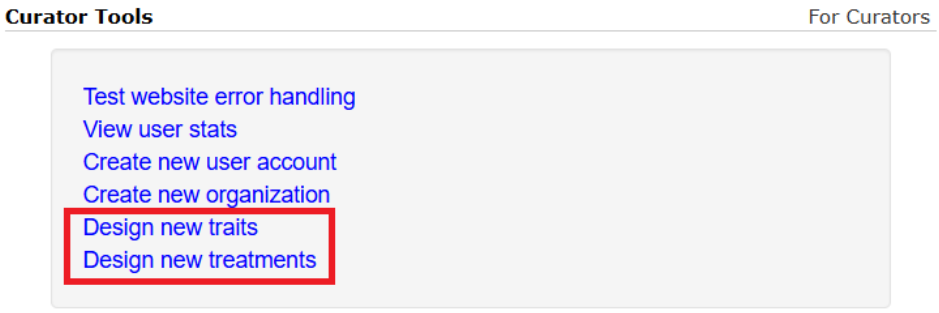



Afterwards, you can use the newly created observation variable ontology term in your phenotyping. The same interface can also be used for treatments, but on the *Compose a new treatment* page.

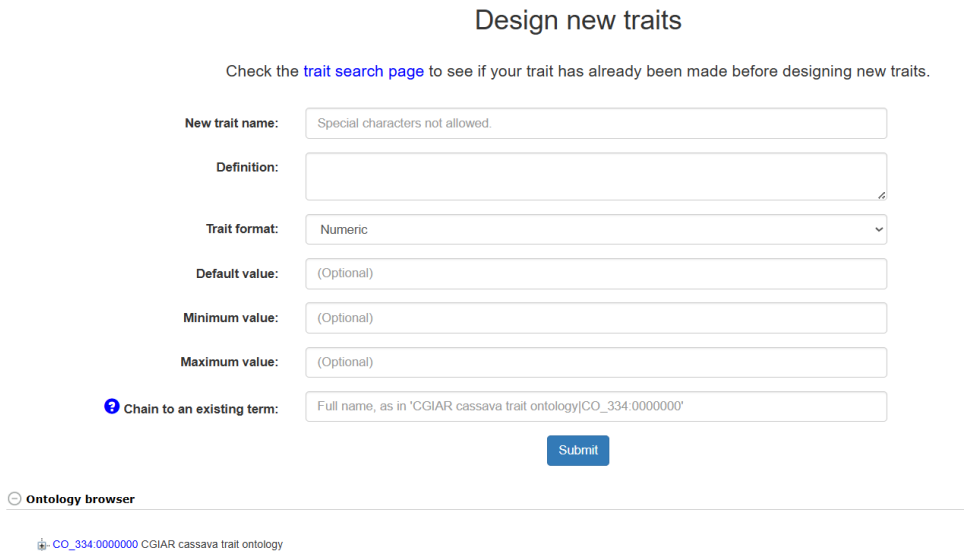
For databases where the user has greater control, we have an interface to allow curators to add new traits or treatments.

If the database you are on allows you to directly add observation variables (contact a server administrator to change this), you will see links in the tools

section of the user profile page.




On the trait design page, you will see a form prompting you for the trait name, definition, and optional data like the default value, minimum, maximum, or categories. At the bottom of the page, the ontology browser is duplicated for your convenience.




The final input box allows you to choose the parent term for the new term. The trait format dropdown allows you to specify numeric traits, categorical traits, or a new ontology. New ontologies allow you to organize your traits into hierarchies. For example, an 'Agronomic trait' sub-ontology could be made to include all agronomic trait measurements in the same group. Note that once a new term has been designated as numeric or categorical, it cannot have more child terms - it is the end of the hierarchy.

Once a new trait has been submitted, the page will refresh and the trait will be available for use in the composition tool.

# Chapter 21

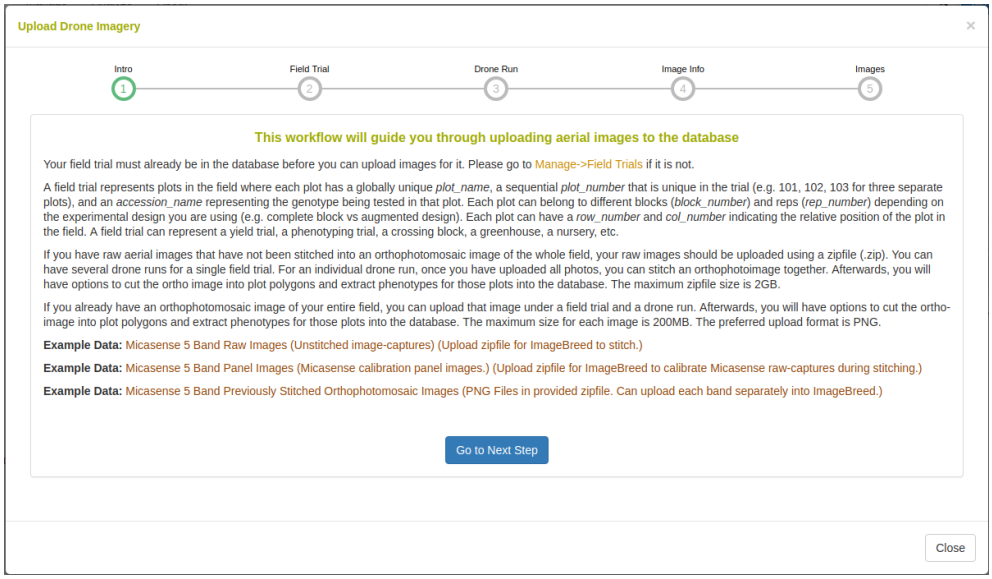
# Managing Drone Imagery

## 21.1 Image-Phenotyping Dashboard

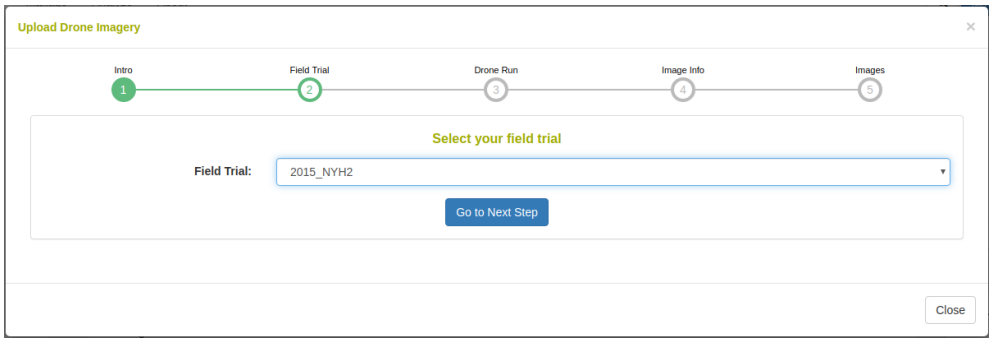
1. Upload raw image-captures in a compressed file (.zip) for orthophotomosaic assembly or upload previously stitched orthophotomosaic raster (.PNG, .JPG) imagery.
2. Dashboard shows all field trials and uploaded imaging events in collapsible sections.
3. Follow standard processes to manually create templates for assignment of plot-polygon images to the field experiment design.
4. All imagery is shown with the spectral category within collapsible sections. Figure shows NIR imagery.
5. Apply Fourier transform filtering, thresholding, and vegetation index masking. Plot-polygon images for all image processes are shown.
6. Extract and export phenotypic values from plot-polygon images for analyses and model training.

## 21.2 Image Input

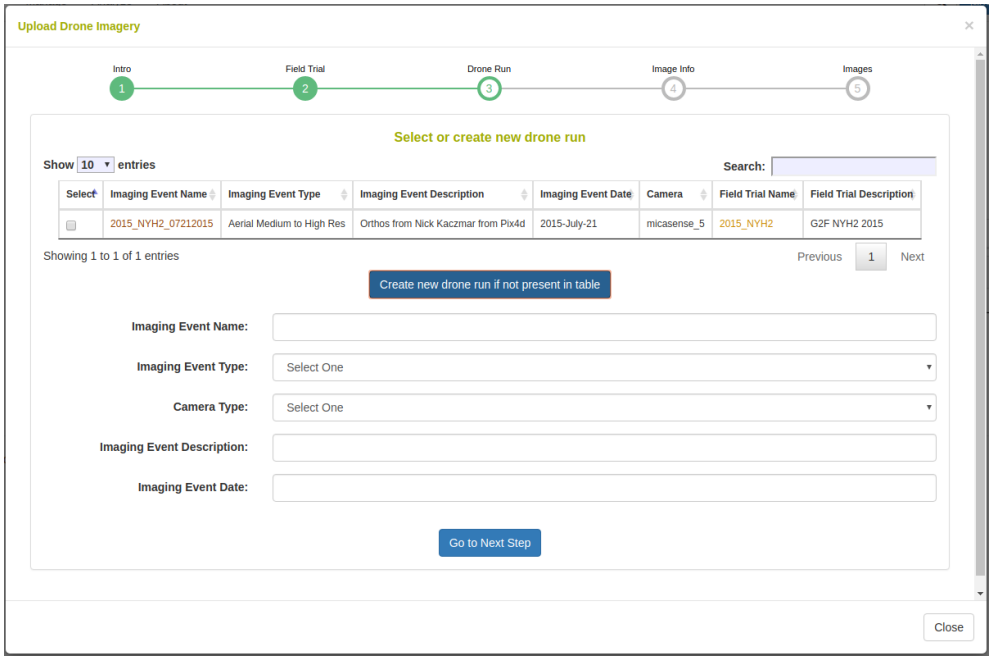
Clicking "Upload Imagery" will open the following dialog.



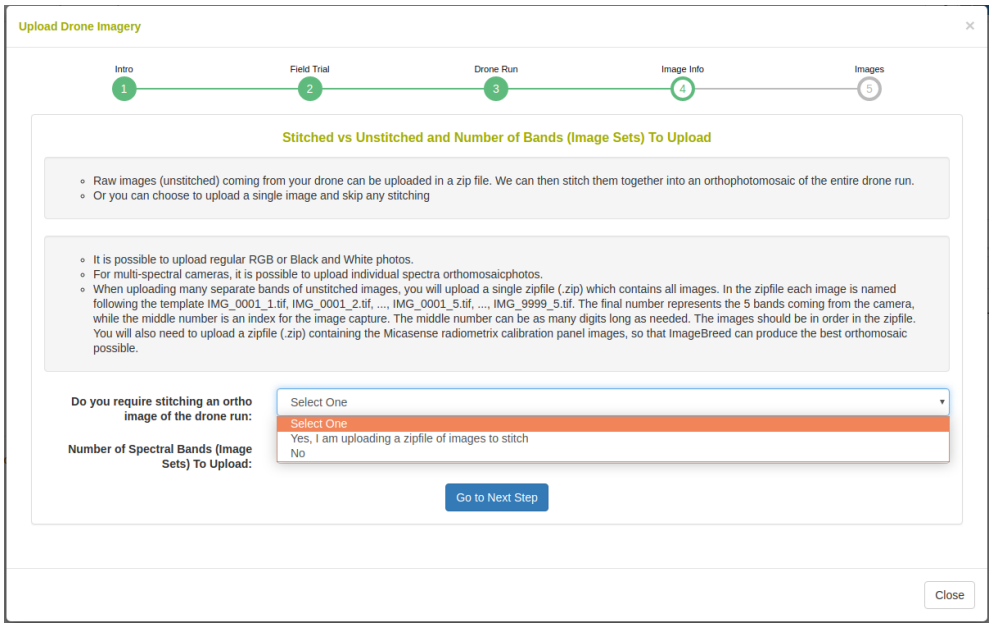

Raw-captures can be uploaded in a compressed (.zip) file so that they can be assembled into an orthophotomosaic. If orthophotomosaic assembly is not required, raster images (.PNG, .JPG) can be uploaded. Example data is given for raw Micasense RedEdge 5-band multispectral captures and for stitched orthophotomosaics.



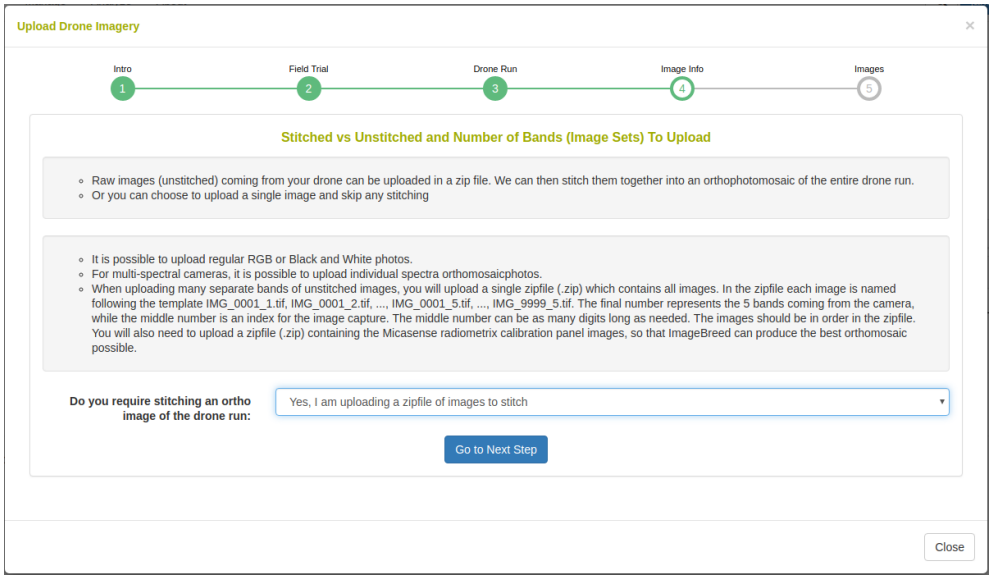

To begin uploading images, a field trial must be selected. The field trial must already be saved in the database. For information about adding a field trial, please read the Field Trial documentation.



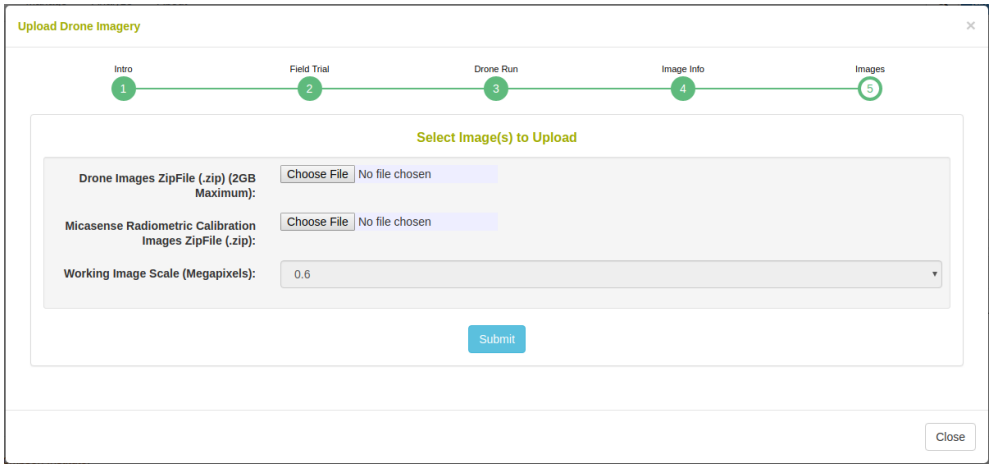

The image data is added to an imaging (drone run) event. Here you can select a previously saved imaging event or you can create a new one by defining a name, description, and date.



The uploaded data can be raw image-captures or complete raster images. Here you can select whether orthophotomosaic stitching is required.

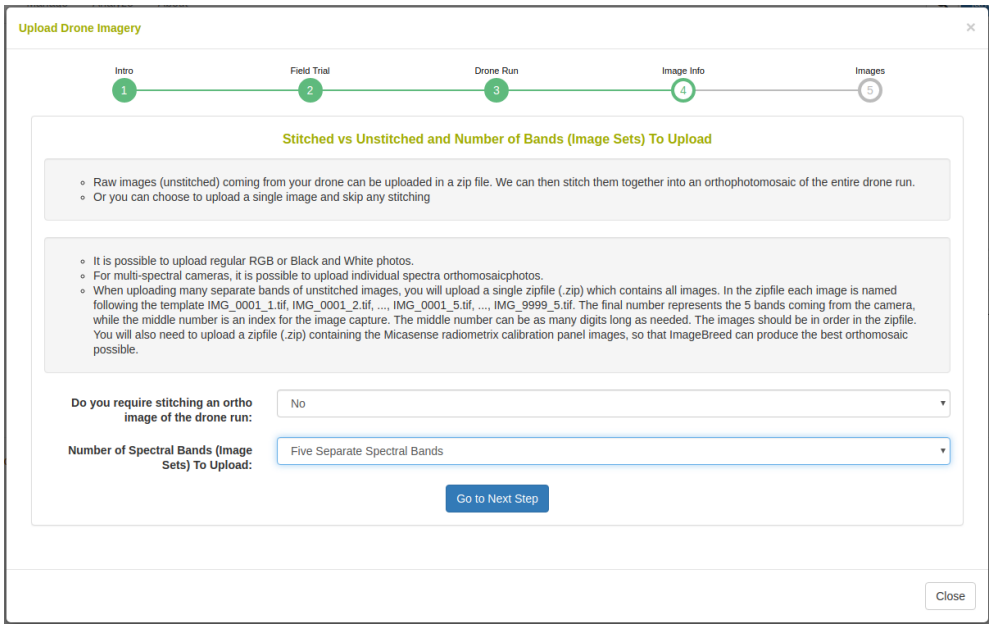



In the case that orthophotomosaic stitching is required, select 'yes'. On the next step you will see the following: Upload a zipfile with the raw-captures. When uploading Micasense RedEdge raw-captures, provide images of the Micasense calibration panels in a zipfile as well.

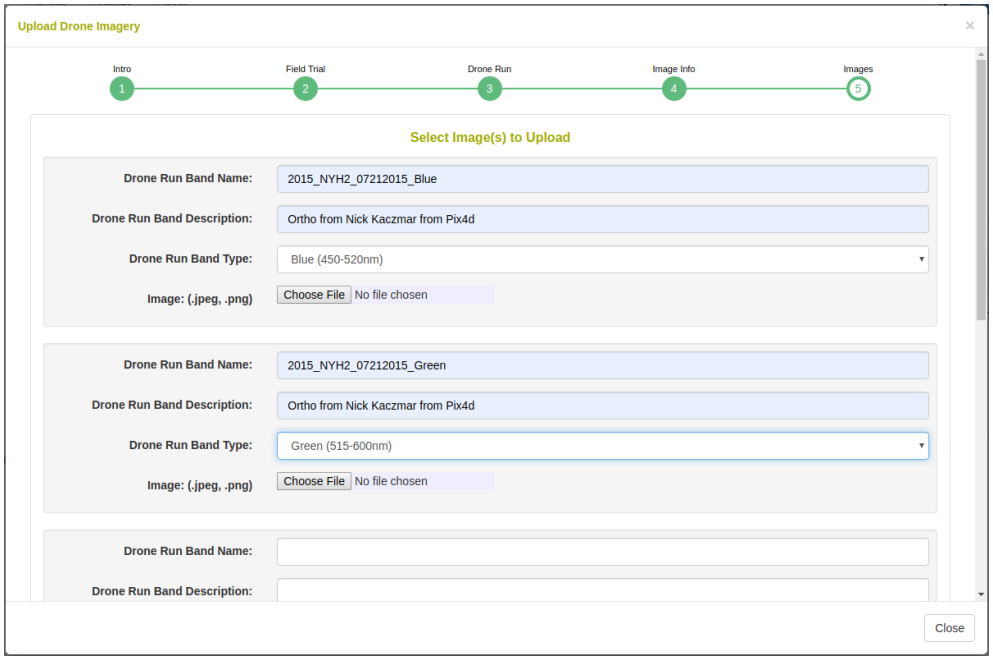



In the case that orthophotomosaic assembly is not required, simple upload the raster images. Select the number of image bands that will be uploaded

e.g. for a five band multispectral camera, select 5.



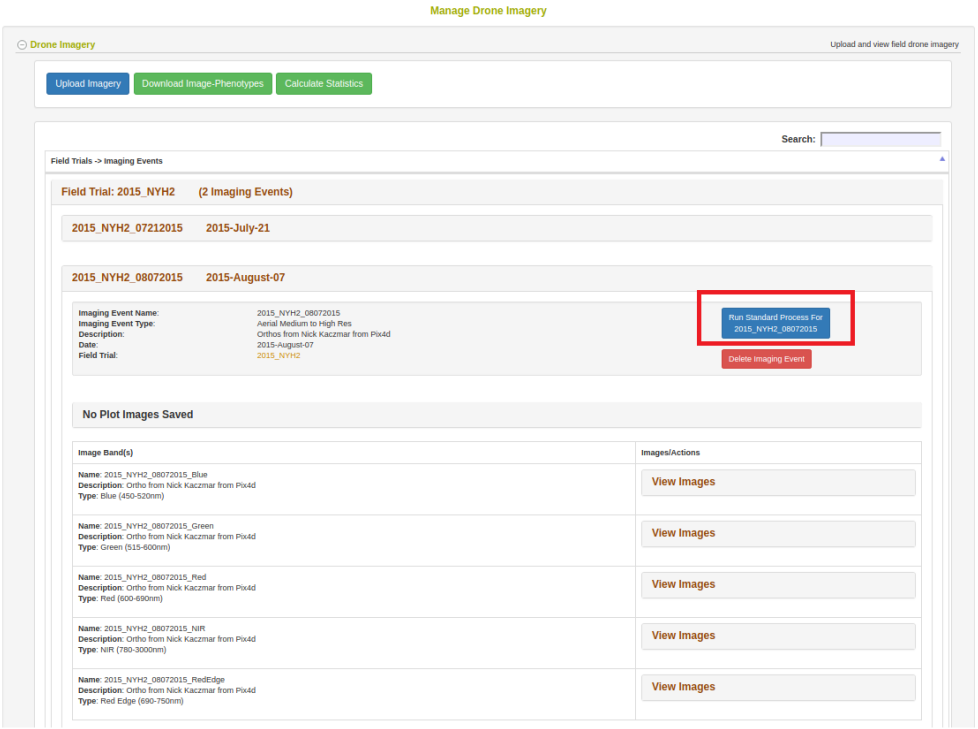

In the caes that orthophotomosaic stitching is not required, select 'no'. On the next step you will see the following:



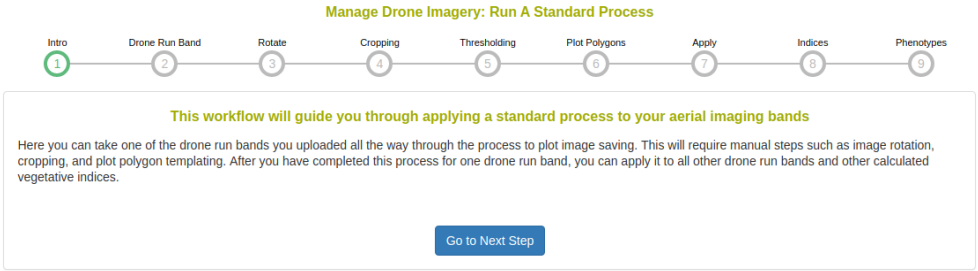
Upload an image at each band with a unique name, description, and spectral type.

## 21.3 Standard Process

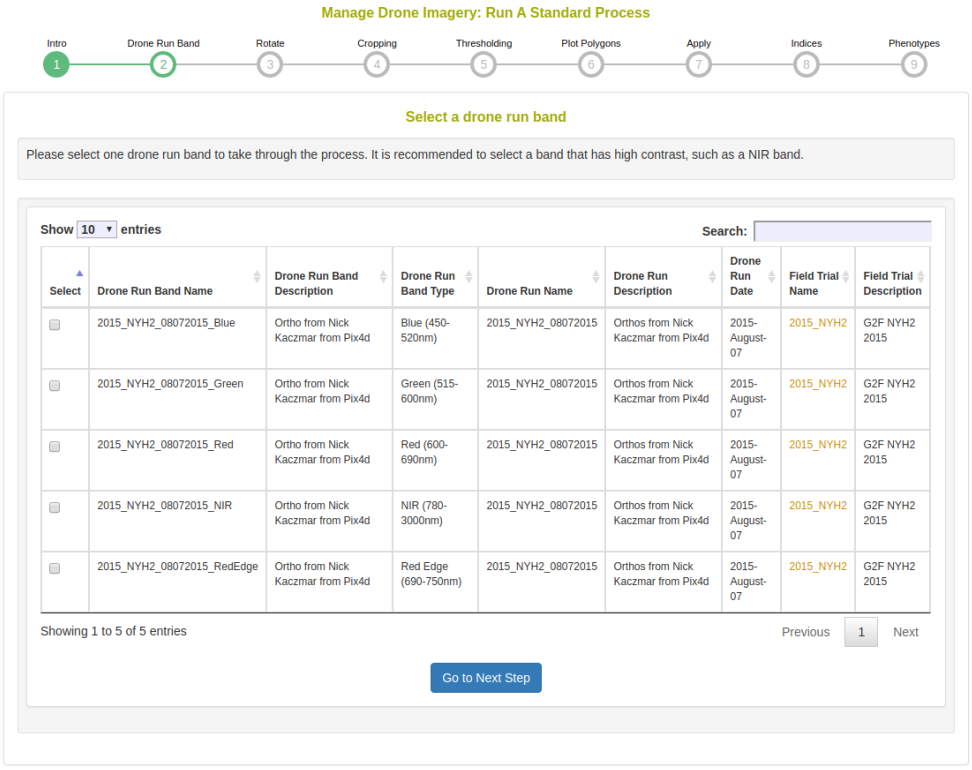
Once imagery is uploaded, it will appear on the dashboard under the field trial. Clicking the "Run Standard Process" button will begin extracting plot-polygon phenotypes from the imagery.



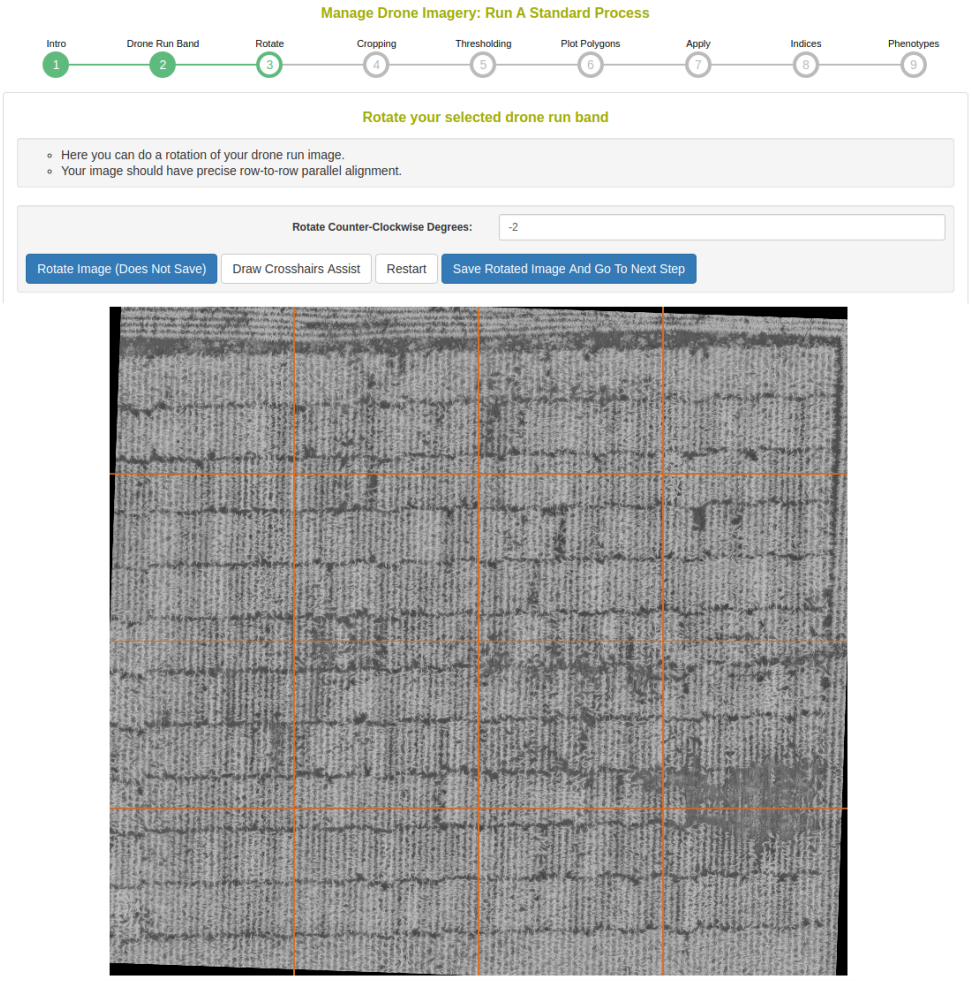

Clicking the button will open the following dialog.



Select a drone run band to use in this process. In the case of the Micasense 5 band multispectral camera there will be 5 bands shown here; select the NIR channel in this case because it has the highest contrast. In the case of standard color images, there will only be the RGB Color Image option here.

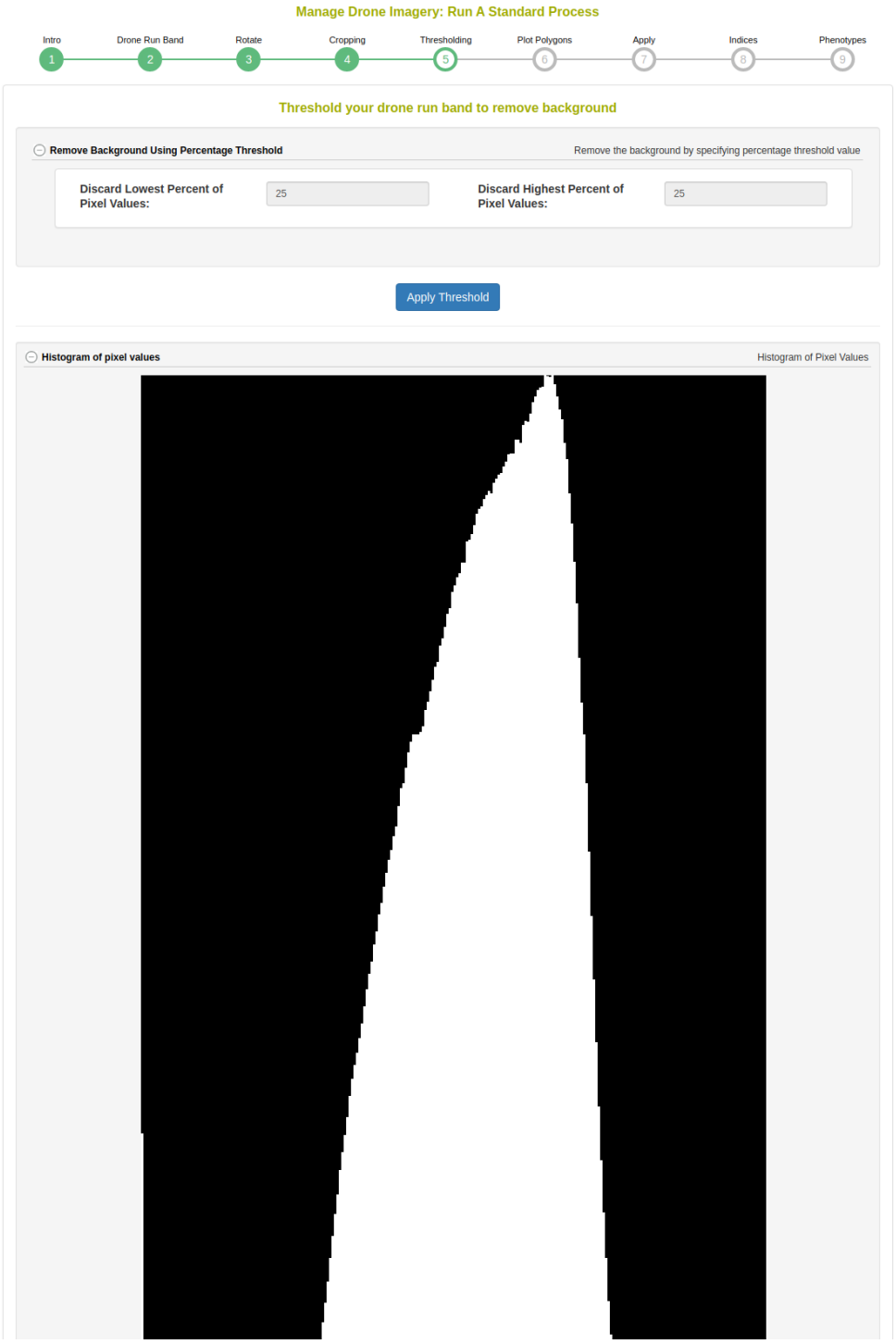



Rotate the image so that there the plots are oriented in a grid fashion. There can be a skew in the field layout, as seen in the following example.




Perform a rough cropping of the image by clicking on the four corners of the field. Cropping is important to remove any extraneous parts of the image.

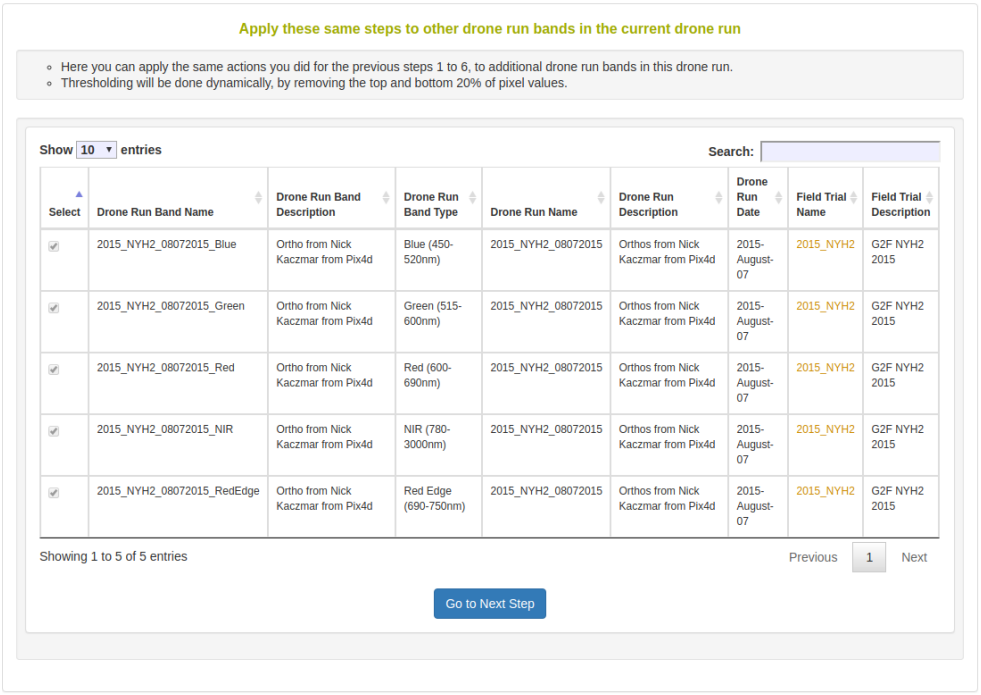
This step shows a histogram of the cropped image. The standard process will magnitude threshold the top and low ends of the distribution.


Manage Drone Imagery: Run A Standard Process
Intro
Drone Run Band
Rotate
Cropping
Thresholding
Plot Polygons
Apply
Indices
Phenotypes
1
2
3
4
5
6
7
8
9
Threshold your drone run band to remove background
Remove Background Using Percentage Threshold
Remove the background by specifying percentage threshold value
Discard Lowest Percent of Pixel Values:
25
Discard Highest Percent of Pixel Values:
25
Apply Threshold
Histogram of pixel values
Histogram of Pixel Values

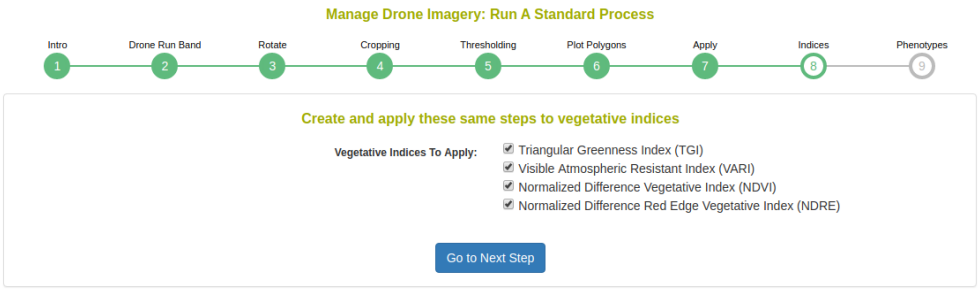
In this step, the template for the plot polygons in the experimental field design are associated to the image. First, defined the number of rows and columns in the field experiment. Then click the four corners of the image, in respect to the top right, top left, botton left, and bottom right positions. Next click on "Draw Plot Polygon Template". Review the template and clear/repeat the process until the template matches well. It is possible to "copy/paste" templates in the case where there are large breaks in the field design. Next, scroll down to the "assign Plot Polygons to Field Trial Entities" section. Select the location of Plot Number 1 as either "top left" or "top right" and whether the field design is serpentine or zigzag. Click on "Generate Assignments" and review that the names of the plots appear correctly in the overlay on the image. Finally, click "Finish and Save Polygons to Plots" when you have have confirmed the assignments.

Manage Drone Imagery: Run A Standard Process
Intro
Drone Run Band
Rotate
Cropping
Thresholding
Plot Polygons
Apply
Indices
Phenotypes
Define plot polygons relative to the field layout
Generate Polygon Template Tool
Overlay a uniform grid over the image.
1
Number of Rows:
10
Number of Columns:
50
Click Top Left Corner
Click Top Right Corner
Click Bottom Left Corner
Click Bottom Right Corner
Template Parameters
View and edit template parameters. These parameters can be filled using the above button clicks
2
Draw Plot Polygon Template (Does not save. Apply multiple templates if needed.)
Previously Used Plot Polygon Templates
View and use previous plot polygon templates used on this field trial
Total Image Width: 3316px. Total Image Height: 2680px.
Clear All Polygons
Clear One Polygon
Find Distance Between Points
Assign Plot Polygons to Field Trial Entities
Location of First Plot (e.g. plot number 1):
Top Left
Second Plot Follows First Plot Going:
Right
3
Plot Number Orientation:
Serpentine
4
5
Generate Assignments (Does Not Save)
Finish and Save Polygons To Plots
Show 10 entries
Search:
plot_name
accession_name
plot_number
block_number
is_a_control
rep_number
row_number
col_number
plot_geo_json
Polygon Assigned
2015_NYH2_plot_1 PHV63/LH195 1 1 null 1 1 45
2015_NYH2_plot_10 CG106/PHZ51 10 1 null 1 1 54
2015_NYH2_plot_100 LH145/LH82 100 4 null 1 2 45
2015_NYH2_plot_101 PHN11_LH145_0002/PHB47 101 5 null 1 3 45
2015_NYH2_plot_102 NYH-081/PHB47 102 5 null 1 3 46
2015_NYH2_plot_103 W1000S_0029/PHB47 103 5 null 1 3 47
2015_NYH2_plot_104 B109/PHZ51 104 5 null 1 3 48
2015_NYH2_plot_105 PH2D7_PHG47-13/PHB47 105 5 null 1 3 49
2015_NYH2_plot_106 PH2D7_PHG47-19/PHB47 106 5 null 1 3 50
2015_NYH2_plot_107 CG65/LH82 107 5 null 1 3 51
Showing 1 to 10 of 500 entries
Previous
1
2
3
4
5
...
50
Next

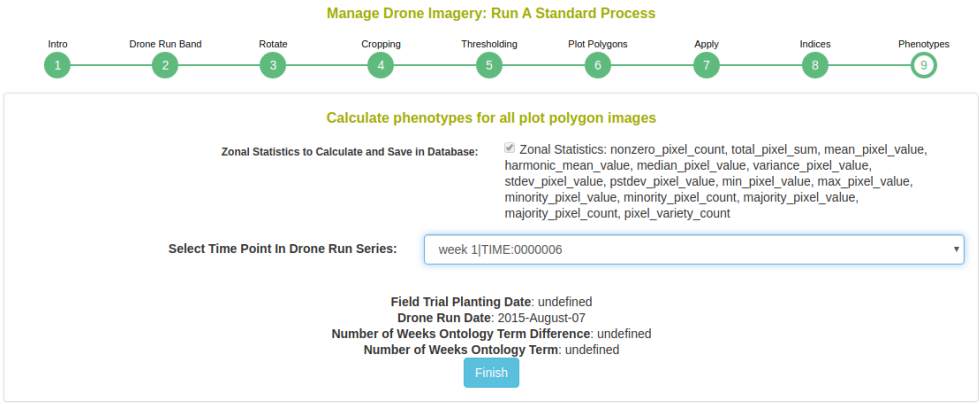
Next, the dialog shows you that the standard process will be repeated for all uploaded image bands.




Next, choose which vegetation indices to apply.




Next, choose the phenotypic values to extract. You must define the time point for which the phenotype is; if the field trial has a planting date, the time point will automatically be populated as image date minus the planting date.




After completing the standard process, the job will continue in the background until it completes. You can check the status of the job from the dashboard.

## 21.4 Ground Control Points

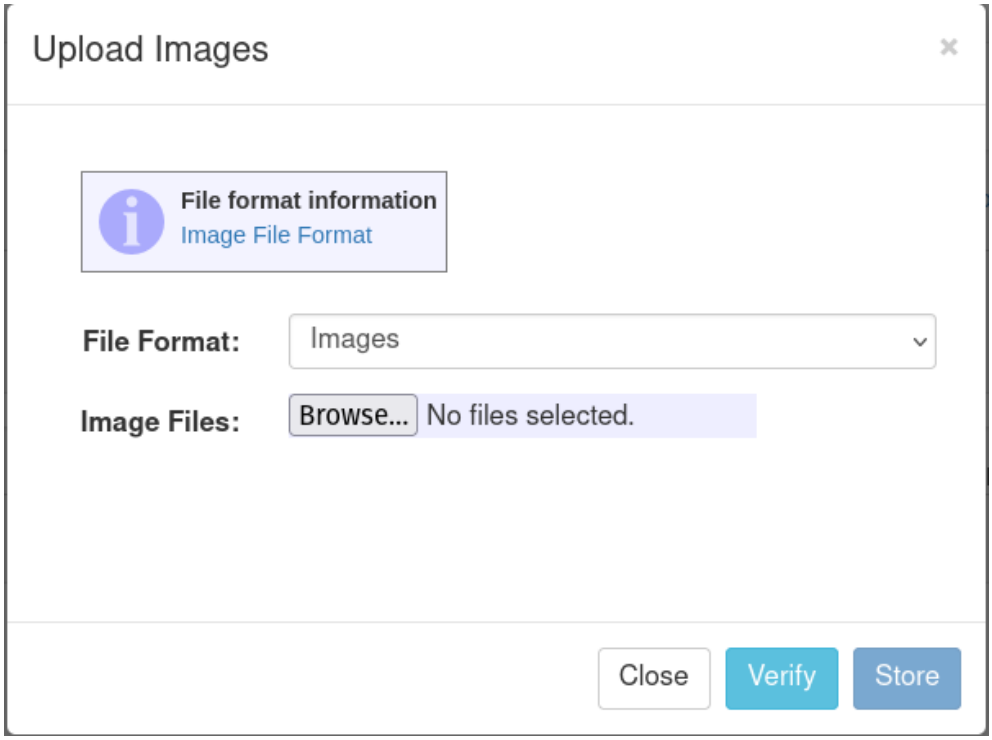
Ground control points can be saved after an imaging event has undergone the standard process on orhomosaics. Ground control points can then be used across imaging events on the same field experiment in order to automate the entire standard process.

# Chapter 22

# Managing Images

## 22.1 Uploading Image files

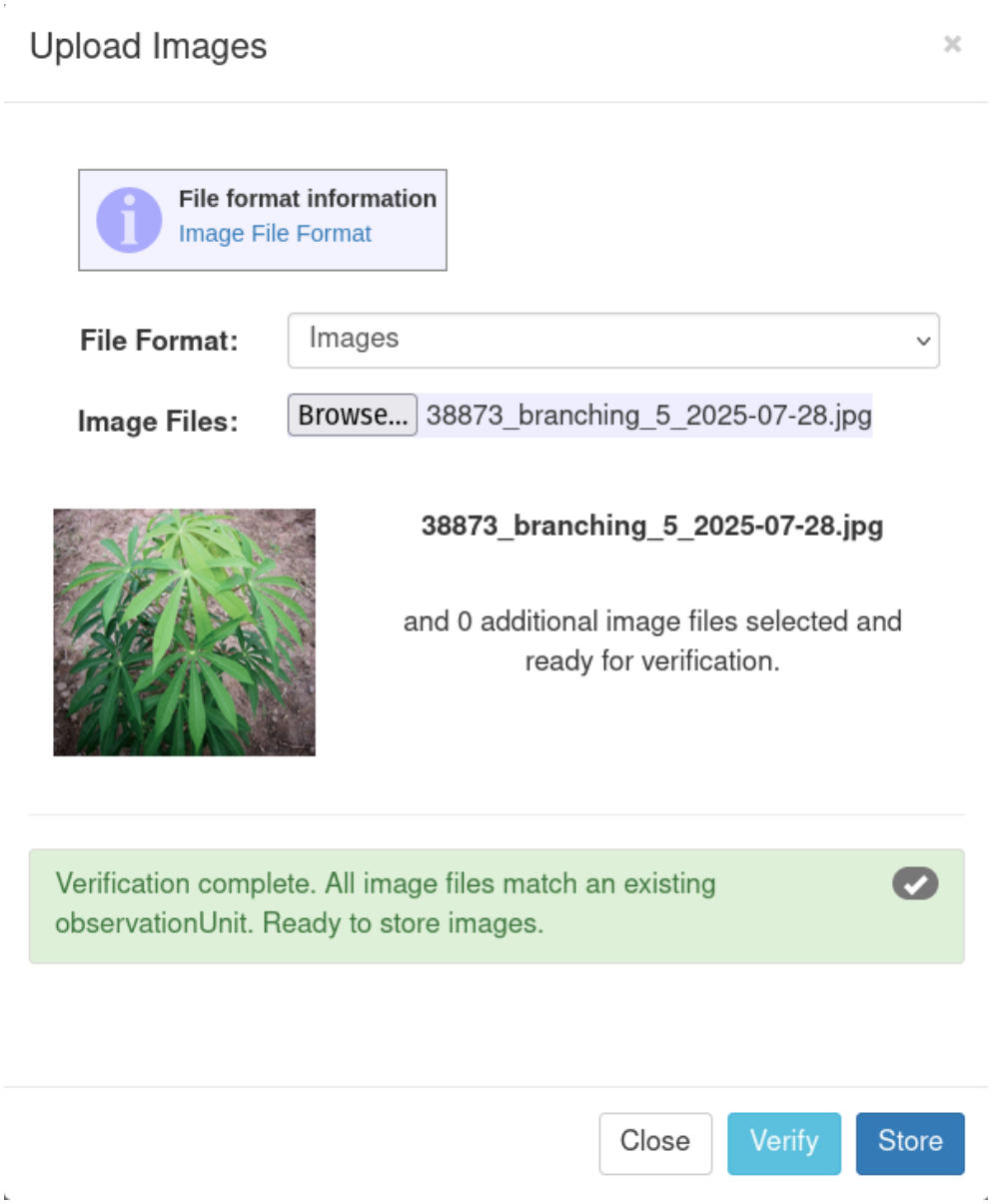
Clicking “Phenotyping” under Manage, and then “Upload Images” will open this dialog:




If uploading images directly, select the file format "Images". A single image or multiple selected images can be uploaded using this file format. Each of these images must have been downloaded from the Fieldbook app, or have a filename that follows this structure: observationUnitName, traitname, number, timestamp joined by underscores. The filename can not include any underscores other than the ones separating each value. For example: 38873_branching_5_2025-07-28.jpg

If the images have been taken in the Fieldbook app, they will have EXIF data that includes their associated stocks and traits. This EXIF data will be automatically parsed and the associations will be made when the images are uploaded.

After selecting your images for upload, clicking verify will check if your filenames or image metadata is correct and corresponds to existing stocks in the database.




If the verification is successful, click store to save these images in the database.

If uploading images with associated phenotypic data, select the file format "Images with Associated Phenotypes". A spreadsheet with the phenotypic values needs to be uploaded alongside a zipfile of images. The spreadsheet

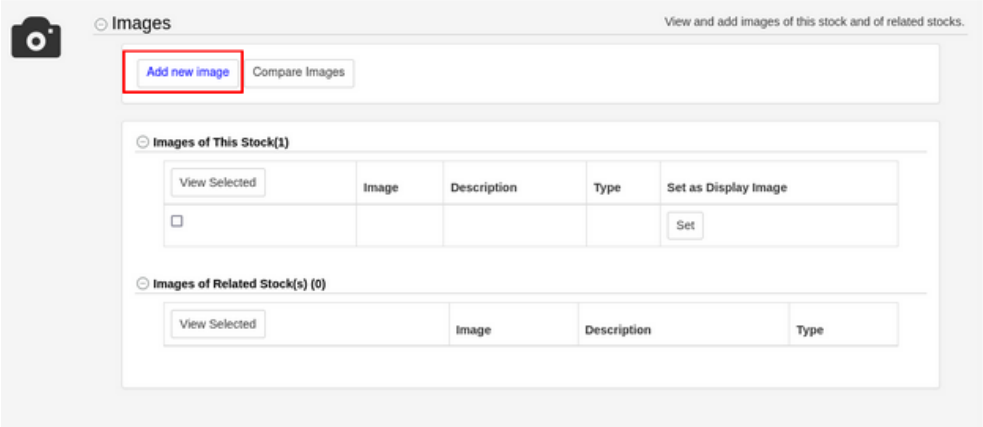
can be in .xls or .xlsx format, and the column headers need to be:

observationUnitName | observationVariableName | value | timestamp | image_name | person

The image_name column should match the file names of the associated images in the zipfile.

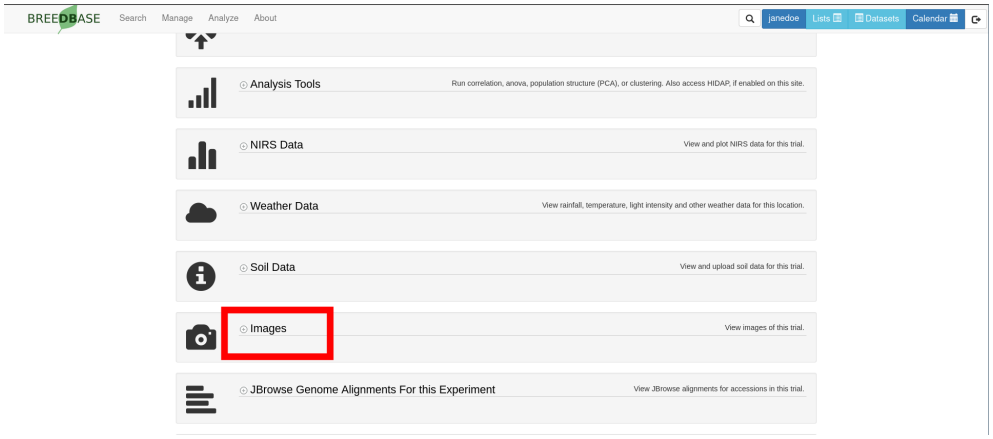
## 22.2 Uploading Images Directly to Stocks

In the Images section on a stock details page you can add a new image associated with that stock by clicking the "Add Image" button. This will associate any images uploaded through this dialog with the corresponding stock.

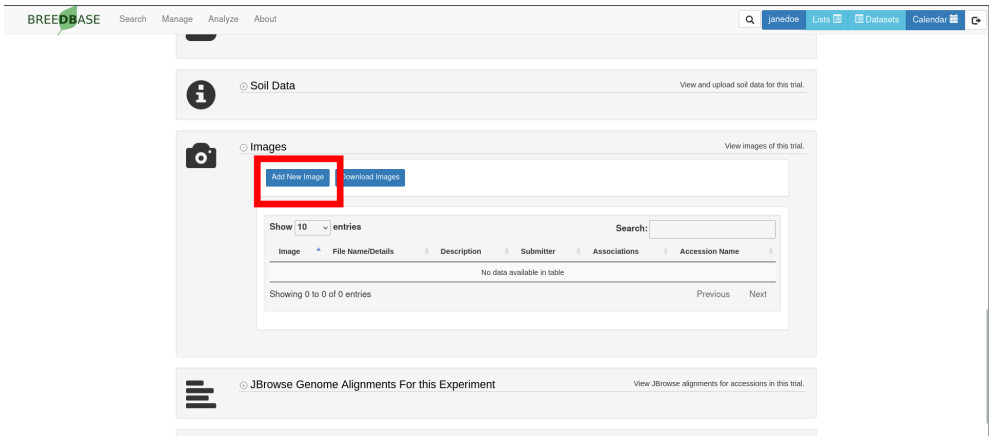



## 22.3 Uploading Images from the Field Trial and Image Analysis Pages

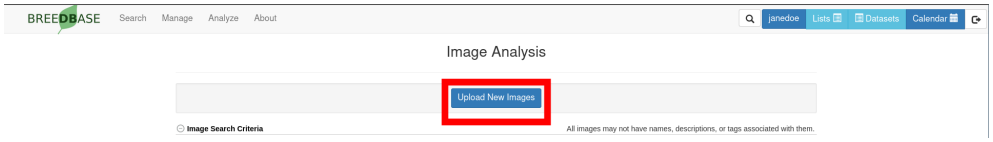
Another way to upload images is through the field trial details page. Navigate to a field trial details page and open the "Images" section.




Then click the "Add New Image" button to open the image upload dialog, where images can be uploaded in the same formats as described previously.




To upload images from the Image Analysis page, click on "Image Analysis" under the "Analyze" menu. Then click the "Upload New Images" button on this page.




This will open the same image upload dialog.

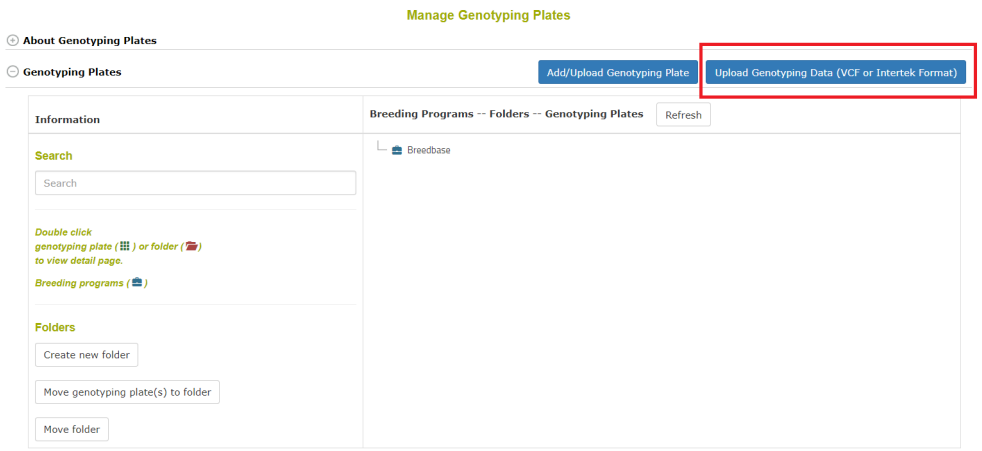
# Chapter 23

# Managing VCF Data

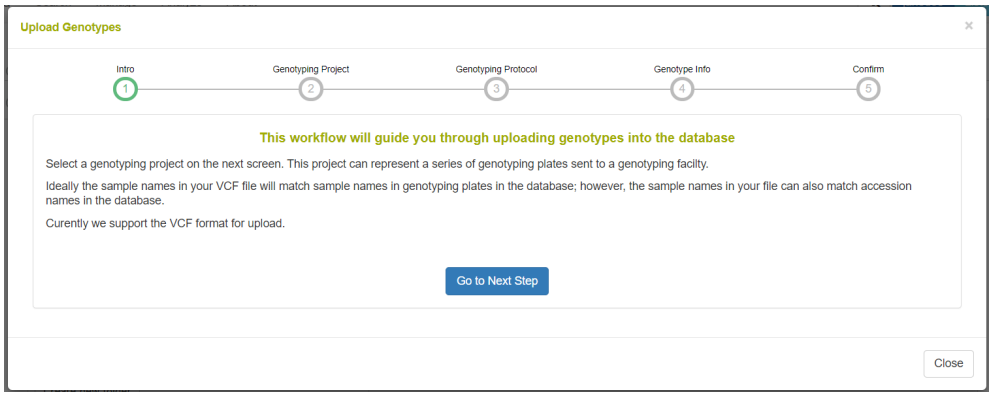
## 23.1 Uploading VCF Data

Genotyping data in VCF can be loaded from the web-interface. Breedbase can store any genotypic variants from a VCF, allowing for polyploids, structural variants, etc. without problems.

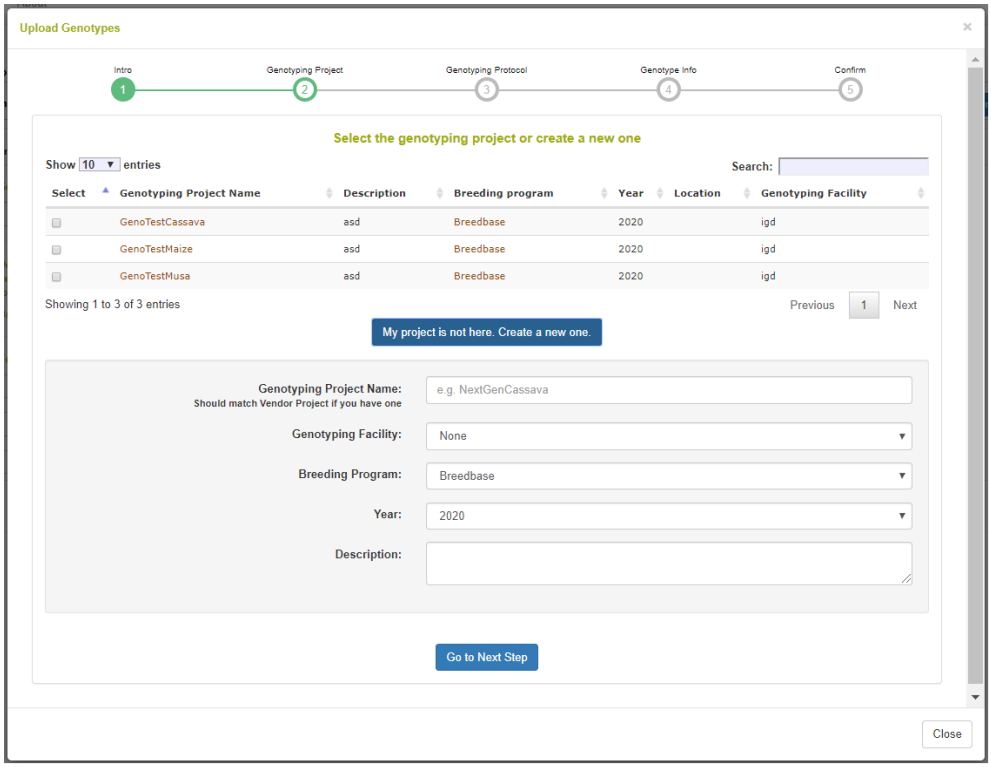
To begin go to Manage->Genotyping Plates and click the button seen below: Note that you do not need to have genotyping plates uploaded to upload VCF data; you may upload genotyping data to accessions or you can upload genotyping data for tissue samples in genotyping plates.



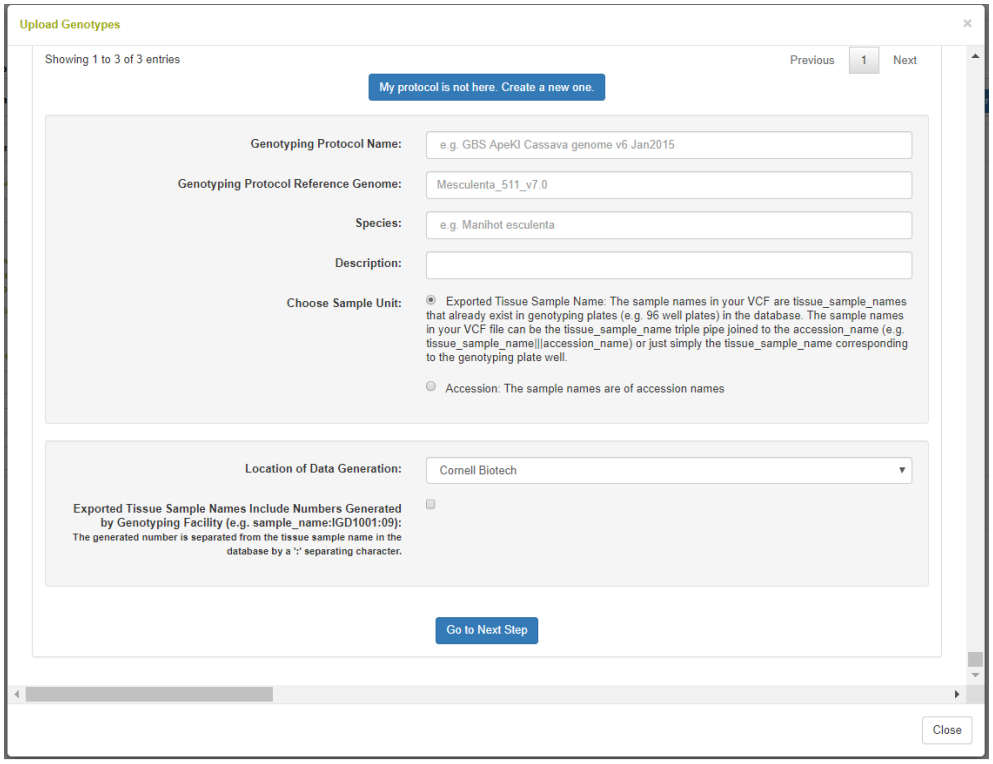
The workflow begins with an intro:



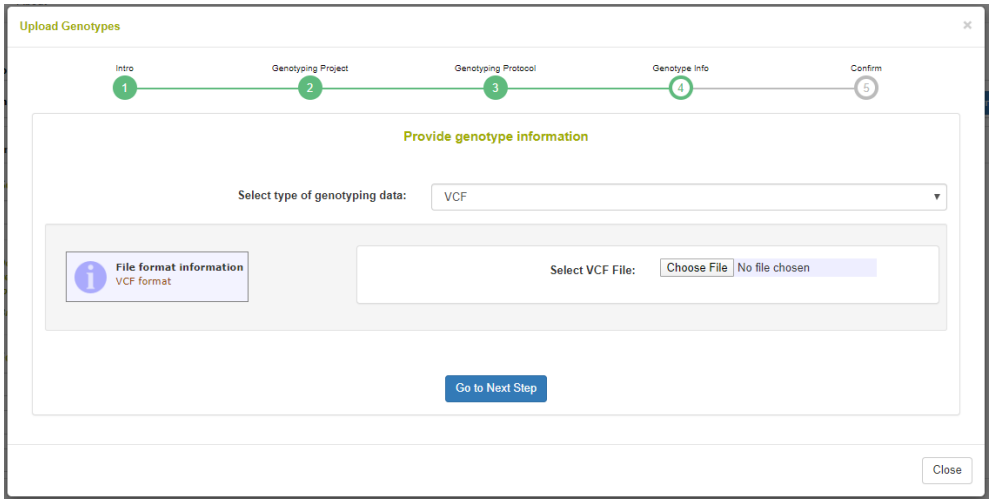

On the following step in the workflow, a genotyping project is defined or selected. A genotyping project is a high-level entity for grouping several genotyping events. It is defined with a name, description, name, breeding program, and genotyping facility (IGD, Intertek, etc.).



The following step is to define or select a genotyping protocol. A genotyping protocol represents the set of markers being called against a specific reference genome. A genotyping protocol is defined with a name, description, reference genome name, species name, and a location of data generation. Note in the picture that you can select whether the samples in your file are accessions or tissue samples in the database; tissue samples are for when a genotyping plate is stored in the database. There is an option to parse the sample names for appended sequencing numbers from IGD, where the sample names are like "accession:igdnumber".



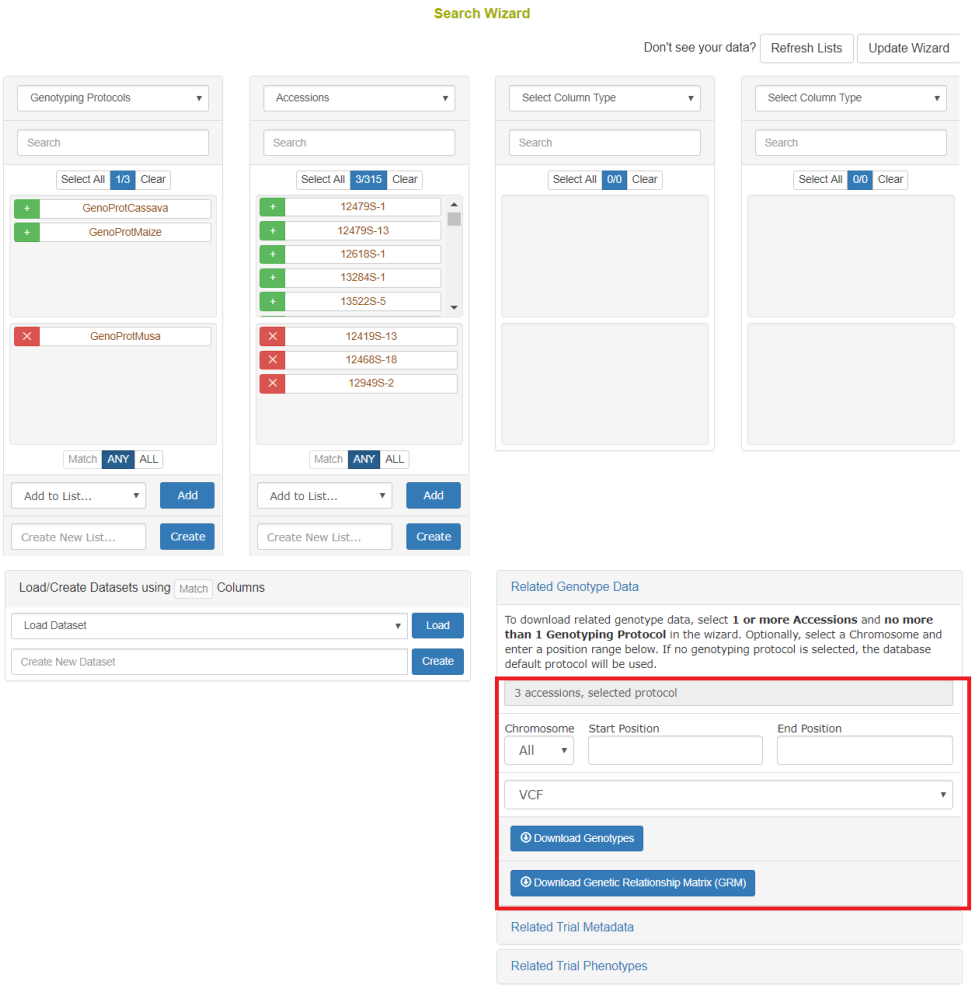

The final step is to select the VCF from your computer and upload it. The web interface can be used to upload files arbitrarily large; it is a NGINX configuration to set this size.




## 23.2 Searching and Downloading VCF Data

The Search Wizard is the primary means of querying data in the database. Go to Search->Wizard to begin.

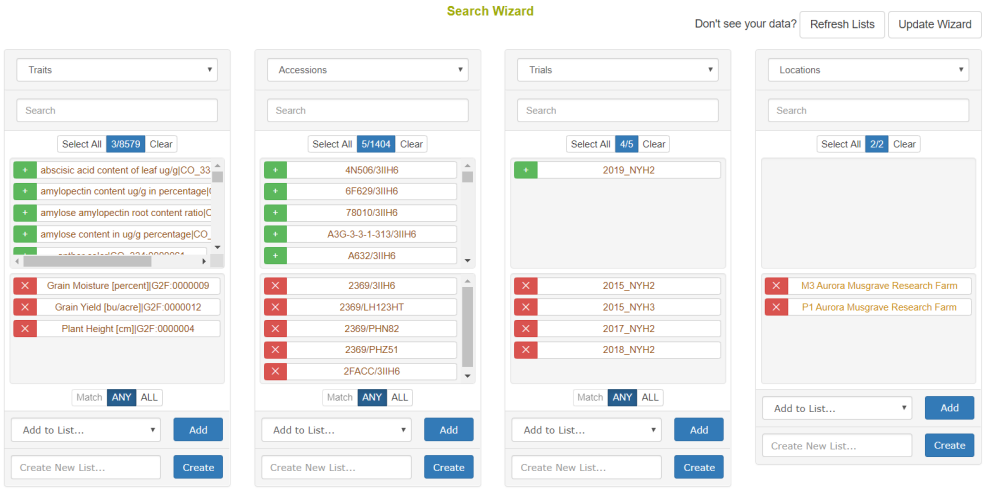
Once genotyping protocols are stored, select Genotyping Protocols from the first dropdown menu. Then if you select one or more and select Accessions from the second dropdown menu, you will see the accessions for which genotypes were stored. As seen in the following picture, there is a section for filtering genotypes by chromosome, start position, and end position. Genotypes can be downloaded in VCF or DosageMatrix formats.



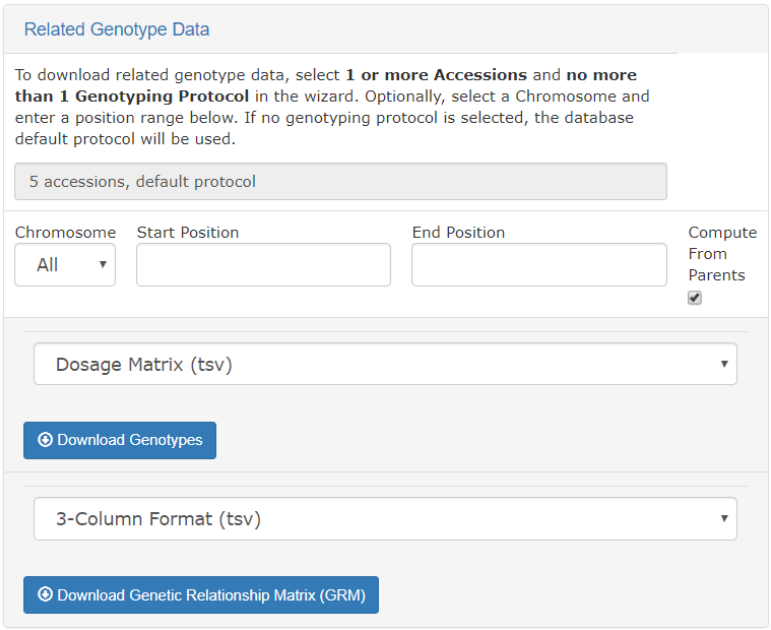

Using the "Default genotyping protocol" which is configured in a system, you can query over field phenotypic evaluations before downloading genotypes and phenotypes.

## 23.3 Searching Protocols

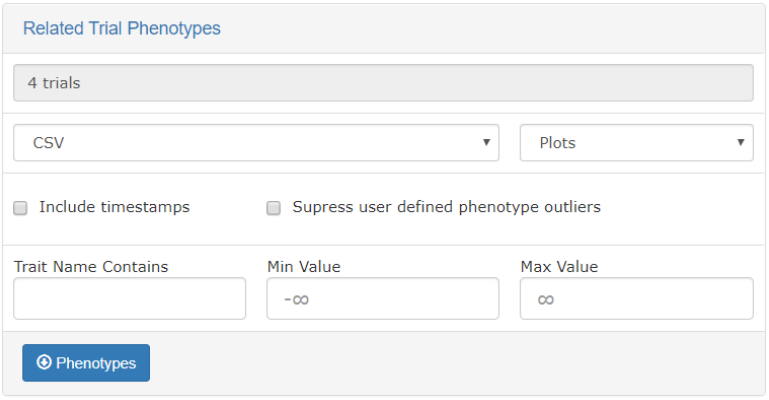
Genotyping protocols can be search by going to Search->Genotyping Protocols. To download genotypes accessions must be selected, though any combination of search criteria can be used to filter and select those accessions. If a genotyping protocol is not selected, then the default genotyping protocol set in the configuration will be used. Genotyping protocols can also be selected in the wizard.



The genotyping download menu on the Search Wizard presents options for filtering by chromosome, start position, and end position. Genotypes can be downloaded in VCF of Dosage Matrix formats. The genomic relationship matrix (GRM) can be downloaded for the selected accessions in a tab-delimited matrix format or in a three-column format that is useful in Asreml. Genotypes can be computed from the parents in the pedigree if those parents are genotyped by clicking on the "compute from parents" checkbox. Additionally, the GRM can be computed using genotypes of parents in the pedigree if the "compute from parents" checkbox is selected.



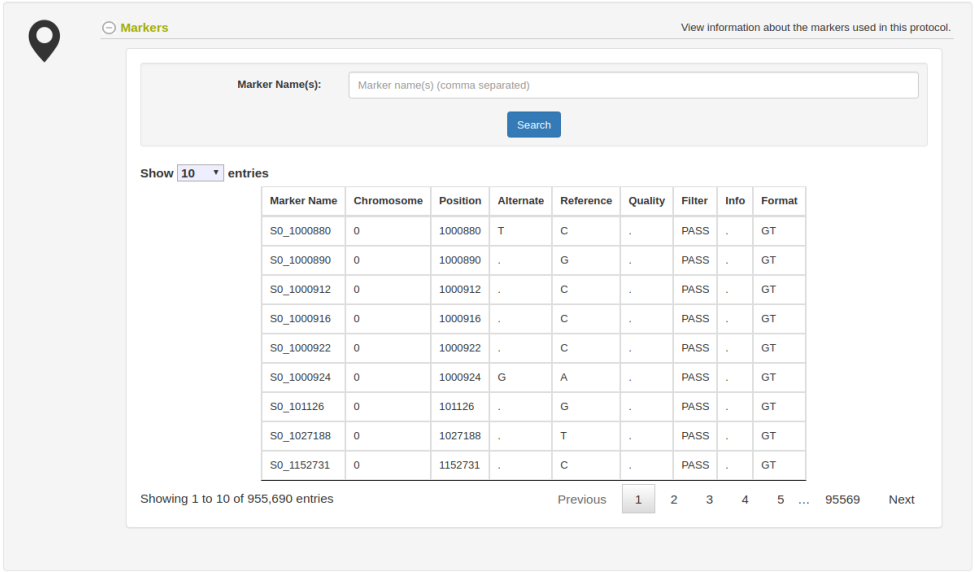

As is described elsewhere, the Search Wizard presents a way to filter phenotypic values by minimum and maximum values, and allow for download in CSV and Excel formats.




## 23.4 Detail Pages and Deletion

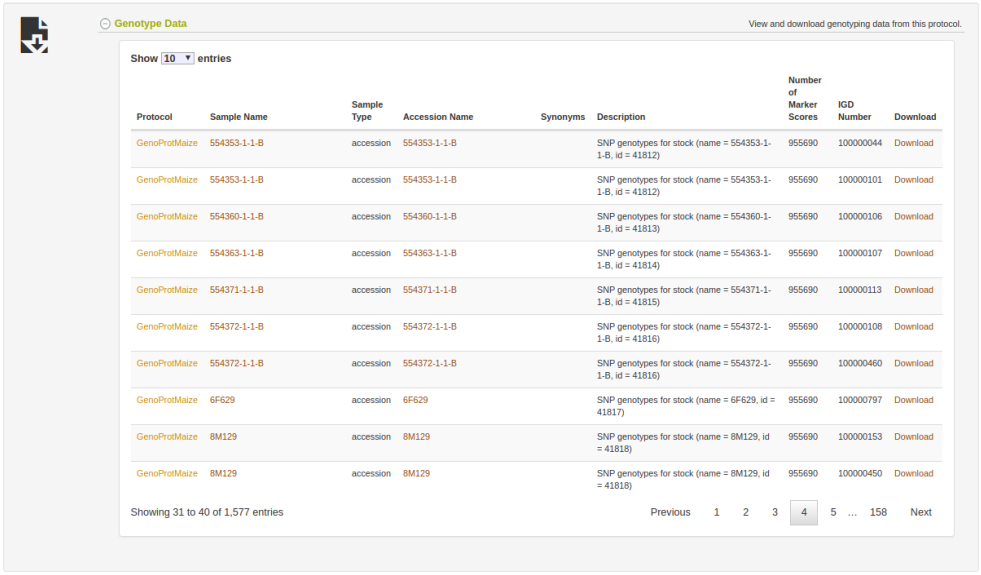
The genotyping protocol detail page will show all information about the protocol such as the reference genome used, the header information lines in the uploaded VCF file, the markers involved, and the samples genotyped.

The markers section will show all markers used and their annotations, such as position, chromosome, alternate allele, reference allele, marker format, etc.

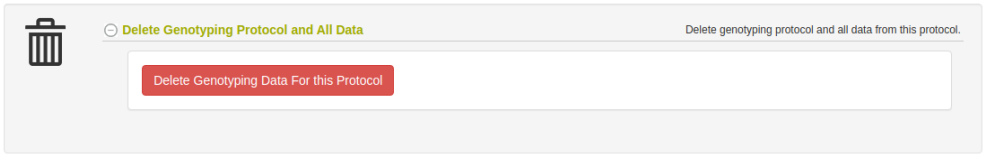



The samples section will show all samples genotyped. Notice the Download links in the table which can be used to easily get the VCF file results for each

genotyped samples with all markers in the genotyping protocol. For getting mulitple samples at once, use the Search Wizard as discussed above.



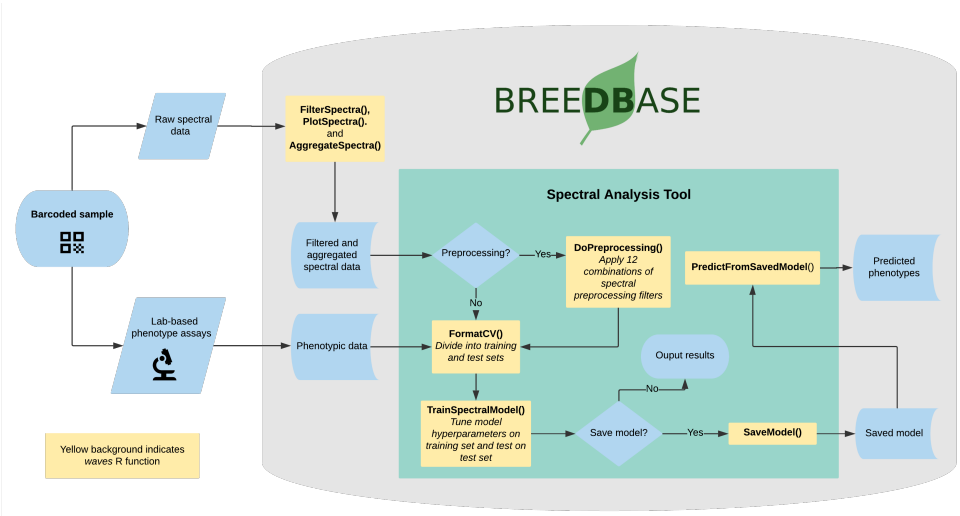

The genotyping protocol and all associated genotyping data can be deleted from the genotyping protocol page.



# Chapter 24

# Managing Spectral Data

Breedbase has implemented a flexible spectral data storage protocol that handles spectral data irrespective of the source spectrometer. Spectral data storage and analysis in Breedbase makes use of the R package *waves* for outlier identification, plotting, sample aggregation, and prediction model training.

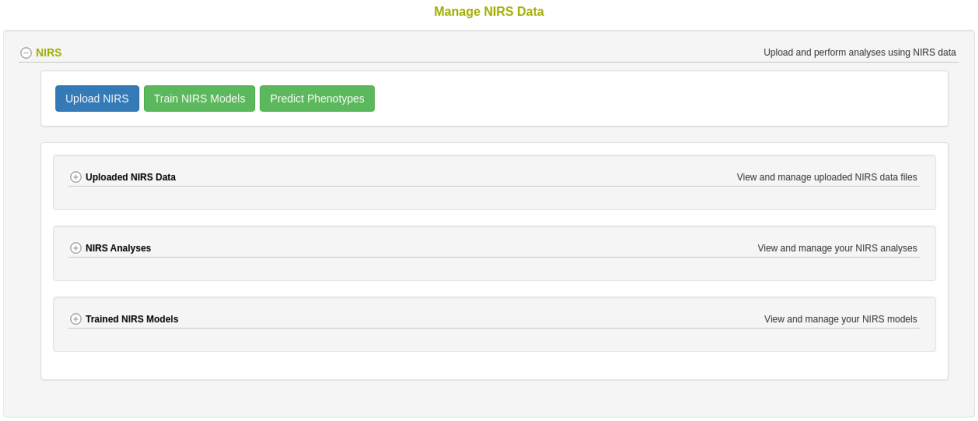


## 24.1 Upload Spectral Data

Spectral data can be added as a CSV file that includes metadata in the leftmost columns followed by one column per spectral measurement to the right. Rows represent a single scan or sample, each with a unique ID that must match to a Breedbase observationUnitName. Future data transfer using BrAPI will allow for interoperability with data collection software.

To upload a spectral dataset, navigate to the 'Manage NIRS Data' page by selecting 'NIRS' in the 'Manage' menu and click the blue 'Upload NIRS' button. This will open an upload workflow. A link to the required file format and an example .csv file can be found by clicking in the light blue info box in this workflow. Another example of the file format is shown below.

- **id**: Optional identifier for each NIRS read. The id must be an integer.
- **sampling_id**: Optional identifier for each sample. Strings are allowed.
- **sampling_date**: Optional field. The format allowed is: YYYY-MM-DD.
- **observationunit_name**: Required field that matches existing data in the database. It can be the plot name, subplots, plant name, or tissue sample, depending how your trial is designed.
- **device_id**: Optional field to identify your device. Strings are allowed.
- **device_type**: Required field. It is possible upload data for a single device type. They can be: SCiO, QST, Foss6500, BunchiN500, or LinkSquare.
- **comments**: Optional field for general comments. All other columns are required wavelengths. You can add how many columns you want upload – there is no limit.




| id | sample_id | sampling_date | observationunit_name | device_id | device_type | comments | 740 | 741 | 742 | 743 | 744 | 745 | 746 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 7a6ac477-d291-4d07-a6 | 2020-6-24 | myTrial20_rep1_acc_001 | 503E4BFC4E923999 | SCIO | | 0.885707958 | 0.88572938 | 0.885590265 | 0.885493457 | 0.885493162 | 0.885572662 | 0.885628732 |
| 2 | 7a6ac477-d291-4d07-a6 | 2020-6-24 | myTrial20_rep1_acc_002 | 503E4BFC4E923999 | SCIO | | 0.909132994 | 0.908724223 | 0.90824451 | 0.907891706 | 0.907697281 | 0.907626115 | 0.90755938 |
| 3 | 7a6ac477-d291-4d07-a6 | 2020-6-24 | myTrial20_rep1_acc_003 | 503E4BFC4E923999 | SCIO | | 0.889220207 | 0.889013119 | 0.888681812 | 0.888431257 | 0.888310362 | 0.888297321 | 0.888284052 |
| 4 | 73c648ca-f5fb-4231-a1 | 2020-6-24 | myTrial20_rep1_acc_004 | 503E4BFC4E923999 | SCIO | | 0.8900087 | 0.889604969 | 0.889191073 | 0.888956654 | 0.88883379 | 0.889072741 | 0.8892472 |
| 5 | 73c648ca-f5fb-4231-a1 | 2020-6-24 | myTrial20_rep1_acc_005 | 503E4BFC4E923999 | SCIO | | 0.938101707 | 0.93868202 | 0.93820132 | 0.937873867 | 0.937742819 | 0.937775129 | 0.93784594 |
| 6 | 73c648ca-f5fb-4231-a1 | 2020-6-24 | myTrial20_rep1_acc_006 | 503E4BFC4E923999 | SCIO | | 0.876289461 | 0.875981159 | 0.875579263 | 0.875289805 | 0.875162225 | 0.875171404 | 0.87520442 |
| 7 | d5b55c93-4c8e-4ef6-9c | 2020-6-24 | myTrial20_rep1_acc_007 | 503E4BFC4E923999 | SCIO | | 0.879217838 | 0.878925781 | 0.878588441 | 0.8783872 | 0.878346547 | 0.878423929 | 0.878495739 |
| 8 | 7a6ac477-d291-4d07-a6 | 2020-6-24 | myTrial20_rep1_acc_008 | 503E4BFC4E923999 | SCIO | | 0.890746588 | 0.890515672 | 0.89016542 | 0.889903562 | 0.889783304 | 0.889782828 | 0.889792154 |
| 9 | 7a6ac477-d291-4d07-a6 | 2020-6-24 | myTrial20_rep1_acc_009 | 503E4BFC4E923999 | SCIO | | 0.850444238 | 0.85032422 | 0.850094039 | 0.8499495 | 0.849942163 | 0.850052119 | 0.850175036 |

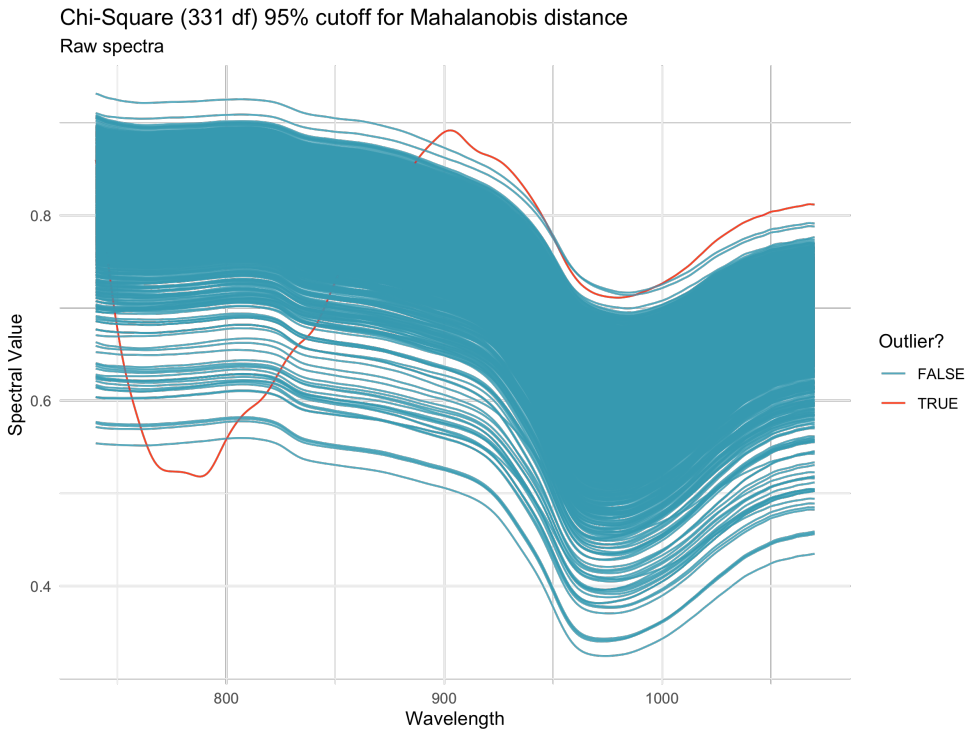
## 24.2 Evaluate and Remove Outliers

Spectral calibration models can be heavily affected by the presence of outliers, whether they come from spectrometer spectral artifacts or user errors. Mahalanobis distance (Mahalanobis, 1936) is a measure of the distance between a single observation and a larger distribution and is commonly used in the identification of outliers in a multivariate space (Des Maesschalck et al, 2000). The *FilterSpectra()* function in the R package *waves* calculates the Mahalanobis distance of each observation in a given spectral matrix using the *stats::mahalanobis()* function. Observations are identified as outliers if the squared distance is greater than the 95th percentile of a $\chi^2$-distribution with $p$ degrees of freedom, where $p$ is the number of columns (wavelengths) in the spectral matrix (Johnson and Wichern, 2007). In Breedbase, this procedure is applied on a per-dataset basis on upload and outliers are given binary tags "Outlier."

## 24.3 Plot Spectra

After outlier identification, a plot is generated using the *PlotSpectra()* function in *waves*. This function uses the filtered spectra and *ggplot2::ggplot()* to

create a line plot with outliers highlighted by color. A list of rows identified as outliers are shown beneath the plot. Plots are saved as .png files and linked to the original input datasets. Plot image files can be downloaded via the "Download Plot" button in the upload workflow.




## 24.4 Aggregate Spectra

To obtain a stable and reliable spectral profile, most spectrometer manufacturers recommend that multiple spectral scans are captured for each sample. While some spectrometers aggregate these scans internally, many do not, requiring the user to do so before analysis can take place. Breedbase handles these cases upon data upload following filtering steps by calling the *AggregateSpectra()* function from *waves*, saving the aggregated scans for future access through the search wizard feature. Scans are aggregated by sample mean (e.g. plot-level basis) according to the provided observationUnitName field. After aggregation, the user exits the upload workflow and the raw data file is saved in the upload archive.

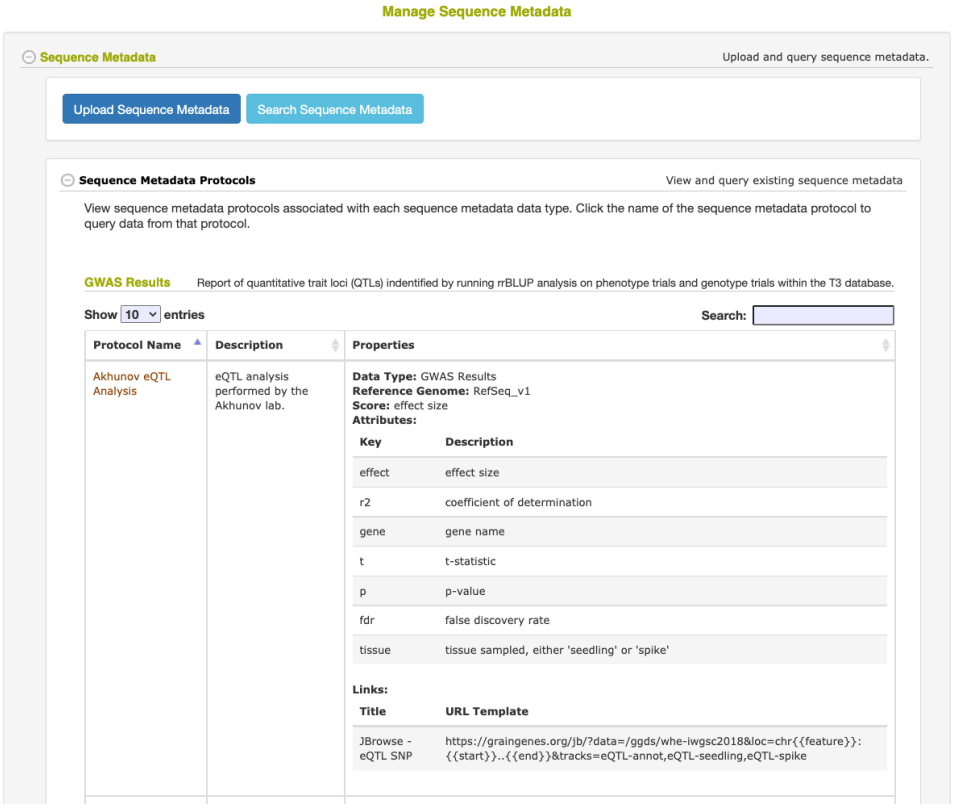
## 24.5 References


- De Maesschalck, R., Jouan-Rimbaud, D., and Massart, D. L. (2000). The Mahalanobis distance. Chemom. Intell. Lab. Syst. 50(1): 1-18.
- Johnson, R. A. & Wichern, D. W. (2007). Applied Multivariate Statistical Analysis (6th Edition). p 773.
- Mahalanobis, P. C. (1936). On the generalized distance in statistics. National Institute of Science of India.


**Analysis tool documentation**

# Chapter 25

# Managing Sequence Metadata



## 25.1 What is Sequence Metadata?

Sequence Metadata is a feature that allows for the efficient storage and retrieval of sequence annotations for a specific region along a reference genome. The annotation data can contain a primary "score" value and any number of secondary key/value attribute data. For example, Sequence Metatadata can store MNase open chromatin scores for every 10 basepairs along the reference genome as well as genome-wide association study (GWAS) statistics, including the trait information associated with the result. This data can then be filtered by position and/or scores/attribute values and even cross-referenced with markers stored in the database.

## 25.2 Loading Sequence Metadata

Sequence Metadata can be loaded into the database using a gff3-formatted file. The following columns are used to load the data:

- **#1 / seqid:** The name of the database feature (ie chromosome) the metadata is associated with (The feature name must already exist as a feature in the database)
- **#4 / start:** The metadata's start position
- **#5 / end:** The metadata's end position
- **#6 / score:** (optional) The primary score attribute of the metadata
- **#9 / attributes:** (optional) Secondary key//value attributes to be saved with the score. These should be formatted using the gff3 standard (key1=value1;key2=value2). The attribute key cannot be either score, start, or end.

To upload the gff3 file:

1. Go to the **Manage > Sequence Metadata** page
2. Click the **Upload Sequence Metadata** button
3. On Step 2 of the Wizard, select the Type of data to be uploaded
    - This groups similar datasets together in the same Data Type category
4. On Step 3 of the Wizard, select an existing Protocol or create a new one
    - The Protocol is used to describe how the data was generated and define the score value and any secondary attributes. Adding the

attributes (and their descriptions) to the Protocol will allow the Sequence Metadata queries to filter the data based on the value of one or more of these attributes. Attributes not defined in the Protocol will still be stored and displayed on retrieval, but will not be able to be used in a search filter.

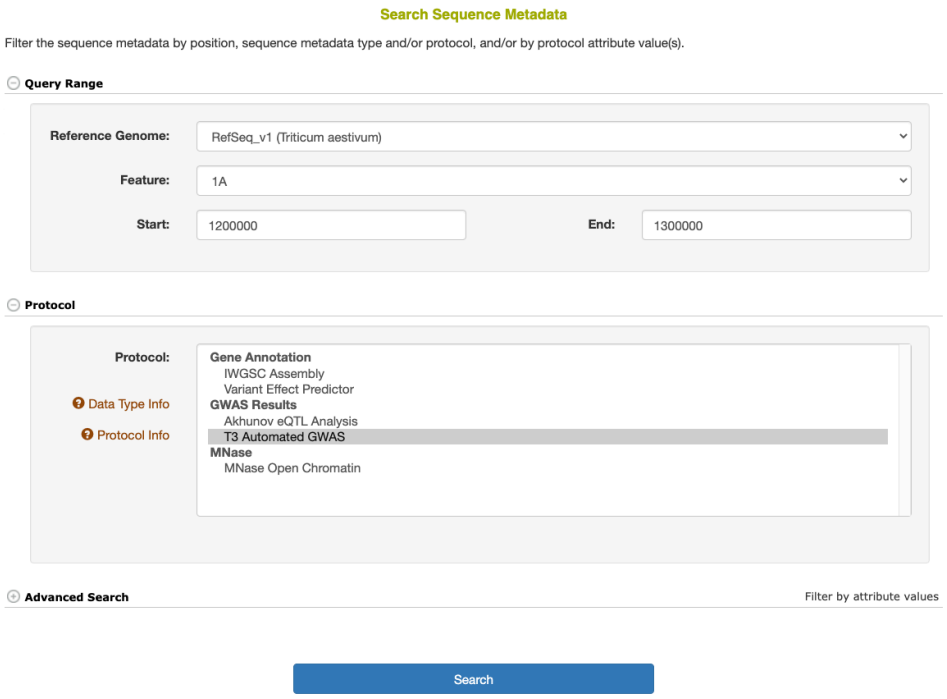
5. Finally, select and upload your gff3 file to the database. The database will verify the format of the file before its contents are stored.

## 25.3 Searching Sequence Metadata

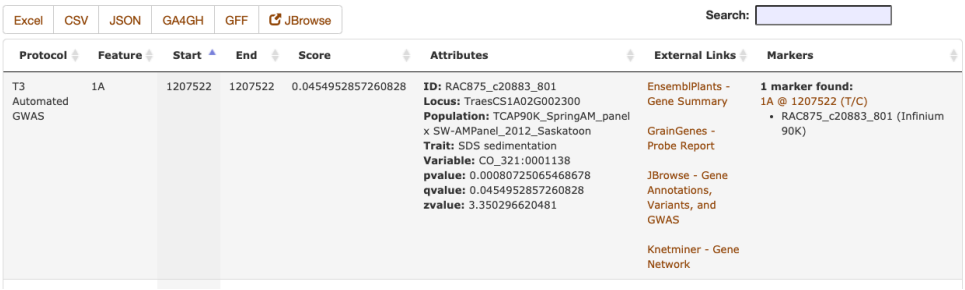
To retrieve stored Sequence Metadata, go to the **Search > Sequence Metadata** page.

### 25.3.1 Basic Search

The basic Sequence Metadata search options include selecting the reference genome and species, the chromosome, and (optionally) the start and/or end position(s) along the reference genome. In addition, one or more specific protocols can be selected to limit the results.

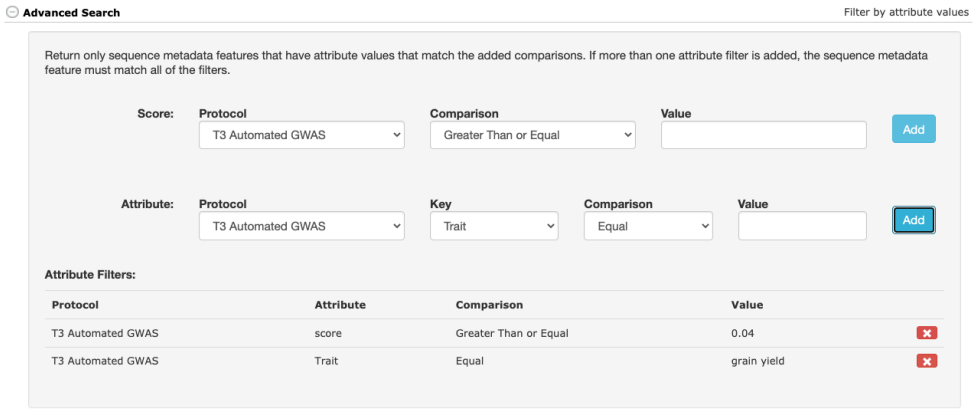



The Sequence Metadata search results are returned as a table, including the chromosome and start/stop positions of the annotation, along with the primary score value and any additional key/value attributes. The markers column will include a list of marker names of any stored markers that are found within the start/stop positions of the Sequence Metadata. The data can be downloaded as a table in an Excel or CSV file or a machine-readable (code-friendly) JSON file. If the Sequence Metadata JBrowse configuration is set, the filtered results can be displayed as a dynamic JBrowse track.



### 25.3.2 Advanced Search

Any number of advanced search filters can be applied to the query. The advanced filters can limit the search results by the value of the primary score and/or any of the secondary attribute values.




## 25.4 Marker Integration

A table of Sequence Metadata annotations are embedded on the Marker/Variant detail page. The table will include any annotations that span the poisiton of the marker (for data of the same reference genome and species).

## 25.5 Sequence Metadata API

A publicly accessible RESTful API (Application Programming Interface) is available to query the database for Sequence Metadata directly from your programming environment (R, python, etc) to be used in analysis. The data is returned in a JSON format. Documentation for the API can be found on the **Manage > Sequence Metadata** page

# Chapter 26

# Managing Datasets

## 26.1 What is a Dataset?

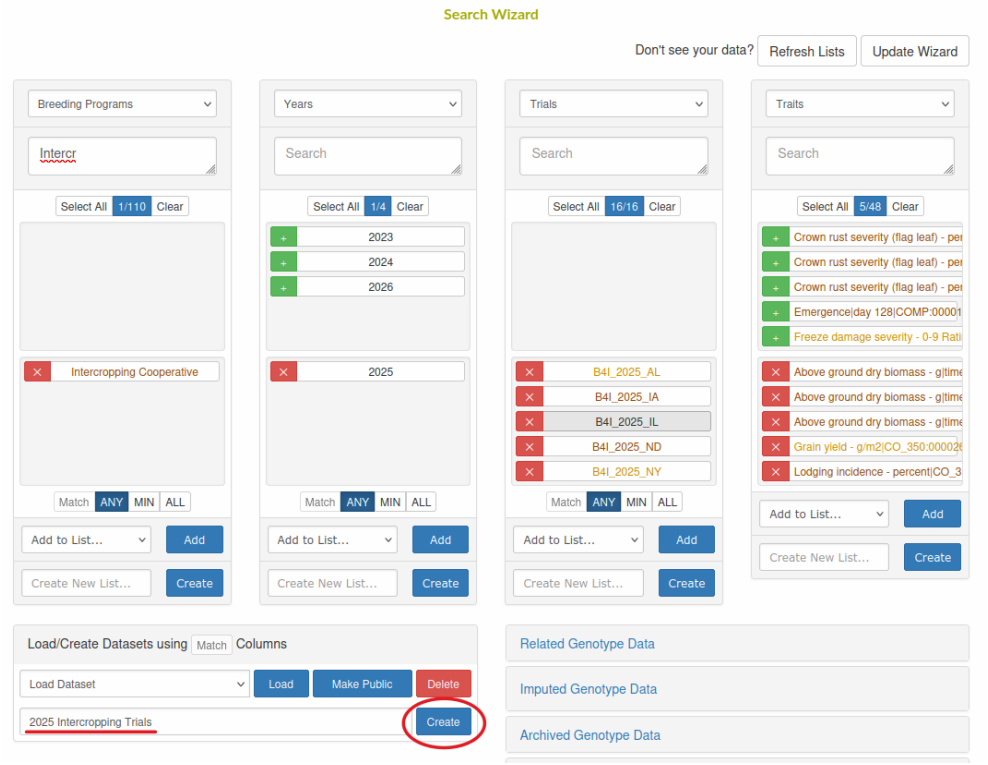
Similar to Lists, Datasets can be used to save selected data in the database to your account to be reused later or shared with other users. While Lists only contain items from a single data type (for example a selection of Accessions), Datasets contain selected items across multiple data types (for example a selection of Accessions, Locations, Phenotyping Trials, and Traits used in an analysis). Datasets can be created to make selections in the Search Wizard reusable, used as inputs in some of the analysis tools, shared with other users by making them public, and exported to FigShare to make the data more easily accessible to external users.

## 26.2 Managing Datasets

### 26.2.1 Dataset Creation

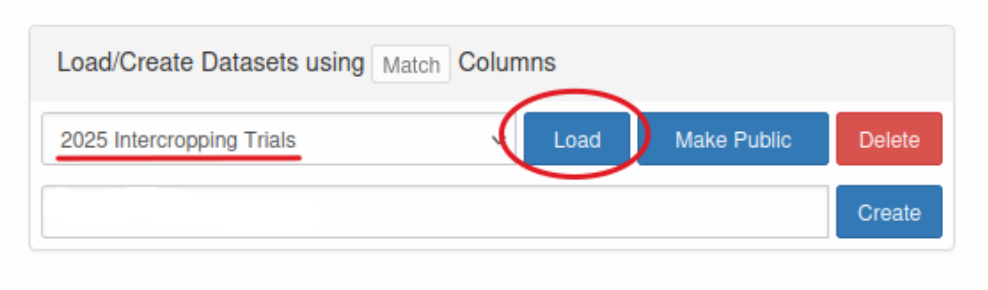
Datasets are created from within the Search Wizard:

1. First, make your selection in the Search Wizard. Select the data types you want saved in the dataset and the specific items to include for each data type.
2. Enter the name of the new Dataset and click the "Create" button to save the dataset.



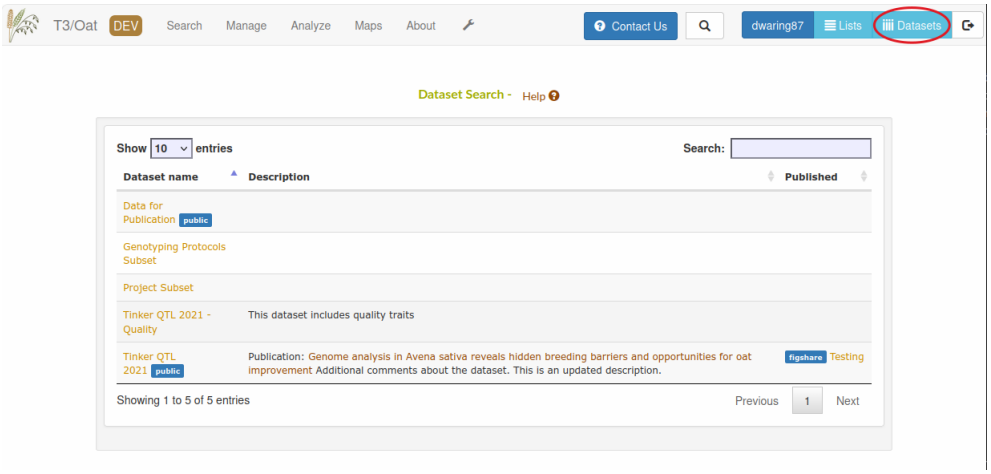

## 26.2.2 Existing Datasets

To use a saved Dataset in the Search Wizard, select the Dataset name in the drop down menu below the Search Wizard columns and click the "Load" button. This will set the Search Wizard column selections to all of the data types and items in the selected Dataset.

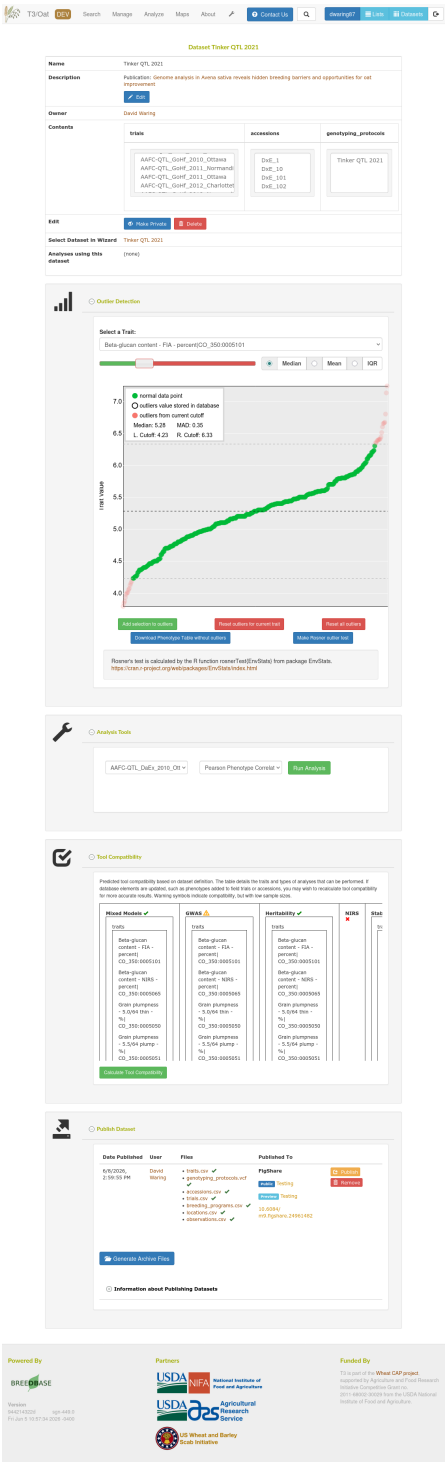



You can also view a table of all of your saved Datasets and any public Datasets from other users by clicking the Datasets button in the top right of

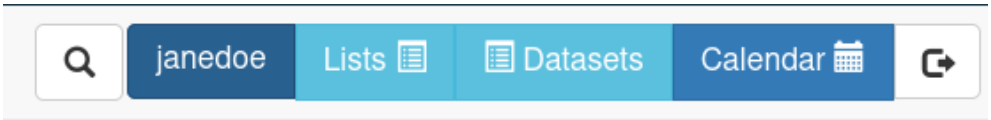
the toolbar.




From this table, you can select an individual Dataset to modify its properties (description, visibility, etc), check for outliers, view the Dataset compatibility for analysis tools, run some analyses, and publish the Dataset to FigShare.


T3/Oat
DEV
Search
Manage
Analyze
Maps
About
Contact Us
dwwaring07
Lists
Datasets
Dataset Tinker QTL 2021
Name
Tinker QTL 2021
Description
Publication: Genome analysis in Avena sativa reveals hidden breeding barriers and opportunities for oat improvement
Edit
Owner
David Waring
Contents
trials
accessions
genotyping_protocols
AAFC-QTL_GxHF_2010_Ottawa
AAFC-QTL_GxHF_2011_Normandi
AAFC-QTL_GxHF_2011_Ottawa
AAFC-QTL_GxHF_2012_Charlottet
DxE_1
DxE_10
DxE_101
DxE_102
Tinker QTL 2021
Edit
Make Private
Delete
Select Dataset in Wizard
Tinker QTL 2021
Analyses using this dataset
(none)
Outlier Detection
Select a Trait:
Beta-glucan content - FIA - percent|CO_350:0005101
Median
Mean
IQR
normal data point
outliers value stored in database
outliers from current cutoff
Median: 5.28
MAD: 0.35
L. Cutoff: 4.23
R. Cutoff: 6.33
TRAIT VALUE
7.0
6.5
6.0
5.5
5.0
4.5
4.0
Add selection to outliers
Reset outliers for current trait
Reset all outliers
Download Phenotype Table without outliers
Make Rosner outlier test
Rosner's test is calculated by the R function rosnerTest(){EnvStats} from package EnvStats.
https://cran.r-project.org/web/packages/EnvStats/index.html
Analysis Tools
AAFC-QTL_DaEx_2010_Ott
Pearson Phenotype Correlat
Run Analysis
Tool Compatibility
Predicted tool compatibility based on dataset definition. The table details the traits and types of analyses that can be performed. If database elements are updated, such as phenotypes added to field trials or accessions, you may wish to recalculate tool compatibility for more accurate results. Warning symbols indicate compatibility, but with low sample sizes.
Mixed Models
GWAS
Heritability
NIRS
traits
Beta-glucan content - FIA - percent| CO_350:0005101
Beta-glucan content - NIRS - percent| CO_350:0005065
Grain plumpness - 5.5/64 thin - %| CO_350:0005050
Grain plumpness - 5.5/64 plump - %| CO_350:0005051
Calculate Tool Compatibility
Publish Dataset
Date Published
User
Files
Published To
6/8/2026, 2:59:55 PM
David Waring
trait_data.csv
genotyping_protocols.vcf
accessions.csv
trials.csv
breeding_programs.csv
locations.csv
observations.csv
FigShare
Publish
Remove
Generate Archive Files
Information about Publishing Datasets
Powered By
BREEDBASE
Partners
USDA NIFA National Institute of Food and Agriculture
USDA ARS Agricultural Research Service
US Wheat and Barley Scab Initiative
Funded By

## 26.3 Outlier Detection




As in step The Search Wizard we can create a dataset.

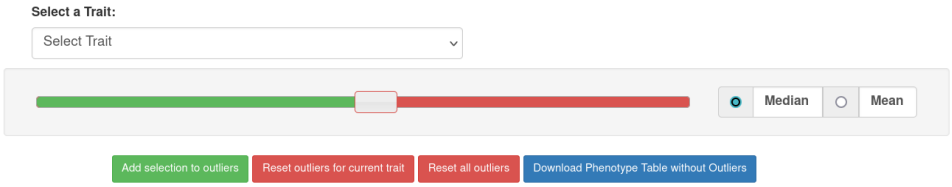
The dataset incorporates a feature to identify outlier points, which we may choose to exclude from a specific dataset. It's important to note that these exclusions only apply at the dataset level, and no data is permanently removed from the database. Additionally, outlier categorization can be modified at any time, and these changes are visible to all other functionalities within the system.

Each dataset stores a wholly unique set of outlier points, completely independent of any other dataset in the database. Outliers are specifically designated for traits within datasets, exclusively encompassing phenotype data. If a particular dataset lacks traits as a part of wizard selection, this functionality is not available.

Each trait has its own set of defined outliers.

### 26.3.1 Accessing Trait Visualization

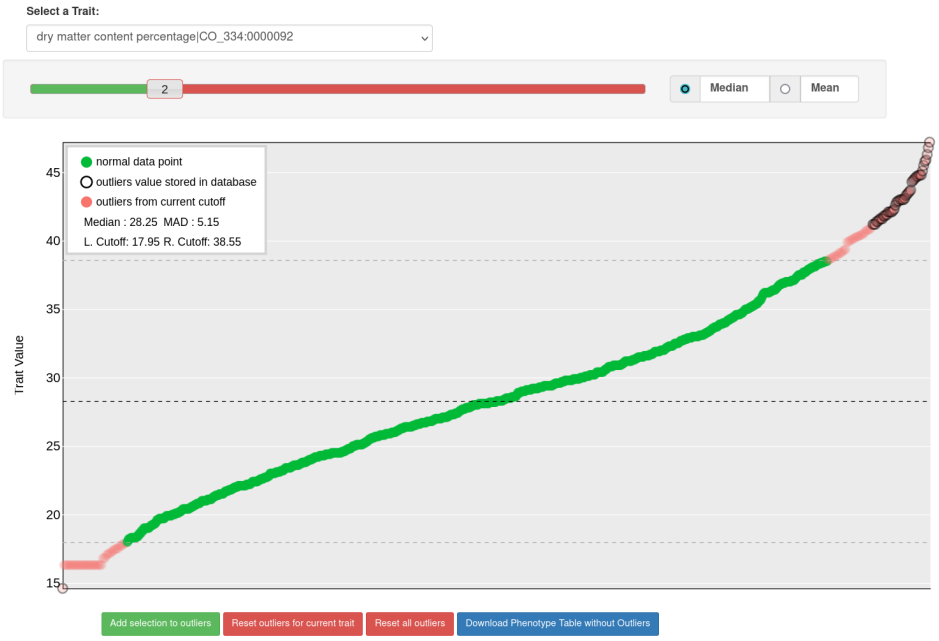
Once you've selected a specific trait, the web application provides access to a visualization of the data points associated with that trait.




### 26.3.2 Interpreting Visual Elements

Once you've selected a specific trait, the web application provides access to a visualization of the data points associated with that trait.

- **Green Points**: As per the legend, represent values for the selected trait that fall below the cut-off point set by the slider. (non-outliers)
- **Black Outlined Points**: These data points are outlined with black borders, indicating that they are currently designated as outliers in the database.
- **Red Points**: The red data points denote the cut-off points established by the slider for the allowable deviation value.




### 26.3.3 Choosing Cut-Off Values

You have two fundamental options for setting cut-off points:

- **Median with MAD**: This option involves using the median (middle value) along with the Mean Absolute Deviation (MAD) as a reference point for determining cut-off values.
- **Mean with Standard Deviation**: Alternatively, you can choose to use the mean (average) in conjunction with the Standard Deviation to set cut-off points.

### 26.3.4 Setting Deviation Multiplier

The slider allows you to specify the deviation multiplier from a central point, which influences the cut-off values.

### 26.3.5 Utilizing Graph Controls

Beneath the graph, you'll find four buttons, each serving a distinct function:

- **Add selection to outliers**: This button enables you to save the current cut-off points to the database for future reference.
- **Reset outliers for current trait**: You can use this option to reset outliers for the selected trait.
- **Reset all outliers**: This button allows you to reset outliers for the entire dataset.
- **Download Phenotype Table without outliers**: You can download the phenotype data table in a comma-separated value format file, using this feature, with outliers excluded for selected dataset.

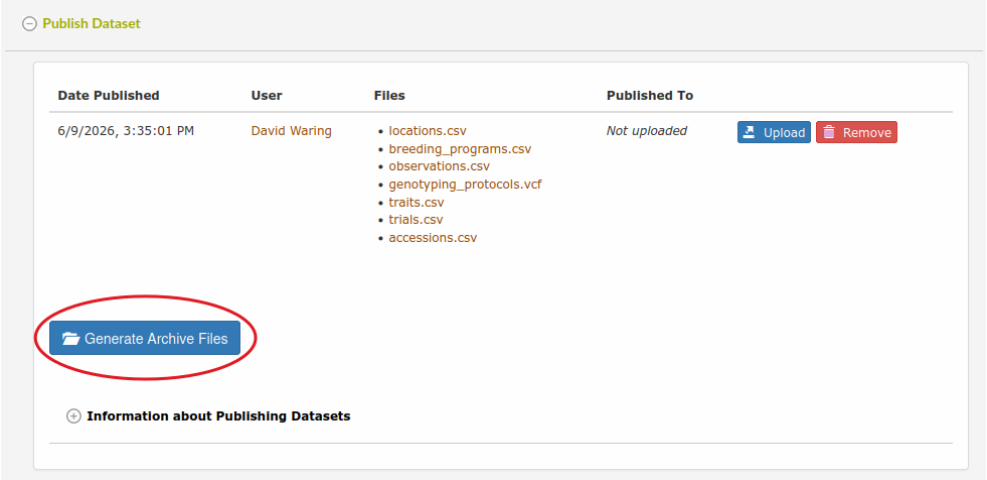
These tools and functions are designed to provide you with control and insights when working with data visualization and outliers.

## 26.4 Publishing a Dataset

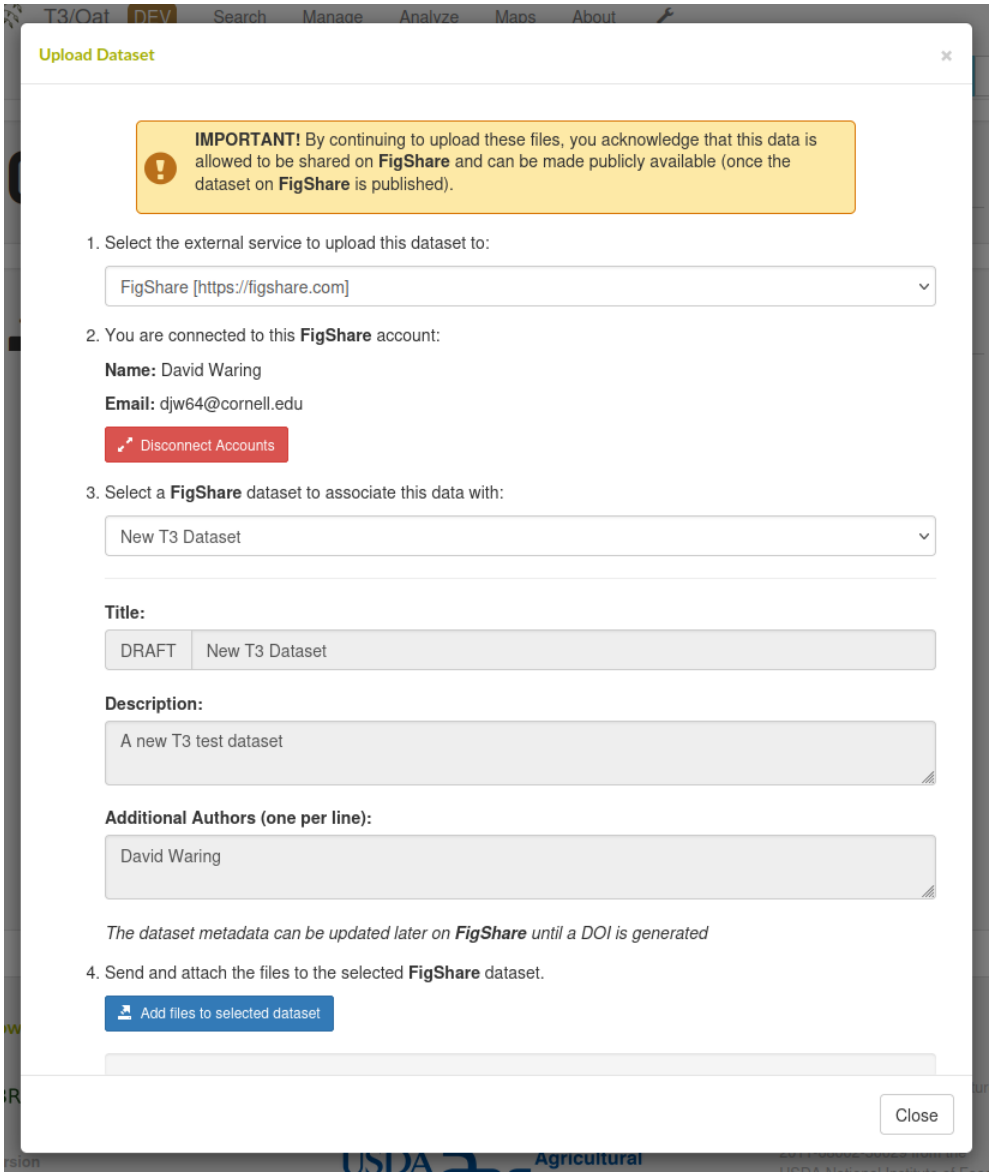
The owner of a Dataset can generate a static snapshot of the data within their Dataset at the point in time when the archive is created. This creates plain-text files that are stored on the breedbase server and stay the same even if the underlying data in the database has changed. When enabled on the database, the owner can also export that snapshot to an external service, such as FigShare, to make the data more easily accessible to external users. In addition, a DOI (digital object identifier) can be generated to create a persistent link to the published data that can be used as a reference in a publication.

To publish a Dataset:

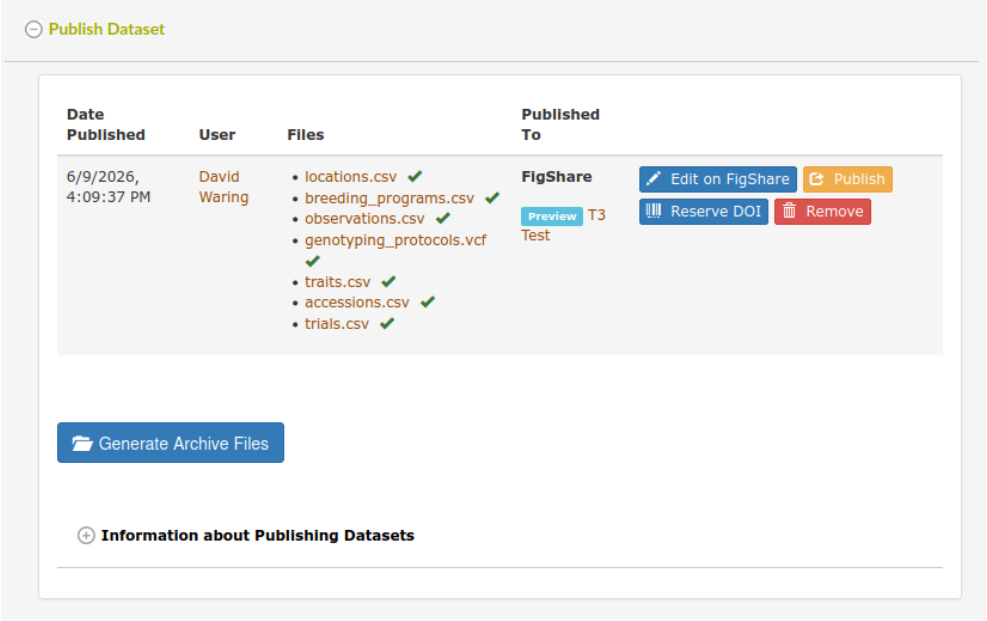
1. **Generate the archived files** - this creates a static snapshot of the data within the dataset at the time when the files are generated. The data in this snapshot will stay the same even if the underlying data in the dataset is modified in the database. The snapshot consists of plain-text files (.csv, .vcf, etc) for all of the data contained within the Dataset. It also extrapolates to include relevant data types that are not explicitly defined by the Dataset. For example, if you have phenotyping trials selected in the Dataset but do not explicitly include locations or traits, the archive will include location and trait information for all of the locations of the selected trials in the datasets and all of the traits observed by the trials in the datasets. If you explicitly include traits in the dataset, then the archive will only include those defined traits.



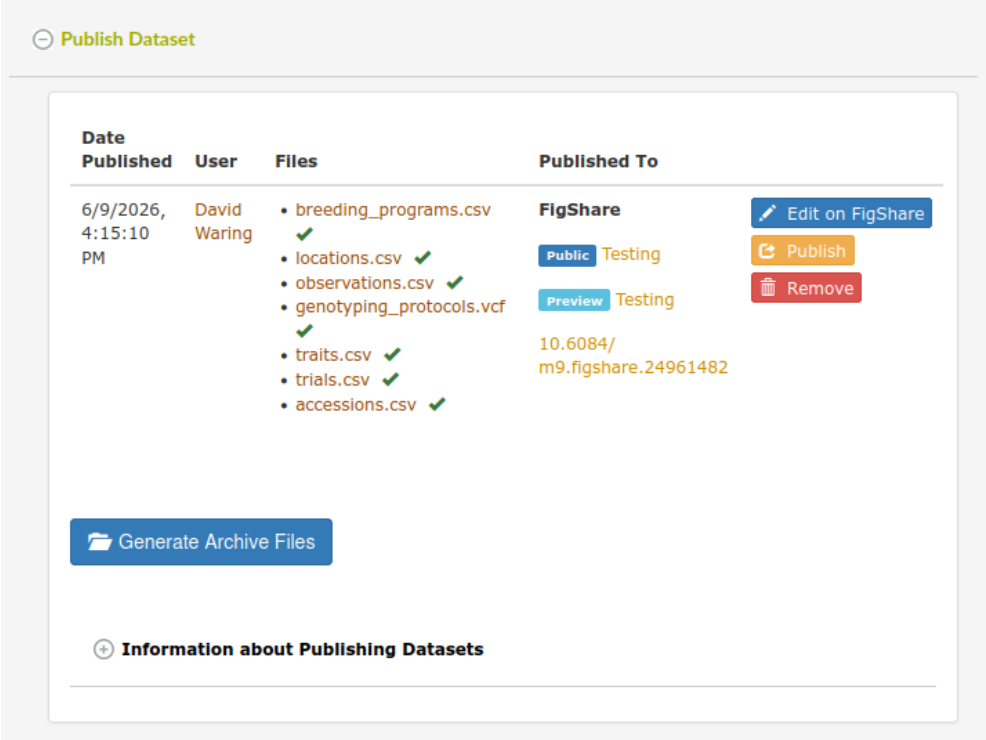

2. **Upload the files to an external service** - this will upload the archived files to the external service. You will be given the option to upload the files to a new dataset or an existing dataset on the external service.




3. **Publish the external dataset** - once the files have been uploaded to an external dataset, you can make the external dataset publicly available for others to find. This will make the data available to anyone that has access to the external service - it does not change any visibility settings on breedbase. **The external dataset cannot be removed onced it is published**.




4. **Reserve a DOI** - generate a Digital Object Identifier (DOI) for the external dataset that can be used as a reference in publications. If the external dataset is still private, the DOI will become active once the external dataset is published and made public.

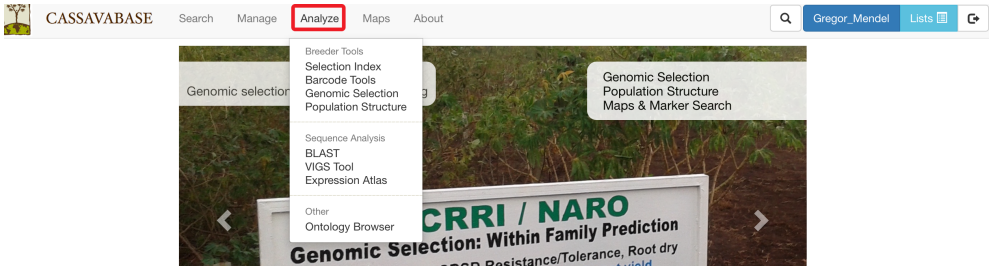



You can make additional changes to the external dataset's metadata and uploaded files *until the external dataset has been published*. There are more detailed edit options on the website for the external service - click the Edit button to open the external service's edit webpage. Once the external dataset has been published, all new changes will be made to a new version of the external dataset, which will have to be published again to make the changes public. Any previously published versions will be preserved and still accessible. **The external dataset cannot be removed onced it is published**.

# Chapter 27

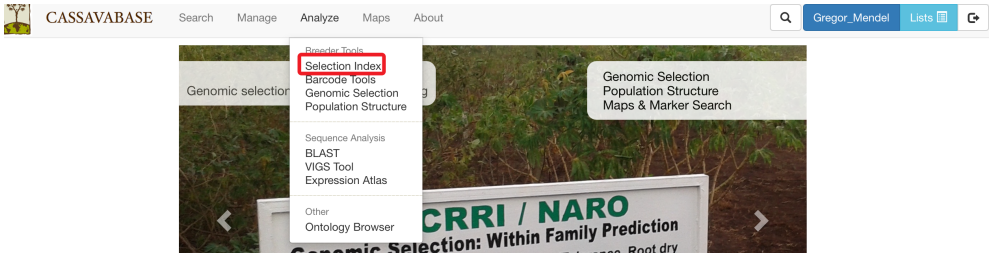
# Data Analysis Tools

SGN databases provides several tools for phenotype data analysis, marker-assisted selection, sequence and expression analyses, as well as ontology browser. These tools can be found in the "Analyze" menu.

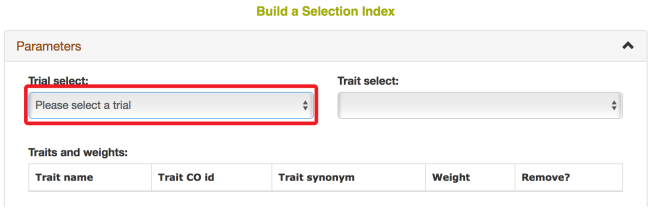



## 27.1 Selection Index

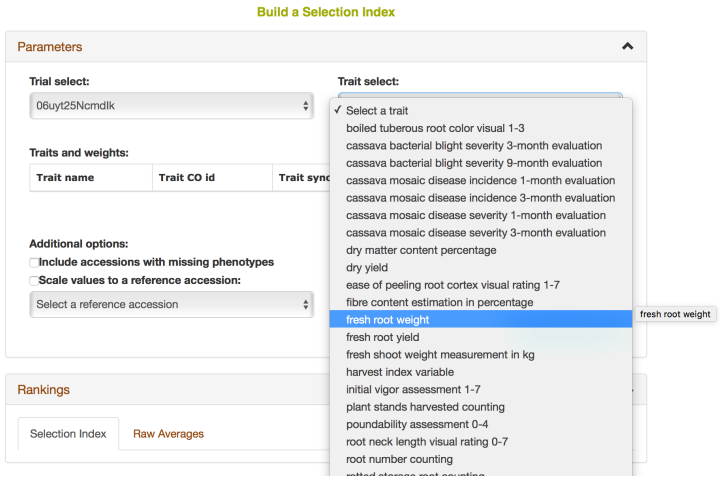
To determine rankings of accessions based on more than one desirable trait, SGN databases provide a "Selection Index" tool that allows you to specify a weighting on each trait. To access the tool, clicking on "Selection Index" in the "Analyze" menu.



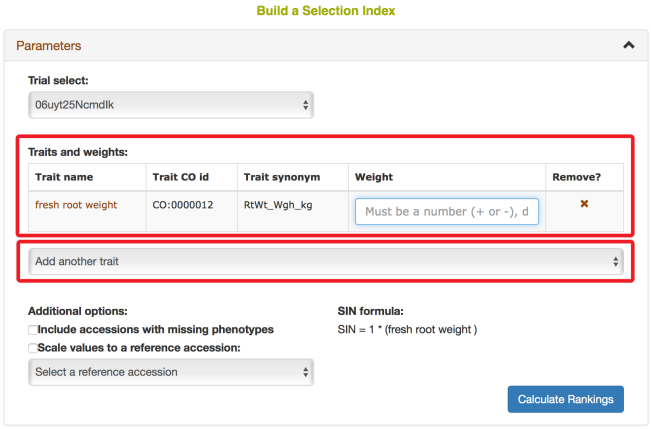

On the Selection Index page, selecting a trial that you want to analyze.



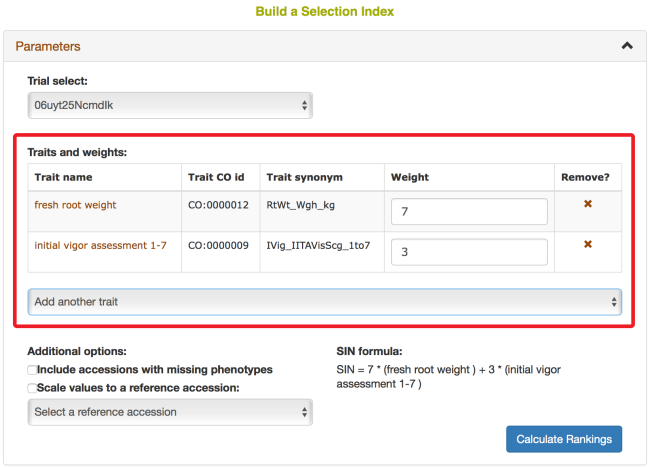

After you selected a trial, you can find traits that were assayed in that trial in the "Trait" box.



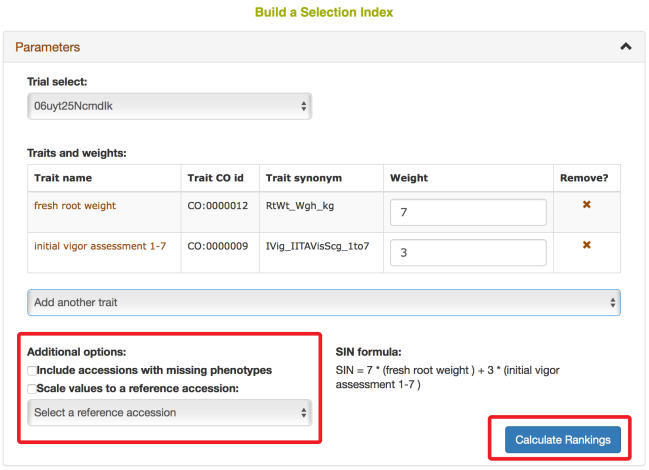

Selecting a trait that you want to include in the analysis will open a new dialogue showing the selected trait and a box that you can assign a "Weight" of that trait. After you are done, you can continue by selecting another trait by clicking on "Add another trait" link.



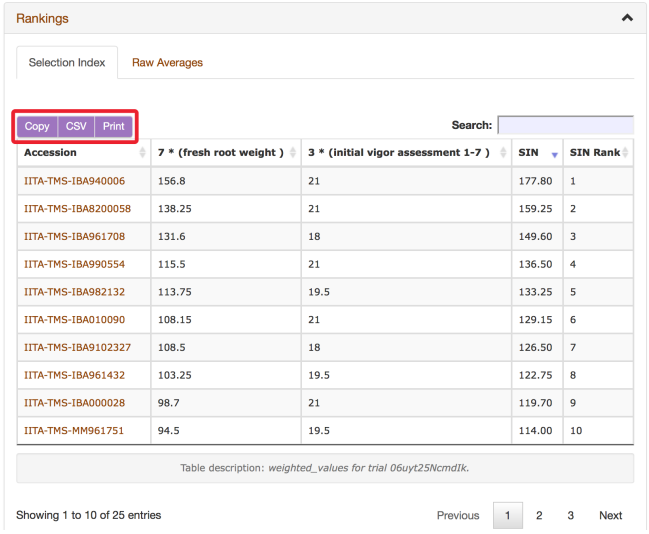

After you selected another trait, this page will automatically update information for you by showing all of the traits that you selected for the analysis.



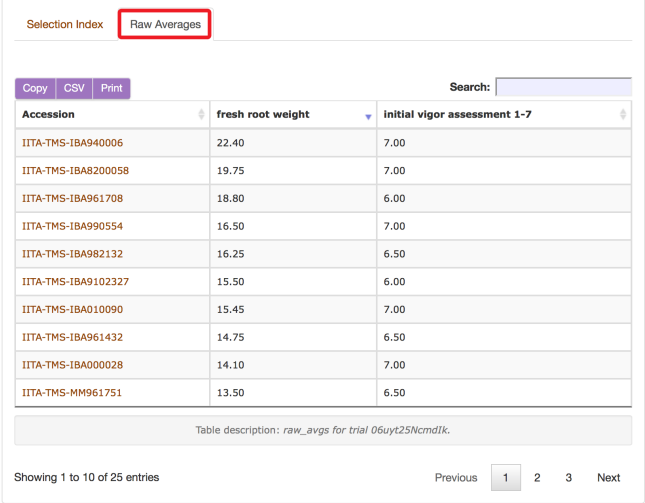

You also have options to choose a reference accession, choose to include accessions with missing phenotypes, scaling values to a reference accession. After you complete your setting, clicking on “Calculate Rankings”



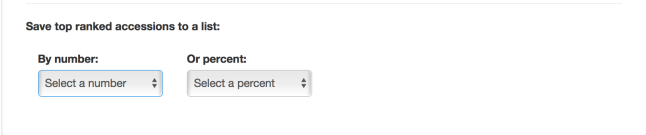

The Selection Index tool will generate rankings of accessions based on the information that you specified. You can copy the results to your system clipboard, convert the table data to CSV format, or print the data.




Clicking on "Raw Average" will display average values of the phenotypes of those ranked accessions.



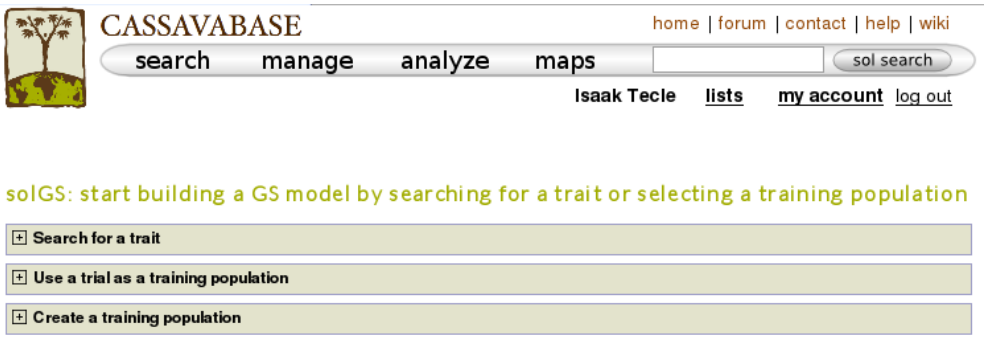

Selection Index tool also allows you to save top ranked accessions directly to "Lists". You can retrieve top ranked accessions by selecting a number or a percent.




## 27.2 Genomic Selection

The prediction of breeding values for a trait is a one step or two steps process, depending on what stage in your breeding cycle you are. The first step is to build a prediction model for a trait using a training population of clones with phenotype and genotype data. If you have yet to select parents for crossing for your first cycle of selection you can use the breeding values of the training population. If you are at later stages of your selection program, you need to do the second step which is applying the prediction model on your selection population. All clones in your training and selection populations must exist in the database.

To use the genomic selection tool, on *cassavabase.org*, select "Genomic Selection" from the "analyze" pull-down menu.



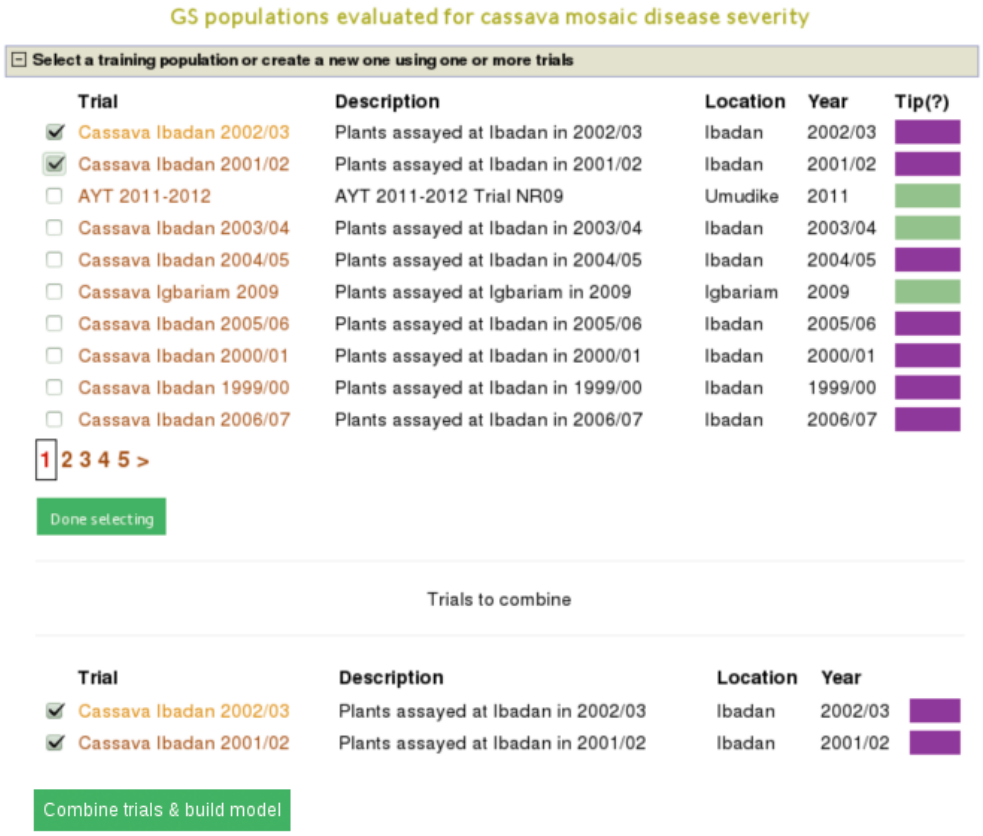

There are three ways to build a model for a trait.

### 27.2.1 Building a Model - Method 1:

One way to build a model is, using a trait name, to search for trials in which the trait was phenotyped and use a trial or a combination of trials to build a model for the trait. For example, if you search for "mosaic disease severity, you will get a list of trials you can use as training populations.

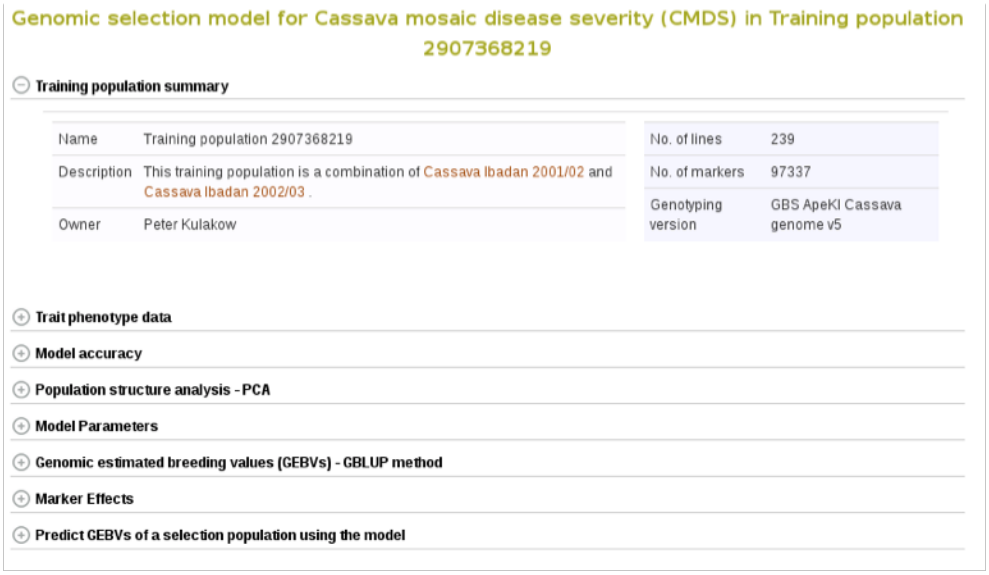
You will get a list of trials (as shown below) in which the trait of your inter-

ested was phenotyped. From the list, you can use a single trial as a training population or combine several trails to form a training population for the prediction model of the trait. Let"s say, you want to create a training population using individuals from trials "cassava ibadan 2001/02" and "cassava ibadan 02/03" and build a model for "cassava mosaic disease severity" using all clones from the training population.




Select the trials to combine (the same coloured), click 'done selecting', click the "combine trials and build model" button, and you will get a model and its output for the trait. On the model detail page, you can view the description of input data used in the model, output from the model and search interface for selection populations the model you can apply to predict their breeding values. The description of the input data for the model includes the number of phenotyped clones, and the number of markers, scatter and frequency distribution plots for the phenotype data, relationship between the phenotype

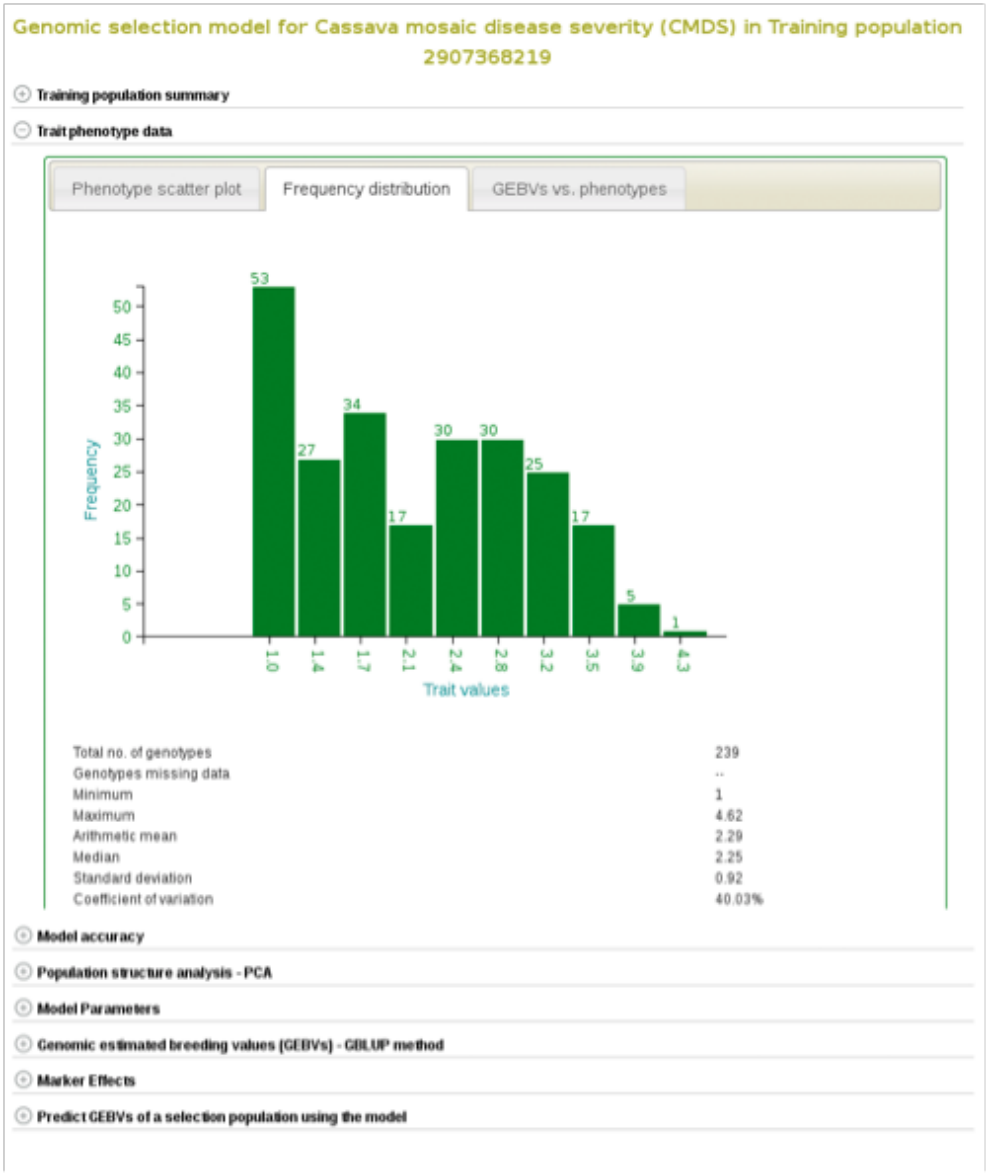
data and GEBVs, population structure. The model output includes model parameters, heritability of the trait , prediction accuracy, GEBVs of the individuals from the training population and marker effects.



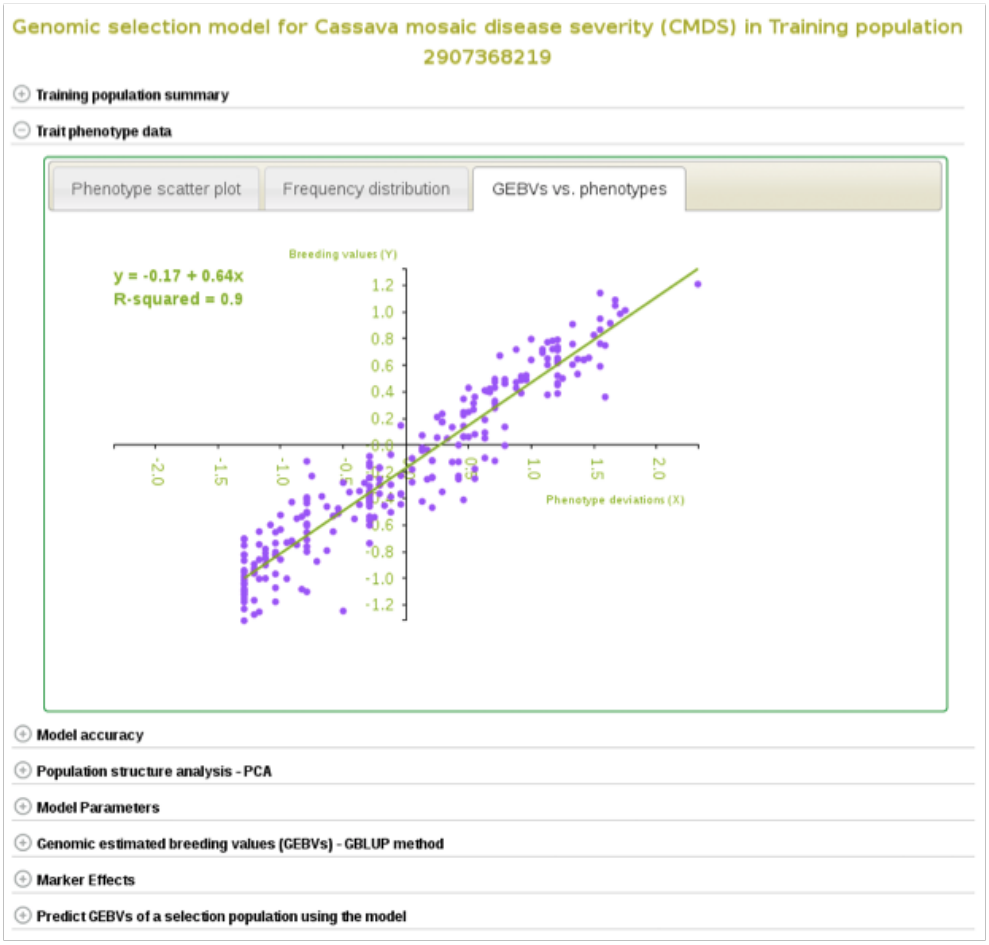

Expand each section to see detailed information.

If you expand the 'Trait phenotype data' section, you will find plots to explore the phenotype data used in the model. You can assess the phenotype data using a scatter and histogram plots and the descriptive statistics.

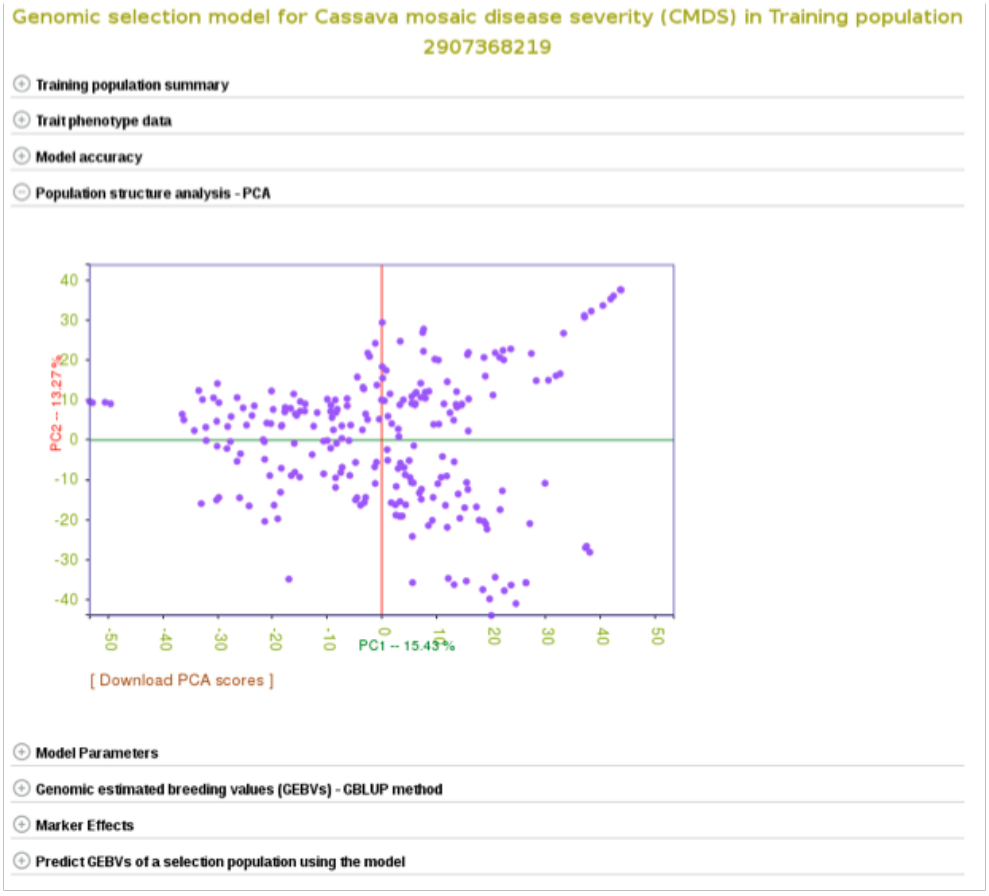
Genomic selection model for Cassava mosaic disease severity (CMDS) in Training population 2907368219
Training population summary
Trait phenotype data
Phenotype scatter plot
Frequency distribution
GEBVs vs. phenotypes
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
Phenotype values
Clones
Model accuracy
Population structure analysis - PCA
Model Parameters
Genomic estimated breeding values (GEBVs) - GBLUP method
Marker Effects
Predict GEBVs of a selection population using the model



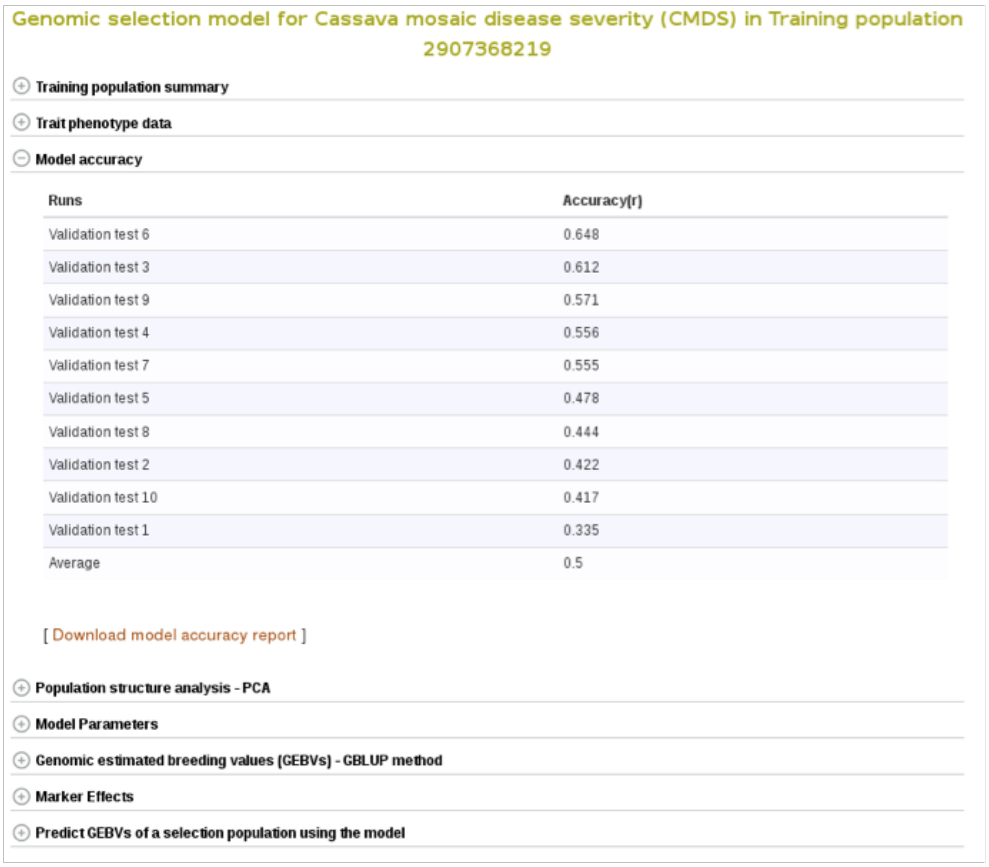

A regression line between observed phenotypes and GEBVs shows the relationship between the two.




You can also explore if there is any sub-clustering in the training population using PCA.




To check the model accuracy, a 10-fold cross-validation test, expand the 'model accuracy' section.



Genomic selection model for Cassava mosaic disease severity (CMDS) in Training population 2907368219

⊕ Training population summary

⊕ Trait phenotype data

⊖ Model accuracy

| Runs | Accuracy(r) |
|---|---|
| Validation test 6 | 0.648 |
| Validation test 3 | 0.612 |
| Validation test 9 | 0.571 |
| Validation test 4 | 0.556 |
| Validation test 7 | 0.555 |
| Validation test 5 | 0.478 |
| Validation test 8 | 0.444 |
| Validation test 2 | 0.422 |
| Validation test 10 | 0.417 |
| Validation test 1 | 0.335 |
| Average | 0.5 |

[ Download model accuracy report ]

⊕ Population structure analysis - PCA

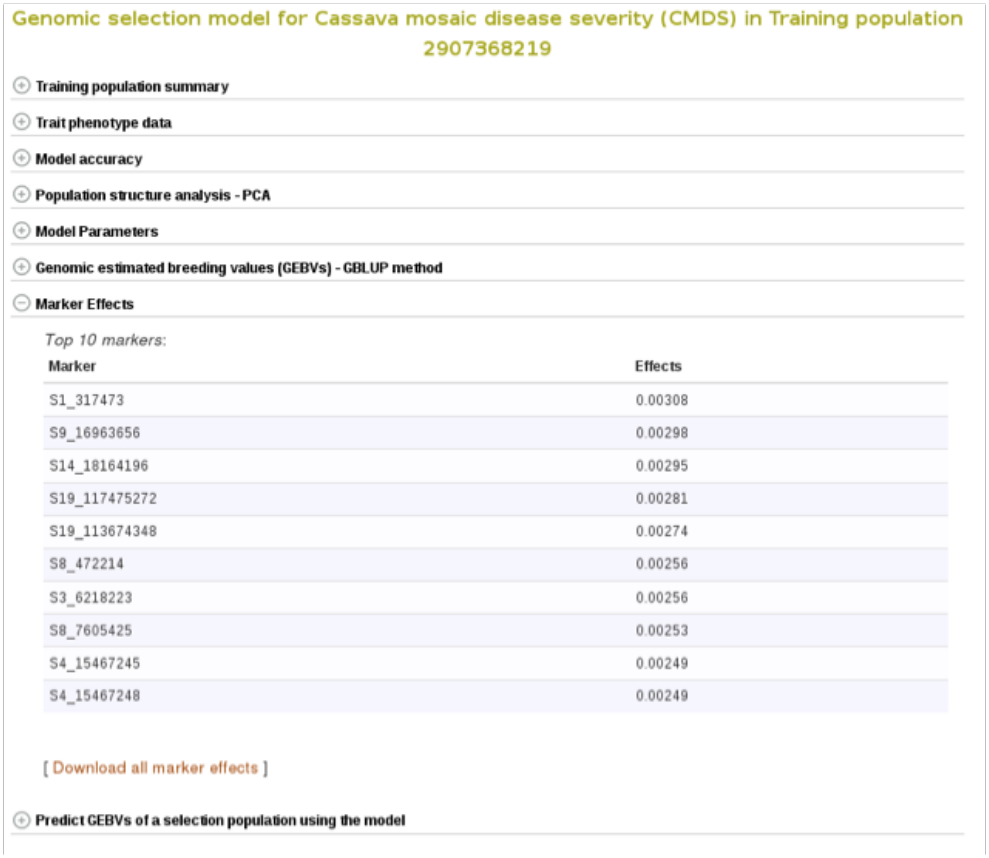
⊕ Model Parameters

⊕ Genomic estimated breeding values (GEBVs) - GBLUP method

⊕ Marker Effects

⊕ Predict GEBVs of a selection population using the model

Marker effects are also available for download. To do so, expanad the 'Marker Effects' section and click the 'Download all marker effects' link and you will get a tab delimited output to save on your computer.



Genomic selection model for Cassava mosaic disease severity (CMDS) in Training population 2907368219

- Training population summary
- Trait phenotype data
- Model accuracy
- Population structure analysis - PCA
- Model Parameters
- Genomic estimated breeding values (GEBVs) - GBLUP method
- Marker Effects

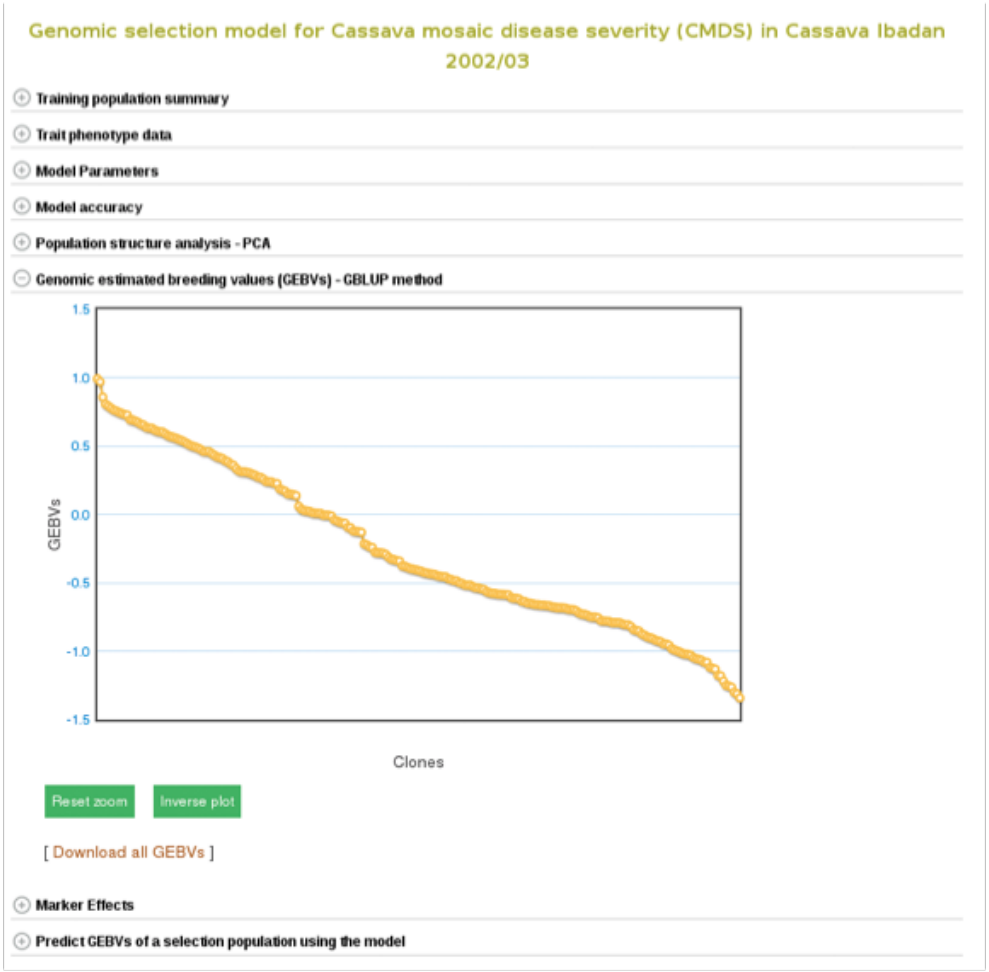
*Top 10 markers:*

| Marker | Effects |
|---|---|
| S1_317473 | 0.00308 |
| S9_16963656 | 0.00298 |
| S14_18164196 | 0.00295 |
| S19_117475272 | 0.00281 |
| S19_113674348 | 0.00274 |
| S8_472214 | 0.00256 |
| S3_6218223 | 0.00256 |
| S8_7605425 | 0.00253 |
| S4_15467245 | 0.00249 |
| S4_15467248 | 0.00249 |

[ Download all marker effects ]

- Predict GEBVs of a selection population using the model

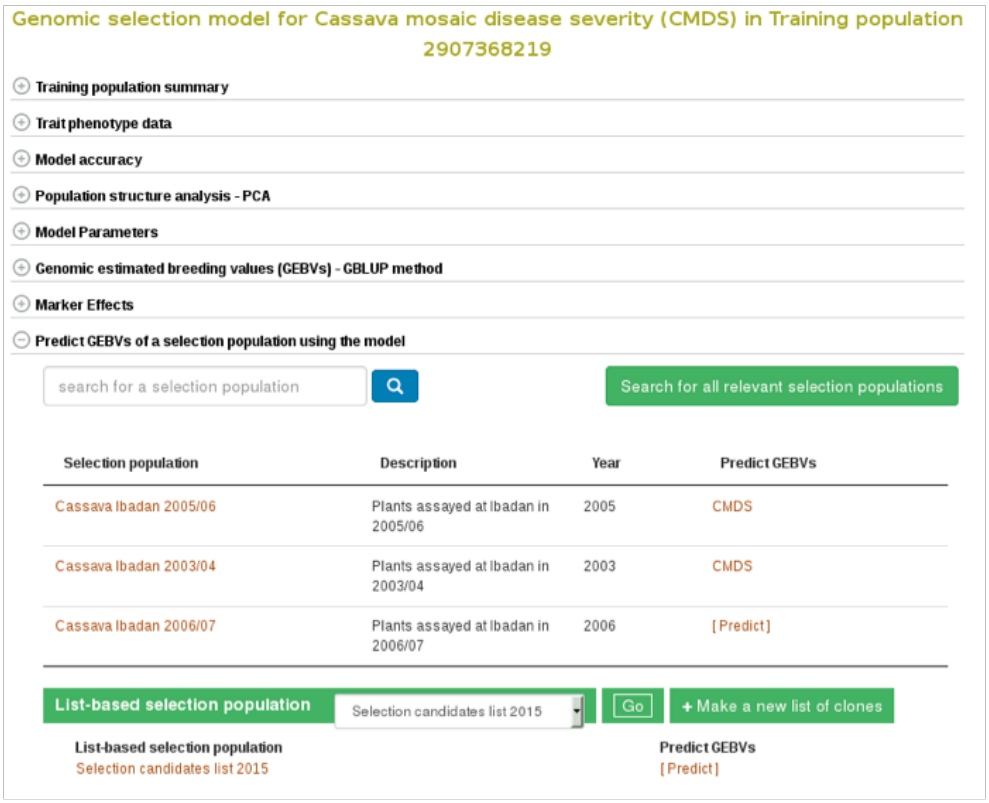
The breeding values of the individuals used in the training population are displayed graphically. Mousing over each data point displays the clone and its breeding value. To examine better, you can zoom in into the plot by selecting an area on the plot. You can download them also by following the "Download all GEBVs" link.




### Estimating breeding values in a selection population

If you already have a selection population (in the database), from the same model page, you can apply the model to the selection population and estimate breeding values for all the clones in the population. You can search for a selection population of clones in the database using the search interface or you can make a custom list of clones using the *list interface*. If you click the "search for all relevant selection populations", you will see all relevant selection populations for that model. However, this option takes long time decause of the large set of populations in the database and the filtering. Therefore, the fastest way is to search for each of your selection populations by name. If you are logged in to the website you will also see a list of your

custom set of genotyped clones.



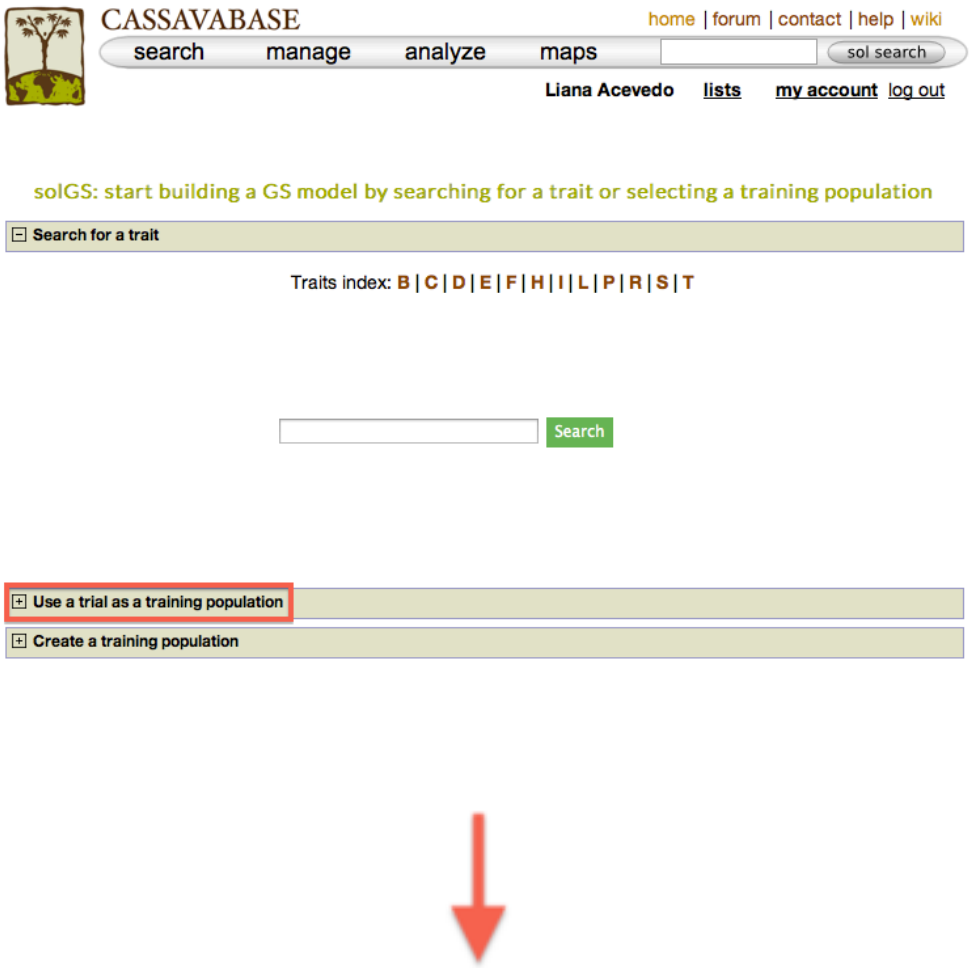

To apply the model to a selection population, simply click your population name or "Predict Now" and you will get the predicted breeding values. When you see a name of (or acronym]) of the trait, follow the link and you will see an interactive plot of the breeding values and a link to download the breeding values of your selection population.

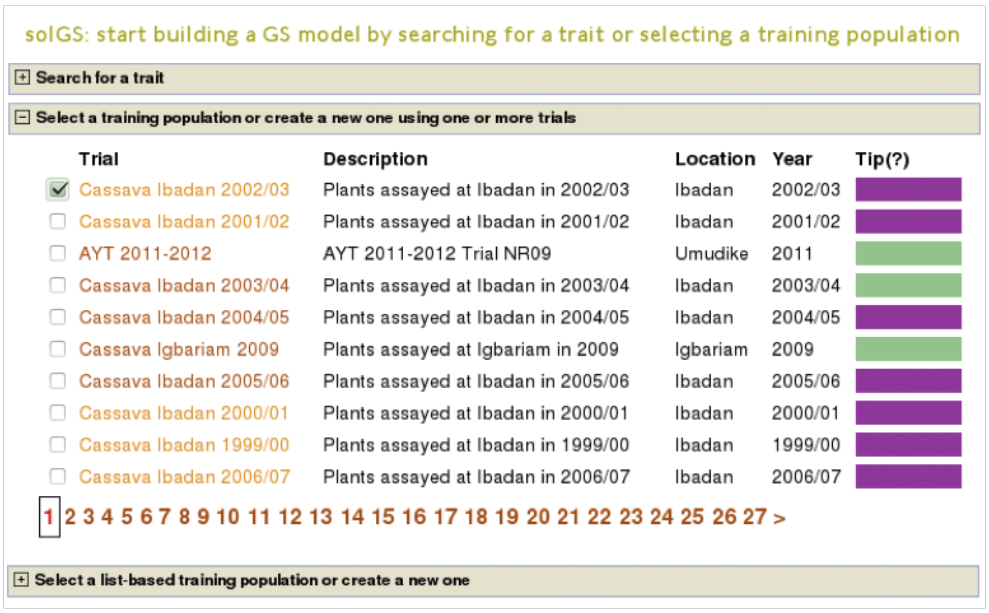
### 27.2.2 Building a Model - Method 2

Another way to build a model is by selecting a trial, instead of selecting and searching for a specific trait. This approach is useful when you know a particular trial that is relevant to the environment you are targeting to breed material for. This method allows you to build models and predict genomic estimated breeding values (GEBVs) for several traits within a single trial at once. You can also calculate selection index for your clones when GEBVs are

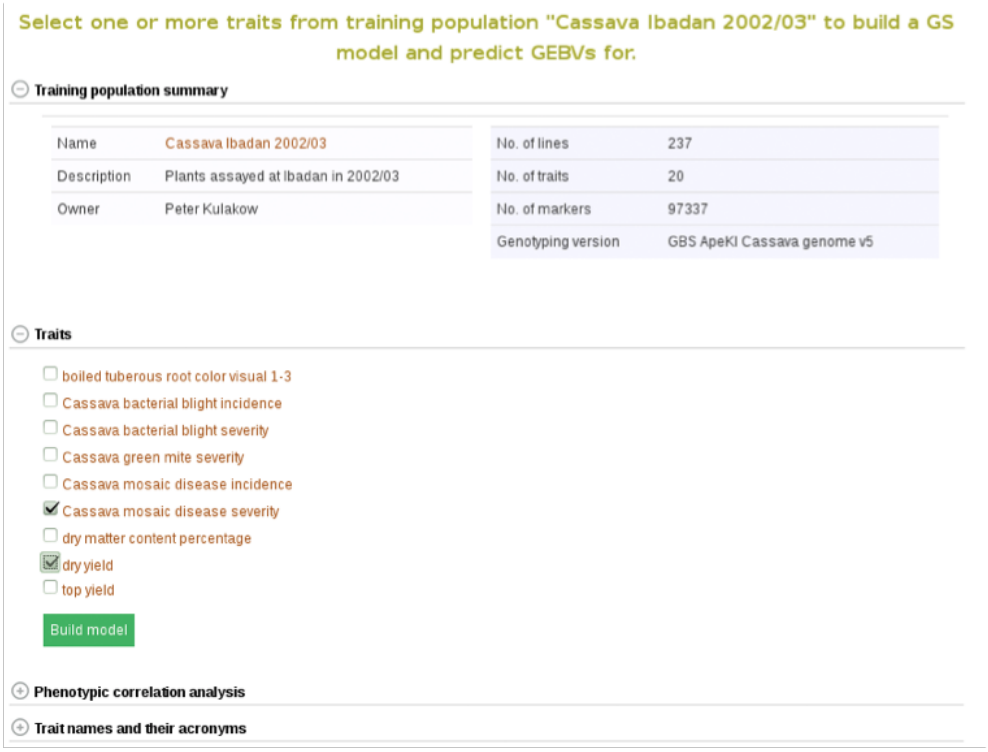
estimated for multiple traits.

To do this select the "Genomic Selection" link found under the "analyze" menu. This will take you to the same home page as used with Method 1. However, instead of entering information to search for in "Search for a trait", click on "Use a trait as a trial population". This will expand a new menu that will show all available trials.

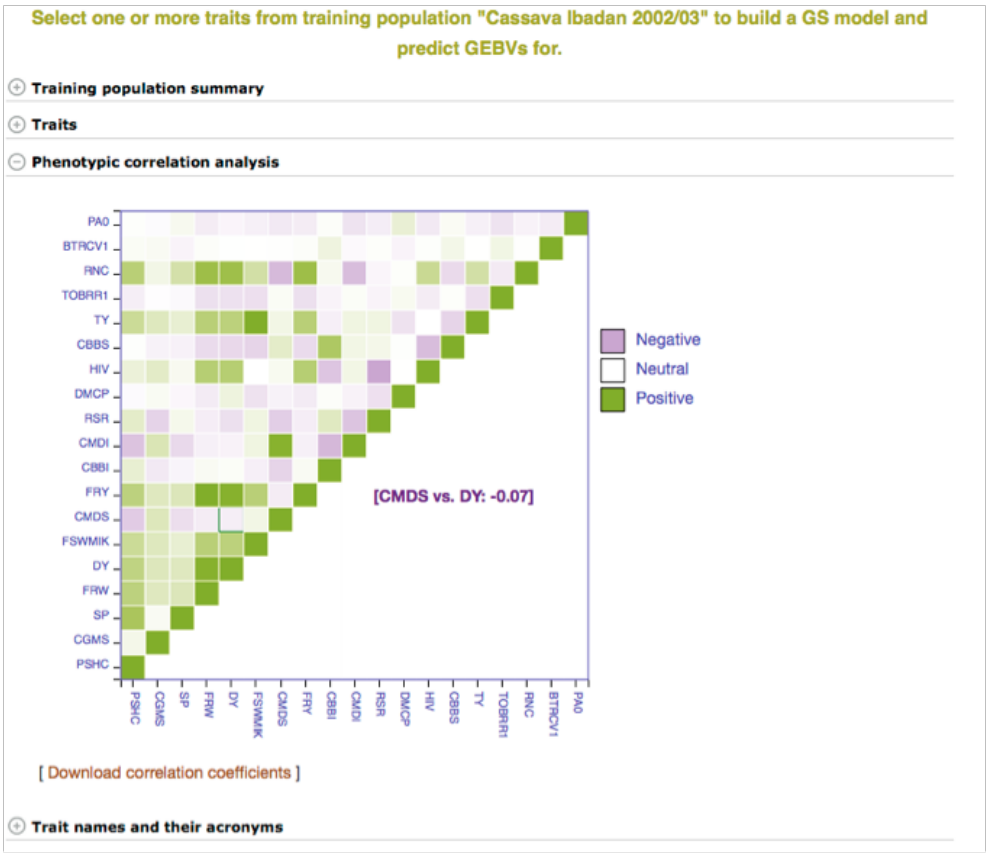




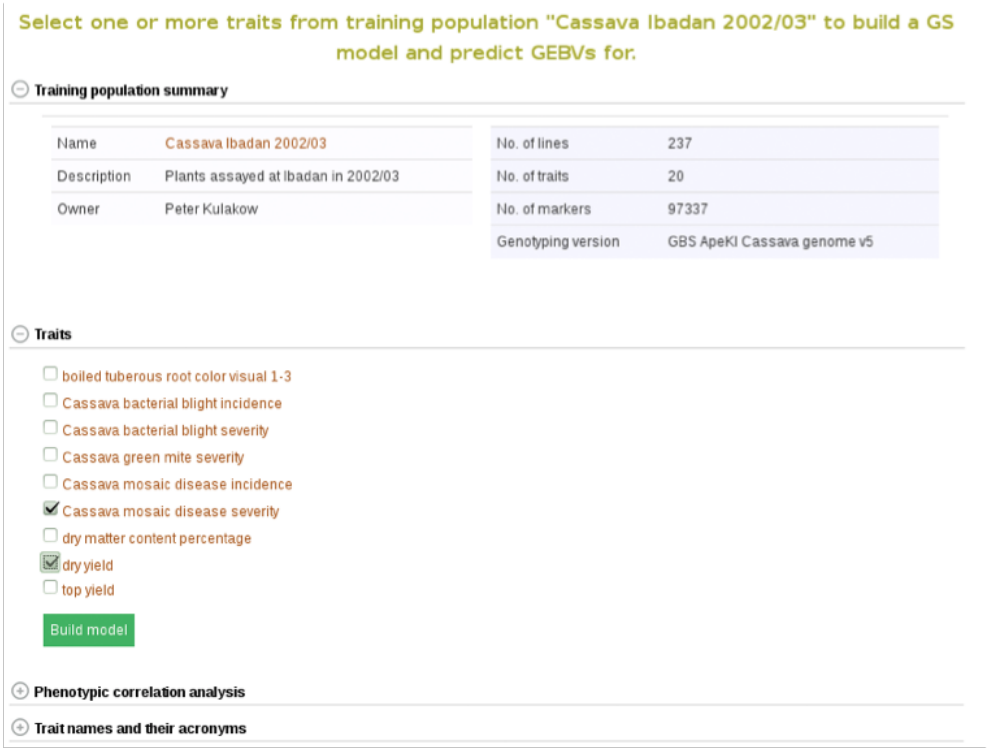

To begin creating the model, select the existing trial that you would like to use. In this example I will be using the trial and trait data from "Cassava Ibadan 2002/03" trial. Clicking on a trial will take you to a page where you can find information such as number of markers and number of phenotypes clones.



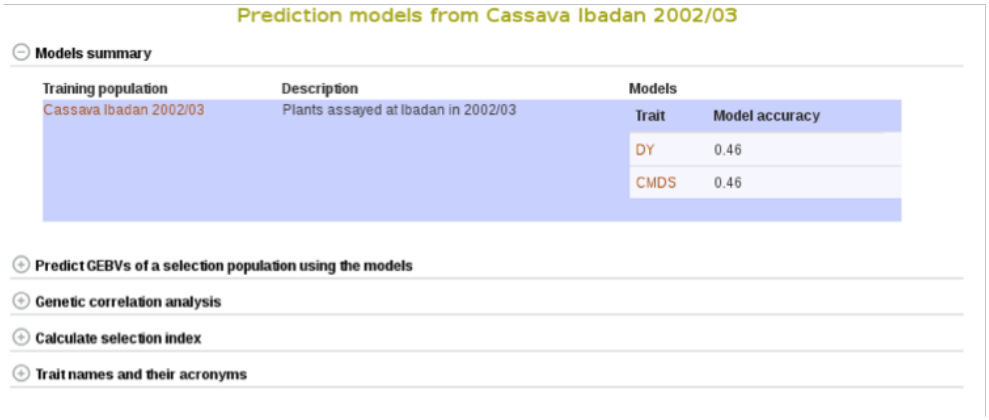

In addition to the number of phenotype clones and number of markers, the main page for the trial selected also has information and graphs on phenotypic correlation for all of the traits. By moving your cursor over the graph you can read the different values for correlation between two traits. A key with all of the trait names of the acronyms used can be found in the tab below the graph.



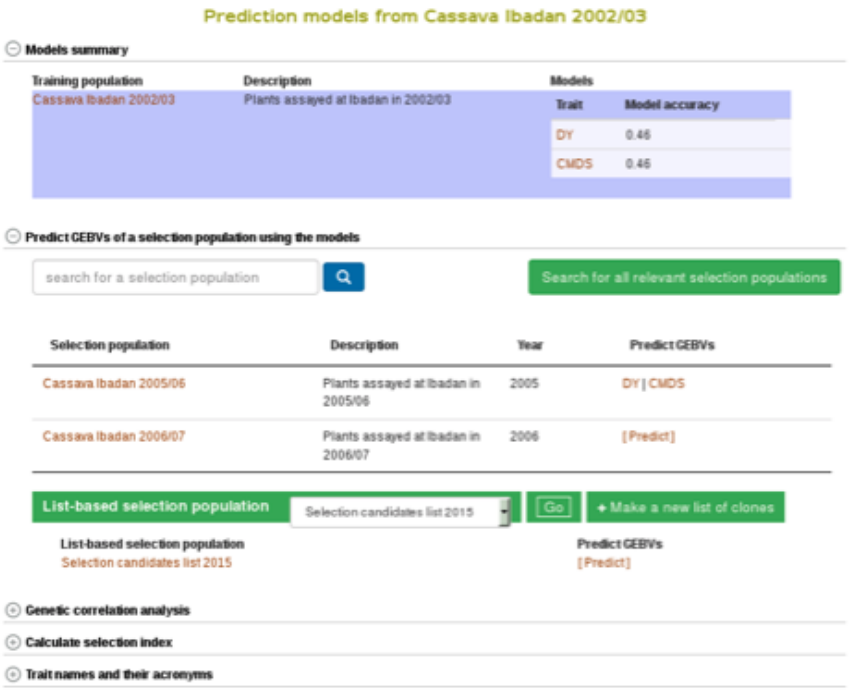

Below the "Training population summary" there is a tab for "Traits". Clicking on this tab will show all available traits for the specific trial. You can create a model by choosing one or multiple traits in the trial and clicking "Build Model". In this example, the traits for "cassava bacterial blight severity" and "cassava mosaic disease severity" have been selected.




Clicking on "Build Model" will take you to a new page with the models outputs for the traits. Under the "Genomic Selection Model Output" tab you can view the model output and the model accuracy. Clicking on any of the traits will take you to a page with information about the model output on that individual trait within the trial. There you can view all of the trait information that was seen in more detail in *Method 1*.



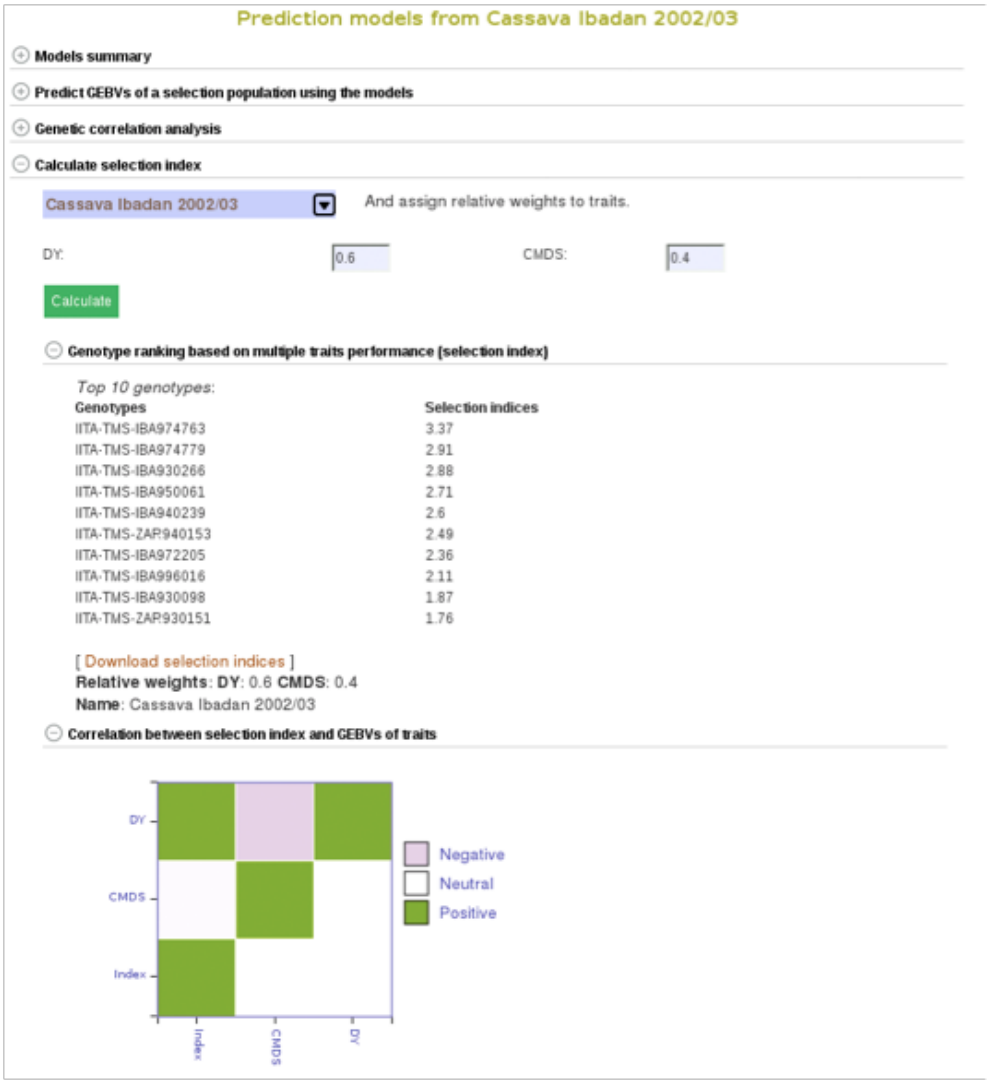
You can apply the models to simultaneously predict GEBVs for respective traits in a selection population by clicking on "Predict Now" or the name of the selection population. You can also apply the models to any set of genotyped clones that you can create using the "lists" feature. For more information on lists, click *here*. Follow the link to the trait name to view and download the predicted GEBVs for the trait in a selection population.




To compare clones based on their performance on multiple traits, you can calculate selection indices using the form below. Choose from the pulldown menu the population with predicted GEBVs for the traits and assign relative weights for each trait. The relative weight of each trait must be between 0 - 1. 0 being of least weight and importance, not wanting to consider that particular trait in selecting a genotype and 1 being a trait that you give highest importance.

In this example we will be using the "Cassava Ibadan 2002/03" population and assigning values to each of the traits. Remember that there is a list of acronyms and trait names at the bottom of the page for reference. After entering whatever values you would like for each trait click on the "Calculate"

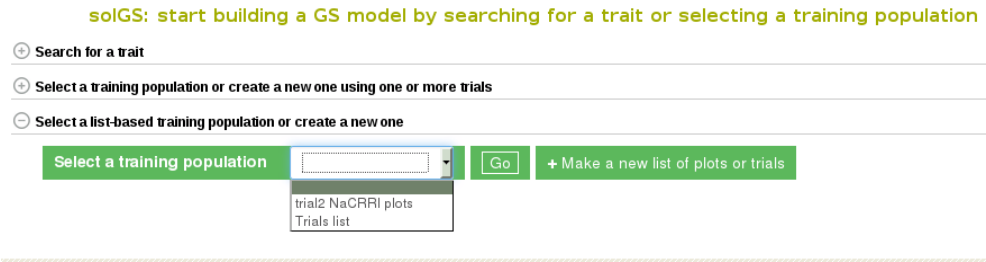
button to generate results. This will create a list of the top 10 genotypes that most closely match the criteria that you entered. The list will be displayed right below the "selection index" tab. This information can also be downloaded onto your computer by clicking on the "Download selection indices" link underneath the listed genotypes and selection indices.



### 27.2.3 Building a Model - Method 3

In addition to creating a model by searching for pre-existing traits or by preexisting trial name, models can also be created by using your own list of clones. This creates a model by using or creating a training population.

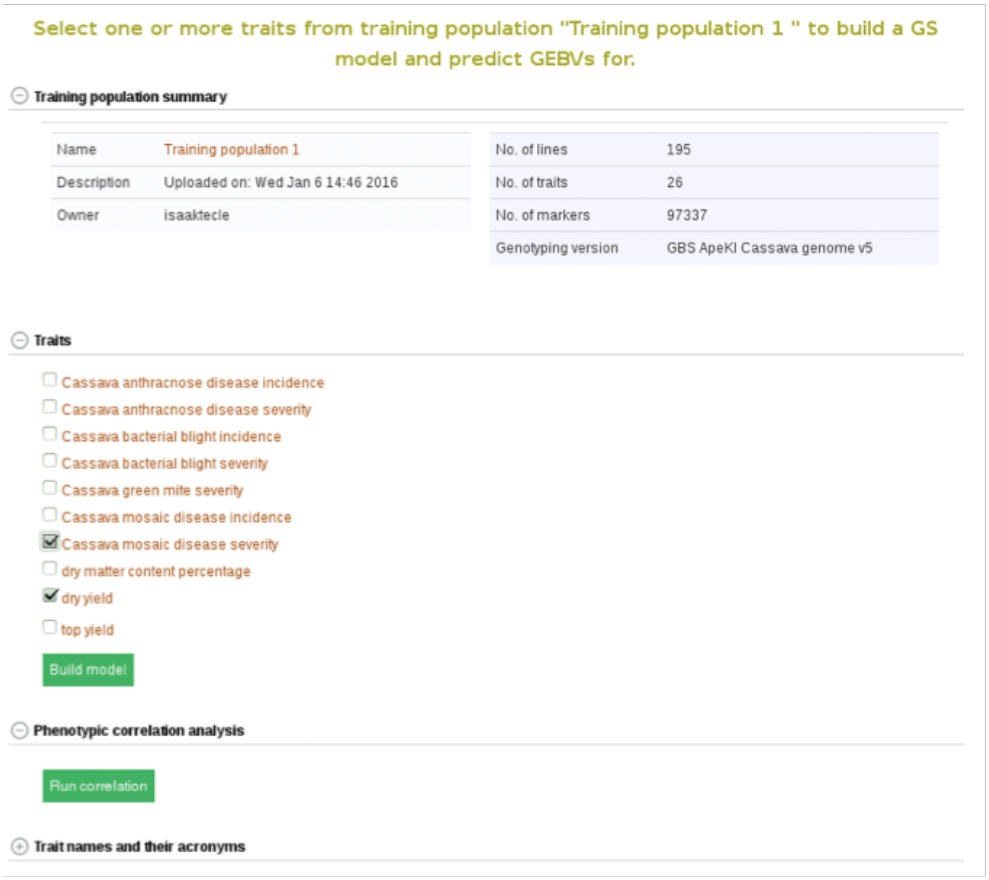
The page to use the third Method for creating a population model is the same as for the other two models. Select "Genomic Selection" from under the "analyze" menu of the main toolbar. This will take you to the Genomic Selection homepage and show you all three available methods to create a model. To see and use Method 3 scroll down and click on the tab labeled "Create a Training Population". This will open a set of tools that will allow you to use pre-existing lists or to create a new list.




Once the "Create a Training Population" tab is opened you have the option to use a pre-existing list or create new one. To learn how to create a list, click *here*. The "Make a new list of plots" link will take you directly to the Search Wizard that is usually used to create lists.

Please note: the only lists that can be used in Method 3 to create a model are lists of plots and trials. If the pre-existing list is not of plots or trials (for example, traits, or locations) it will not show up and cannot be used as a training population. When you create you use a list of trials, the trials data will be combined to create a training data set.

To use your custom list of plots or trials as a training population, select the list and click "Go". This will take you to a detail page for the training population.




From here on you can build models and predict breeding values as described in *Method 2*.

## 27.3 Genome Browsing

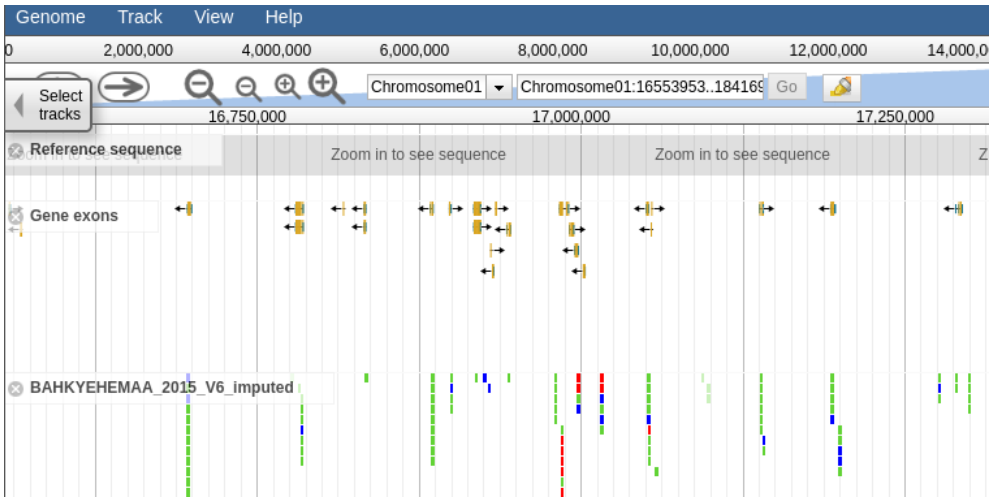
There are two ways to evaluate genotype information within the browser, from an accession detail page or a trial detail page.

### 27.3.1 Browsing Genotype data by Accession

If you are interested in browsing genotype information for a single accession, for example 'BAHKYEHEMAA', navigate to the accession detail page.

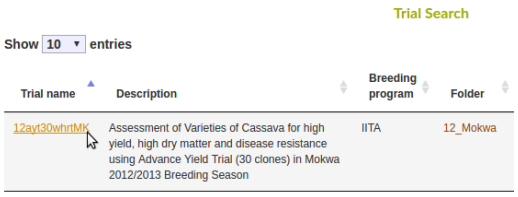
Near the bottom of the detail page is a collapsible section called "Accession Jbrowse".

This section will contain a link to the accession jbrowse page if the necessary genotype data is available. Clicking the link should take you to a page that looks like this, a which point you can browsre the genotype data in the form of a vcf track aligned to the latest build of the genome.

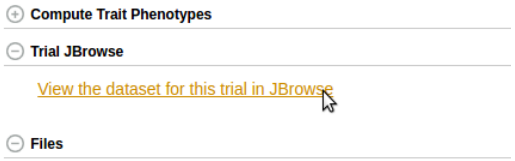



## 27.3.2 Browsing Genotype data by Trial

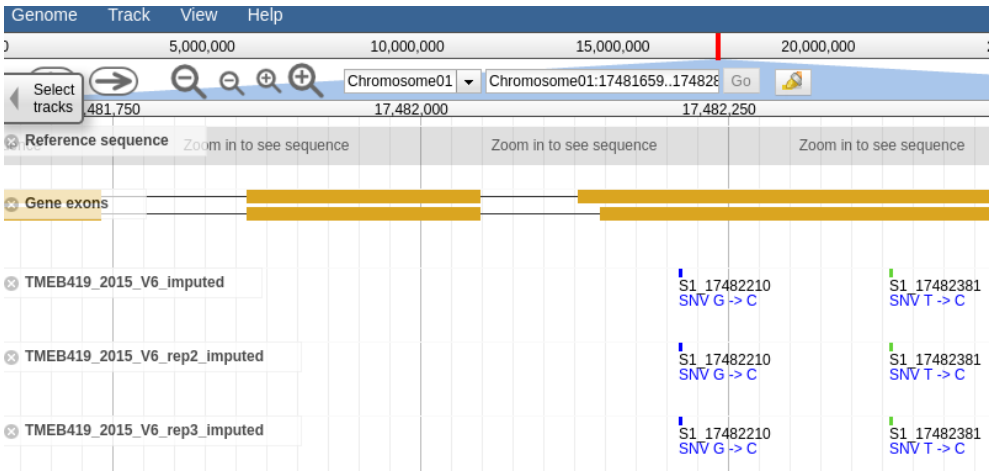
If you are interested in browsing genotype information for the accessions within a given trial, navigate to the trial detail page.




Halfway down the page is a collapsible section called “Trial Jbrowse”. This section will contain a link to the trial jbrowse page if the necessary genotype data for at least two accessions planted in the trial is available.




Clicking the link should take you to a page that looks like this, a which point you can browse the genotype data in the form of vcf tracks aligned to the latest build of the genome.




## 27.4 Principal Component Analysis (PCA)

Principal component analysis helps estimate and visualize if there is sub-grouping of individuals within a dataset based on a number of variables. Currently, you can use marker data to run PCA on datasets.

You can run PCA from multiple places on the website. To do PCA on

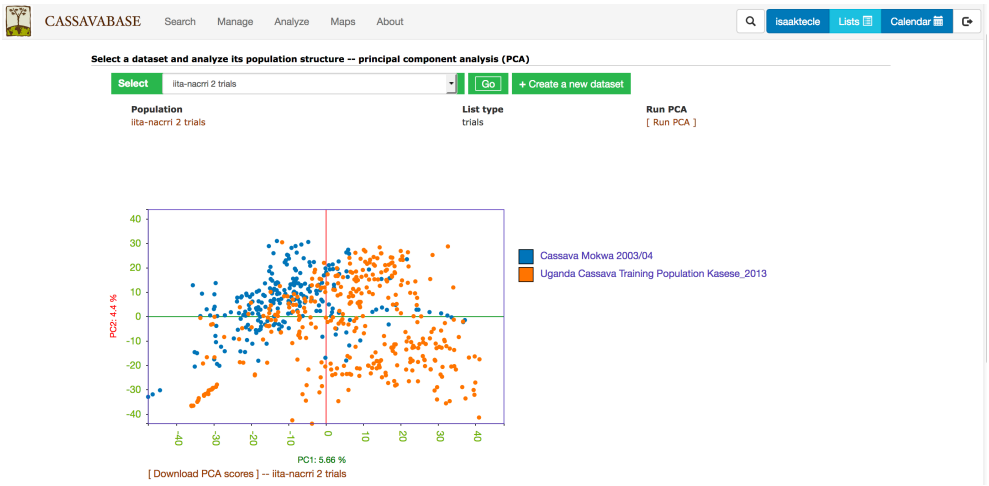
(1) individuals from a trial, go to the trial detail page and find the PCA tool under the "Analysis tools" section.

(2) individuals from a training population you used in a GS modeling, do your modeling and find the PCA tool in the model output page.

(3) individuals in a training population and selection population you applied the training model, do your modeling, apply the model on the selection population and find the PCA tool on the selection population prediction output page.

(4) individuals in a list of accessions you created, for example using the search wizard, go to the "Analyze" menu and select the "Population Structure", select your list of individuals and run PCA.

(5) individuals from multiple trials, create a list of the trials using the search wizard, go to the "Analyze" menu and select the "Population Structure", select your list of trials and run PCA.




With all the options, you will get a interactive plot of the two PCs (shown below) that explain the largest variance. Point the cursor at any data point and you will see the individual name with its corresponding PCs scores. By clicking the 'Download all PCs', you can also download the 10 PCs scores in the text format.

## 27.5 ANOVA

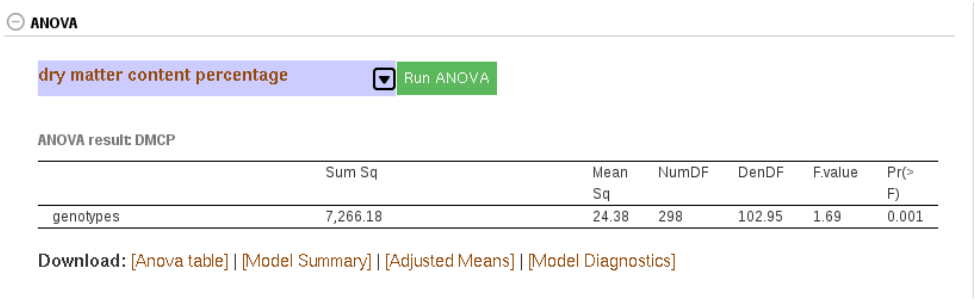
Currently, ANOVA is implemented for a single trial (single year and single location). You can do ANOVA for RCBD, CRD, Alpha and Augmented trial designs. ANOVA is done using linear mixed effects model, where the genotypes is fixed effect and the replications and blocks are random effects. Fixed effect significance level is computed using "lmer" from "lmeTest" R package.

You can do ANOVA from two places: trial detail and training population detail. In both cases, if the phenotype data was from the supported trial designs,

– Go to the ANOVA section down in the trial or training population page

– Select the trait of you want to perform ANOVA

– Click the "Run ANOVA" and wait for the result




## 27.6 Clustering (K-Means, Hierarchical)

The K-Means method allows you to partition a dataset into groups (K number). The hierarchical clustering, agglomerative, allows you to explore underlying similarity and visualize in a tree structure (dendrogram) the different levels of similarities (clusters) among samples. You can do clustering based on marker data, phenotype data and GEBVs. When you use phenotype data, first clone averages for each trait are calculated. Both methods use Euclidean distance as a measure of similarity. For the hierachical clustering, the complete-linkage (farthest neighbour) method is used to link up clusters.

There are three pathways to using this tool.

(1) When you have data in the form of a list or dataset from the search wizard:

(A) – go to the "Analyze" menu and select the clustering option

(B) – make sure you are logged in

(C) – Select the relevant genotyping protocol, if you are clustering using genotype data

(D) – select your list or dataset, click "Go"

(E) – select clustering type

(F) – select the data type to use

(G) – If you are running K-Means clustering, provide the number of partitions (K). If left blank it will partition the data set into optimal numbers for the dataset.

(H) – click the "Run Cluster" and wait for the analysis to finish or queue the request and wait for an email with the analysis result.

(I) – You can download the outputs following the download links.

(2) From the trial detail page:

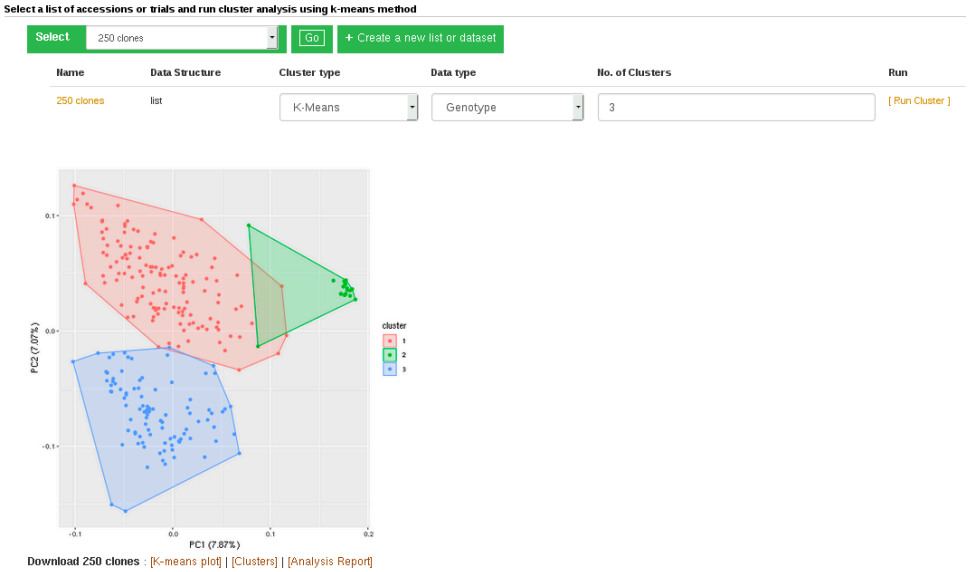
(A) – Go to the "Analysis Tools" section

(B) – Follow steps D to G in (1)

(3) In the solGS pipeline:

(A) – Once you you are in a model output put page, you will see a section where you can do clustering in the same way as above (option 2).

K-Means clustering:




Hierarchical clustering:

Download 34 clones : Dendrogram | Newick tree format | Analysis Report

## 27.7 Genetic Gain

You can check for genetic gain by comparing the the GEBVs of a training and a selection population. You can do this in the solGS pipepline once you build a model and apply the model to predict the GEBVs of a selection population. Once at that stage, you will see a section "Check Genetic Gain". Select a selection population to compare with the training population and click the "Check Genetic Gain" button. The genetic gain will be visualized in boxplots. You can download the boxplot(s) as well as the GEBVs data used for the plot(s).

## 27.8 Kinship and Inbreeding Coefficients

This tool allows you to estimate genetic relatedness between a pair of individuals (kinship), homozygousity across loci in an individual (inbreeding coefficient), and genetic similarity of an individual relative to the rest of the population (averge kinship).

There are three pathways to using this tool.

(1) When you have a list or dataset clones, created from the search wizard:

(A) – go to the "Analyze" menu and select the kinship and inbreeding

(B) – make sure you are logged in

(C) – Select the genotypic protocol for the marker data

(D) – select your list or dataset of clones, click "Go"

(E) – click the "Run Kinship" and wait for the analysis to finish, depending on the data size this may take minutes. You can choose to submit the analysis and wait for an email notice to view the results or wait for it to complete.

(F) – You can download the output following the download links.

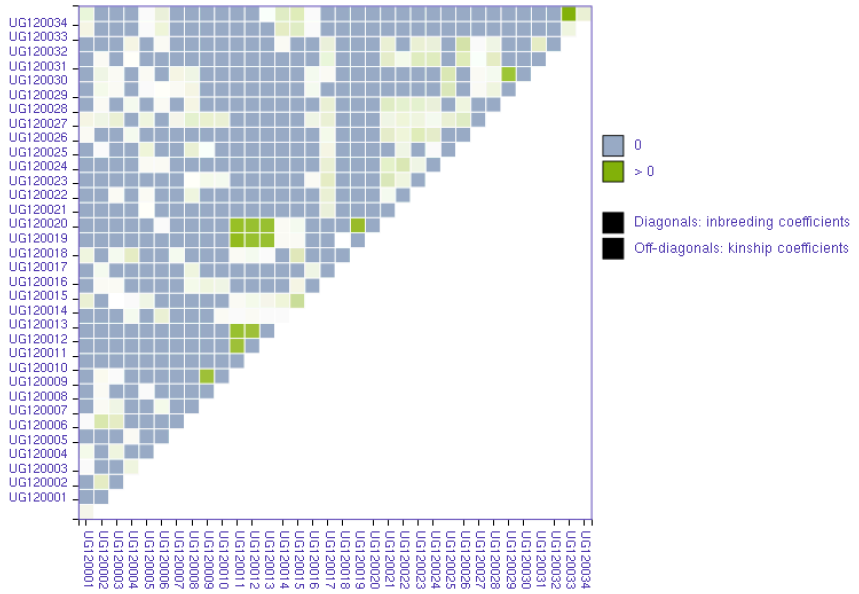
(2) From the trial detail page:

(A) – Go to the "Analysis Tools" section

(B) – Follow steps C to G in (1)

(3) In the solGS pipeline:

(A) – Once you you are in a model output put page, scroll down to the "Kinship and Inbreeding" section and run kinship.




**Download:** 34 clones Kinship matrix | Average kinship | Inbreeding coefficients

## 27.9 Creating Crossing Groups

If you calculate selection index based on GEBVs of multiple traits, and you want to select a certain proportion of the indexed individuals (e.g. top 10%, or bottom 10%) and then you want to partition the selected individuals into a number of groups based on their genotypes, you can use the k-means clustering method.

The procedure is:

(1) predict GEBVs for 2 or more traits

(2) In the models output page, calculate selection indices. Note the name of the selection index data.

(3) Go to the clustering section,

– select the selection index data,

– select “K-means”,

– select “Genotype”,

– in the K-numbers textbox, fill in the number of groups you want to create,

– in the selection proportion textbox, fill in the proportion of the indexed individuals you want to select, e.g. for the top 15 percent, 15. if you wish to select bottom performing, prefix the number with minus sign (e.g. -15)

– then run cluster and wait for the result.

Calculate selection index

Kasese solgs trial And assign relative weights to traits.

DMCP 2 FRW 3

Calculate

Genotype ranking based on multiple traits performance (selection index)

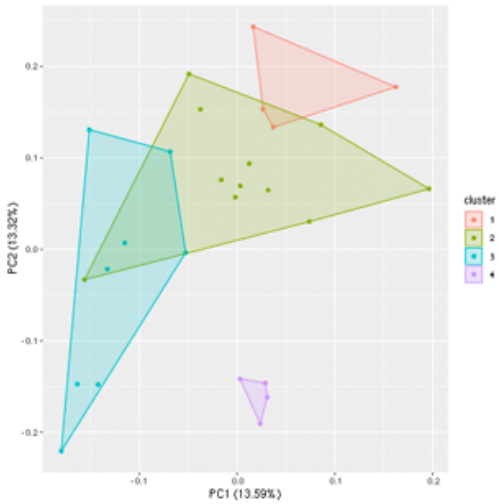
*Top 10 genotypes:*

| Genotypes | Selection indices |
|---|---|
| UG120221 | 19.43 |
| UG120001 | 15.56 |
| UG120193 | 14.72 |
| UG120071 | 14.3 |
| UG120057 | 11.43 |
| UG120072 | 11.2 |
| UG120006 | 9.97 |
| UG120264 | 9.88 |
| UG120278 | 9.49 |
| UG120184 | 9.35 |

Download selection indices | **Index Name:** 139-DMCP-2-FRW-3
**Population name:** Kasese solgs trial | **Relative weights: DMCP**: 2 **FRW**: 3

Clustering

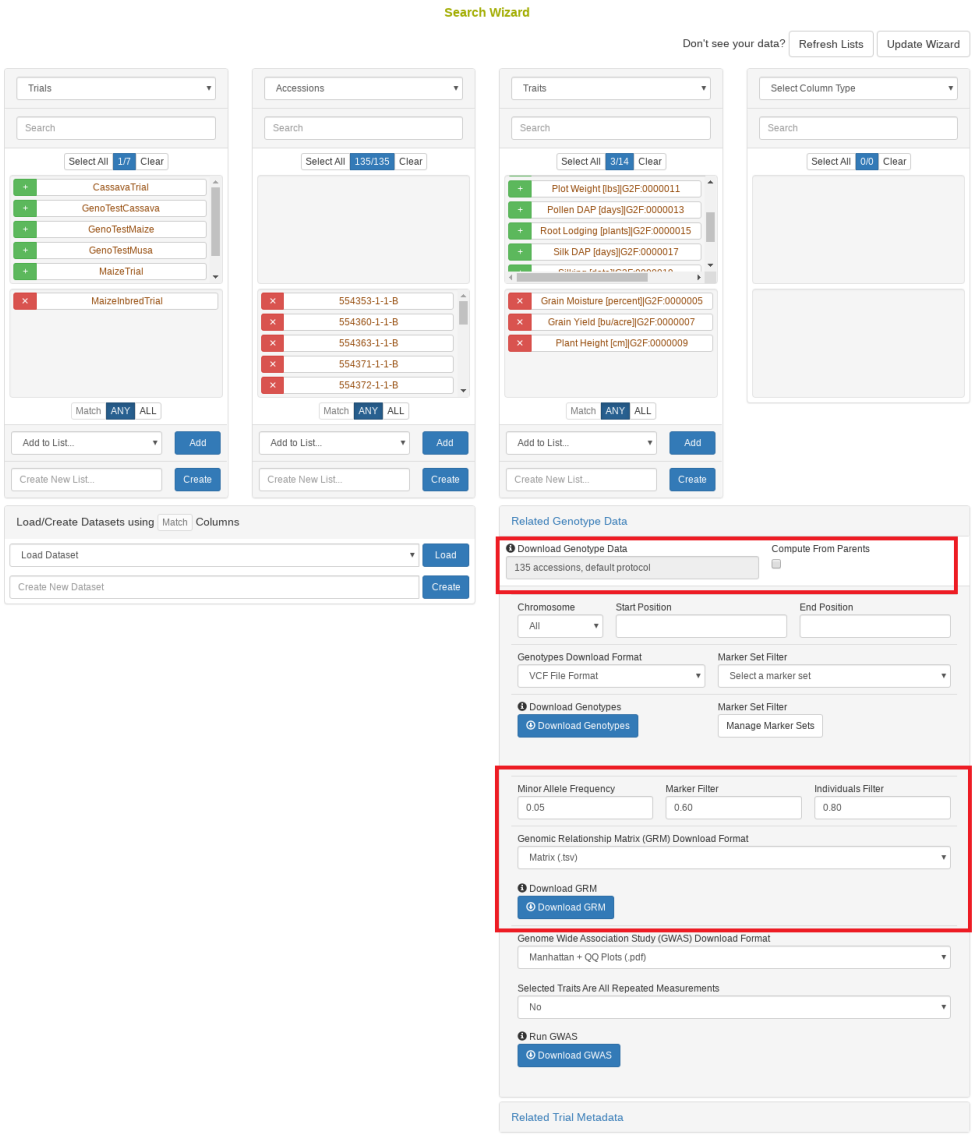
139-DMCP-2-FRW-3 K-Means Genotype 4 15 Run




Download 139-DMCP-2-FRW-3 : [K-means plot] | [Clusters] | [Analysis Report]

## 27.10 Search Wizard Genomic Relationship Matrix (GRM) Download

The genomic relationship matrix (GRM) is useful for understanding underlying structure in your population. Breedbase can compute the GRM using rrBLUP. First, select accessions in the search wizard and optionally select a genotyping protocol. If no genotyping protocol is selected, the default genotyping protocol in your system is used (as defined in sgn_local.conf). Specify the minor allele frequency, missing marker data, and missing individuals data filters to apply. The GRM can be returned in a matrix format (.tsv) which shows all pairwise relationships between the selected accessions and is useful for visualization; alternatively, the GRM can be returned in a three-column format (.tsv) which is useful for programs like ASReml outside of Breedbase. The GRM can also be returned as a simple correlation heatmap image (.pdf). The GRM can be computed from parents of the selected accessions granted the parents were genotyped, by clicking the checkbox "compute from parents"; this is useful for programs where parental lines are genotyped and then hybrids are created and evaluated in the field.

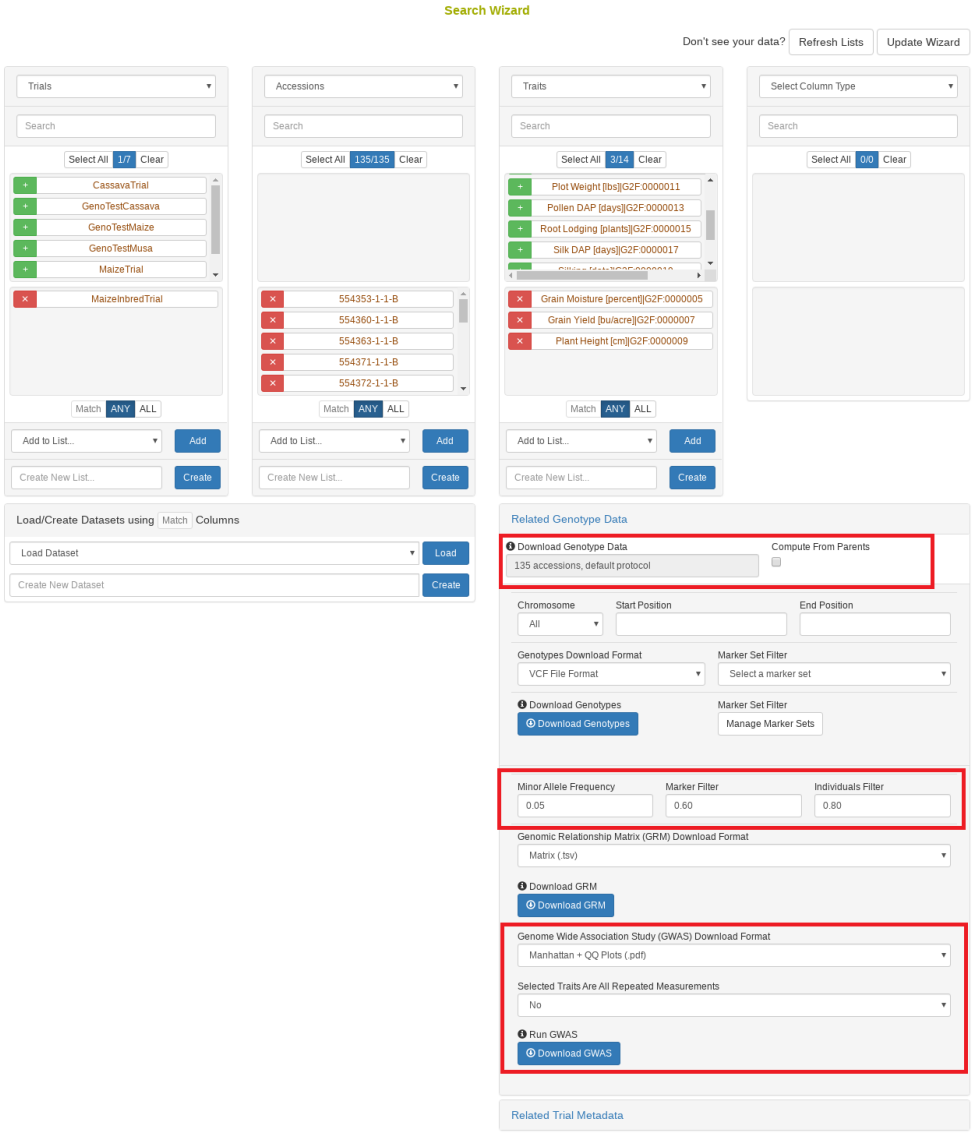


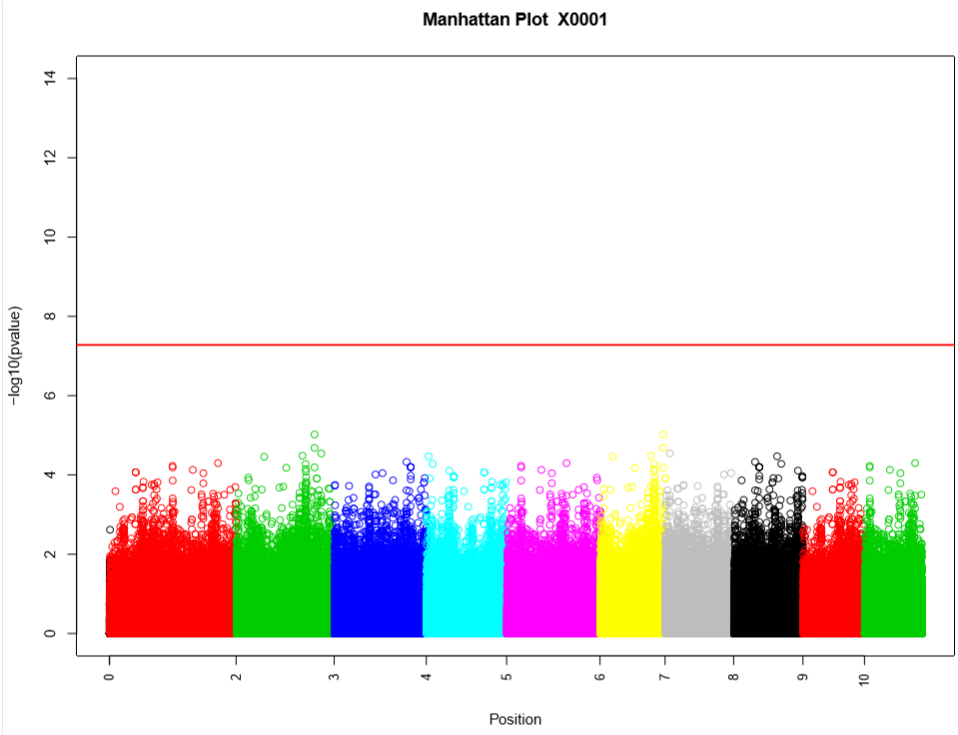

## 27.11 Search Wizard Genome Wide Association Study (GWAS)

Performing a genome wide association study (GWAS) can determine genotypic markers which are significantly correlated to phenotypic traits. Breedbase can compute GWAS using rrBLUP. First, select accessions and trait(s)

in the search wizard, and optionally select a genotyping protocol. If no genotyping protocol is selected, the default genotyping protocol in your system is used (as defined in sgn_local.conf). Several traits can be selected in the search wizard; if the traits are not to be treated as repeated measurements then select "no" in the select box and this will tell Breedbase to return GWAS results independently for the selected traits. If the selected traits are indeed all repeated measurements then select "yes" in the select box and Breedbase will return as single GWAS analysis across all the phenotypic records. Specify the minor allele frequency, missing marker data, and missing individuals data filters to apply. GWAS results can be returned in a tabular format (.tsv) where the -log10(p-values) for the selected traits are returned; alternatively, the GWAS results can be returned as Manhattan and QQ plots for the selected traits. The GWAS can be computed from parents of the selected accessions granted the parents were genotyped, by clicking the checkbox "compute from parents"; this is useful for programs where parental lines are genotyped and then hybrids are created and evaluated in the field.

The GWAS will filter the data by the input MAF and missing data filters provided. After filtering the data is imputed using an "EM" method in rrBLUP. The Kinship matrix (GRM) is computed from the imputed genotypic data and used in the GWAS model. The GWAS uses fixed effects for different field trials and replicates in the phenotypic data.


Search Wizard
Don't see your data?
Refresh Lists
Update Wizard
Trials
Search
Select All 1/7 Clear
CassavaTrial
GenoTestCassava
GenoTestMaize
GenoTestMusa
MaizeTrial
MaizeInbredTrial
Match ANY ALL
Add to List...
Add
Create New List...
Create
Accessions
Search
Select All 135/135 Clear
554353-1-1-B
554360-1-1-B
554363-1-1-B
554371-1-1-B
554372-1-1-B
Match ANY ALL
Add to List...
Add
Create New List...
Create
Traits
Search
Select All 3/14 Clear
Plot Weight [lbs]|G2F:0000011
Pollen DAP [days]|G2F:0000013
Root Lodging [plants]|G2F:0000015
Silk DAP [days]|G2F:0000017
Grain Moisture [percent]|G2F:0000005
Grain Yield [bu/acre]|G2F:0000007
Plant Height [cm]|G2F:0000009
Match ANY ALL
Add to List...
Add
Create New List...
Create
Select Column Type
Search
Select All 0/0 Clear
Load/Create Datasets using Match Columns
Load Dataset
Load
Create New Dataset
Create
Related Genotype Data
Download Genotype Data
135 accessions, default protocol
Compute From Parents
Chromosome
All
Start Position
End Position
Genotypes Download Format
VCF File Format
Marker Set Filter
Select a marker set
Download Genotypes
Download Genotypes
Marker Set Filter
Manage Marker Sets
Minor Allele Frequency
0.05
Marker Filter
0.60
Individuals Filter
0.80
Genomic Relationship Matrix (GRM) Download Format
Matrix (.tsv)
Download GRM
Download GRM
Genome Wide Association Study (GWAS) Download Format
Manhattan + QQ Plots (.pdf)
Selected Traits Are All Repeated Measurements
No
Run GWAS
Download GWAS
Related Trial Metadata


Manhattan Plot X0001
−log10(pvalue)
0
2
4
6
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12
14
0
2
3
4
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7
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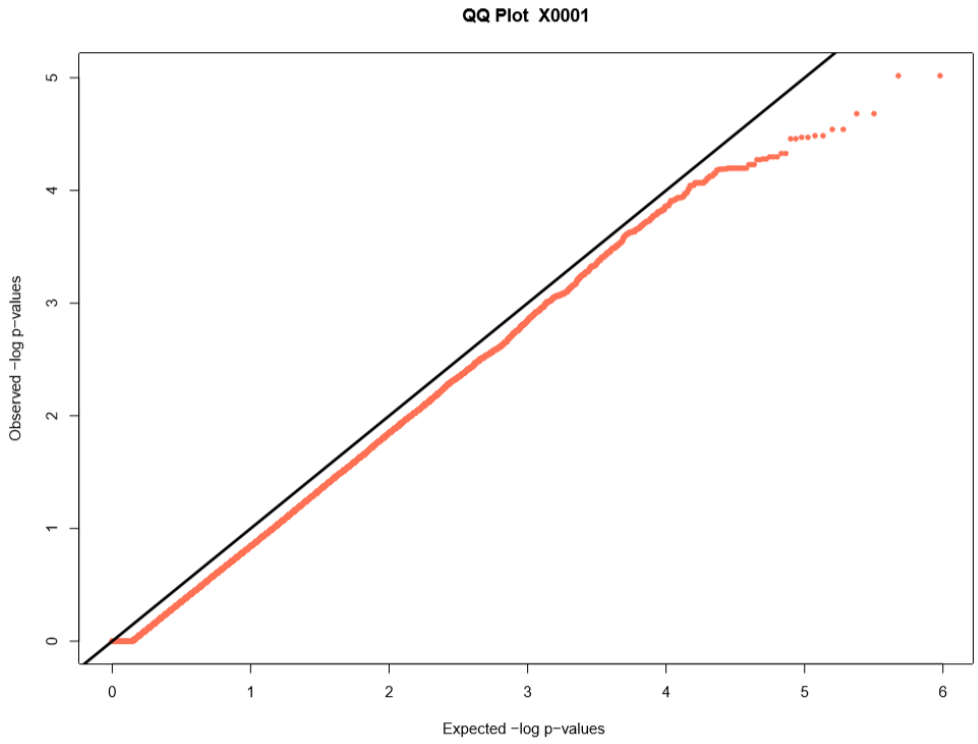
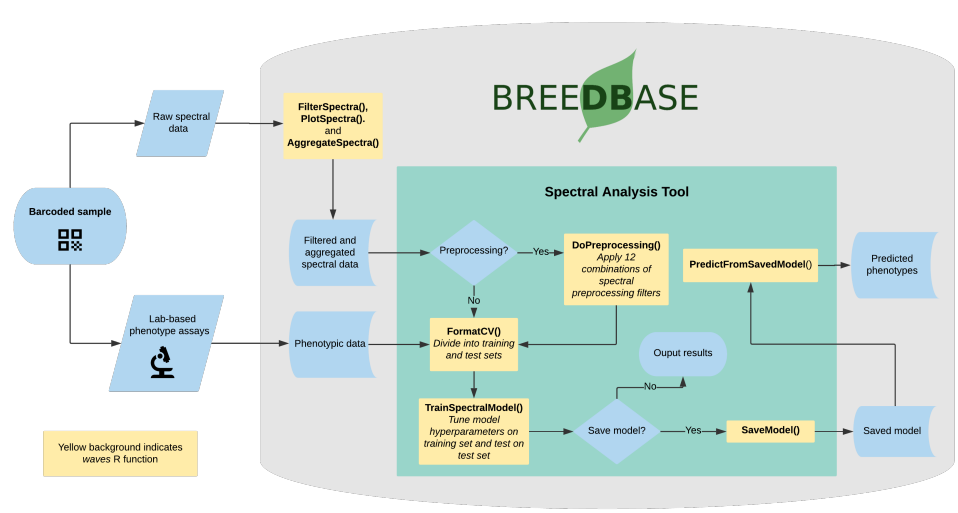
10
Position



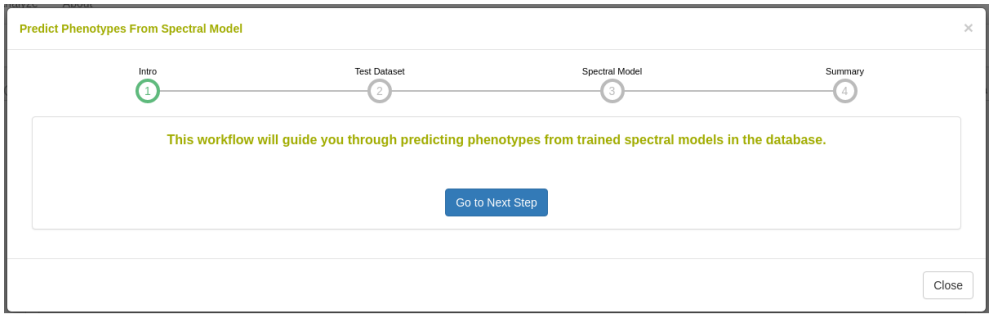

## 27.12 Spectral Analysis

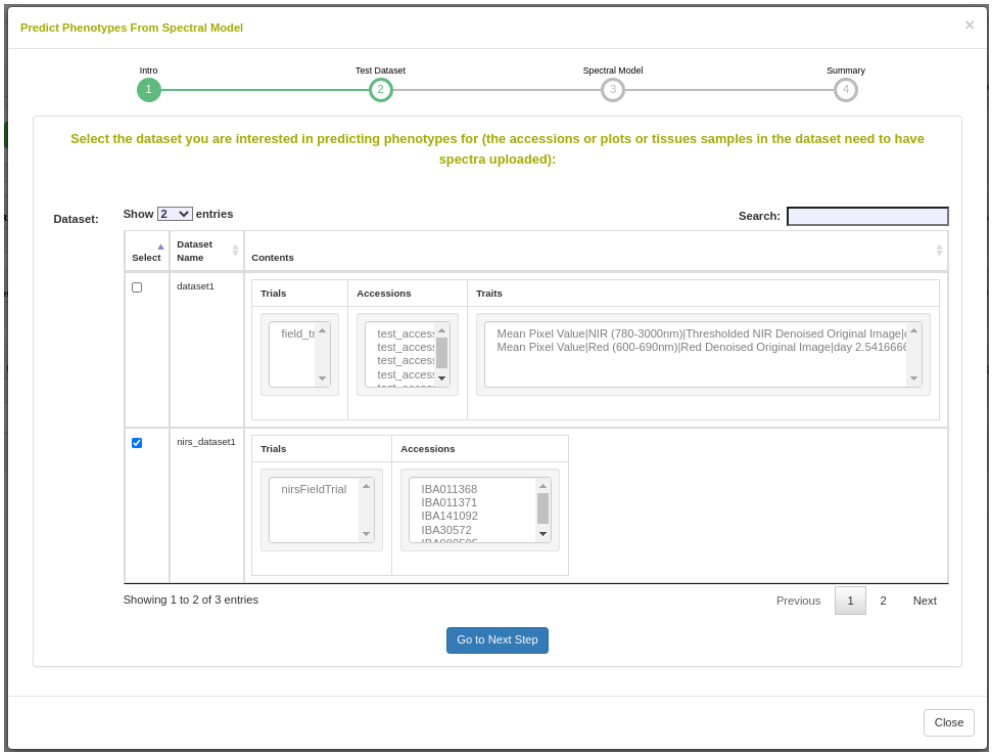
Visible and near-infrared spectroscopy (vis-NIRS) can be related to reference phenotypes through statistical models to produce accurate phenotypic predictions for unobserved samples, increasing phenotyping throughput. This technique is commonly used for predicting traits such as total starch, protein, carotenoid, and water content in many plant breeding programs. Breedbase implements the R package *waves* to offer training, evaluation, storage, and use of vis-NIRS prediction models for a wide range of spectrometers and phenotypes.




## 27.12.1 Dataset selection

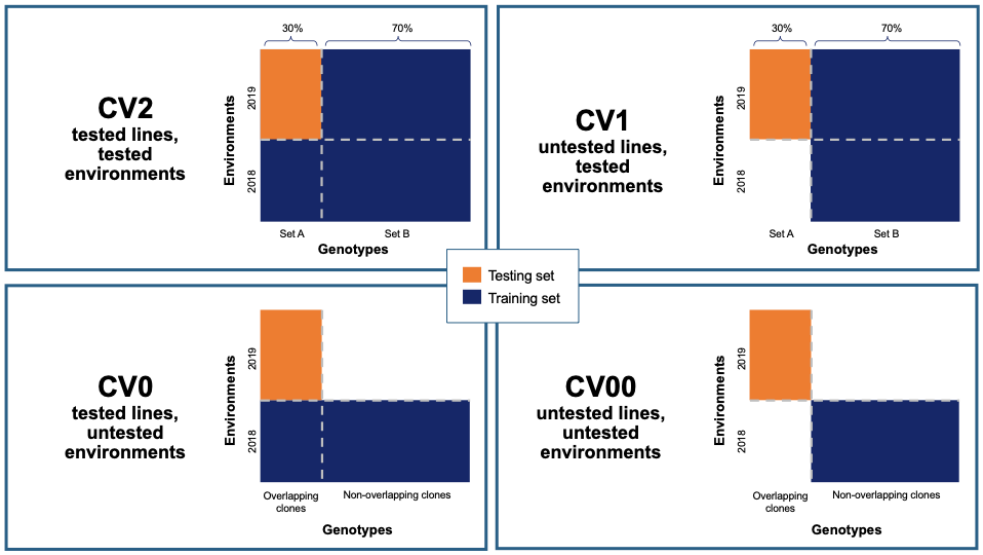
In order to initiate an analysis, the user must select one or more datasets using 2.1. A dataset in Breedbase can contain observationUnit-level (plot-, plant-, or sample-level) trial metadata and phenotypic data from one or more trials. After navigating to the "NIRS" webpage under the "Manage" tab in Breedbase, the user can initiate an analysis and select one of these datasets as input for model training. An optional test dataset can be selected in the second step of the workflow.






## 27.12.2 Cross-validation

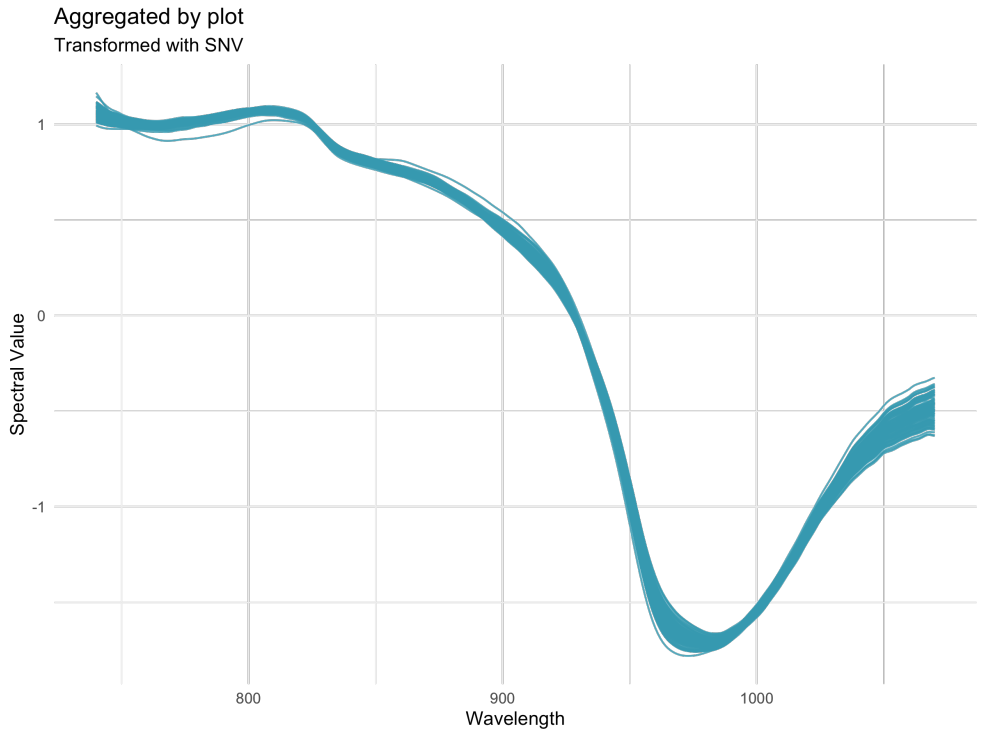
Five cross-validation schemes that represent scenarios common in plant breeding are available for this analysis. These include CV1, CV2, CV0, and CV00 as outlined below and described in depth by Jarquín et al. (2017) as well as random and stratified random sampling with a 70% training and 30% validation split. For those schemes from Jarquín et al. (2017), specific input datasets must be chosen based on genotype and environment relatedness. Cross-validation choices: * **Random sampling** (70% training / 30% validation) * **Stratified random sampling**, stratified based on phenotype (70% training / 30% validation) * **CV1**, untested lines in tested environments * **CV2**, tested lines in tested environments * **CV0**, tested lines in untested environments * **CV00**, untested lines in untested environments




### 27.12.3 Preprocessing

Preprocessing, also known as pretreatment, is often used to increase the signal to noise ratio in vis-NIR datasets. The *waves* function *DoPreprocessing()* applies functions from the *stats* and *prospectr* packages for common spectral preprocessing methods with the following options: * Raw data (default) * First derivative * Second derivative * Gap segment derivative * Standard normal variate (SNV; Barnes et al., 1989) * Savitzky-Golay polynomial smoothing (Savitzky and Golay, 1964)

For more information on preprocessing methods and implementation, see the *waves* manual, available through CRAN: waves.pdf




### 27.12.4 Algorithms

Several algorithms are available for calibration model development in Breedbase via the *waves* package. The *TrainSpectralModel()* function in waves performs hyperparameter tuning as applicable using these algorithms in combination with cross validation and train functions from the package *caret*. Currently, only regression algorithms are available, but classification algorithms such as PLS-DA and SVM clasification are under development. * **Partial least squares regression** (PLSR; Wold et al., 1982; Wold et al., 1984) is a popular method for spectral calibrations, as it can handle datasets with high levels of collinearity, reducing the dimensionality of these data into orthogonal latent variables (components) that are then related to the response variable through a linear model (reviewed in Wold et al., 2001). To avoid overfitting, the number of these components included in the final model must be tuned for each use case. The PLSR algorithm from the *pls* package is implemented by waves. * **Random Forest regression** (RF; Ho, 1995) is a machine learning algorithm based on a series of decision trees. The num-

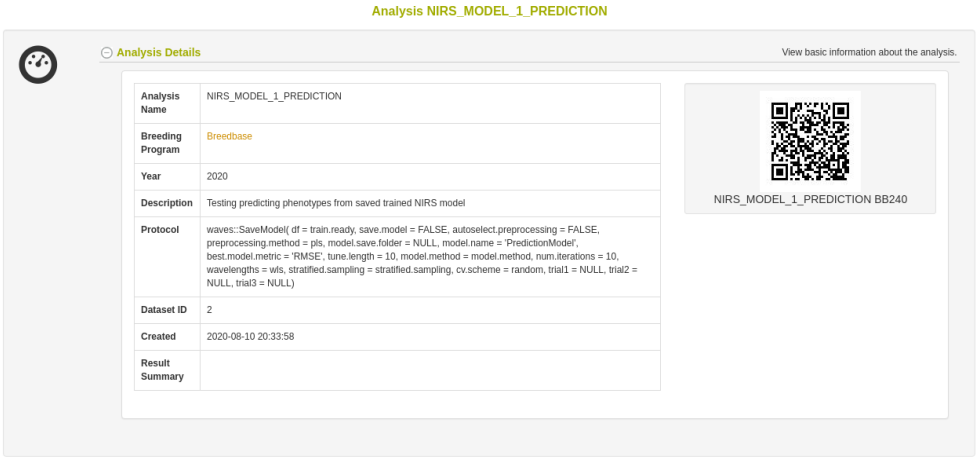
ber of trees and decisions at each junction are hyperparameters that must be tuned for each model. Another feature of this algorithm is the ability to extract variable importance measures from a fitted model (Breiman, 2001). In Breedbase, this option is made available through implementation of the RF algorithm from the package randomForest in the waves function TrainSpectralModel(). This function outputs both model performance statistics and a downloadable table of importance values for each wavelength. It is worth noting that this algorithm is computationally intensive, so the user should not be alarmed if results do not come right away. Breedbase will continue to work in the background and will display results when the analysis is finished. * **Support vector machine regression** (SVM; Vapnik, 2000) is another useful algorithm for working with high-dimension datasets consisting of non-linear data, with applications in both classification and regression. The package waves implements SVM with both linear and radial basis function kernels using the kernlab package.

### 27.12.5 Output: common model summary statistics

After training, model performance statistics are both displayed on a results webpage and made available for download in .csv format. These statistics are calculated by the *TrainSpectralModel()* function in *waves* using the *caret* and *spectacles* packages. Reported statistics include: * Tuned parameters depending on the model algoritm * **Best.n.comp**, the best number of components to be included in a PLSR model * **Best.ntree**, the best number of trees in an RF model * **Best.mtry**, the best number of variables to include at every decision point in an RF model * **RMSECV**, the root mean squared error of cross-validation * **R2cv**, the coefficient of multiple determination of cross-validation for PLSR models * **RMSEP**, the root mean squared error of prediction * **R2p**, the squared Pearson's correlation between predicted and observed test set values * **RPD**, the ratio of standard deviation of observed test set values to RMSEP * **RPIQ**, the ratio of performance to interquartile distance * **CCC**, the concordance correlation coefficient * **Bias**, the average difference between the predicted and observed values * **SEP**, the standard error of prediction * **R2sp**, the squared Spearman"s rank correlation between predicted and observed test set values

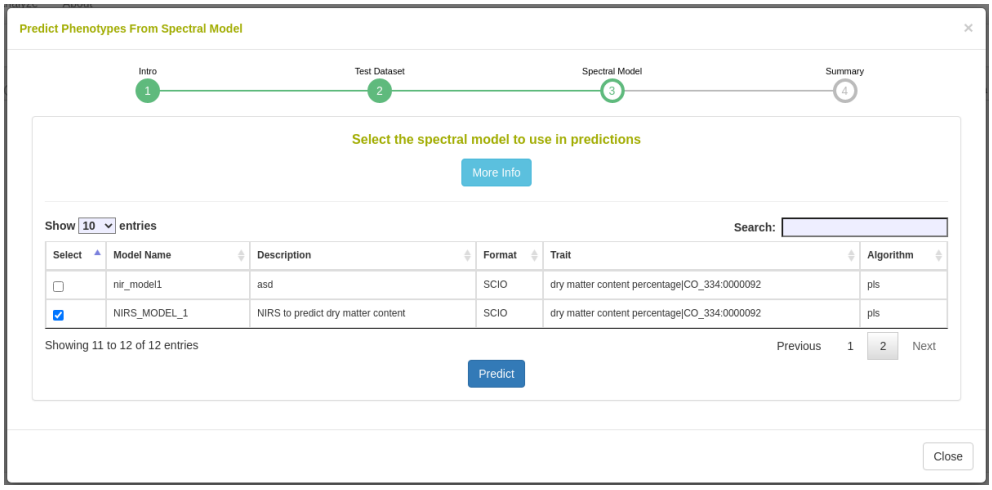
### 27.12.6 Export model for later use

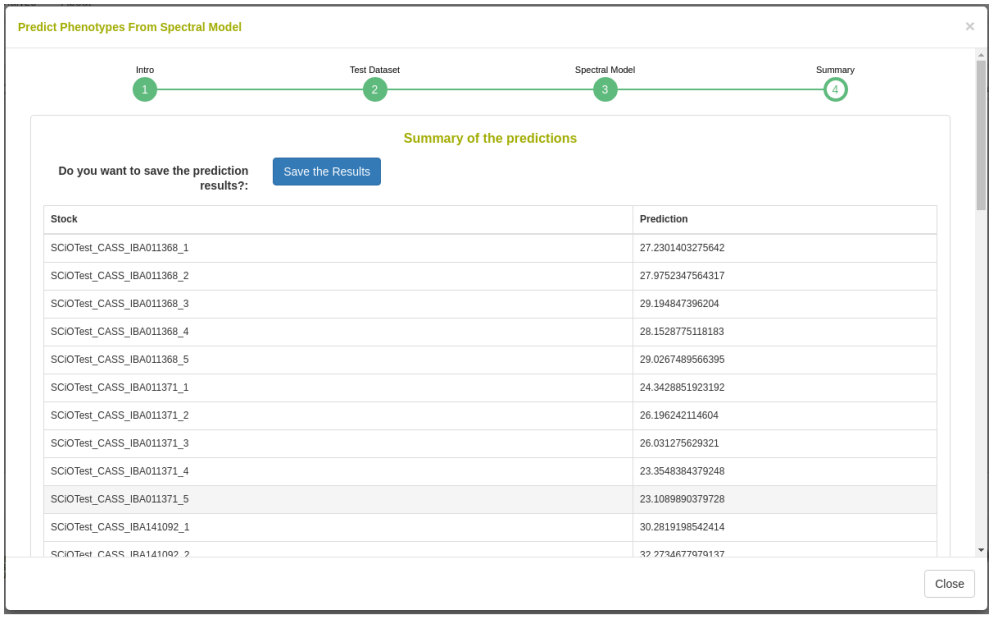
Once a model has been trained, it can be stored for later use. This action calls the *SaveModel()* function from *waves*. Metadata regarding the training dataset and other parameters specified by the user upon training initialization are stored alongside the model object itself in the database.




### 27.12.7 Predict phenotypes from an exported model (routine use)

For phenotype predictions, users select a dataset and can then choose from models in the database that were trained using the same spectrometer type as the spectral data in the chosen dataset. Predicted phenotypes are stored as such in the database and are tagged with an ontology term specifying that they are predicted and not directly measured. Metadata regarding the model used for prediction is stored alongside the predicted value in the database. Predicted phenotypes can then be used as normal in other Breedbase analysis tools such as the Selection Index and GWAS.




## 27.12.8 FAQ

The Breedbase Spectral Analysis Tool does not allow for prediction models involving data from multiple spectrometer types at once.

References * Barnes, R.J., M.S. Dhanoa, and S.J. Lister. 1989. Standard normal variate transformation and de-trending of near-infrared diffuse reflectance spectra. Appl. Spectrosc. 43(5): 772-777. doi:

10.1366/0003702894202201. * Breiman, L. 2001. Random forests. Mach. Learn. 45: 5-32. doi: 10.1201/9780429469275-8. * Ho, T.K. 1995. Random decision forests. Proc. Int. Conf. Doc. Anal. Recognition, ICDAR 1: 278-282. doi: 10.1109/ICDAR.1995.598994. * Jarquín, D., C. Lemes da Silva, R.C. Gaynor, J. Poland, A. Fritz, et al. 2017. Increasing Genomic-Enabled Prediction Accuracy by Modeling Genotype x Environment Interactions in Kansas Wheat. Plant Genome 10(2): plantgenome2016.12.0130. doi: 10.3835/plantgenome2016.12.0130. * Johnson, R.A., and D.W. Wichern. 2007. Applied Multivariate Statistical Analysis (6th Edition). De Maesschalck, R., D. Jouan-Rimbaud, and D.L. Massart. 2000. The Mahalanobis distance. Chemom. Intell. Lab. Syst. 50(1): 1-18. doi: 10.1016/S0169-7439(99)00047-7. * Mahalanobis, P.C. 1936. On the generalized distance in statistics. Natl. Inst. Sci. India. * Savitzky, A., and M.J.E. Golay. 1964. Smoothing and Differentiation of Data by Simplified Least Squares Procedures. Anal. Chem. 36(8): 1627-1639. doi: 10.1021/ac60214a047. * Shrestha, R., L. Matteis, M. Skofic, A. Portugal, G. McLaren, et al. 2012. Bridging the phenotypic and genetic data useful for integrated breeding through a data annotation using the Crop Ontology developed by the crop communities of practice. Front. Physiol. 3 AUG(August): 1-10. doi: 10.3389/fphys.2012.00326. * Vapnik, V.N. 2000. The Nature of Statistical Learning Theory. Springer New York, New York, NY. * Wold, S., A. Ruhe, H. Wold, and W.J. Dunn, III. 1984. The Collinearity Problem in Linear Regression. The Partial Least Squares (PLS) Approach to Generalized Inverses. SIAM J. Sci. Stat. Comput. 5(3): 735-743. doi: 10.1137/0905052. * Wold, S., M. Sjöström, and L. Eriksson. 2001. PLS-regression: a basic tool of chemometrics. Chemom. Intell. Lab. Syst. 58(2): 109-130. doi: 10.1016/S0169-7439(01)00155-1.

## 27.13 General Mixed Model Tool

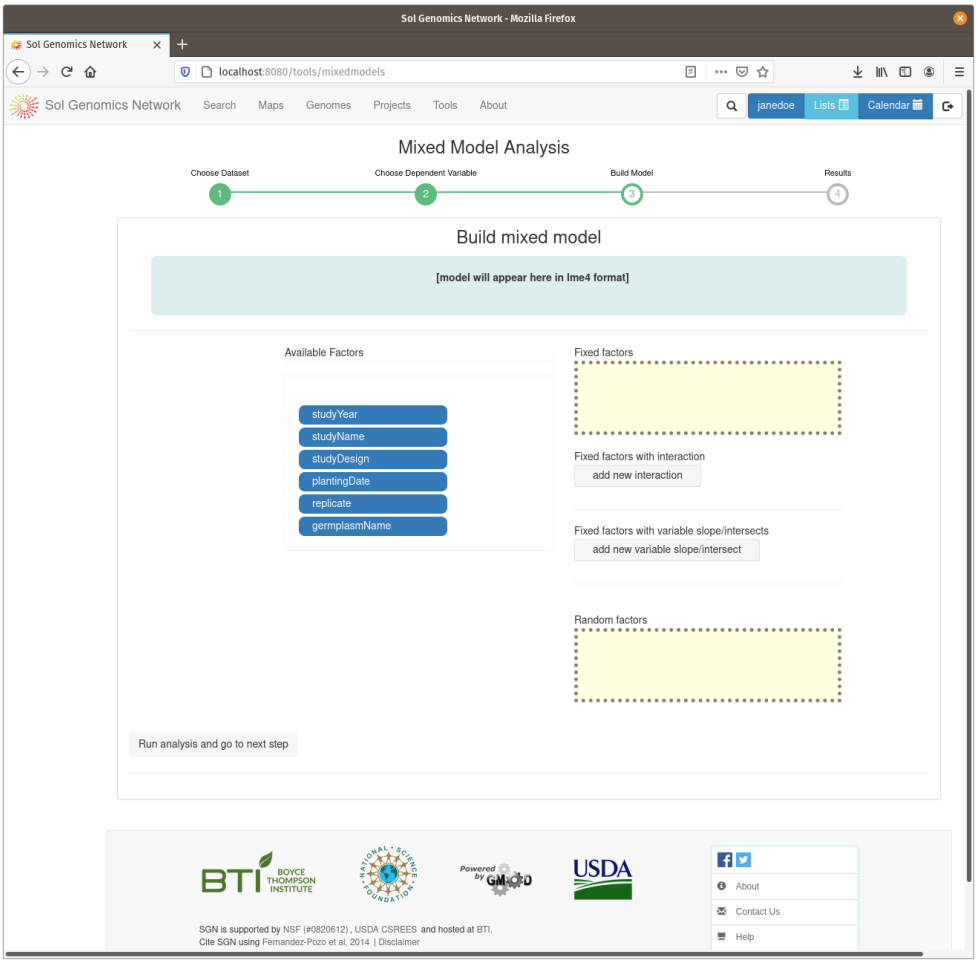
The general mixed model tool is available at /tools/mixedmodels and a link is provided from the Analyze menu.

To use the mixed model tool, first create dataset using the Wizard containing the data that you would like to analyze.

Select the Mixed Model tool from the Analyze menu.

You are presented with a workflow. On the first step of the workflow, select the dataset that you wish to analyze, click on "Choose dataset" to continue.

The second part of the workflow presents you with the traits in the dataset; you can select one or more traits from the lists using the select buttons. If you selected one trait, a bargraph of the trait distribution will be shown. Click the "Next step" button to move to the next screen.




On the model build screen, all the factors are displayed that are contained within the dataset. The factors are presented as a list of blue buttons that can be dragged using the mouse to areas on the screen which build a mixed model equation. The areas correspond to fixed factors, random factors, and optionally to more complex factors, such as fixed factors with interaction

and fixe factors with vriable slope/intersects. Drag the available factors to the corresponding area. To calculate BLUPs for germplasm, drag the germplasmName button to the "Random factors" area. To calculate BLUEs, drag it to the "Fixed factors" area. The factors need to have different levels contained within them, for example, if there is only one trial in the dataset, it cannot be used as one of the factors. Click on "Run analysis and got to next step" to run the mixed model and display the results.

The result view contains two tabs, one with the raw data, either BLUPS or BLUEs, and the other the adjusted means from the raw data.

The results can be stored in the database as an analysis, by clicking the button provided on the top of the data.

## 27.14 Genomic Prediction of Cross Performance (GPCP)

The GPCP tool is available at /tools/gcpc and a link is provided from the Analyze menu. The GCPC tool implements genomic prediction with additive and directional dominance in the linear mixed model to predict for cross performance.

Before using the tool, first create a dataset using the Wizard containing the data that you would like to analyze. (The dataset should have genotyping_protocols). Second, create Selection Indices for your traits using Selection Index in Analyze Menu.

To use the tool, Select the GPCP tool from the Analyze menu.

Then, select the dataset with genotyping_protocols that you wish to analyze, click on "Proceed to Factor Selection" to load available factors that can be included in the model.

Select the factors you wish to include in the model either as Fixed or Random. Click "None" for factors that you don"t want to include in the model. Note that the "germplasmName" is factored as Random by default.

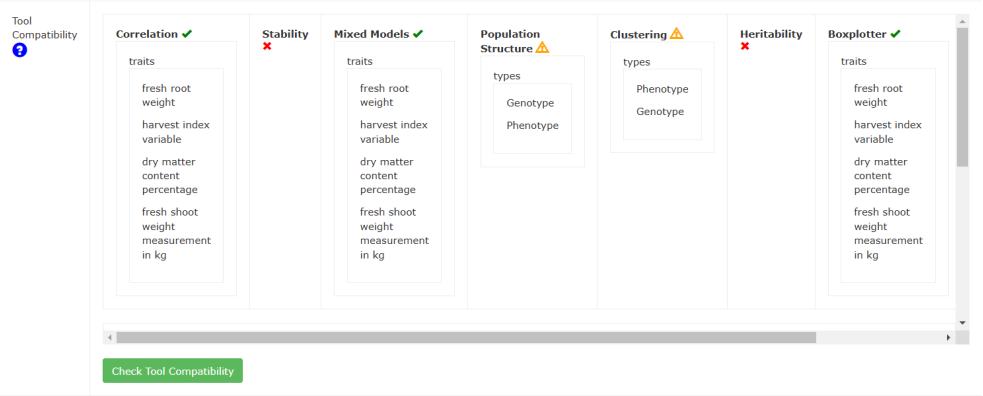
The next step is to select the selection index for your traits on the dropdown menu.

Once you are through, click “Run GPCP” to run the model. The output will be presented in form of a table with “ID”, “Parent1”, “Parent2” and their cross prediction merit organized in descending order. The results will also have sex information based on whether the dataset has plant sexes available in the database.

## 27.15 Tool Compatibility

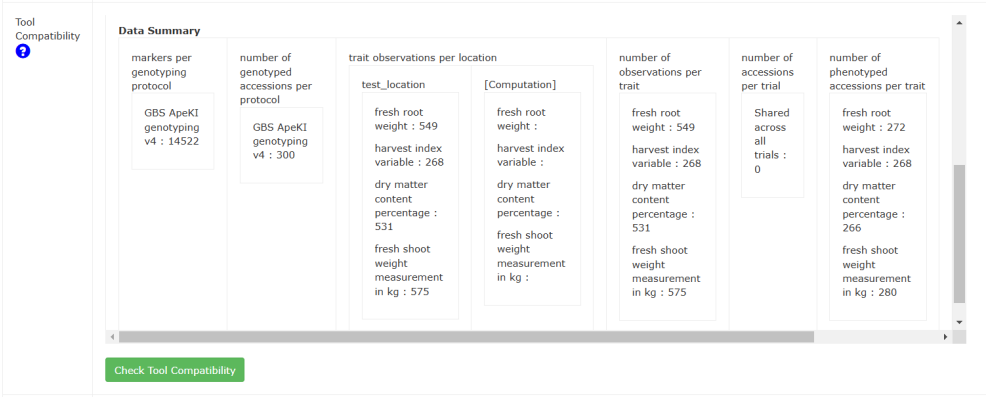
The dataset definition enables one to predict whether the dataset can be used in various analysis tools.

Upon creating a dataset, the site will automatically predict its compatibility with the available analysis tools and report these values on the dataset details page.




In the table, each tool will report to the user which traits are available to be analyzed based on phenotype data, and if different types of analyses are available, these will also be reported to the user. Some tools may give a warning sign to indicate that this dataset is compatible, but with potentially low sample sizes. Hover over the warning symbol to get a readout of the reason for the warning.

Below the table, there is a button that enables the user to re-calculate tool compatibility. This can be useful if a dataset is created before phenotypes are uploaded to a trial, since phenotype data is used in determining dataset compatibility. Even if the page appears to hang, do not worry; the compatibility check will continue in the background, and you can check later.




Below the tool compatibilities, there is also a summary of the data encompassed by the dataset and the criteria used for determining tool compatibility. Those criteria are used in the following way: - Correlation: A dataset can be used in a correlation analysis if there are many phenotype measurements for different traits made on the same accession. - Population Structure (PCA): A genotype PCA can be run if there are many accessions all genotyped with the same protocol. A phenotype PCA can be run if many accessions all have measurements on many traits. - Clustering: Like a PCA, clustering can be done in both phenotype and genotype modes. They have the same requirements as PCA. - Kinship & Inbreeding: A dataset with many accessions genotyped with the same protocol can be used for kinship analyses. - Stability: A dataset containing many accessions with the same trait measured across multiple locations can be used in stability analyses. - Heritability: This requires one or more trials with the same trait measured on the same accession across those trial(s). - Mixed Models: This requires sufficient accession numbers, trait measurements, and trial designs. - GWAS: A dataset is compatible with GWAS if there are many accessions genotyped for the same genotyping protocol, and the genotyping protocol has enough markers to run a GWAS. In addition, each accession needs to be phenotyped for a trait. - Boxplotter: There must be sufficient trait measurements to make a boxplot of the trait.

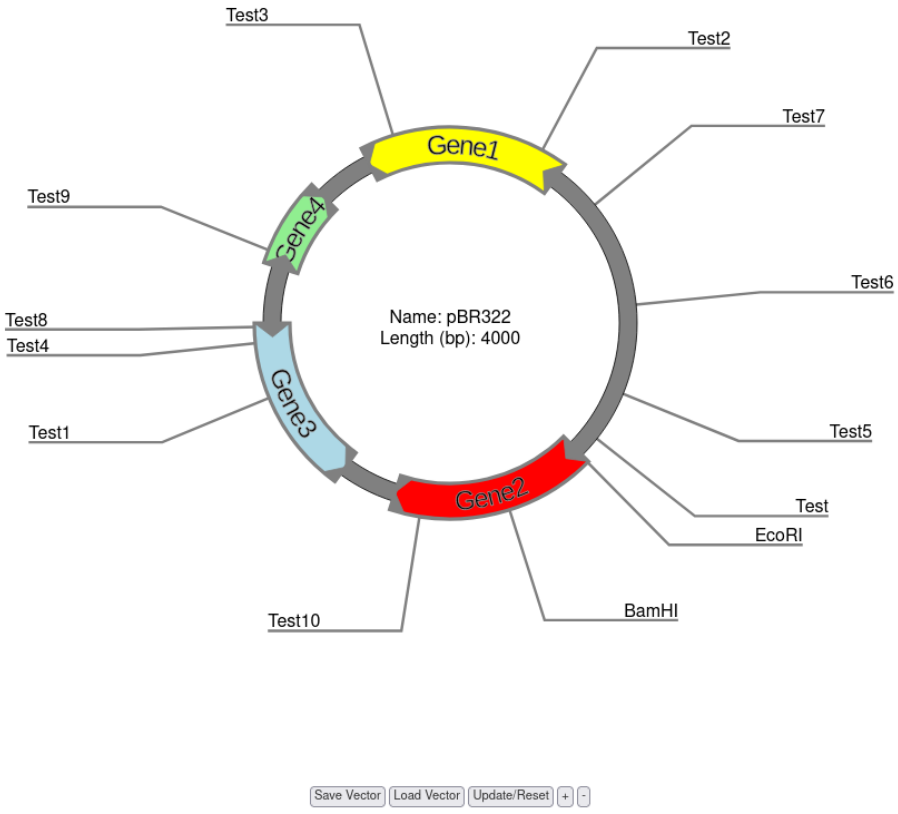
In addition to being on the dataset details page, tool compatibilities may be listed on the dataset selection screens for analysis tools. The compatibilities are non-blocking; you may always try using a dataset in an analysis even if there are warnings or if it is deemed non-compatible. As before, you can hover over the warning symbols to see why a dataset may not have statistical

power. For analyses with multiple modes, such as clustering and PCA, you can also hover over the compatibliity checkmark to see what types (phenotype or genotype) the dataset is compatible with.

# Chapter 28

# Vector Viewer

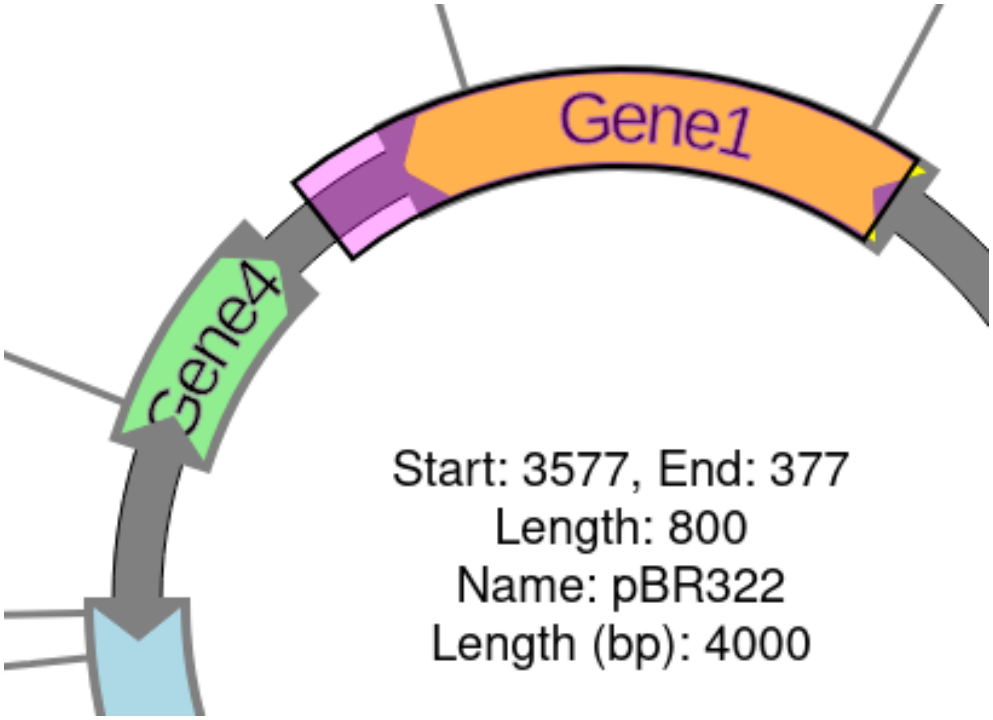
BreedBase provides a vector viewer tool that can generate, save, or upload vectors onto BreedBase.



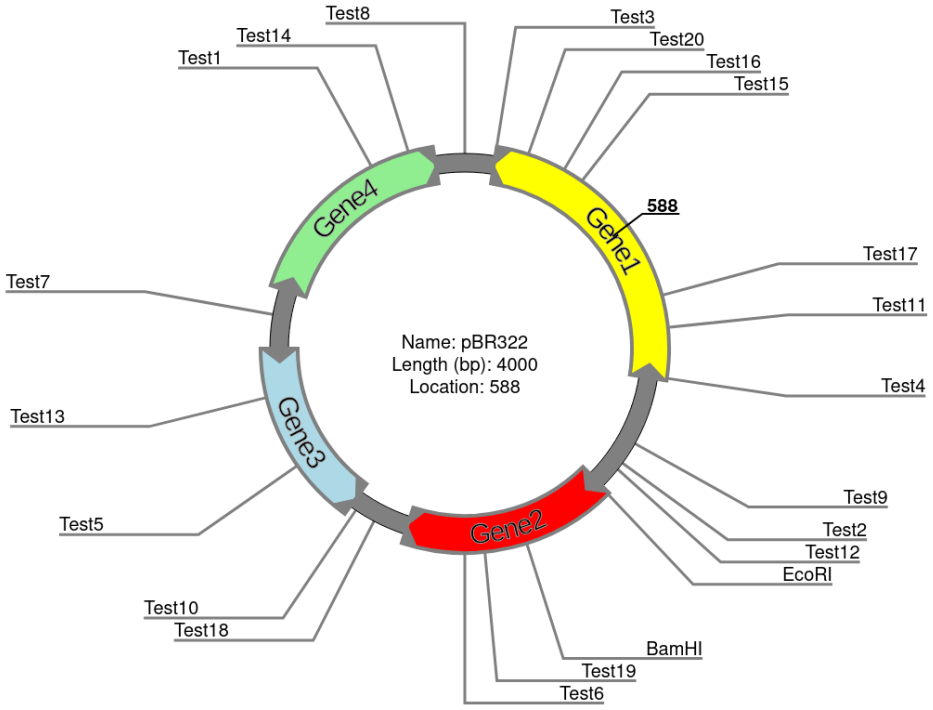

The vector viewer tool comes with many interactive features:

- Highlighting
- Location display
- Zooming in / panning out
- A feature and restriction enzyme table that support real time editing, deleting, or adding

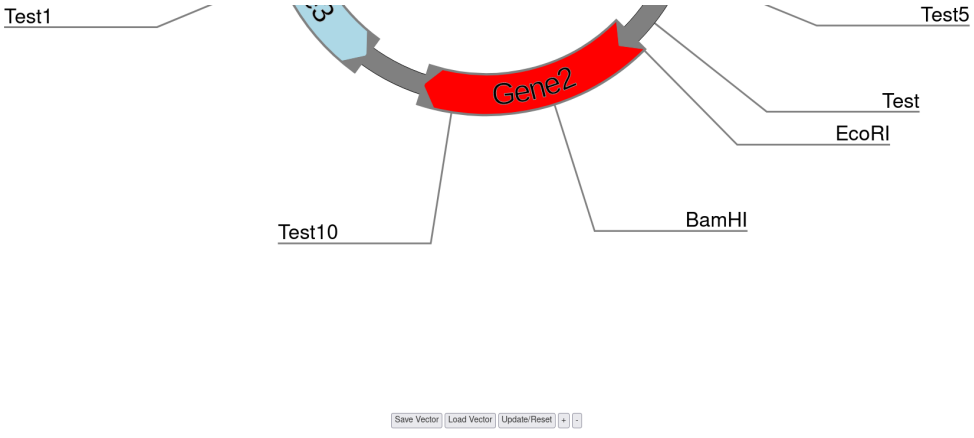
To highlight a region of the vector viewer, click and drag your cursor clockwise. To delete the new highlight, either start a new highlighted segment by clicking on the vector, or click the “Update/Reset” button.

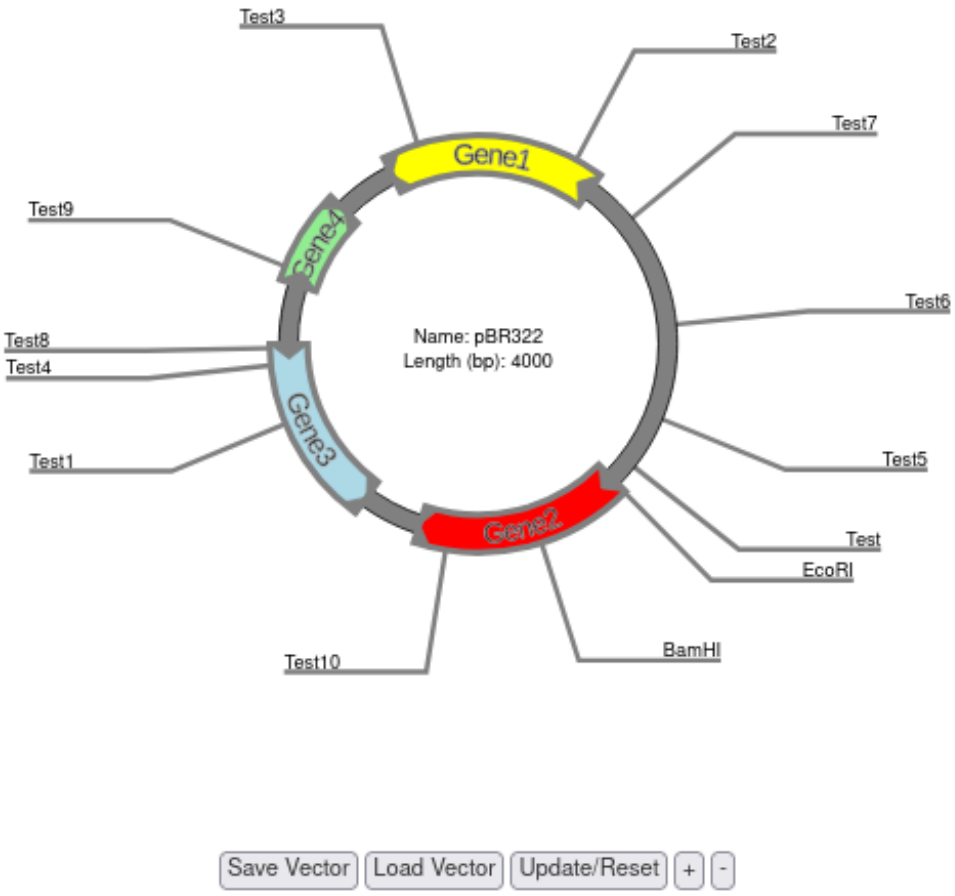



To find the location of a spot on the vector, hover your cursor over the vector and a number will appear. That is the location, in base pairs, of your cursor.




To zoom in and pan out, click the "+" or the "-" button underneath the vector.






To add a feature or restriction site, click "Add". To edit a feature or restriction site, select the feature or restriction site of choice, click "Edit", and make any wanted changes. To delete a feature or restriction site, selected the feature or restriction site that you want to delete and then click "Delete".